Tools to Design a Biomimetic Humanoid Robot Actuated by Pneumatic Artificial Muscles and a Synthetic Neural Network

by

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in
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Abstract

Biomimetic robots offer a platform for studying neuromechanical control through experiments that would be impractical or unethical with living subjects. This dissertation advances a long-running effort at Portland State University to build a humanoid robot, from the trunk down, actuated by pneumatic artificial muscles and controlled by a synthetic nervous system. The isometric force of Festo braided pneumatic actuators (BPAs) was characterized over a wide range of resting lengths, contractions, and pressures using a custom test jig. The analysis produced novel equations for the maximum isometric force as a function of resting length and a three-coefficient nondimensional force surface that predicts isometric force from relative strain and relative pressure. The actuator model was carried to the joint level through isometric knee torque experiments on two test stands: a simplified pinned knee and a biomimetic knee with a migrating instantaneous center of rotation. This approach isolated the sources of prediction error, revealing constant length offsets, series compliance in brackets and artificial tendons, and a loss of usable actuator length when BPAs wrap around the joint. The redesigned flexor and cam-extensor routes met or exceeded their selected human-muscle reference torques at all measured angles; these discrete isometric measurements do not establish dynamic performance or a continuous measured torque envelope. Tools were created for curating relevant neuromechanical research, as well as simulating neuromechanical models based on human

and robot biomechanical models.

Dedication

*

Darian

azīzam, jānam,

nūr-e cheshmam — the light of my eyes

qorbānat beram

Bābā kheyli dūset dāre

↘ ↘ ↘

"Los corazones obreros que laten

por nuestra causa, felices serán.

Si entusiasmados y unidos combaten,

de la victoria, la palma obtendrán."

Acknowledgments

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Finally, I thank my family for their support throughout this long process.

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1 Introduction

1.1 Motivation

Biomimetic robots provide a platform for investigating how mechanical structure and control interact to generate movement, offering complementary insights to simulation-based studies [Andrews, 1985, Millard et al., 2013, Arnold et al., 2010]. Wheeled robots remain the platform of choice in structured environments, where their speed, efficiency, and payload capacity are difficult to match; in the unstructured terrain of the natural and built world, legged machines hold the advantage, and biomimetic legs that reproduce human force capabilities have the potential to excel where wheels cannot [Frey et al., 2026, Ha et al., 2025]. Musculoskeletal humanoids and limb systems driven by tendon-like actuators have been shown to support biologically meaningful experiments on motion generation, internal force transmission, and neural control hypotheses [Asano et al., 2017, 2019, Hitzmann et al., 2018, Liu et al., 2018, Chen et al., 2019]. The dual perspective of using robots to learn about biology, and using biology to design more adaptive robots, has been emphasized in both biomimetic robotics and biomechanics [Ijspeert, 2020, Asano et al., 2017]. With such robotic systems, experiments can be performed that would not be practical or ethical with human test subjects, such as removing a specific muscle from the system or blocking a sensory feedback pathway. Experiments of this kind allow direct testing of how the components of the neuromechanical system interact, and they offer new opportunities for testing neural control theories that are faster and cheaper than human

observation studies [Shin et al., 2018, Goldsmith et al., 2021, Schilling et al., 2022].

Artificial muscles are a key enabling technology for this class of robot. Among the available actuation technologies, McKibben-style braided pneumatic actuators (BPAs) combine low mass, high force-to-weight ratio, and intrinsic compliance with a force–length curve that qualitatively resembles that of skeletal muscle [Liang et al., 2020, Hunt et al., 2017a]. BPAs have been used in quadrupedal robots [Aschenbeck et al., 2006, Hunt et al., 2017a, Scharzenberger, 2019], musculoskeletal humanoids [Asano et al., 2017, Hitzmann et al., 2018], and dexterous manipulation systems [Chen et al., 2019]. However, BPAs also differ from biological muscle in important ways: their maximum tensile force occurs at resting length rather than at an optimal fiber length, their maximum contraction ratio is substantially lower than that of human muscle, and their force output is a highly nonlinear function of internal pressure, contraction, and loading state [Hunt et al., 2017a, Sárosi et al., 2017, Liang et al., 2020]. My experience with 10 mm actuators made this gap concrete: manufacturer specifications overstate the force and contraction available at the short resting lengths a leg design requires, and designs based on those specifications delivered less range of motion and joint torque than intended. Nor could the published empirical models fill in the gaps, because each is fitted to measurements at a single resting length [Sárosi et al., 2017, Martens and Boblan, 2017, Hunt et al., 2017a]; none accurately predicts how force changes when a designer chooses a different actuator length. Because muscle resting length is one of the primary degrees of freedom available to the robot designer, the absence of a force model valid at arbitrary resting length blocked accurate actuator selection. Designing a biomimetic robot around BPAs therefore requires quantitative, predictive models of the actuator and of the joint torques it can produce.

Actuator force alone, however, does not determine the torque a joint produces. The robot knee transmits BPA force to the skeleton through mounting brackets, artificial tendons, and paths that wrap over the joint, and each of these elements departs from the rigid-body idealization: brackets and fixtures deflect under load, tendons stretch, and a wrapping actuator loses usable length as the knee flexes. At the start of this work it was unknown how much these transmission effects reduce the torque delivered at the knee relative to the rigid-body prediction, and no method existed for correcting that prediction in a form that would transfer between joint configurations. The earlier robot routes were selected from predicted muscle lengths, moment arms, and torque profiles [Bolen and Hunt, 2019]. The placement searches developed during this dissertation extended that design approach [Morrow et al., 2020], but their rigid-body predictions still required experimental validation at the knee.

Researchers implementing neural control models face two related problems. Sensory-afferent findings are distributed across species, preparations, and experimental protocols, so selecting a feedback pathway requires locating its measured properties and the conditions under which they were observed [Van de Crommert et al., 1998, Windhorst, 2007, 2022]. I addressed that evidence-access problem with the Sensory Afferent Database. A second problem arises when a published neural circuit is coupled to a different mechanical model: its behavior depends on the actuator mechanics, sensory signals, support conditions, and numerical implementation as well as on the circuit itself. Published architectures and software supply essential components [Rybak et al., 2006, McCrea and Rybak, 2008, Markin et al., 2010, Nourse et al., 2023], but the behavior of their particular combination must be established. I developed coupled human and artificial-muscle models, graphical

neural blocks, and controlled evaluation procedures to distinguish these contributions. Representative tests establish circuit agreement across implementations, corrective reflex behavior, and the dependence of a supported stepping configuration on selected feedback pathways.

The dissertation therefore addresses four connected needs: an actuator-force model that accounts for resting length, corrections for transmission losses at the knee, a curated sensory-afferent resource, and tools for developing and checking neuromechanical simulations. Together, they support the design and evaluation of the robot; achieving unsupported walking remains a separate control and integration problem.

1.2 The Biomimetic Humanoid Robot Project

This dissertation is part of a long-running effort at Portland State University to build a biomimetic humanoid robot from the trunk down, actuated by pneumatic artificial muscles and controlled by a synthetic nervous system (SNS). The project is deliberately iterative. Knowledge from biomechanics and neuroscience guides the design of robots intended to perform complex tasks in unstructured environments with improved efficiency, balance, adaptability, and resilience. Those robots then become physical models with which biological hypotheses can be tested, validated, and refined through controlled changes that would be impractical or unethical in living subjects. The resulting biological insight informs the next generation of robot design, and lessons from both directions can be translated into improved prosthetic and orthotic devices.

The present work builds directly on a series of studies by me and my collaborators. In

my master's project, the kinematics and dynamics of a biomimetic robot leg were matched against the Gait2392 OpenSim model, artificial muscle attachment and wrapping locations were determined for the robot legs, and MATLAB software was developed for evaluating muscle lengths, moment arms, and isometric joint torques over the full range of motion [Bolen, 2019, Bolen and Hunt, 2019]. That work showed that simply transplanting human muscle attachment locations onto a BPA-actuated robot does not reproduce human torque profiles; the placement optimization developed during this dissertation addresses this limitation and is introduced in Section 7.1. The rest of the platform advanced in parallel, and this dissertation builds on its hardware and demonstrations: the biomimetic four-bar knee linkage with a migrating instantaneous center of rotation, designed and validated by Steele [Steele et al., 2017, Steele, 2018, Steele et al., 2018], is the knee of the robot leg whose torque transmission Chapter 4 measures, and the canine-inspired quadruped platform Muscle Mutt, formerly called DoggyDeux, demonstrated antagonistic BPA pairs under synthetic nervous system control on a physical robot [Scharzenberger, 2019], extending an earlier AARL demonstration of central pattern generator control of a pneumatically actuated robot [Hunt et al., 2017b]. The humanoid leg of this dissertation carries that line forward.

1.3 Research Objectives

This dissertation investigates actuator force and knee torque experimentally, consolidates sensory-afferent evidence in the Sensory Afferent Database [Bolen et al., 2023, Bolen and Hunt, 2024], and develops neural-model and simulation interfaces with representative

verification checks. The objectives are:

1. **Characterize the isometric force of Festo BPAs** with uninflated diameters of 10 mm and 20 mm over a wide range of resting lengths, contractions, and pressures; derive improved maximum-force and normalized force–strain–pressure models; and compare them against existing models (Chapter 3).

2. **Validate and improve the prediction of isometric knee torque** produced by BPAs on pinned-joint and biomimetic four-bar knees; identify and model losses caused by constant length offsets, series compliance, and actuator wrapping; and compare the resulting torque envelopes with human capabilities (Chapter 4).

3. **Consolidate the sensory afferent literature into a structured afferent resource**, the Sensory Afferent Database, by auditing and curating the published record of afferent properties across animal species and their role in motor control, and by evaluating AI-assisted literature triage for expanding the corpus (Chapter 5).

4. **Develop and demonstrate neuromechanical modeling tools** by implementing the Li–Deng walker, Gait2392-based models with alternative neural controllers and artificial-muscle actuation, SNS-Simscape blocks for neural pathways, and staged tuning and experiment procedures; verify representative circuit responses, implementation agreement, and neural–mechanical signal exchange (Chapters 6 and 7).

1.4 Dissertation Organization

Chapter 2 introduces the robot, BPA actuation, musculoskeletal design, locomotor control, and the scientific rationale for robotic models. Chapter 3 presents the isometric force characterization and actuator models, adapted from my publication [Bolen et al., 2026]. Chapter 4 tests the joint-torque predictions, identifies compliance and wrapping corrections, and uses them to redesign actuator routes.

Chapter 5 presents the Sensory Afferent Database, its curation, and its use limits. Chapter 6 begins with the AnimatLab attempt to combine the Deng CPG and Li walker, identifies the development requirements exposed by that attempt, and presents the tools and representative checks built to address them. Chapter 7 discusses the contributions together and develops the future work; Chapter 8 states the conclusions. Figure 1.1 summarizes these connections.

Appendices A–C provide the muscle calculations, neural equations, model geometry, and supporting mechanical results. Appendix D collects the larger simulation evaluations, and Appendix E provides the database coverage and pathway tables.

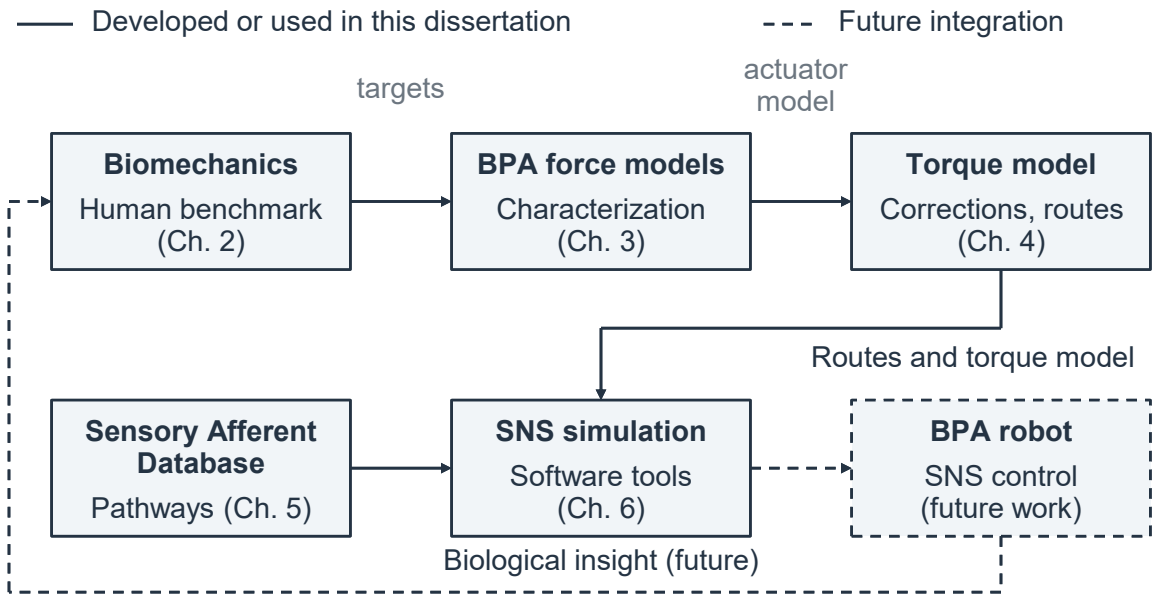


Figure 1.1: Dissertation roadmap. Biomechanical targets guide actuator characterization and route design; the sensory-afferent resource informs feedback selection; and simulation tools support controller evaluation. Dashed connections indicate future integration and the intended return of biological insight.

2 Background

This chapter reviews the established knowledge against which the dissertation's contributions are defined: biomimetic and musculoskeletal robotics, the actuation technologies available to them, the human lower-limb biomechanics they imitate, the neuroscience of locomotor control they implement, and the simulation and ethical context in which they are tested. Work from the author's own program that goes beyond this background, including the prior AnimatLab bipedal walker, the Sensory Afferent Database, and the neuromechanical simulation pipeline, is presented as dissertation work in Chapters 5 and 6 rather than repeated here.

2.1 Biomimetic and Musculoskeletal Robots

Biomimetic robots offer many potential applications, ranging from assisting in factories and environmental exploration to serving as physical models of human movement. By developing high-fidelity biomimetic robots, we can perform tests that help improve our understanding of both biomechanics and the underlying neuromechanical systems that control movement [Shin et al., 2018, Asano et al., 2019, Ijspeert, 2020, Goldsmith et al., 2021, Hunt et al., 2017b, Schilling et al., 2022]. A few such biomimetic robots exist, such as the impressive Kengoro and Kenshiro robots of Asano et al. [2017] and Asano et al. [2019], which have proportionally human bodies, anatomically correct musculotendon arrangements, rolling-sliding joints, and joint ranges and output power comparable with human-measured values; their artificial muscles are implemented with brushless DC

motors, wires as tendons, gears, and integrated tension measurement. Kengoro, the more recent of the two, stands 1670 mm tall, weighs 56.5 kg, and carries 174 degrees of freedom when the end effectors are included [Asano et al., 2017]. A dog-like robot developed by Aschenbeck et al. [2006] and actuated with Festo BPAs was designed from a planar kinematic model scaled to an adult greyhound, with three sagittal degrees of freedom per leg driven by antagonistic BPA pairs; its front-leg pair alone lifted a 13.5 kg payload, almost double the complete robot's predicted mass. That platform was used to show how central pattern generators provide robust locomotion patterns that are adaptive to perturbations in ground speed [Hunt et al., 2017b].

Robots built with artificial muscles thus provide opportunities to investigate how muscles are controlled to produce desired motions; however, artificial muscles do not exactly replicate muscle properties, and existing robots do not capture all of the degrees of freedom and joint torques of the animal limbs they mimic. The biomimetic leg developed and tested in this dissertation sits squarely in this lineage: a human-proportioned skeleton, muscle-like actuators, and the goal of human-matched joint torques.

Figure 2.1 illustrates the human-scale skeletal structure and tendon routing of this approach.

2.2 Robot Actuation Technologies

2.2.1 Overview

A wide range of actuation technologies have been explored in robotics, including electric motors, hydraulic systems, smart-material actuators, and artificial muscles [He and Gao,

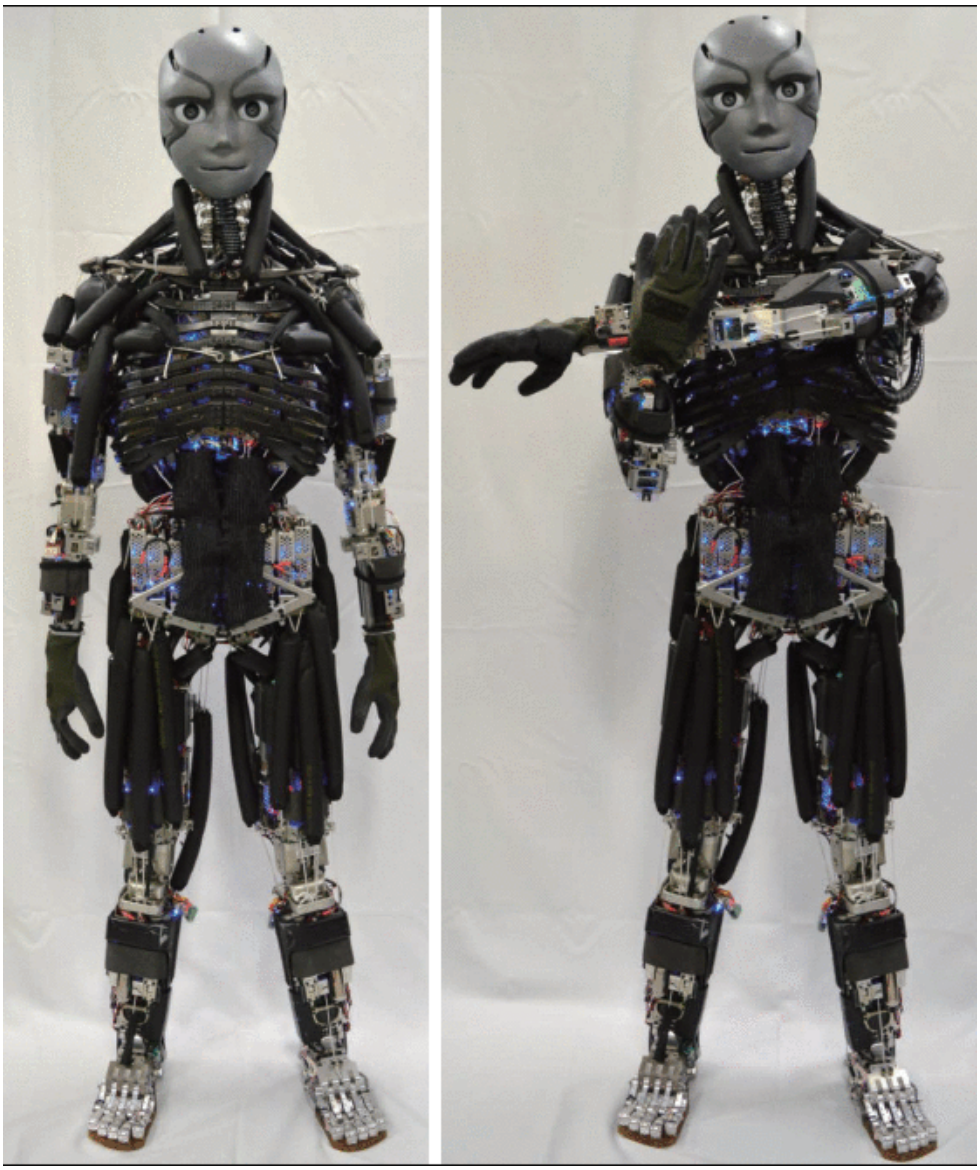


Figure 2.1: Kengoro human-mimetic musculoskeletal humanoid in neutral and upper-limb poses, illustrating its human-scale skeletal structure and densely routed tendon-driven musculature [Asano et al., 2017].

2020, Liang et al., 2020]. Electric motors are widely used due to their controllability and maturity, but they are limited by thermal constraints, gearing requirements, and reduced compliance compared to muscle-like actuators [Veale and Xie, 2016, Suzumori and Faudzi, 2018]. Hydraulic actuators offer exceptionally high power density and have enabled highly dynamic legged systems, but they remain relatively heavy, less efficient than biological muscle, and require a dedicated fluid infrastructure [Suzumori and Faudzi, 2018].

Among smart-material actuators, dielectric elastomer actuators (DEAs), a class of electronic electroactive polymers, contract by a different mechanism: an applied voltage squeezes a soft polymer film between compliant electrodes through electrostatic stress, and the thinning and radial spreading of the film produces strains that can exceed biological muscle [Madden et al., 2004]. Comparative studies report high specific work and good bandwidth for DEAs [Liang et al., 2020], but kilovolt-range driving voltages, electromechanical instability, and the compliant-electrode requirement have kept them out of legged robots. Thermally activated artificial muscles reach the most extreme performance figures of any technology surveyed by Liang et al. [2020]: twisted-fiber muscles support loads more than one hundred times those of a biological muscle of the same length and weight, with output stresses of 22 MPa. Their activation route, however, is shared with shape-memory alloys, whose heating is fast but whose cooling is limited by heat transfer away from the element, and that asymmetry sets the cycle time.

Closer to muscle, series-elastic and pneumatic actuators trade peak force and bandwidth for compliance, a trade that wearable and rehabilitation robotics exploit because force control and backdrivability matter most when a machine interacts with a human [Aksöz et al., 2019, Leibach, 2020, Li et al., 2020]. A recent review of pneumatic artificial

muscle technology documents continued progress in actuator design and in assistive applications [Zhagiparova et al., 2025]. None of these technologies yet combines muscle-like force, strain, and compliance in one actuator, which is the combination a biomimetic humanoid leg demands and that braided pneumatic actuators approximate most closely.

Figure 2.2 compares the operating mechanisms of these actuator technologies.

Figure 2.2 shows representative operating mechanisms for robotic actuators. **(A)** An electric torque motor produces controlled shaft rotation. **(B)** A braided pneumatic actuator shortens as internal pressure expands its sleeve radially. **(C)** A hydraulic cylinder converts fluid pressure into linear piston force. **(D)** A dielectric elastomer thins and expands laterally under an applied electric field. The panels use a consistent schematic style to emphasize the distinct energy-conversion mechanisms and design tradeoffs summarized in the text [Suzumori and Faudzi, 2018, Liang et al., 2020].

2.2.2 Braided Pneumatic Actuators

In contrast to the technologies above, McKibben-style braided pneumatic actuators (BPAs) combine low mass, high force-to-weight ratio, and intrinsic compliance with a force–length curve that qualitatively resembles that of skeletal muscle [Liang et al., 2020, Hunt et al., 2017a]. A BPA consists of an inflatable bladder inside a braided sleeve; pressurizing the bladder expands the bladder radially and contracts the sleeve axially, producing tensile force. As shown by Chou and Hannaford, the force–length relationship of McKibben actuators is governed by geometric constraints imposed by the braided shell, leading to a nonlinear dependence of force on actuator length through the braid angle [Chou and

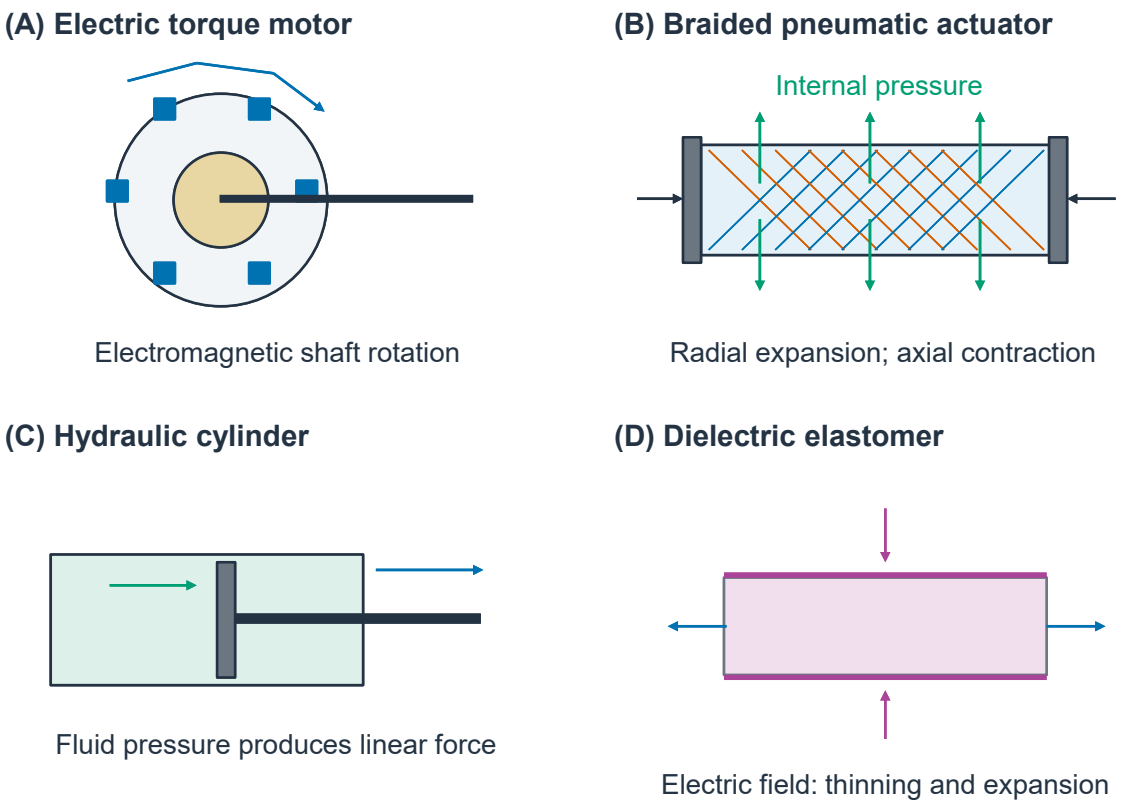


Figure 2.2: Representative actuator mechanisms: electric motor (A), braided pneumatic actuator (B), hydraulic cylinder (C), and dielectric elastomer (D) [Suzumori and Faudzi, 2018, Liang et al., 2020].

Hannaford, 1996]: actuator force is proportional to pressure and to a nonlinear geometric term $(3\cos^2\theta - 1)$, which governs both the maximum contraction limit and the curvature of the force–length relationship. Predicting their force as a function of length and pressure is therefore the first improvement of the torque tools developed in this dissertation, and the state of that prediction is the subject of the next subsection.

Figure 2.3 shows the change in actuator shape with inflation.

However, BPAs exhibit several important differences from biological muscle. First, BPA maximum tensile force occurs at resting (i.e., uninflated) length (l_0) rather than at an optimal fiber length [Liang et al., 2020]. Second, their maximum contraction ratio is substantially lower than that of human muscle, presenting challenges for achieving biomimetic joint ranges of motion [Hunt et al., 2017a, Sárosi et al., 2017, Liang et al., 2020]. Third, BPAs have a highly nonlinear dependence on internal pressure (P), contraction (ϵ), and loading state [Hunt et al., 2017a, Martens and Boblan, 2017], which complicates control and motivates high-fidelity force–length–pressure modeling. Despite these differences, their muscle-like properties have made them attractive for biomimetic quadrupeds [Aschenbeck et al., 2006, Scharzenberger, 2019], humanoids [Asano et al., 2017, Hitzmann et al., 2018], and musculoskeletal arms [Kurumaya et al., 2016, Chen et al., 2019].

2.2.3 BPA Force–Length–Pressure Models

Previous studies have generated empirical models of the force–length–pressure relationship of BPAs. Sárosi et al. [2017] discuss several high-fidelity BPA force models. They present a static force model with a 21-coefficient polynomial function for a Festo MAS-



Figure 2.3: Festo BPAs in the test fixture: the inflated actuator is wider and shorter than the deflated actuator beside it.

20-200N (i.e., ∅20 mm, $l_0 = 200$ mm) with an impressive $R^2 = 0.9994$, as well as Sárosi's static force model for a Festo DMSP-20-400N-RM-RM (i.e., ∅20 mm, $l_0 = 400$ mm), with six coefficients and a $R^2 = 0.9995$ [Sárosi, 2012, Sárosi et al., 2013]. Martens and Boblan present an even more accurate model, in terms of absolute error, using five coefficients for a DMSP-10-250 (i.e., ∅10 mm, $l_0 = 250$ mm) and a DMSP-20-300 (i.e., ∅20 mm, $l_0 = 300$ mm) [Martens and Boblan, 2017]. However, as demonstrated in this work, different resting lengths produce different force–length curves, preventing the use of these single-length models for designing systems with different resting lengths.

Hunt et al. [2017a] looked at six resting lengths of ∅10 mm Festo BPAs, accounting for differences in maximum contractile percentages, and elucidated the force-length-pressure relationship. In particular, for a given contraction percent and force F (in Newtons), the scalar pressure P (in kPa) required to produce this force for a ∅10 mm Festo artificial muscle can be determined by solving the equation:

$$P = P_0 + k_F F + k_S S + P_t \tan\left(c_a \left[\frac{\epsilon}{\epsilon_{\max} - k_E F} - c_b \right] \right) \tag{2.1}$$

where F is the required force, ϵ is the contraction, S encodes hysteresis ($S = 1$ shortening, $S = -1$ lengthening, $S = 0$ static), and the seven coefficients are defined in Table 2.1. The coefficients have been updated with corrected values, as those reported in Hunt et al. [2017a] contained typographical errors. The contraction ϵ and the maximum free-load contraction at 620 kPa, which appears as $\epsilon_{\max}$ in Equation (2.1), are defined in Section 3.1.1, together with the relative contraction $\epsilon^* = \epsilon / \epsilon_{620}$ on which the force surface developed in this dissertation is built. Because Equation (2.1) expresses pressure in closed form as

a function of force and contraction, it can also be solved onto a grid of pressures and contractions to build a force–pressure–relative-contraction lookup table for design studies.

Table 2.1: Coefficients of the 10 mm Festo BPA pressure model of Equation (2.1), corrected from Hunt et al. [2017a].

Symbol	Value	Role
P_0	254 kPa	pressure offset
k_F	1.23 kPa N^{-1}	linear force-to-pressure gain
k_S	15.6 kPa	hysteresis offset (sign of S)
P_t	192 kPa	tangent-term amplitude
c_a	2.03	contraction-shape coefficient
c_b	0.46	contraction-shift coefficient
k_E	0.331×10^{-3} N^{-1}	force-dependent reduction of usable contraction

Festo provides a manufacturer tool, MuscleSIM, for predicting force as a function of pressure and contraction [Lang, Robert, 2005], but experimental comparisons indicate that it does not adequately account for variation in l_0 or actuator-specific contraction limits. Comparisons with experimental data indicate that none of the existing models effectively take into account differences in maximum force due to initial actuator length, and the Hunt model was taken with low external forces ($\leqslant$106 N), so it is unclear how well it captures actuator behavior at higher forces. Chapter 3 compares these models with data collected for this dissertation and develops the improved characterization on which the torque and controller work of the later chapters depends.

2.3 Human Lower-Limb Biomechanics

Designing a biomimetic leg requires a quantitative benchmark of human muscle–joint behavior. This dissertation uses OpenSim [Delp et al., 2007, Seth et al., 2011, 2018] and the

Gait2392 model, named for its 23 degrees of freedom and 92 musculotendon actuators. Gait2392 descends from the interactive lower-extremity model of Delp et al. [1990] and includes musculoskeletal geometry from Hoy et al. [1990] and planar knee kinematics from Yamaguchi and Zajac [1989]; related lower-limb models of the same family are characterized and validated by Arnold et al. [2010]. Muscle moment arms, force-generating properties, and functional roles in such models have been characterized extensively [Zajac and Gordon, 1989, Brand et al., 1982, Menegaldo et al., 2004, Millard et al., 2013, Sherman et al., 2013, Suderman and Vasavada, 2012], and standard reporting conventions for kinematics and joint mechanics are established by the International Society of Biomechanics [Wu and Cavanagh, 1995, Wu et al., 2002, 2005, Derrick et al., 2020].

Moment arms relate muscle force to joint torque. For a fixed-axis joint, the signed moment arm is $r_{\mathrm{m}} = \hat{\mathbf{a}} \cdot (\mathbf{r} \times \hat{\mathbf{u}})$, where $\hat{\mathbf{a}}$ is the joint-axis direction, $\mathbf{r}$ locates a point on the muscle's line of action relative to the joint, and $\hat{\mathbf{u}}$ is the force direction [Brand et al., 1982]. The tendon-excursion method instead evaluates the sensitivity of musculotendon length to the joint coordinate, with sign determined by the coordinate and force conventions [Menegaldo et al., 2004]. For multiarticular muscles, the result also depends on adjacent-joint configurations [Zajac and Gordon, 1989, Young et al., 2019]. OpenSim uses length sensitivities, so the reference moment arms in this dissertation are properties of the Gait2392-derived geometry [Delp et al., 2007, Seth et al., 2018]. Chapter 4 compares the robot's resulting torque predictions with test-stand measurements.

Knee modeling requires particular care because the instantaneous center of rotation changes with flexion. Four-bar descriptions reproduce this behavior through linkage geometry [Romero et al., 2020, Steele, 2018]. Gait2392 instead prescribes planar translations as

functions of knee flexion [Yamaguchi and Zajac, 1989]. These different kinematics produce different moment arms for the same nominal attachment locations. Joint-coordinate conventions must therefore be stated consistently when comparing models and measurements [Wu et al., 2002]. Gait2392 provides the human reference in this dissertation, while Steele's four-bar mechanism supplies the physical robot knee.

2.4 Biomimetic Knee and Bipedal Robot Model

Steele's biomimetic knee joint supplies the physical knee used in this dissertation [Steele et al., 2017, Steele, 2018, Steele et al., 2018]. It is a one-degree-of-freedom four-bar linkage whose crossed links reproduce the posterior migration of the human knee's instantaneous center of rotation (ICR) during flexion. The rolling-and-sliding behavior of the human joint therefore emerges from the link geometry itself rather than from cams or gear profiles, and the design was calculated to reproduce an average sliding-to-rolling ratio of 0.079 over a 151° range of motion [Steele et al., 2018]. Because the ICR migrates, muscle moment arms about the knee change with flexion angle in a human-like way, and a torque prediction tool for this knee must capture that variation rather than assume a constant arm. The biomimetic knee tests of Chapter 4 therefore exercise the corrected torque model against this geometry, and Figure 2.4 summarizes the mechanism.

In my master's work, muscle lengths, moment arms, and isometric joint torques were computed over the range of motion for both the Gait2392 model and a PAM-actuated robot leg model, demonstrating that several robot muscle torque profiles could closely match the human model while many fell far short for part of the RoM [Bolen, 2019, Bolen and Hunt,

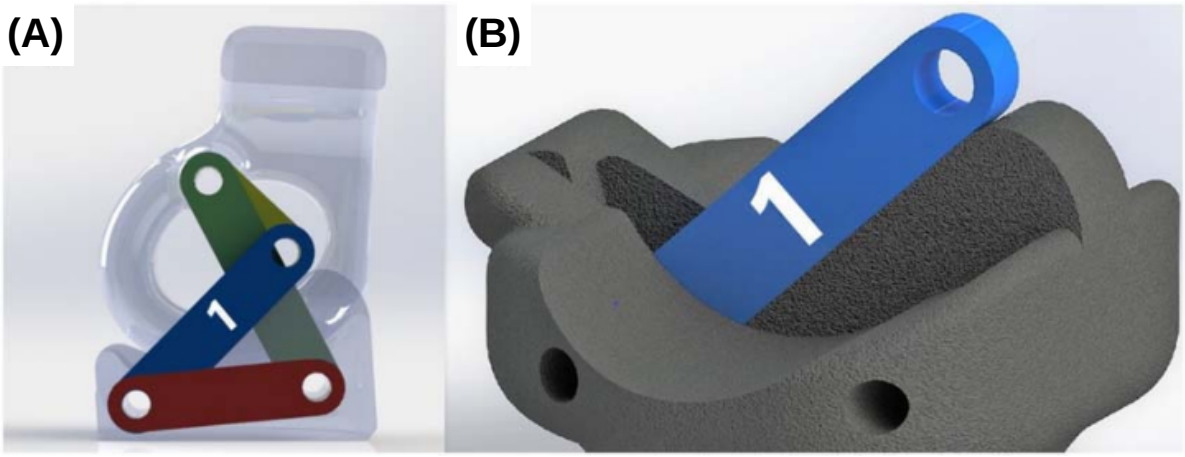


Figure 2.4: Biomimetic knee mechanism. **(A)** Sagittal cutaway at full extension showing the crossed four-bar linkage. **(B)** Exploded view of the mechanical stop. Reproduced from Steele et al. [2018, Fig. 6] under CC BY 4.0.

2019]. Hand placement of muscle origins and insertions was a painstaking and iterative task. This resulted in the creation of a placement optimization algorithm [Morrow et al., 2020], discussed in Chapter 4. Figure 2.5 places the laboratory skeleton beside the robot PAM routes and human muscle paths used in that comparison.

The comparison evaluated muscle-path lengths, moment arms, and isometric torques across the human and robot models [Bolen, 2019]. I adjusted BPA origins, insertions, and via-point locations to respect usable contraction and avoid interference. The biceps femoris long head illustrates the remaining mismatch: the robot path and its knee- and hip-torque surfaces differ substantially from the human reference. These discrepancies motivated placement optimization [Morrow et al., 2020]. Figures 2.6 and 2.7 show the moment-arm construction and the torque comparison.

(A) **(B)** **(C)**

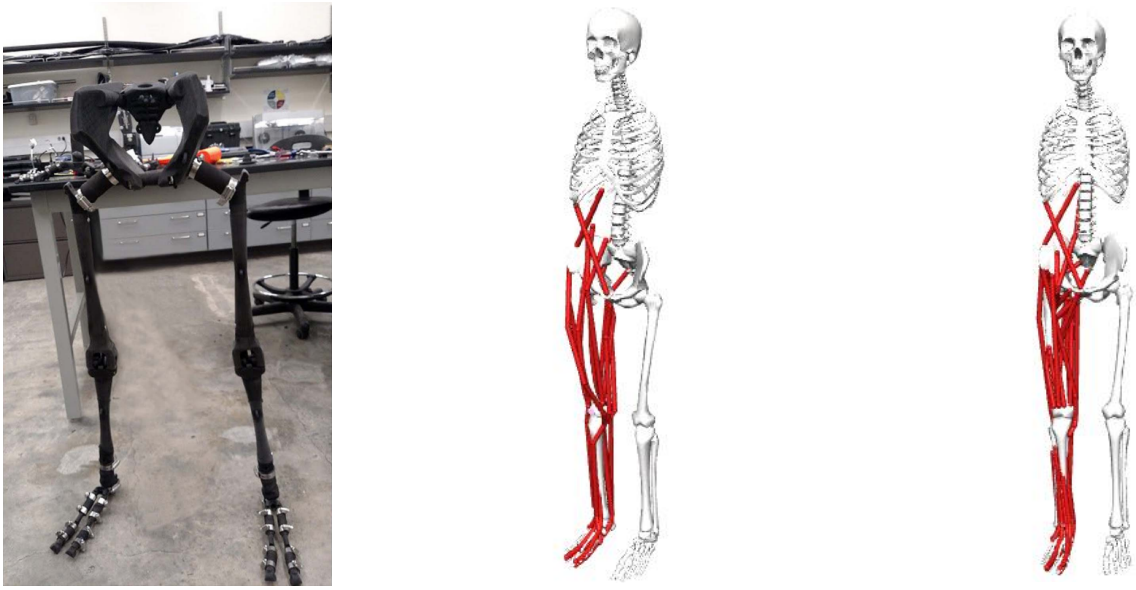


Figure 2.5: **(A)** Laboratory robot skeleton. **(B)** Robot PAM routes. **(C)** Human Gait2392 muscle paths used as design references [Bolen, 2019, Morrow et al., 2020].

2.5 Neural Control of Locomotion: Central Pattern Generators

Locomotion in mammals is driven by spinal central pattern generators (CPGs), neural circuits that can produce alternating rhythmic activity of flexor and extensor motoneurons in the absence of rhythmic input and proprioceptive feedback [Brown, 1911, Rybak et al., 2006]. The founding hypothesis is Brown's half-center: alternating flexor and extensor activity arises from two reciprocally inhibitory central populations with opposing bias [Brown, 1911]. Modern instantiations of the half-center endow each neuron with persistent sodium channels, whose slow inactivation terminates each burst so that reciprocal inhibition switches the populations without any phasic input [Deng et al., 2019]; rhythmogenesis, in this tradition, is a property of the intrinsic membrane dynamics, not of a clock elsewhere

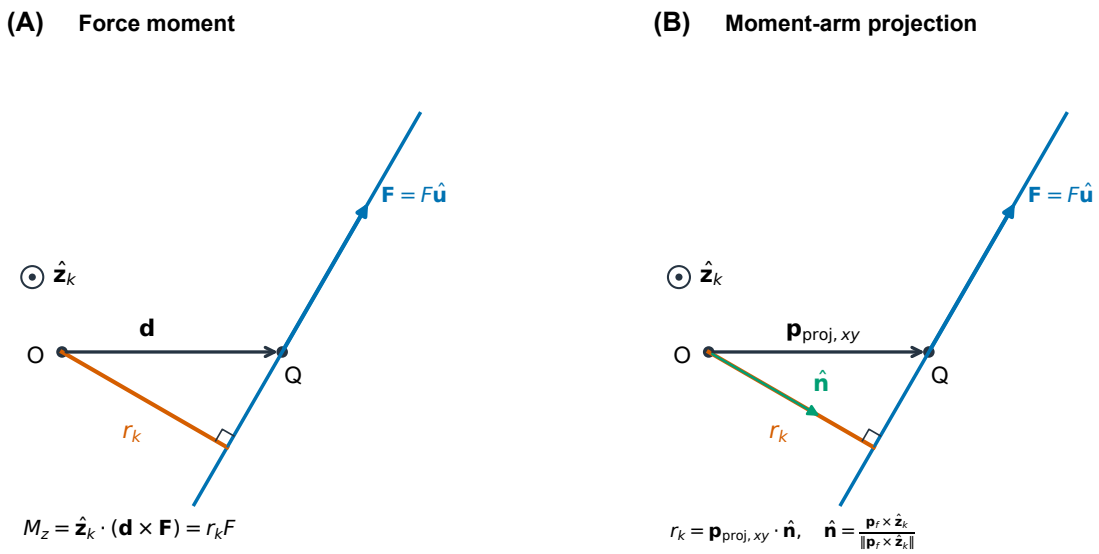


Figure 2.6: Equivalent fixed-axis constructions. **(A)** The cross product gives the moment of a force about the joint. **(B)** Projecting a joint-to-path vector onto the force-path normal gives the signed moment arm used in Equation (4.2). The joint axis points out of the page.

in the network. The modern architectural understanding of mammalian locomotor CPGs comes largely from the modeling work of the Rybak group and collaborators. Rybak et al. [2006] proposed that the locomotor CPG consists of two levels: a half-center rhythm generator (RG) and a pattern formation (PF) network, with reciprocal inhibitory interactions between antagonist neural populations at each level. This two-level organization explains, among other phenomena, spontaneous deletions of motoneuron activity during fictive locomotion in which the locomotor period and phase are maintained, and it permits independent control of the level of motoneuron activity and of step-cycle timing. McCrea and Rybak [2008] reviewed the electrophysiological support for this organization, and Rybak et al. [2015] related the model architecture to genetically identified spinal interneuron classes. Left–right coordination is provided by commissural pathways between the two

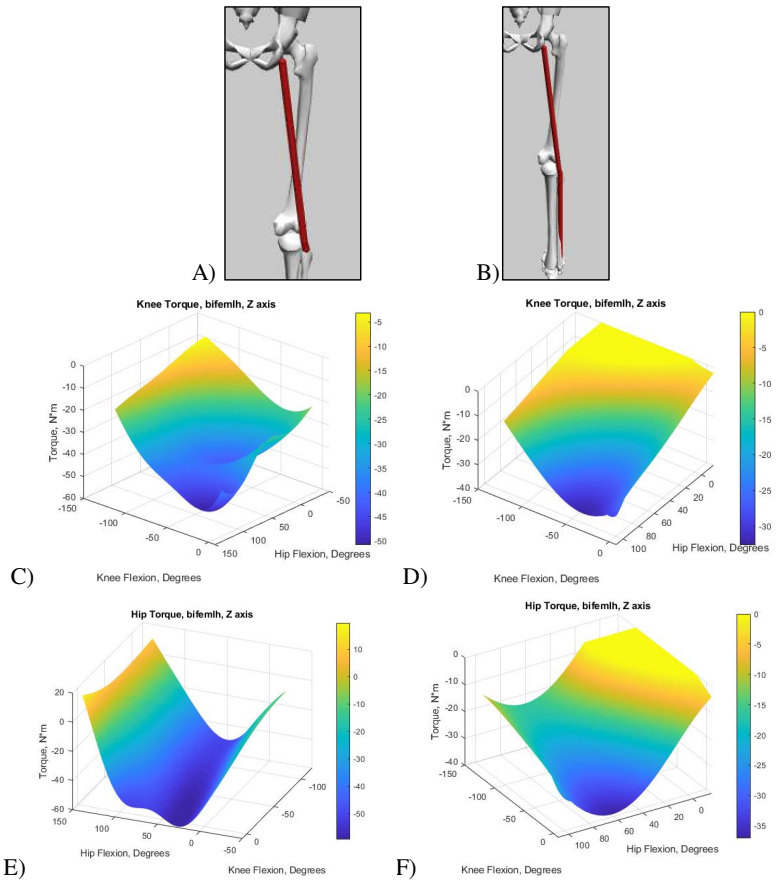


Figure 2.7: Biceps femoris long-head paths and torque surfaces in OpenSim (A, C, E) and the robot model (B, D, F). Panels C–D show knee torque; E–F show hip torque over the hip–knee range [Bolen, 2019, Fig. 8].

half-centers [Molkov et al., 2015, Danner et al., 2016], and the rhythmogenic core of the CPG continues to be refined experimentally [Rancic and Gosgnach, 2021].

The two-level CPG organization is the current working architecture, and its models account for specific observations beyond the deletions that motivated the split: the genetically identified interneuron classes mapped onto the model [Rybak et al., 2015] and the organization of circuits controlling two-joint muscles [Shevtsova et al., 2016], a convergence problem a robot muscle routed across two joints also presents. Coupling such circuits to musculoskeletal models shows how afferent feedback shapes CPG output [Markin et al., 2010] and how biomechanics and sensory feedback regularize the behavior of different CPG architectures [Deng et al., 2022]. Physiologically plausible models of human locomotion use the same machinery to dissect the relative contributions of reflexes and CPGs, with locomotor primitives, Ia, II, and Ib pathways, and Renshaw cells represented explicitly [Di Russo et al., 2023], and recent models address interlimb coordination under abrupt perturbations such as the sudden loss of ground support [Shinohara et al., 2025]. How this architecture is expressed as simulable circuit models is the subject of Section 2.7; the locomotor controller developed in Chapter 6 is built directly on this two-level organization.

2.6 Sensory Feedback in Locomotion

Sensory feedback is not a passive correction layer but an integral part of locomotor pattern generation. Reviews of afferent control of the CPG [Van de Crommert et al., 1998] and of muscle proprioceptive feedback and spinal networks [Windhorst, 2007, 2022] summarize the classical picture: group Ia afferents from muscle spindles signal muscle length and

velocity, group Ib afferents from Golgi tendon organs signal force, and group II afferents signal length, with load-related feedback playing a role distinct from length-related feedback during locomotion [Dietz and Duysens, 2000]. Removing proprioceptive feedback degrades locomotor patterns in mice [Akay et al., 2014]; proprioceptive input regulates and adapts locomotor activity [Lam and Pearson, 2002]; and the relative contributions of proprioceptive and vestibular systems to locomotion are the subject of ongoing study [Akay and Murray, 2021, Akay, 2020]. Tonic central and sensory stimuli have been shown to facilitate involuntary air-stepping in humans [Selionov et al., 2009], demonstrating that simple graded stimulation of motoneuronal and rhythmic circuits can elicit stepping-like output, an experiment readily reproduced in simulation by tonic stimulation of pattern-formation and rhythm-generator populations. The proprioceptive and contact feedback channels designed in Chapter 6 follow this afferent organization.

Three published circuits, reproduced in Figures 2.8, 2.9, and 2.10, place these afferent channels in their circuit context. The rat hindlimb controller of Deng et al. [2019] is organized exactly as Section 2.5 describes: a rhythm-generator half-center and a pattern-formation layer sit above a sensory–motor network in which extensor and flexor muscle receptors feed back through interneurons and Renshaw cells onto the same motoneurons the CPG drives. The human-scale model of Di Russo et al. [2023] carries the idea to a full lower limb: each agonist–antagonist muscle pair carries Ia, II, and Ib reflex populations with Renshaw cells, and pattern-formation nodes from the locomotor controller terminate on the same motoneuron pools. The cat interlimb model of Shinohara et al. [2025] adds the left–right half of the problem, coupling the two rhythm generators through V3 and commissural interneuron populations while force, length, and velocity feedback returns

from every muscle. The schematics differ in animal, scale, and emphasis, but they wire the same afferent classes of this section into the circuit at the level each hypothesis concerns, whether that is the rhythm generator, the pattern-formation layer, or the motor pool.

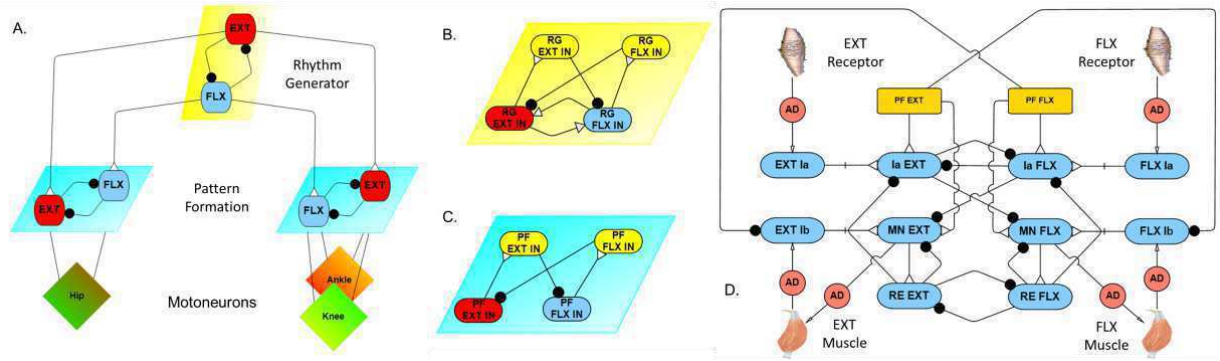


Figure 2.8: Synthetic nervous system of the rat hindlimb model: **(A)** single limb control hierarchy, **(B)** rhythm-generator configuration, **(C)** pattern-formation network, and **(D)** sensory–motor network. Reproduced from Deng et al. [2019, Fig. 4] under CC BY 4.0.

The two-level organization these circuits share descends from the modeling program of the Rybak group, and its foundational schematics are reproduced in Figures 2.11 and 2.12. In the concept model of Rybak et al. [2006], afferent feedback ascends to both the rhythm generator and the pattern-formation network, so a sensory perturbation can reset the rhythm at the generator or delete motoneuron activity at the pattern level without necessarily doing both, precisely the deletion phenomenology that motivated the two-level split. The extended model of Shevtsova et al. [2016] carries the same organization to the two-joint muscles, embedding dedicated pattern-formation populations for the bifunctional posterior biceps–semitendinosus and rectus femoris pools, and that biarticular convergence problem is the one a robot muscle routed across two joints presents to the controllers developed in this dissertation.

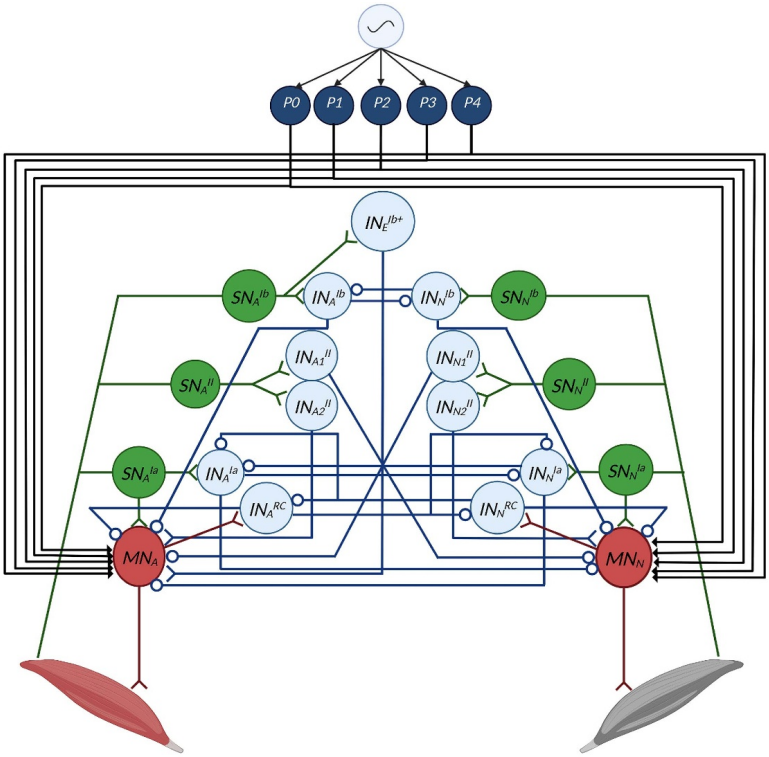


Figure 2.9: Antagonist-muscle spinal network with Ia, II, and Ib reflex pathways, Renshaw cells, and locomotor-pattern input. Reproduced from Di Russo et al. [2023, Fig. 5] under CC BY 4.0.

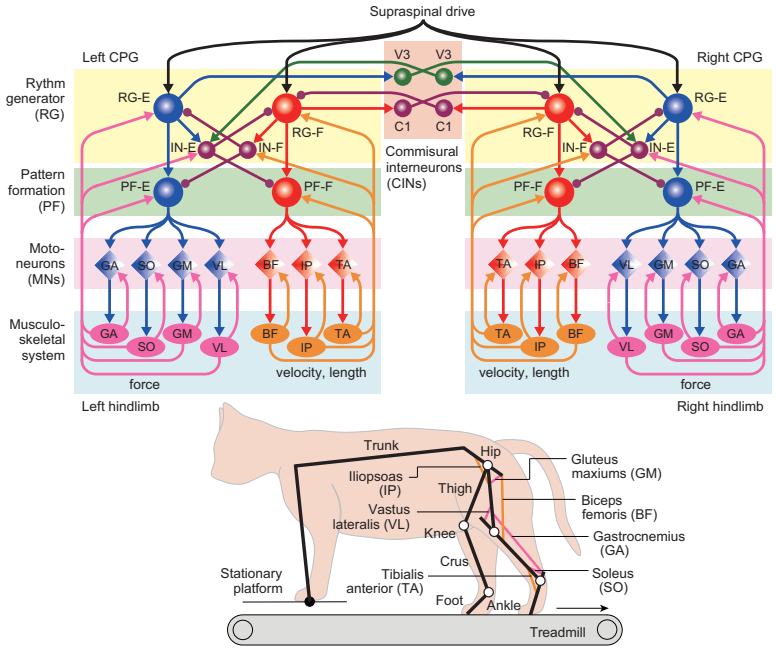


Figure 2.10: Neuromusculoskeletal cat model of Shinohara et al. [2025]: bilateral rhythm-generator and pattern-formation layers coupled by V3 and commissural interneurons, motoneuron pools for seven muscles per hindlimb, and force, length, and velocity afferent feedback. Reproduced from Shinohara et al. [2025, Fig. 1] under CC BY 4.0.

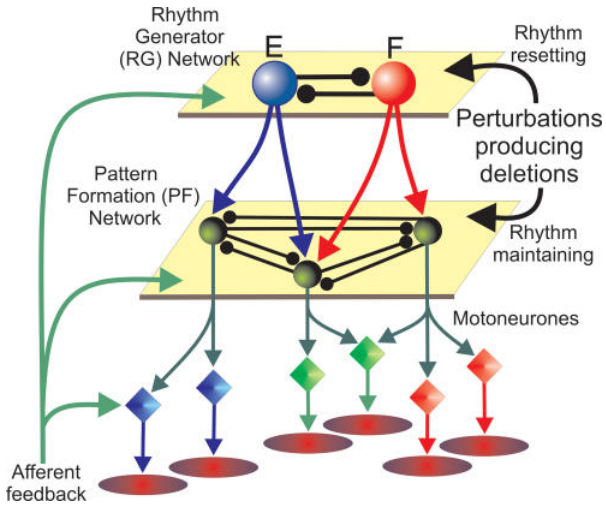


Figure 2.11: Two-level CPG: rhythm generation (RG), pattern formation (PF), motor output, and afferent feedback. Reproduced from Rybak et al. [2006, Fig. 1].

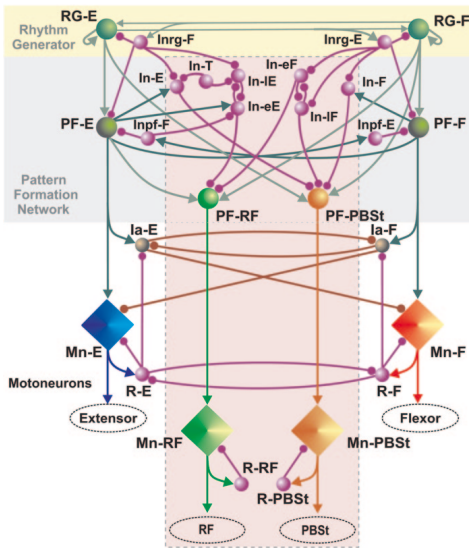


Figure 2.12: Extended CPG with pattern-formation and motor populations for the two-joint posterior biceps–semitendinosus (PBSt) and rectus femoris (RF). Reproduced from Shevtsova et al. [2016, Fig. 5.8].

2.7 Computational Neuroscience

Computational neuroscience expresses hypotheses about neural circuitry as executable models. In conductance-based models, membrane voltages evolve through capacitance, leak, synaptic, and applied currents, and connections represent identifiable pathways [Nourse et al., 2023]. Model predictions can be compared with recordings or with responses to stimulation and pathway removal. This provides a way to test a proposed mechanism rather than infer it from a wiring diagram alone. Synthetic nervous systems use this approach to control physical and simulated bodies.

2.8 Synthetic Nervous Systems

For robots, these biological architectures can be implemented as synthetic nervous systems (SNSs): networks of conductance-based spiking and non-spiking neurons that are small enough to run in real time yet biologically grounded enough to be compared against animal data. The approach has been pursued most persistently by the insect-biorobotics lineage of the Quinn laboratory at Case Western Reserve University and by laboratories closely connected to it, the author's own included. Works in this lineage span several platforms: CPG-based neural control of a pneumatically actuated dog robot [Hunt et al., 2017b], analytically designed dynamic neural networks for human balance control [Hilts et al., 2019], the full-muscle rat hindlimb model and its moment-arm analysis [Young et al., 2019], the two-layer CPG neuromechanical walker built on it [Deng et al., 2019], which supplies this dissertation's controller architecture, and the Muscle Mutt quadruped, a canine-inspired BPA robot designed at the author's laboratory, the Agile and Adaptive

Robotics Laboratory (AARL) at Portland State University, for synthetic nervous system control [Scharzenberger, 2019]. Related controllers from the wider neuromechanical community include sensory-entrained CPG control of bipedal walking [Li et al., 2017], inter-leg coordination analyses in stick-insect-inspired controllers [Nourse et al., 2019], reinforcement-reflex hybrids in insects [Goldsmith et al., 2021], and adaptive hindlimb walking controllers [Schilling et al., 2022].

The SNS-Toolbox [Nourse et al., 2023] makes such networks practical: it is an open-source Python package that simulates hundreds to thousands of spiking and non-spiking neurons in real time or faster, supports multiple software and hardware backends including GPUs and embedded platforms, and couples directly to physics simulators such as MuJoCo for musculoskeletal control. The field it serves is active and growing: the perspective review of Szczecinski et al. [2023] traces neuromorphic legged-locomotion control from its insect-robot beginnings through current platforms to open problems, and physical testbeds in this lineage now extend from insect-scale robots to a robotic cat hindlimb [Armstead et al., 2025].

The boundary between background and contribution runs through this section as well. The prior AnimatLab walker with which the author's program first applied these architectures to a biped [Bolen and Morrow, 2020] is presented as prior work of this dissertation in Section 6.1, the Sensory Afferent Database that organizes the afferent data such networks require [Bolen et al., 2023] is the subject of Chapter 5, and the simulation pipeline that couples the controller to the robot model is built in Chapter 6. None of it is repeated as background here.

2.9 Neuromechanical Simulation

Neuromechanical simulation, the coupling of a musculoskeletal model to a model of the neural controller that drives it, has advanced along three partially overlapping lines. The first line embeds reflex physiology directly into the controller. The muscle-reflex walking model of Geyer and Herr [2010] demonstrated that positive force, length, and velocity feedback around each muscle, organized with simple interlimb coordination, is sufficient to produce self-stabilizing human-like walking dynamics and realistic muscle activations without any central pattern generator. That model became the reference point for a generation of reflex-based neuromechanical controllers, including the cat hindlimb models of Section 2.7 [Markin et al., 2010, Shevtsova et al., 2016] and their extensions. Hybrid work makes the complementarity of the two strategies explicit: Haeufle et al. [2018] combined feed-forward muscle activations with neuronal reflex feedback in simulated human walking and found that the robustness benefit of each depends on which muscle carries it.

The second line replaces hand-designed circuitry with machine learning. Song et al. [2021] trained deep reinforcement learning policies on a physiologically realistic musculoskeletal model in OpenSim, reproducing human kinematics and muscle activations across locomotion tasks, with the learned policy re-usable for experiments that would be difficult in the laboratory. This approach inherits the fidelity of the musculoskeletal model but encodes control in opaque function approximations: the trained policy can reproduce behavior, but the individual "neurons" of the controller do not correspond to identifiable physiological circuits, so the class of scientific questions it can answer, such as lesions,

deletions, and pathway stimulation, is narrower than its behavioral repertoire. Recent perspectives on predictive musculoskeletal simulation [De Groote and Falisse, 2021] place both learned and optimal-control approaches in the same frame: the models are mature, and the open problems are the cost of solving them and the biological interpretability of the control they find.

The third line, in which this dissertation sits, uses synthetic nervous systems whose neurons and synapses explicitly represent candidate physiological circuits. SNS-Toolbox supports conductance-based networks at speeds suitable for selected real-time robotic applications [Nourse et al., 2023]. Related work includes motor-module bipedal walkers [Ichimura and Yamazaki, 2019], two-layer rhythm-generator and pattern-formation control of a musculoskeletal macaque model [Nishizaki et al., 2026], and CPG control of muscle-driven robotic hind legs [Fukuoka et al., 2021]. These studies establish the value of coupled neural and mechanical models. For the humanoid design studied here, the remaining methodological question is how to implement and verify candidate spinal circuits with human musculotendon mechanics and custom BPA actuation, while distinguishing neural effects from conversion or coupling errors. Chapter 6 addresses that question through specific model implementations, graphical neural blocks, and controlled verification and feedback tests.

2.9.1 Simulation Tools

All three lines depend on a software substrate that is itself part of the method: elaborate musculoskeletal geometry and complex neural models. The tools used in this dissertation

are the following.

OpenSim [Delp et al., 2007, Seth et al., 2018], distributed through the SimTK (Simulation Toolkit) repository, provides the biomechanical modeling side: musculoskeletal models with detailed bone geometry, muscle routing, and force-generating properties, along with tools for analyzing moment arms and isometric torque. These models are approximations with known limits rather than exact anatomies: muscle paths are interpolations over discrete attachment and wrapping data [Bolen, 2019]. Their geometry is accurate for a purpose, namely moment arms and torque predictions over the functional range of motion, and that is the standard against which the robot model of this program is compared.

The same SimTK infrastructure hosts SCONE, an open-source package for predictive simulation and optimization of neuromuscular control of biological motion [Geijtenbeek, 2019].

AnimatLab [Cofer et al., 2010] provides a graphical environment in which physical bodies and synthetic neural systems are constructed and co-simulated; the AARL's prior walker (Section 6.1) was built in it.

MuJoCo [Todorov et al., 2012] provides the physics-engine side: fast, accurate simulation of rigid-body dynamics with contacts, along with muscle-like actuators that include activation dynamics.

MyoConverter bridges the biomechanical and physics-engine worlds by converting OpenSim musculoskeletal models to MuJoCo format with optimized muscle kinematics and kinetics [Caggiano et al., 2022, Wang et al., 2022].

SNS-Toolbox supplies the neuron and synapse models used to simulate the controller [Nourse et al., 2023].

I used these tools to develop a robotic OpenSim leg model, convert it to MuJoCo, and couple it to a synthetic nervous system. Chapter 6 describes the assembled interfaces and their evaluation. The coupled simulation supports repeated parameter, stimulation, and pathway-removal tests under controlled conditions. Whether those tests support a biological hypothesis depends on the model's validation and the experimental comparison, as well as simulation speed.

2.10 Testing Neuromechanical Hypotheses on Robots: The Ethical Case

The experiments this research program is designed to enable, including deleting a muscle from a limb, blocking a sensory pathway, lesioning part of a rhythm generator, and applying tonic stimulation to spinal circuits, cannot be performed on healthy humans. Invasive intervention on the human neuromuscular system is limited to clinical populations, is tightly constrained by ethics review, and permits neither the within-subject control nor the reversibility that neuromechanical hypotheses demand. The same experiments in animals also carry real moral cost, and their justification is governed by the Three Rs framework: replacement of animal subjects with non-animal methods, reduction of the numbers required, and refinement of the remaining procedures [Russell and Burch, 1959, Tannenbaum and Bennett, 2015]. Robotic and computational models allow proposed neural interventions to be tested under controlled conditions before corresponding animal experiments are considered. The same model and initial conditions can be reused across interventions to isolate the effect of a changed pathway. Repeating a model does not create independent biological subjects, and its relevance to animal experiments depends

on validation against biological observations.

Schematics of decerebrate-cat locomotion emphasize the neural circuits under study, but often leave the surgical preparation out of view. A simplified drawing of a walking cat can make the animal appear intact, alert, or even contented. That appearance should not be confused with the condition of the experimental animal: a brainstem transection has disconnected the forebrain, while neural circuits below the transection can still generate treadmill stepping [Brown, 1911, Severin et al., 1967]. Figure 2.13, taken from archival footage, shows the physical preparation behind the circuit diagrams [WhiteCoatWaste, n.d.]. This distinction makes the ethical meaning of replacement concrete: the robotic experiments discussed here are intended, in part, to reproduce selected neural mechanisms without repeating these invasive preparations in living animals.

The procedures behind that record are documented in films that remain publicly available. A 50 min classroom recording posted by the UTHealth Houston School of Dentistry first shows an intact cat playing with its handler, righting itself when lowered and placing its paws accurately on the edge of a table, and then documents the decerebration performed for the class [UTHealth Houston School of Dentistry, 2025]: the animal is anesthetized with open-drop ether, more than the horse in the same demonstration series received, the carotid arteries are ligated and a tracheal cannula installed, and the skull is opened with a trephine before the brainstem is transected between the superior and inferior colliculi. The treadmill preparations that established locomotor neurophysiology are likewise preserved in the historical literature: Stuart [2007] reproduce the arrangement of the high-decerebrate cat treadmill experiments, with electromyographic recording from the hindlimb muscles, from the original Russian-language reports [Severin et al., 1967].

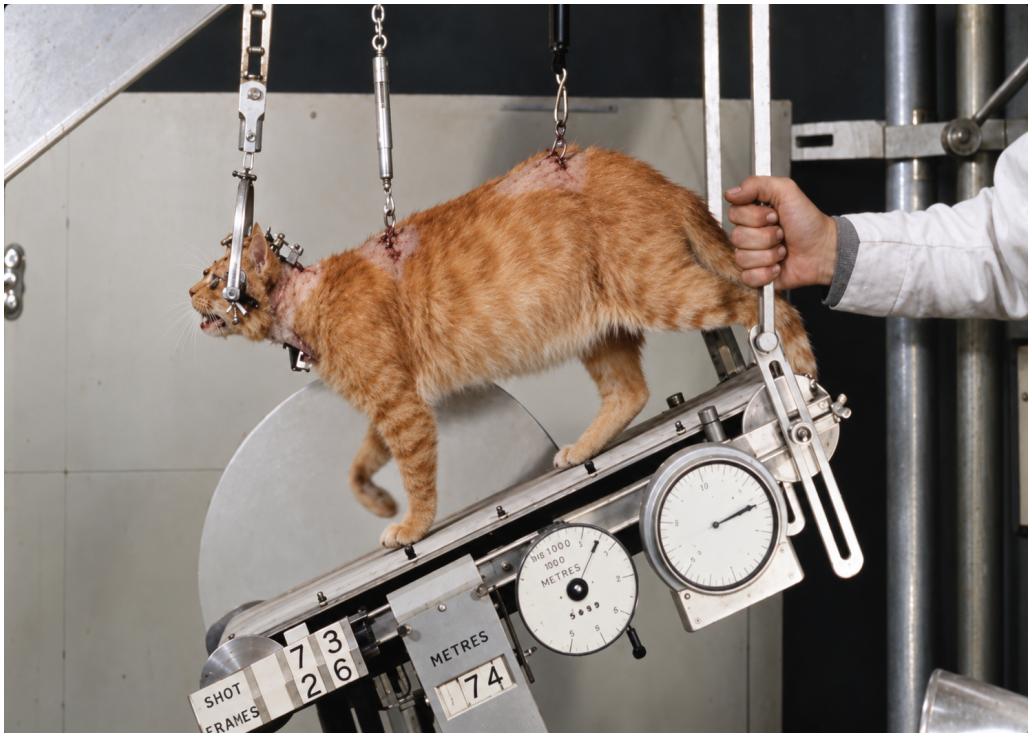


Figure 2.13: Archival decerebrate-cat treadmill experiment [WhiteCoatWaste, n.d.]. The frame is restored and machine colorized; color and fine detail are inferred. It provides documentary context, not gait evidence.

The honest accounting is murkier than the Three Rs framing alone suggests, because the models a robot implements are themselves products of animal experimentation. The half-center organization, the two-level rhythm and pattern architecture, and the length, force, and load signaling of the afferents were each established in preparations like the ones described above [Brown, 1911, Severin et al., 1967, Rybak et al., 2006], and the databases of afferent properties that parameterize sensory feedback are compiled from animal recordings [Bolen et al., 2023]. A synthetic nervous system therefore replaces future experiments; it does not replace, and cannot repay, the animal work on which its architecture is built. Recognizing this inheritance sharpens rather than undermines the case for robotic subjects: the knowledge purchased at that cost is precisely what makes a robotic replacement scientifically meaningful, and the robot is the instrument that keeps that knowledge productive without adding to the ledger.

For a robot to serve as a replacement subject, however, it must first earn standing as a scientific model. Webb's influential analyses of bio-robotic methodology [Webb, 2001, 2002] argue that a robot is a good model of biological behavior when the mechanism it implements is tested at the same level of organization as the target phenomenon, matching dynamics, control structure, and sensory ecology rather than surface appearance. The validation chain of this dissertation is constructed to that standard: the actuator force model is validated against bench data (Section 3.2.1), the joint torque model against isometric measurements and human torque envelopes (Section 4.2), and selected controller responses are compared with published tonic stimulation and deletion phenomena [Selionov et al., 2009, Rybak et al., 2006], the verification step that precedes locomotion tuning in this program. Only then can a robot experiment be read as evidence about the hypothesized

biological circuit rather than about the robot.

The argument is ultimately bidirectional, completing the loop described in Section 1.2. Experiments that are unethical, impractical, or statistically unreachable in humans and animals become routine on the robot; the resulting insights into neuromechanical control feed back into biology, into the design of the next robot iteration, and directly into prosthetic and orthotic devices that face the same problem of controlling muscle-like actuators for human users, whose central pattern generator controllers are an active topic in lower-limb exoskeleton research [Duan et al., 2024, Pery et al., 2026]. The ethical case and the scientific case are the same case: the robot is the instrument that makes these experiments possible.

3 Isometric Force Characterization of Braided Pneumatic Actuators

Publication note: Sections 3.1.1–3.1.2, Section 3.2.1, and Sections 3.3.1–3.3.5, with their associated figures and tables, appeared in Bolen et al. [2026] (Bolen, Elzein, Pang, and Hunt, *Actuators*, 2026). For that work, B.B. performed the conceptualization, methodology, software, validation, formal analysis, investigation, data curation, visualization, and original draft preparation; M.E. and L.P. contributed methodology, software, investigation, and data curation; and A.H. supervised the work, administered the project, and acquired funding. All co-authors contributed to writing, review, and editing.

This chapter develops and validates the actuator force models on which the rest of the dissertation builds, and it reads as one self-contained study, from methods through results to discussion. Sections 3.1.1 and 3.1.2 present the methods: the force equations used throughout this work, and the test jig and measurement procedure used to characterize Festo BPAs with 10 mm and 20 mm uninflated diameters over a wide range of resting lengths, contractions, and pressures. Section 3.2.1 reports the resulting maximum-force and normalized force–strain–pressure models and compares them with the manufacturer's tool. The remaining sections discuss the effects of resting length, the nondimensionalized force prediction, the comparison with other published BPA models, the implications for real-time control and simulation, and the analogy between these artificial muscles and biological muscle. The validated force model is the one used for the knee-torque predictions of Chapter 4 and for the BPA actuation of the neuromechanical simulations in

Chapter 6.

3.1 Methods

3.1.1 Actuator Force Model

The actuator force model used throughout this dissertation is described here. First, the amount of contraction ε in a BPA is calculated as

$$\varepsilon = \frac{(l_0 - l_{\mathrm{m}})}{l_0} \tag{3.1}$$

where l_0 is uninflated resting length of the BPA measured from hose clamp to hose clamp. Current muscle length is l_{m}.

Similarly, ε_{620} is the amount of contraction in a BPA without external load when inflated at 620 kPa, calculated as

$$\varepsilon_{620} = \frac{(l_0 - l_{620})}{l_0} \tag{3.2}$$

where l_{620} is the muscle length measured at 620 kPa and free from external loads.

The amount of relative contraction of a BPA ε^* can then be calculated by dividing equation (3.1) by (3.2), such that

$$\varepsilon^* = \frac{\varepsilon}{\varepsilon_{620}} \tag{3.3}$$

describes the BPA's current contraction as a normalized percentage of its total.

The amount of normalized isometric force F^* in a Festo BPA is then calculated as

$$F^*(\varepsilon^*, P^*) = \begin{cases} c_0 \cdot \left(\exp(-c_1 \cdot \varepsilon^*) - 1\right) + P^* \cdot \exp\left(-c_2(\varepsilon^*)^2\right) & \text{for } F^* > 0 \\ 0 & \text{for } F^* \leqslant 0 \end{cases} \tag{3.4}$$

where P^* is pressure P normalized by 620 kPa. BPA diameter dependent coefficients are determined by the actuator characterization experiments reported in Section 3.2.1 (Table 3.2).

The maximum force F_{max} in a BPA is calculated as

$$F_{max}(l_0, P_{\max}) = a_1 \cdot P \cdot \arctan\left(a_2 \cdot P \cdot (l_0 - 0.0075)\right) \tag{3.5}$$

with coefficients determined by the actuator characterization experiments reported in Section 3.2.1 (Table 3.1). When maximum system pressure is set to $P = 620$ kPa, $F_{\max}(l_0, 620) = F_{620}(l_0)$. Multiplying (3.4) by (3.5) yields

$$F(\varepsilon^*, P^*, l_0) = F^*(\varepsilon^*, P^*) \cdot F_{620}(l_0) \tag{3.6}$$

as the equation for isometric force F in a Festo BPA.

3.1.2 BPA Isometric Force Characterization Experiment

To measure artificial muscle force as a function of length and pressure, we built a test jig from extruded aluminum and 3D printed parts. Festo BPAs with 10 mm and 20 mm diameter were tested with various resting lengths and different amounts of contraction.

These data are used to develop an improved model that more accurately predicts maximum muscle force based on l_0, current muscle length l_m, and pressure P. The results are compared with existing models and the new data are used to create an updated model.

A test jig was made of 80/20 brand 1515 series extruded aluminum (Figure 3.1). Artificial muscles were placed vertically in the jig one at a time. The upper end was attached to a load cell. Force data was collected using a CALT DYLY-103 100 kg S-shaped load cell for the ∅10 mm BPAs and a Loadstar RAS1-2HKS-S for the ∅20 mm BPAs. The load cell was used in conjunction with a HX711 Load Cell Amplifier. Pressure data was obtained from a Freescale MPX5700 GP 5 V pressure sensor. Building air supply pressure was controlled with two pressure regulators in series. The first was a Parker model 20R113GC $0 - 120$ psi pressure regulator. The second was a Husky 3/8 in. High Performance Air Regulator HDA72200. A Festo VTUG-10-MRCR-S1T-26V20-T516LA-UL-532S-8K valve manifold VTUG-G was used to deliver air from the pressure regulator to the actuator. This manifold is comprised of eight two-in-one bidirectional normally closed Festo VUVG-S10-T32C-AZT-M5-1T1L valves.

Pressure and load cell amplifier data were sent to MATLAB 2021b via an Arduino Uno style Sparkfun BlackBoard C microcontroller. The computers that used MATLAB were running either Windows 10 or Windows 11. During phases when the Arduino was collecting force and pressure data to send to MATLAB, the Arduino would also trigger (via MATLAB) an Onsemi 2N4401 NPN transistor to make the valve manifold open or close the valve. For other data collection, the valve was manually opened and closed.

The inner distance between the hose clamps on each BPA was measured with a FANUC tape measure to determine the muscle's length l and resting length (l_0). This is how

Festo AG & Co. KG [2026] defines l_0, although in Hunt et al. [2017a] it was measured to also include end cap length. We then inflated each BPA to $P_{620} = 620\,\text{kPa}$, with one end allowed to move freely in the axial Degree of Freedom (DoF), and measured the length l_{620} to calculate maximum contraction at 620 kPa using Equation (3.2). The distance between the crossmembers was then controlled to obtain different amounts of contraction, the muscles were inflated to various pressures, and the contractile force was recorded at the pressure–contraction pairs. This was done for ∅10 mm BPA resting lengths (l_0) of 112 mm–518 mm. For ∅20 mm BPAs, resting lengths (l_0) of 300 mm–509 mm were used.

3.2 Results

3.2.1 BPA Force Characterization

This section reports the results of the isometric force characterization experiments described in Section 3.1.2. First, maximum isometric force at 620 kPa is modeled as a function of resting length. The remaining data are then normalized to build a predictive force surface, and the resulting model is compared with the manufacturer's tool and with other published BPA models.

Maximum Force at 620 kPa

Data of the maximum force from BPA characterization tests of the 10 mm and 20 mm diameters at 620 kPa show a dependency on resting length (Figure 3.2). This is a previously unreported characteristic of these artificial muscles. Detailed analysis of the Festo

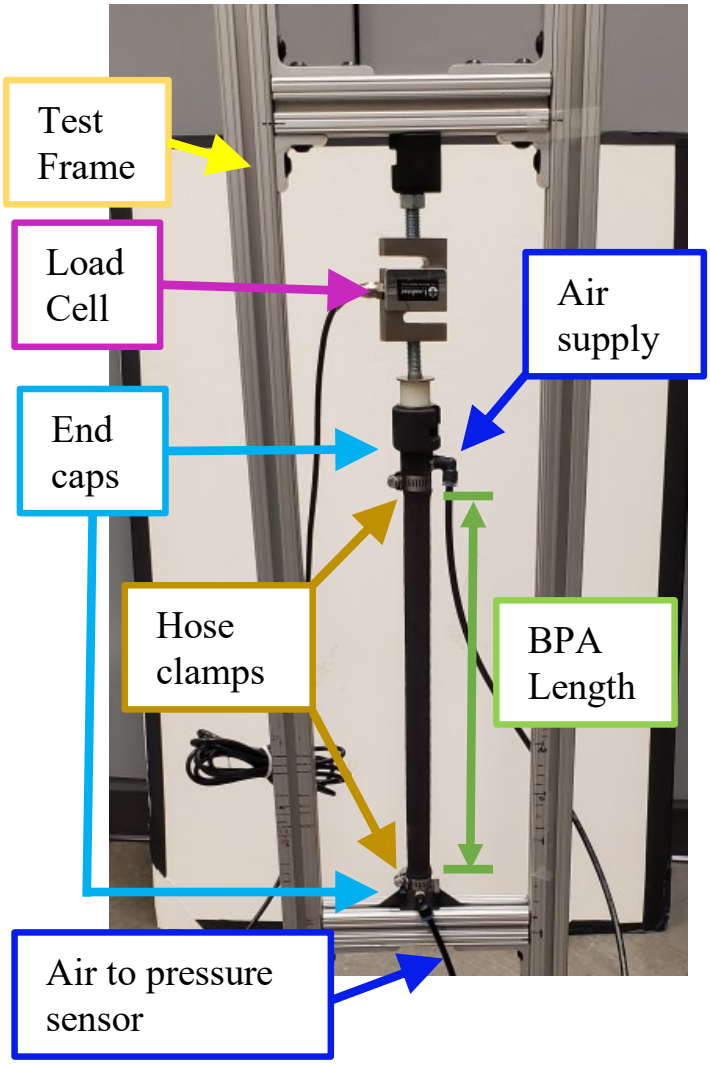


Figure 3.1: A picture of a BPA in the isometric force test stand with components labeled.

tool [Festo AG & Co. KG, 2026] does in fact predict a change in maximum force with the resting length; however, the Festo tool predicts increased force with shorter lengths, while our collected data indicate decreasing force with shorter lengths. The data show a force response resembling an arctan curve along the l_0 dimension. Using the Nonlinear Least Squares method and a Least Absolute Residual robustness, we fit an arctan curve to the data to get the maximum force at 620 kPa given a resting length as:

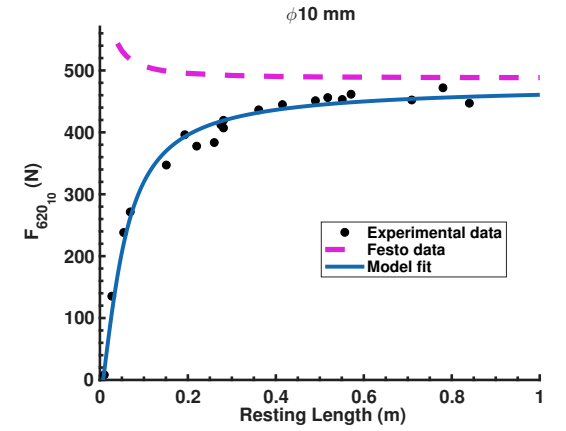
$$F_{620_{10}} = 303.5\,\mathrm{N} \cdot \arctan\left(19.03\,\mathrm{m}^{-1} \cdot (l_0 - 0.0075)\right) \tag{3.7}$$

$$F_{620_{20}} = 922.4\,\mathrm{N} \cdot \arctan\left(15.37\,\mathrm{m}^{-1} \cdot (l_0 - 0.013)\right) \tag{3.8}$$

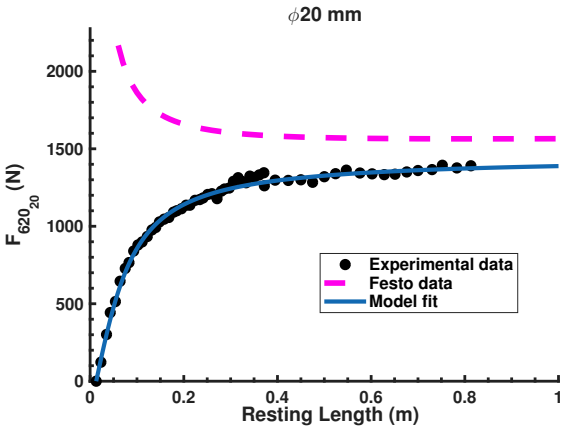
The length is offset by 0.0075 m and 0.013 m because solid modeling showed that the end caps contact each other at these lengths. At these lengths, the actuator would not be able to contract to produce force.

Force as a Function of Pressure and Resting Length

Data from BPA characterization tests of the 10 mm muscles resulted in force–pressure pairings for different muscle resting lengths (l_0) (Figure 3.3a). Similar to the maximum force data, these data show a force response resembling an arctan curve along the l_0 dimension with a more linear response to changes in pressure. Using the Nonlinear Least Squares method and a Least Absolute Residual robustness, we fit an arctan curve to the data to get the maximum force given a resting length and pressure as described by the model form given in Section 3.1.1, where $a_1 = 0.4895\,\mathrm{N\,kPa}^{-1}$ and $a_2 = 0.030\,68\,\mathrm{kPa}^{-1}\,\mathrm{m}^{-1}$



(**a**)



(**b**)

Figure 3.2: Results for finding the relationship between F_{620} and l_0. (**a**) $F_{620_{10}}$ as a function of l_0, at P_{620}. Dashed line is the $F_{620_{10}}$ data from Festo. Solid line is the fit from Equation (3.7). (**b**) $F_{620_{20}}$ as a function of l_0 at P_{620}. Dashed line is $F_{620_{20}}$ data from Festo. Solid line is the fit from Equation (3.8).

for the 10 mm actuator, and $a_1 = 1.49\,\mathrm{N\,kPa^{-1}}$, $a_2 = 0.0248\,\mathrm{kPa^{-1}\,m^{-1}}$ for the 20 mm actuator. Goodness-of-fit measures are given in Table 3.1; throughout this dissertation, goodness of fit is reported as the root-mean-square error (RMSE), the adjusted fraction of variance unexplained (FVU, for which $R^2 = 1 - \mathrm{FVU}$), and the maximum absolute residual, as defined alongside the reference implementation in Appendix A.

Maximum Contraction at 620 kPa

Figure 3.3(b) shows an attempt at a linear fit for maximum contraction at 620 kPa (ϵ_{620}) as a function of resting length (l_0). Error bars have been included to show the effect of ± 1 mm accuracy with measurements of l_0 and l_{620}. There was a large amount of variance in the data, with the linear fit giving an adjusted $\mathrm{R}^2 = 0.4124$ and an RMSE $= 0.0083$. Since there is no direct, predicable relationship between maximum contraction and l_0, this value should be recorded in each muscle used on a robot to best predict the force it may produce at different pressures and contraction.

Force as a Function of Pressure and Contraction

In addition to being a function of the pressure and resting length, the force produced by the actuator is also a function of the amount of actuator contraction, with less force being applied as the actuator contracts. To build this model, the collected data of force, pressure, and contraction are normalized by dividing by F_{620}, P_{620}, and ϵ_{620}, respectively. This had the effect of compressing the data into a 3D surface ranging from 0 to 1 on all axes. Normalized data are compared with pressure isoclines of force predicted by the

Table 3.1: Maximum force equation coefficient values with confidence intervals (CI) and goodness-of-fit measures. Equation (3.5) is compared to data taken at 620 kPa. Equations (3.7) and (3.8) are compared against maximum force from using the Festo tool. It was not necessary to fit an adjusted R^2 in these cases.

Equation	Coefficient	CI (95%)	Our Model			Festo Tool		
			Adj. R^2	RMSE	Max. Error	Adj. R^2	RMSE	Max. Error
(3.5)	$a_1 = 0.4895\,\mathrm{N\,kPa^{-1}}$ $a_2 = 0.030\,68\,\mathrm{kPa^{-1}\,m^{-1}}$	(0.4822, 0.4968) (0.0282, 0.03317)	0.9945	11.61 N	55.3 N	0.9854	14.7 N	30.9 N
(3.7)	$b_1 = 303.5\,\mathrm{N}$ $b_2 = 19.03\,\mathrm{m^{-1}}$	(300, 308) (17.48, 20.57)	0.9854	14.72 N	30.9 N	–	189.9 N	375.6 N
(3.8)	$b_1 = 922.4\,\mathrm{N}$ $b_2 = 15.37\,\mathrm{m^{-1}}$	(914.2, 930.7) (14.75, 15.98)	0.9945	23.83 N	62.1 N	–	668.4 N	1590.1 N

Table 3.2: Normalized isometric BPA force equation (Equation (3.4)) coefficient values and goodness-of-fit measures.

BPA	Coefficient	CI (95%)	Model			Validation		
			Adj. R^2	RMSE	Max. Error	Adj. R^2	RMSE	Max. Error
∅10 mm	$c_0 = 0.5682$	(0.5584, 0.578)	0.9998	0.005118	10.3%	0.9994	0.0245409	10.6%
	$c_1 = 4.254$	(4.126, 4.383)						
	$c_2 = 0.5597$	(0.5429, 0.5766)						
∅20 mm	$c_0 = 0.2579$	(0.2401, 0.2756)	0.992	0.02294	6.8%	0.9943	0.0231303	5.7%
	$c_1 = 6.477$	(5.558, 7.396)						
	$c_2 = 1.321$	(1.239, 1.403)						

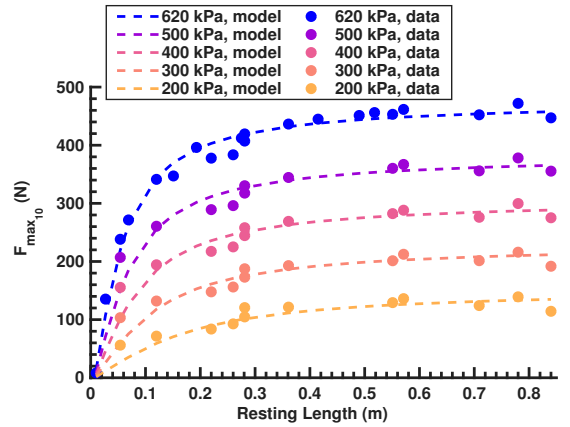
Festo tool divided by the maximum force equation described in the previous section in Figure 3.4. With this comparison, it is clear that the Festo tool over-predicts the expected force, especially at lower pressures, contraction, and resting lengths.

We therefore derived our own equation for isometric force in the BPA as a function of pressure and contraction. Visual analysis of the experimental data shows an exponential relationship between ϵ^* and F^*, and a linear relationship between P^* and F. We fit a surface to the data using Nonlinear Least Squares and a Least Absolute Residual robustness such that it is described by Equation (3.4). For the ∅10 mm BPAs, the result of the improved fit can be seen in Figure 3.4a. The adjusted $R^2 = 0.9998$, a RMSE $= 0.004537$, and a maximum absolute residual of 10.6%. Solving (3.4) for the ∅20 mm BPA, yields different coefficients, and results are seen in Figure 3.4b. Coefficient values and goodness-of-fit statistics are found in Table 3.2.

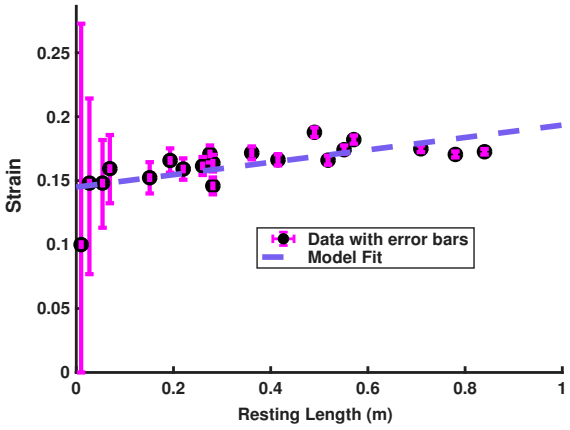
3.3 Discussion

3.3.1 Effects of Resting Length

We have developed new F_{620} equations for 10 mm and 20 mm diameter Festo BPAs that capture the change in maximum force produced by the actuators at 620 kPa as a function of their resting length. When examining Equation (3.7) and measured data in Figure 3.2b, it can be seen that as l_0 goes to infinity, $F_{620_{10}}$ goes to 470 N. When the Festo tool [Lang, Robert, 2005] was queried at $l_0 = 1$ m as an approximation of infinity, it predicts the ∅10 mm BPA can produce $F = 490$ N at P_{620}. This is within the 10% variability that Festo states may occur in manufacturing tolerances. Similarly, the Festo data predict a ∅20 mm

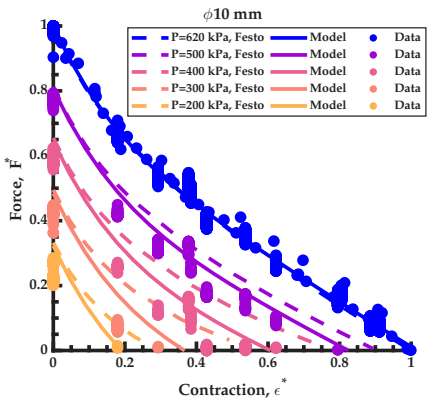


(**a**)

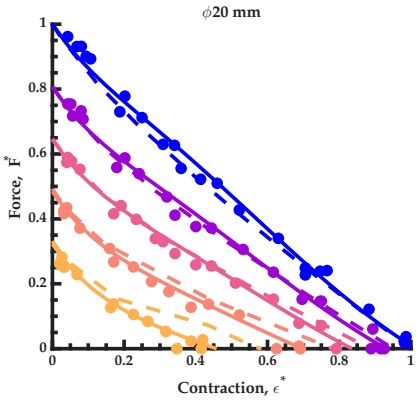


(**b**)

Figure 3.3: Results for finding the relationship between F_{620}, l_0, P_{620}, and $\epsilon_{620_{10}}$. (**a**) Isoclines of the surface fit for $F_{620_{10}}(l_0, P_{620})$. (**b**) $\epsilon_{620_{10}}$ versus l_0 at P_{620}. Although there is a general trend of longer resting lengths producing more contraction, no conclusive relationship between ϵ_{620} and l_0 could be deduced from this experiment. The error bars show the effect of being ± 1 mm in measuring l_0 and l_{620}.



(**a**)



(**b**)

Figure 3.4: Surface fit for $F^*(\epsilon^*, P^*)$. Solid lines are from our model. Dashed lines from Festo supplied data. In total, 537 data points were collected and used to fit the 3D surfaces (321 data points for ∅10 mm BPAs and 216 data points for ∅20 mm BPAs). In total, 80% of these data were used for training, and 20% were used for validation. For figure clarity, not all data is included and plotted circles represent collected data at ± 10 kPa of the stated pressure. (**a**) Fit data for ∅10 mm Festo BPAs. (**b**) Fit data for ∅20 mm Festo BPAs.

BPA will produce $F = 1570\,\text{N}$. Our results for Equation (3.8) show that $F_{620_{20}}$ goes to 1460 N as l_0 approaches infinity. Although the difference is much larger, it is also within the 10% manufacturing tolerance.

Our measured data differ significantly from the Festo model as l_0 goes to zero. For the measured data, as the length gets smaller, the force does as well, whereas the Festo tool predicts exponential force increase (see Figure 3.2). As a specific example, when specifying $l_0 = 112\,\text{mm}$ for the 10 mm diameter BPAs, the Festo predicted F_{620} is 498.6 N. However, using Equation (3.7), F_{620} is calculated to be 335.3 N. Actual force in the $l_0 = 112\,\text{mm}$ was measured at 350.9 N. Therefore, the error in predicted F_{620} for the ∅10 mm BPA is 4.5% for our model and 42.1% using the Festo tool. Researchers using BPA resting lengths under 300 mm should take note of these results.

Festo AG & Co. KG [2026] (p. 12) states that force will be reduced by approximately 10% for a minimum nominal length. In Festo AG & Co. KG [2018] (p. 14) they say the force can deviate by 10% based on factors such as production variation, material variation, and deviation from nominal length (l_0). Festo AG & Co. KG [2013] (p. 13) notes that testing was done with BPAs that are ten times the uninflated diameter, and that theoretical force can increase by up to 10%. In each datasheet, the characteristic curves appear to be the same. As demonstrated above, the 100 mm ∅10 mm BPA should have much less force than what is shown in the curve. This is our experience and the reason for the testing. It is unclear why the MuscleSIM tool [Lang, Robert, 2005] then predicts higher force as resting length decreases. It could be a reversed sign coefficient in their internal curve-fit model. We have shown here that as the nominal length goes to zero, so does force.

BPA models commonly predict higher forces than are measured experimentally because

they do not account for end effects in regions of the pressure vessel where those effects dominate. Nominally, the clamped ends fix the inner and outer diameter while allowing axial and radial displacement. In the isometric case, there is only radial displacement. The boundary conditions force steep gradients in stress and strain near the ends. Near the boundary conditions, axial, radial, and circumferential strains deviate from analytic solutions for a cylindrical model. For a constant pressure, the end effect region appears to have the same length. Therefore, in shorter BPAs, the end effects dominate. The axial length scale of the boundary section is comparable to l_0, and the active force differs substantially from that of an assumed perfect cylinder.

Manufacturing variations seem to play an important role in the behavior of these actuators. It remains to be seen how ϵ_{620} can be known a priori. We still suspect that it is a function of l_0, but it might also be a function of other factors such as product batch number or UV light exposure [Festo AG & Co. KG, 2026] (p. 15); [Festo AG & Co. KG, 2018] (p. 6); [Festo AG & Co. KG, 2013] (p. 13). We found there to be larger variations in the 10 mm BPAs than in the 20 mm BPAs. Modeling work on ideal McKibben actuators describes how the relationship between contraction and initial length is a function of the total number of twists and the angle of twist in the usable muscle area [Chou and Hannaford, 1996]. It is unclear if there are variations in the angle and number of twists in the Festo BPAs due to manufacturing. Uncovering the relationship between ϵ_{620} and l_0 for the Festo DMSP artificial muscles, if it exists in a meaningful way, will require additional controlled tests.

Similarly, it is not clear how the maximum force might change with manufacturing variations. This was reasonably corrected for by normalizing the equations to each individual's maximum contraction, but more nuanced details might emerge with more

data collection. Not knowing these relationships a priori means there is still a need for robots and other systems to go through an iterative design stage if it is discovered that the system does not produce the torque or have the RoM that the design team expects. However, the data and models provided here should enable faster redesigns with fewer additional measurements.

3.3.2 Nondimensionalized Force Prediction

We have also introduced the concept of nondimensionalized isometric force that is a function of relative strain and relative pressure (Equation (3.4)). This elegant equation has only three coefficients with low error (see Table 3.2). In the work presented here, we introduce the relative pressure term, $P^* = P/P_{620}$. Our data show that by normalizing force, contraction, and pressure, we are able to create a simplified force equation as a function of contraction and pressure that scales well with initial actuator length, as expressed by Equation (3.6) in Section 3.1.1.

For example, given $l_0 = 54\,\text{mm}$ and $P = 300\,\text{kPa}$, the measured force was $F_{actual} = 103\,\text{N}$. Force prediction using only the Festo tool predicts the force to be $F_{predict} = 270\,\text{N}$. This is 161% greater than the measured force. Using Equations (3.7) and (3.4) with Equation (3.6) gives $F_{predict} = F^*(0, 0.48) \cdot F_{620}(0.054) = 0.49 \cdot 230\,\text{N} = 112\,\text{N}$. This is an error of 8%, which is much more accurate than using the Festo tool only.

The presented work is capable of predicting force output for 10 mm and 20 mm muscles with different initial muscle lengths, internal pressure, and contraction amount with minimal initial data collection. However, it remains to be seen if these relationships

are maintained over the full lifecycle of the muscles. There are not yet any peer reviewed studies investigating the long-term behavior of these artificial muscles, though the Festo datasheet provides some indication of what can be expected [Festo AG & Co. KG, 2026] (p.19). They report that the 40 mm BPA can complete 1 million cycles at 6000 N (maximum rated load), or 10 million cycles at 4000 N. The service life of the other size BPAs is not reported; however, it is mentioned that thermal stressing may reduce the lifespan and they should be inspected every 500,000 cycles for signs of aging [Festo AG & Co. KG, 2018] (pp. 10–11). Future research should look at the behavior of these muscles over time.

In our work, $P_{620} = 620\,\text{kPa}$ is the maximum supply pressure for our system, and other users of this actuator may use a different supply pressure. However, this supply pressure is not required, and normalizing by this value was chosen as a method for improving the optimization techniques by reducing the sensitivity of the term associated with pressure. This equation will work for any pressure supplies below 620 kPa. We also anticipate the equation will work for pressures up to 800 kPa (the maximum rated pressure for Festo ∅10 mm BPAs); however, we do not have the data to confirm this.

3.3.3 Comparison with Other Published BPA Models

The proposed model was compared with the formulations by Sárosi [2012], Sárosi and Fabulya [2012], and Martens and Boblan [2017], to evaluate their predictive performance across actuator sizes and operating conditions (see Table 3.3 and Figure 3.5). For the ∅20 mm BPA, parameters from [Sárosi, 2012] were used, corresponding to actuators with resting lengths of 200 mm and 400 mm. These were compared against the experimental

data at $l_0 = 300\,\text{mm}$ and $l_0 = 450\,\text{mm}$, respectively. For the $\varnothing 10\,\text{mm}$ BPA with resting lengths of 257 mm and 233 mm, parameters from [Sárosi and Fabulya, 2012] were applied. In each case, the maximum contraction parameter (k_{max}) was determined numerically by solving $F(6.2\,\text{bar}, k_{max}) = 0$, ensuring consistency with the experimental maximum pressure condition.

While the Sárosi [2012] and Sárosi and Fabulya [2012] models capture general trends in force–contraction behavior, discrepancies were observed when applying parameters derived from different actuator lengths. In particular, the model tends to over-predict force at higher pressures and low amounts of contraction, suggesting sensitivity to geometric scaling effects not explicitly accounted for in the formulation.

The Martens and Boblan [2017] model is derived from a geometric and material-based framework and provides a more physically grounded representation of actuator behavior. However, its accuracy depends strongly on precise geometric parameters, including braid angle, wall thickness, and fiber length. Small deviations in these parameters can lead to significant differences in predicted force, particularly at higher pressures.

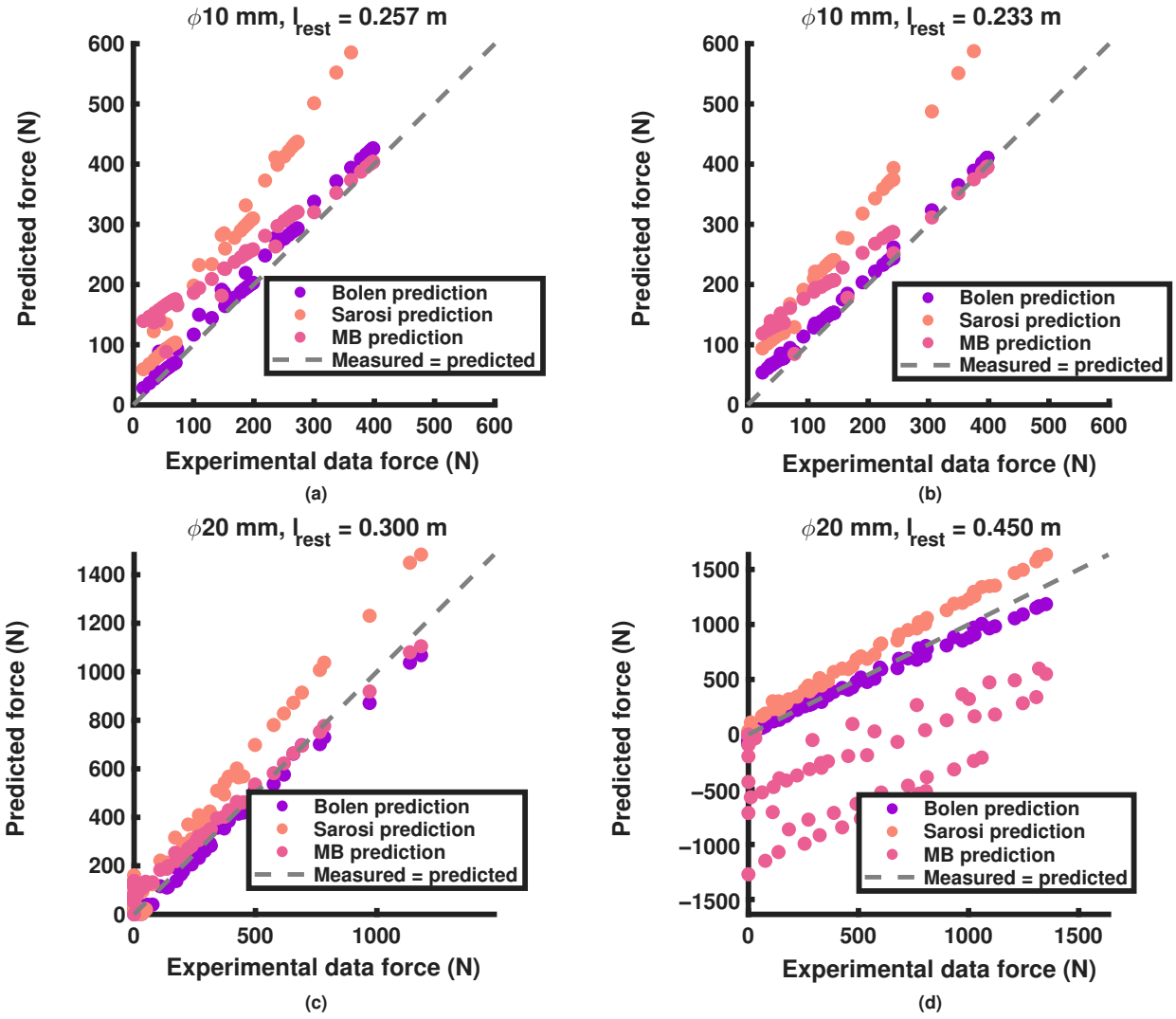


Figure 3.5: Comparison between measured and predicted BPA force at varying amounts of contraction and pressure. A $y = x$ perfect fit line is added for reference. The BPAs compared are: (**a**) ∅10 mm, $l_0 = 257$ mm, (**b**) ∅10 mm, $l_0 = 233$ mm, (**c**) ∅20 mm, $l_0 = 300$ mm, and (**d**) ∅20 mm, $l_0 = 450$ mm.

The Martens and Boblan model was first evaluated at the same pressure–contraction points as in their experimental data [Martens and Boblan, 2017] (Tables A1 and A2),

indicating that it reproduces the general published force map shape but is sensitive to the assumed initial geometry. Examining their ∅20 mm BPA, for example, using the explicitly stated $l_0 = 300\,\text{mm}$ and initial diameter $D_0 = 20\,\text{mm}$ significantly under-predicted the results compared to using the implied $l_0 = 296\,\text{mm}$ ([Martens and Boblan, 2017] (Table A1)) and $D_0 + H_0 = 21.8\,\text{mm}$, where $H_0 = 1.8\,\text{mm}$ is the material thickness. The diameter adjustment can be inferred from [Chou and Hannaford, 1996]. This adjusted diameter was also included for the ∅10 mm BPA to give a better fit. As can be seen in Figure 3.5 and Table 3.3, the highest accuracy is observed when ∅20 mm and $l_0 = 300\,\text{mm}$, which is at or very near their experimental measurements. For the ∅10 mm BPAs the accuracy decreases even though their experimentally measured $l_0 = 250\,\text{mm}$ is close to our $l_0 = 257\,\text{mm}$ and $l_0 = 233\,\text{mm}$. The largest error occurs at ∅20 mm and $l_0 = 450\,\text{mm}$ and the model starts to predict negative force. We have calculated that the $F_{PE} \cdot \frac{dD}{DL}$ term becomes very large compared to the others [Martens and Boblan, 2017] (p. 5). This suggests that the model is sensitive to geometric parameter selection and interpretation, particularly with respect to initial length and diameter definitions.

In comparison, the present normalized model demonstrates improved consistency across actuator lengths by incorporating experimentally derived scaling through F_{620} and ϵ_{620}. This normalization reduces sensitivity to geometric variation and enables a unified representation of actuator behavior, while maintaining predictive accuracy over the tested range of pressures and contractions.

3.3.4 Implications for Real-Time Control and Simulation

Although pneumatic muscle actuators exhibit hysteresis, thermodynamic effects, and other dynamic behavior, accurate and high-performance controller designs can still be achieved using a static force map approach when the underlying static force model is accurate. Martens and Boblan explicitly note that BPA dynamics are often dominated by the air dynamics inside the actuator and the mass flow behavior of the valve, such that precise modeling of the static force characteristic remains crucial for torque control accuracy [Martens and Boblan, 2017]. This view is consistent with more recent BPA control work, including a sensor-less torque control interface for BPA-driven joints [Martens et al., 2018], stable force/position/stiffness control of antagonistic pneumatic-muscle joints [Ugurlu et al., 2019, Hunt et al., 2017b], and sensor-less angle stiffness control of antagonistic BPA systems [Shin and Kogiso, 2022].

In this context, the present model is well suited for real-time use because it reduces the static isometric force prediction to a low-order normalized surface, $F^*(\epsilon^*, P^*)$, scaled by the resting-length-dependent term $F_{620}(l_0)$. This structure is computationally inexpensive to evaluate online and is also suitable for lookup-table implementation. While the present work does not model the full actuator–valve dynamics, it provides the static force relationship needed as a feedforward or inner-model component in real-time torque control, simulation, and preliminary design workflows. It is anticipated that improved real-time control for high-speed and high-precision systems will benefit more from improved modeling of airflow dynamics than from modeling actuator hysteresis, damping, and other dynamic behavior. Characterizing the actuators under dynamic operating conditions,

such as isokinetic (constant-velocity), isobaric, and isotonic (constant-load) contractions, is a natural next step beyond the isometric characterization reported here, for which the pressure–force dynamics model of Elzein et al. [2025] provides the modeling foundation.

3.3.5 Biomimetic Artificial Muscle Analogy

BPA artificial muscles are often said to be analogous to biological muscles because they have force–length curves and can produce force only in tension. This analogy is worth closer inspection. For example, the passive force–length curve has been described as an exponential function arising from connective tissue elements, while the active force–length relationship is typically modeled as a bell-shaped curve reflecting actin–myosin overlap [Winters, 1990]. These formulations emphasize that muscle force generation is governed by nonlinear geometric and material interactions rather than simple linear elasticity. McKibben-type pneumatic artificial muscles (of which Festo is an example) exhibit analogous nonlinear behavior, though arising from fundamentally different physical mechanisms. As shown by Chou and Hannaford, the force–length relationship of McKibben actuators is derived from geometric constraints imposed by the braided shell, leading to a nonlinear dependence of force on actuator length through the braid angle [Chou and Hannaford, 1996]. In their formulation, actuator force is proportional to pressure and a nonlinear geometric term $(3\cos^2\theta - 1)$, which governs both the maximum contraction limit and the curvature of the force–length relationship. The work presented here demonstrates that this nonlinear curve can be well approximated by an exponential as well.

However, the behavior of these artificial muscles differs distinctly from that of biological muscles. For example, the optimal fiber length of a biological muscle, l_{OFL}, is not equivalent to l_0 in an artificial muscle. Artificial muscles produce their maximum active force at their resting length, which is also their maximum length. In contrast, the maximum active force that a biological muscle can produce is somewhere in the middle of the total contractile length of the muscle.

Furthermore, biological muscles can adapt the optimal operating length through structural remodeling (e.g., sarcomere addition or tendon compliance) while BPAs have a fixed geometric configuration. This allows biological systems to self-optimize for the particular tasks the animal is subjected to. In order to re-optimize a robot for new tasks, new actuators would have to be swapped into the system. This distinction highlights a key limitation of BPAs in biomimetic applications, while also reinforcing the importance of accurate nonlinear modeling for control and design.

The analogy also suggests two design observations. First, the force–length penalty is shared by both families: force falls as the actuator operates away from its optimum, so a limb design must route each actuator such that its highest-force operating range coincides with the joint angles where torque is needed most, because the force lost at the extremes of the range of motion cannot be recovered by pressure alone. Second, the two families gain moment arm in different ways. Skeletal muscle attaches to the skeleton itself, through tendons or directly over broad areas of bone, so the moment arms of large animals follow from skeletal geometry. The robot's BPAs instead act through brackets that protrude from the limb segments; that hardware is mechanically exposed, and its series compliance is one of the loss mechanisms quantified in Chapter 4.

The analogy finally raises a question that this dissertation can pose but not settle. Every muscle-like actuator loses force as it operates away from its optimal length, and a soft actuator working against soft structure has no rigid skeleton from which to gain a moment arm. Given those two facts, it may be no accident that nearly all large land animals are built on rigid skeletons and that large soft-bodied land animals are essentially absent. Whether the force–length penalty contributes to that absence, or whether it reflects other constraints such as support and protection, is a question for comparative biomechanics rather than for a single actuator study. For the designer, the lesson holds regardless of the answer: some moment arm must come from somewhere, and on this robot it comes from protruding brackets whose behavior must be modeled rather than ignored.

Table 3.3: BPA force-model comparison. Diameter and resting length are in mm; RMSE and maximum absolute error are in N. FVU is dimensionless.

Diameter	Length	Model	RMSE	FVU	Max. error
10 mm	257	Bolen	22.9	0.0353	46.0
10 mm	257	S'arosi	143.8	1.3903	235.8
10 mm	257	Martens and Boblan	73.8	0.3668	122.9
10 mm	233	Bolen	16.8	0.0187	30.3
10 mm	233	S'arosi	132.7	1.1655	222.6
10 mm	233	Martens and Boblan	62.6	0.2590	101.2
20 mm	300	Bolen	35.6	0.0144	114.0
20 mm	300	S'arosi	132.1	0.1992	313.4
20 mm	300	Martens and Boblan	72.8	0.0605	137.8
20 mm	450	Bolen	67.6	0.0258	166.6
20 mm	450	S'arosi	177.7	0.1785	294.8
20 mm	450	Martens and Boblan	862.1	4.2022	1321.1

4 Isometric Knee Torque

Publication note: This chapter is the subject of an unpublished manuscript in preparation by Bolen, Burgess, Morrow, Elzein, Pang, Scharzenberger, and Hunt [Bolen et al., 2025]. For that work, B.B. performed the conceptualization, methodology, software, validation, formal analysis, investigation, data curation, visualization, and original draft preparation; M.E., L.P., L.B., and C.M. contributed methodology, software, investigation, and data curation; and A.H. supervised the work, administered the project, and acquired funding. All co-authors contributed to writing, review, and editing.

This chapter connects the actuator force characterization of Chapter 3 to torque about the robot knee. The earlier torque model assumed inextensible muscle paths on rigid links. Isometric tests on pinned and biomimetic knees show where those assumptions fail. The methods develop corrections for length offsets, attachment compliance, and wrapping loss, then explain how actuator-placement searches use the corrected prediction. Held-out tests evaluate the correction terms, and physical tests assess whether the redesigned routes meet selected human-muscle torque targets.

4.1 Methods

4.1.1 Robot Leg Model and Knee Test Stands

The primary components of the robot leg are the knee joint, femur, tibia, BPA assemblies, and artificial tendons (Fig. 4.1**A–C**). The artificial bone components are 3D printed using

a combination of Onyx, carbon fiber, and high-strength high-temperature fiberglass on Markforged Onyx One and Mark Two printers, as well as machined links for the knee mechanism. Onyx is a proprietary Markforged material that consists of chopped carbon fiber in nylon. Artificial tendons are made with Shimano bicycle brake cable.

Figure 4.1 shows the test apparatus. Panel **(A)** presents a CAD solid model assembly of the revolute joint knee in the test stand. **(B)** 1-DoF pinned-joint robot knee in the test apparatus with important components labeled. Leg is shown with 30 degrees flexion, i.e. $\theta_{\mathbf{k}} = -30^{\circ}$. **(C)** Picture from isometric torque testing. Setup shows the revolute knee configured for a test of the extensor BPA. The knee is flexed with $\theta_{\mathbf{k}} = -120^{\circ}$. Also note the compressed shape of the inflated BPA on the knee during this high degree of flexion.

The leg assembly geometry and dimensions are based on bones scans of a 6 ft (1.83 m) tall person [Bolen and Hunt, 2019, Seth et al., 2018, Delp et al., 1990, 2007]. Two knee joints were tested, one pure rotational joint, and one biomimetic joint with rotation and translation. Each joint is driven by two antagonistic Festo actuators. Both robot joints have the same center of rotation in the femur frame at $\theta_{\mathbf{k}} = 0\,\text{deg}$.

Details on the biomimetic robot knee used in this study can be found in [Steele, 2018, Steele et al., 2017, 2018]. In brief, this 1-DoF joint (Fig. 4.2) uses a four-bar linkage to change the effective moment arm during rotation. The intersection point of the internal crossed links define the instantaneous center of rotation (ICR). This causes the joint ICR location in the femur frame to translate posteriorly in the X direction and distally in the Y direction except for a brief proximal move near full flexion. Reference International Society of Biomechanics (ISB) coordinate frames are given by Wu and Cavanagh [1995], Wu et al. [2002]. This is similar to how the human knee behaves [Bolen and Hunt, 2019,

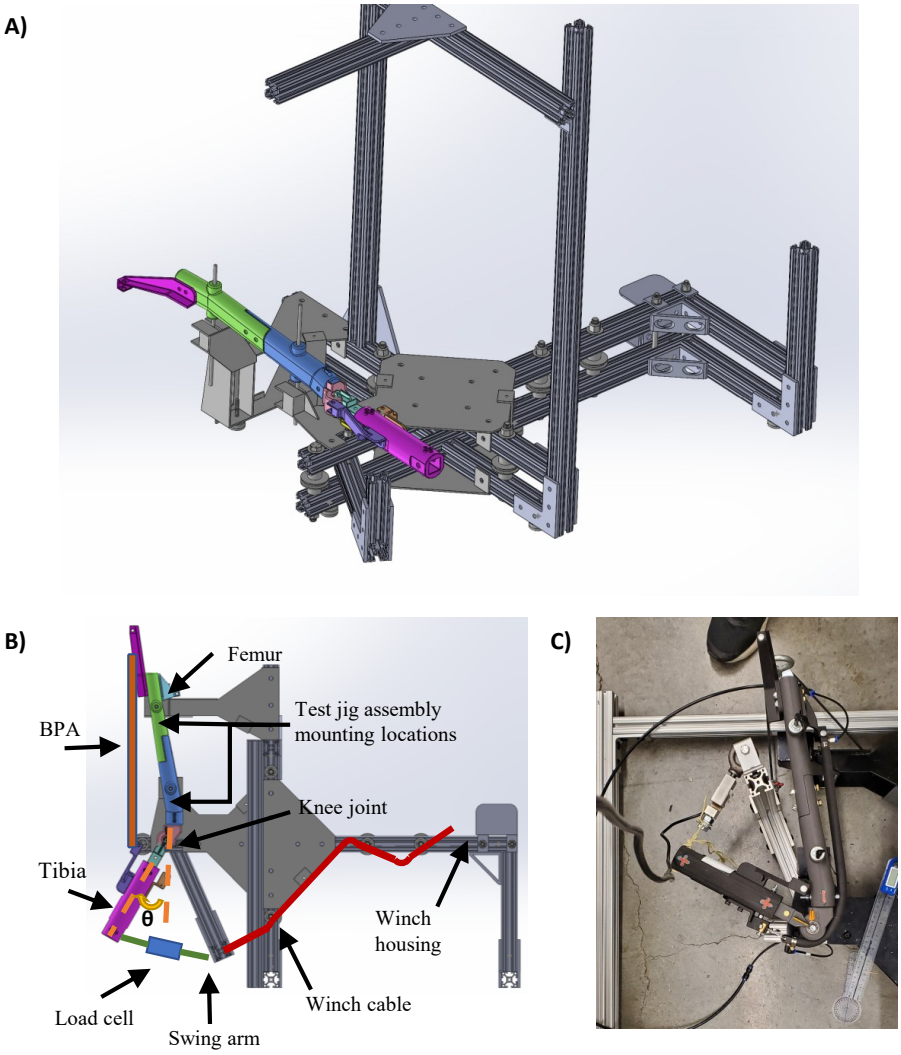


Figure 4.1: Pinned-knee test apparatus: CAD assembly (A), labeled mechanism at -30° (B), and extensor isometric test at -120° (C).

Morrow et al., 2020, Seth et al., 2018, Delp et al., 2007, 1990]. The correct location of the ICR is important for both theoretical and experimental torque calculations.

The measured points in Figure 4.2 use 15° increments from -150° to -15°, 5° increments from -15° to 0°, and additional measurements at 2°, 5°, and 10°.

In the pinned knee joint, uniarticular knee extensor and knee flexor are used to actuate the joint. Each knee joint type was tested with a single ∅10 mm Festo BPA per muscle. The biomimetic knee was additionally tested with ∅20 mm Festo BPAs in a paired configuration: the extensor with two BPAs in parallel, and the flexor with two BPAs whose pathways are mirrored about the Z axis. Each artificial muscle is pinned at the muscle origin location on the femur while the other end is either pinned or attached to the muscle insertion location on the tibia via an artificial tendon.

Torque about the knee is generated by the moment of the BPA force about the joint's instantaneous center of rotation (ICR). Two vectors enter that moment: the position vector $\mathbf{d}$, drawn from the ICR to the point where the actuator force is applied, and the force vector $\mathbf{F} = \|\mathbf{F}\|\hat{\mathbf{u}}$, whose direction $\hat{\mathbf{u}}$ is the line of action of the actuator. The moment of the force about the joint is $\mathbf{M} = \mathbf{d} \times \mathbf{F}$, and the signed knee-axis torque is $M_{\mathbf{k}} = \hat{\mathbf{z}}_{\mathbf{k}} \cdot (\mathbf{d} \times \mathbf{F})$. For force acting in the plane normal to the knee axis, the moment-arm magnitude is

$$r = \|\mathbf{d} \times \hat{\mathbf{u}}\| \tag{4.1}$$

This planar moment arm is a scalar distance and can be computed from the vectors or measured directly. For an out-of-plane force, the knee-axis moment must instead use the axis projection above; the magnitude of the full three-dimensional moment is not generally

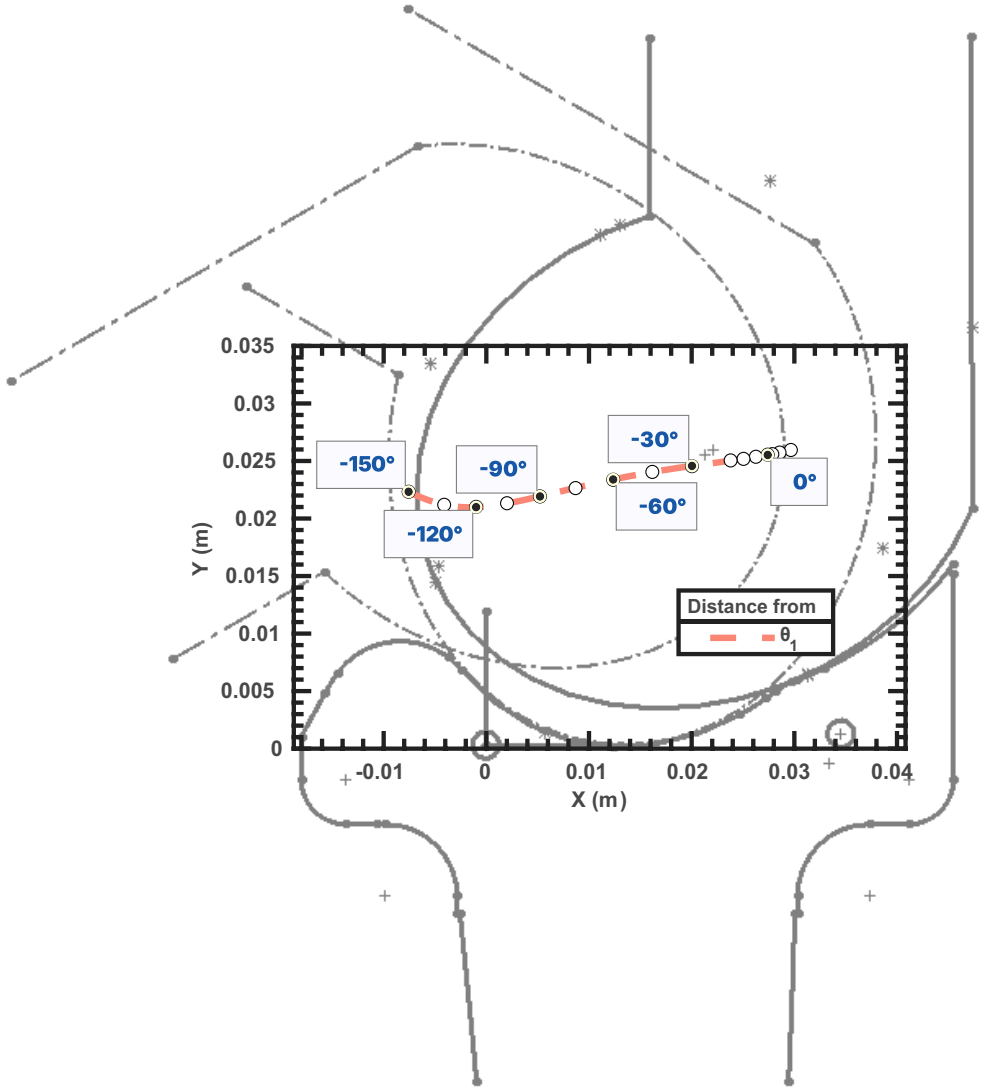


Figure 4.2: Biomimetic-knee instantaneous center of rotation relative to the tibial frame. White circles are measured points; knee outlines show 0° (solid), -60°, and -120° (dash-dot).

the knee torque. In this study it is calculated using the method specified by Young et al. [2019]. The scalar moment arm $r_{\mathbf{k}}$ about the joint axis $\hat{\mathbf{z}}_{\mathbf{k}}$ (i.e., the Z axis in the knee frame) is calculated as

$$r_{\mathbf{k}} = \mathbf{p}_{proj,xy} \cdot \frac{\mathbf{p}_f \times \hat{\mathbf{z}}_{\mathbf{k}}}{\|\mathbf{p}_f \times \hat{\mathbf{z}}_{\mathbf{k}}\|} \tag{4.2}$$

A muscle path may be modeled as a single segment running from origin to insertion, or as a chain of segments joined at via points. The segment that crosses from one reference frame into the other, here from the femur frame to the tibia frame, is the free muscle segment: because its endpoints ride on different bodies, its length and its direction respond directly to joint rotation, whereas a segment contained within one body's frame rotates rigidly with that body and changes neither. Here $\mathbf{p}_f$ is the force-path segment vector, and $\mathbf{p}_{proj,xy}$ locates a point on that line relative to the joint, projected onto the plane normal to the joint axis. The normalized cross product in Equation (4.2) is the in-plane normal to the force path. Its dot product with the projected position gives the signed moment arm of the projected force path (Figure 2.6). Multiplying this arm by the in-plane force magnitude gives the knee-axis torque; the full wrench calculation retains all three force components. Moment arm is calculated in the tibial body frame.

Details of the existing model and the torque calculation methodology in the modeled robot are given by Bolen and Hunt [2019], Morrow et al. [2020].

4.1.2 Knee Torque Data Collection and Equipment

A test jig was built to collect knee torque measurements at different knee angles over its range of motion. The test stand frame is made predominantly out of 80/20 1515 series extruded aluminum components. The knee joint is free to rotate the tibia while the femur is fixed to the frame. A force sensor has one end connected to the tibia and the other is connected to a swing arm. The swing arm is tied to a winch with 3/16 inch Kevlar rope from Quality Nylon Rope routed over intermediate idler pulleys (see Fig. 4.1**B** and 4.1**C**). At each measured position of $\theta_{\mathbf{k}}$ the winch would be locked and the BPA would be inflated and then deflated. This test procedure kept the measured results as isometric as possible. The knee joint is co-axial with gravity to eliminate gravitational torque (see Fig. 4.1**A** and Fig. 4.1**C**).

Force data was collected using one of two different sensors. The first is a MODERN STEP 300 kg digital crane scale. The second force sensor is a CALT DYLY-103 100 kg S shaped load cell. The load cell is used in conjunction with a HX711 Load Cell Amplifier. Pressure is measured with a Freescale MPX5700 GP 5 V pressure sensor. Building air supply pressure is controlled with two pressure regulators in series. The first is a Parker model 20R113GC $0 - 120$ psi pressure regulator. The second is a Husky 3/8 inch High Performance Air Regulator HDA72200. A Festo VTUG-10-MRCR-S1T-26V20-T516LA-UL-T532S-8K valve manifold was used to supply air to the actuator. Pressure and load cell amplifier data are sent to MATLAB via an Arduino Uno style Sparkfun BlackBoard C microcontroller.

The length measurements were made using a tape measure. When there was sufficient

curvature in the BPA during knee extensor torque tests, a flexible piece of string was used to mark the axial length and then transferred to the tape measure. Angle measurements were taken with a Medigauge digital electronic goniometer or with MALENOO analog goniometers of 6, 8, or 12 inch lengths.

4.1.3 Hybrid Torque Calculation Method

A Hybrid Torque calculation method was developed to help uncover inconsistencies between the model and recorded torque values. This method measures the actual muscle length and angles of the robot to calculate force and joint torque. With this method, it is easier to determine which parts of the torque equation did not achieve their predicted value and provide a correction to that part of the equation. The classical mechanics way to calculate the torque about a joint is to take the cross product of the position vector **d** and the force vector **F**, both defined in Section 4.1.1,

$$\mathbf{M} = \mathbf{d} \times \mathbf{F} \tag{4.3}$$

Knee torque is the scalar component of this moment about the joint axis, written $M_{\mathbf{k}}$ to make its axis explicit. For the planar hybrid calculation, its magnitude is the measured perpendicular arm multiplied by the actuator-force magnitude,

$$|M_{\mathbf{k}}| = r\,F \tag{4.4}$$

where r is the scalar moment arm of Eq. (4.1), here measured directly with a tape

measure as the perpendicular distance from the knee axis to the actuator's line of action, and F is the scalar force magnitude, calculated by measuring the BPA's P, l_0, l_{620} and l_{m} values and using the model fit equations determined in the following section. The hybrid method therefore replaces the model-predicted moment arm $r_{\mathbf{k}}$ of Eq. (4.2) and the model-predicted muscle length with their measured counterparts, leaving the force model as the only predicted quantity. The flexor and extensor signs are then assigned using the adopted knee convention.

4.1.4 Improved Torque Model

The classical kinematic torque model assumed rigid links and inextensible muscle paths. Data collected during the first flexor experiments (Section 4.2) showed that this assumption does not hold for a pneumatic system: measured muscle lengths were consistently shorter than predicted, and measured torque deviated from prediction in a load-dependent manner. This section describes the corrections that were developed to account for a constant length offset, series compliance in the attachment hardware and artificial tendons, and the loss of usable muscle length when a BPA wraps around the joint.

The correction terms represent the combined effects of the attachment brackets, fixtures, artificial tendons, and test-stand loading path. They are identified from measured torque because the torque prediction requires their aggregate effect on actuator length, rather than a separate structural model of every component. The constant term χ_0 corrects the unloaded length discrepancy, χ_1 and χ_2 describe effective stiffness in the local bracket frame, and χ_3 scales the wrapping-length loss. The compliance calculation couples force

and displacement: force deflects the attachments, and the resulting change in muscle length changes force. The multiobjective procedure of Section 4.1.4 evaluates candidate terms on fitting and validation tests.

Length Offset and System Stiffness

Analysis of measured muscle length data shows that it was consistently shorter than the predicted muscle length at all joint angles. This shorter length results in less muscle force, a shifted RoM, and less torque in the experimental setup. That this shorter length occurs when $\varepsilon = 1$ and $F = 0$ indicates that a constant length offset should be used. Therefore, to produce an improved torque prediction model, l_{m} was recalculated as

$$l_{\mathrm{m}} = l_{LMT} - 2\, l_{fitting} - \chi_0 \tag{4.5}$$

where l_{LMT} is the path length of the musculotendon from origin to insertion point, $l_{fitting}$ is the length of the end caps, and χ_0 is a constant length offset.

Additionally, as the muscles produce more force, compliant components in the limb segments and attachment hardware deform. This deformation changes the effective braided pneumatic actuator (BPA) length, which alters contraction and therefore changes the force predicted by the BPA model. This effect is most apparent at high forces and low contractions. The purpose of the compliance model developed next is predictive, not structural: it exists so that the torque calculation can correct the actuator length for load-dependent deformation, and it makes no claim about the stiffness of any one component in isolation. Accordingly, the stiffness values that populate the model, χ_1 and χ_2, are not

measured on the hardware directly; they are identified by the optimization of Section 4.1.4 from the measured torque, and every stiffness value quoted later in this chapter is an optimizer output. The combined hardware compliance is approximated using Hooke's law in the bracket's principal directions.

A local bracket frame was established at the attachment bracket. The $\hat{\mathbf{x}}_{br}$ axis was aligned with the vector from the bracket reference point to the BPA attachment point and represents the stiffest direction of the bracket. The $\hat{\mathbf{y}}_{br}$ axis was chosen to point in the posterior direction with respect to the tibia frame, corresponding to the primary cantilever bending direction (see Fig. 4.4), while the $\hat{\mathbf{z}}_{br}$ axis completes the right-handed frame. In this frame the bracket is modeled as three independent linear springs along the principal axes, with stiffnesses k_x, k_y, and k_z in the x_{br}, y_{br}, and z_{br} directions.

For the two-rotation evaluator, the bracket frame was constructed from the vector joining the bracket origin to attachment point $\mathbf{p}_A$ or $\mathbf{p}_B$. The first rotation, through θ_z about the fixed space-frame axis $\hat{\mathbf{z}}_s$, points the intermediate axis $\hat{\mathbf{x}}_1$ toward the sagittal-plane projection, $\mathbf{p}_A^{\text{sag}}$ for the origin point $\mathbf{p}_A$ and $\mathbf{p}_B^{\text{sag}}$ for the insertion point $\mathbf{p}_B$. The second rotation, through $-\theta_y$ about the current axis $\hat{\mathbf{y}}_1 = \hat{\mathbf{y}}_{br}$, brings the final bracket axis $\hat{\mathbf{x}}_{br}$ into alignment with the full three-dimensional point vector. Figure 4.3 shows these geometric targets. The resulting orientation and homogeneous transform are

$$\mathbf{R}_{sb} = \mathbf{R}_z(\theta_z)\mathbf{R}_y(\theta_y)^T, \qquad \mathbf{T}_{sb} = \begin{bmatrix} \mathbf{R}_{sb} & \mathbf{p}_{br} \\ \mathbf{0}^T & 1 \end{bmatrix}. \tag{4.6}$$

$$\mathbf{K}_{br} = \operatorname{diag}\big(k_x,\ k_y,\ k_z\big) \tag{4.7}$$

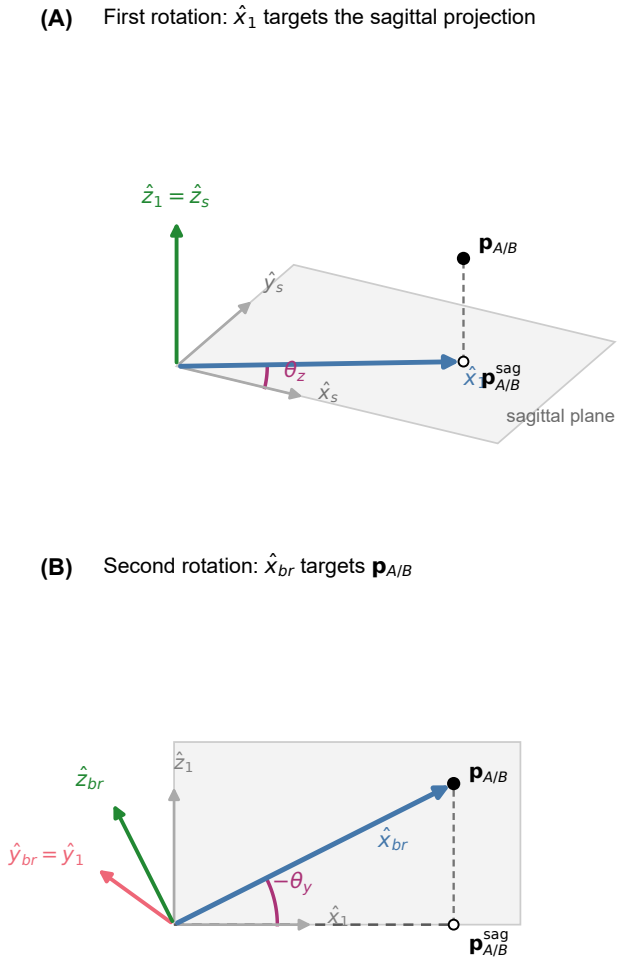


Figure 4.3: Bracket-frame construction from attachment point $\mathbf{p}_A$ or $\mathbf{p}_B$: yaw about fixed $\hat{\mathbf{z}}_s$ aligns the sagittal projection (A); pitch about current $\hat{\mathbf{y}}_{br}$ aligns the full point vector (B).

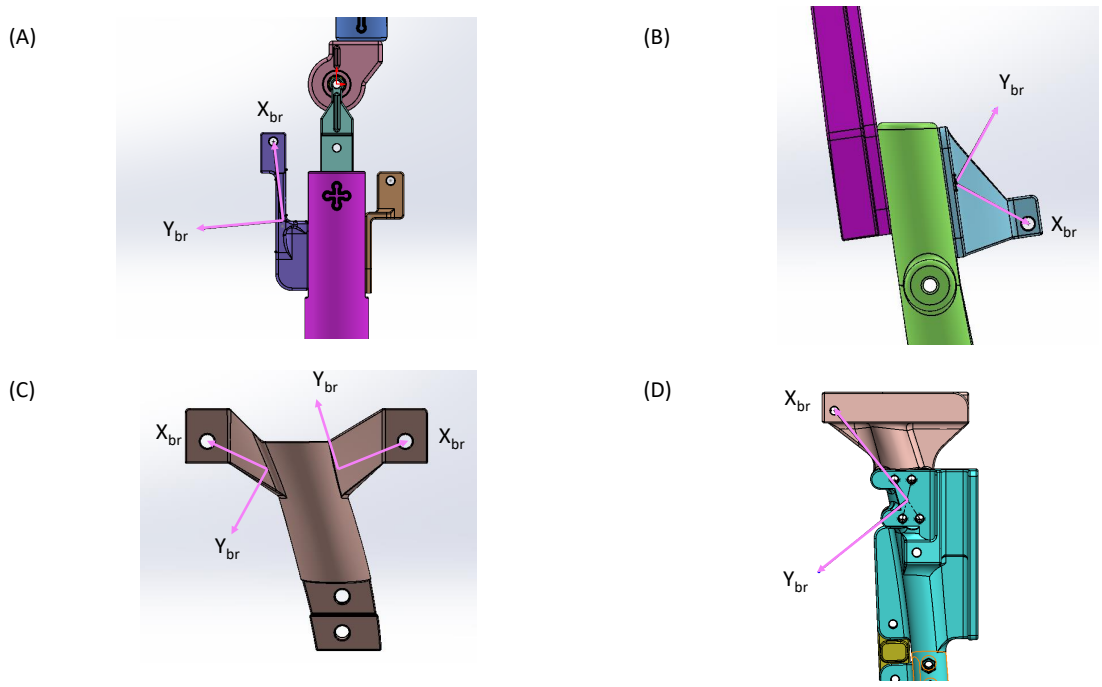


Figure 4.4: Bracket frame orientations for: **(A)** flexor configuration on the pinned knee, **(B)** extensor configuration on the pinned knee, **(C)** flexor and extensor configurations on the biomimetic knee with ∅10 mm BPAs, and **(D)** flexor configuration on the biomimetic knee with the ∅20 mm BPA.

The principal stiffnesses are parameterized by the two fitted values χ_1 and χ_2, with axis assignments fixed by configuration (Table C6). For the pinned-flexor insertion bracket, the assignment is $k_x = k_z = \chi_1$ and $k_y = \chi_2$. This y-symmetric array is invariant to a change of frame orientation about $\hat{\mathbf{y}}_{br}$, so the one- and two-rotation constructions give the same compliance for this assignment. That invariance does not hold for the other assignments. In particular, the pinned-flexor origin bracket uses $[\chi_1, \chi_1, \chi_2]$, the pinned extensor and biomimetic flexor use $[\chi_1, \chi_2, \chi_2]$, and the biomimetic extensor uses the insertion-bracket assignment. Both pinned-flexor brackets use the one-rotation frame in the evaluator of record; the pinned extensor uses the two-rotation frame. The general projected-compliance calculation below applies to each assignment, with the appropriate local matrix. For the insertion bracket,

$$\mathbf{K}_{br} = \begin{bmatrix} \chi_1 & 0 & 0 \\ 0 & \chi_2 & 0 \\ 0 & 0 & \chi_1 \end{bmatrix} \tag{4.8}$$

The corresponding local compliance matrix is

$$\mathbf{C}_{br} = \mathbf{K}_{br}^{-1} \tag{4.9}$$

The force applied to the bracket was expressed in the bracket frame as $\mathbf{F}_{br}$. This force vector can be written in terms of its magnitude and unit direction as

$$\mathbf{F}_{br} = F\,\hat{\mathbf{u}} \tag{4.10}$$

where $F = \|\mathbf{F}\|$ is the scalar force magnitude of Eq. (4.4) and $\hat{\mathbf{u}} = [u_x, u_y, u_z]^T$ is the BPA force direction expressed in the bracket frame. The bracket deflection vector, $\boldsymbol{\epsilon}_{br}$, was calculated from Hooke's law as

$$\mathbf{F}_{br} = \mathbf{K}_{br}\, \boldsymbol{\epsilon}_{br} \tag{4.11}$$

Solving Eq. 4.11 for the bracket deflection gives

$$\boldsymbol{\epsilon}_{br} = \mathbf{K}_{br}^{-1}\, \mathbf{F}_{br} = \mathbf{C}_{br}\, \mathbf{F}_{br} \tag{4.12}$$

Substituting Eq. 4.10 into Eq. 4.12 gives

$$\boldsymbol{\epsilon}_{br} = F\, \mathbf{K}_{br}^{-1}\, \hat{\mathbf{u}} \tag{4.13}$$

For the diagonal stiffness matrix in Eq. 4.8, Eq. 4.12 is equivalent to the component-wise expression

$$\boldsymbol{\epsilon}_{br} = \begin{bmatrix} F\, u_x / \chi_1 \\ F\, u_y / \chi_2 \\ F\, u_z / \chi_1 \end{bmatrix} \tag{4.14}$$

Although $\boldsymbol{\epsilon}_{br}$ is a three-dimensional deflection vector, the BPA force model depends on the scalar change in effective BPA length (δ). Therefore, only the component of $\boldsymbol{\epsilon}_{br}$ along the BPA line of action directly changes the effective BPA length. This scalar bracket deflection is the projection of $\boldsymbol{\epsilon}_{br}$ onto $\hat{\mathbf{u}}$:

$$\delta_{br} = \hat{\mathbf{u}}^T \boldsymbol{\epsilon}_{br} \tag{4.15}$$

Substituting Eq. 4.12 into Eq. 4.15 gives

$$\delta_{br} = \hat{\mathbf{u}}^T \mathbf{K}_{br}^{-1} \mathbf{F}_{br} \tag{4.16}$$

Using Eq. 4.10, Eq. 4.16 becomes

$$\delta_{br} = F\, \hat{\mathbf{u}}^T \mathbf{K}_{br}^{-1} \hat{\mathbf{u}} \tag{4.17}$$

The scalar stiffness along the BPA line of action due to the bracket compliance is therefore

$$k_{\hat{u},br} = \frac{F}{\delta_{br}} = \left(\hat{\mathbf{u}}^T \mathbf{K}_{br}^{-1} \hat{\mathbf{u}} \right)^{-1} \tag{4.18}$$

This projected-compliance form is used because the force is applied along $\hat{\mathbf{u}}$, while the bracket is allowed to deflect independently along its local $\hat{\mathbf{x}}_{br}$, $\hat{\mathbf{y}}_{br}$, and $\hat{\mathbf{z}}_{br}$ directions. The reduced-order Hooke's law expression for the bracket deflection along the BPA line of action is therefore

$$F = k_{eq}\, \delta \tag{4.19}$$

Here k_{eq} is the equivalent stiffness of every compliant element in the actuator path, and the bracket-only case is the special case with no tendon. When the BPA attaches directly to the bracket, k_{eq} excludes the tendon and equals $k_{\hat{u},br}$ of Eq. (4.18). When an artificial

tendon was used, it was modeled as an axial spring with stiffness k_{ten}: bracket deflection and tendon stretch both shorten the effective BPA length along the actuator line of action, the two elements carry the same force, and their scalar compliances add in series,

$$k_{eq} = \left(\frac{1}{k_{\hat{u},br}} + \frac{1}{k_{ten}} \right)^{-1} \tag{4.20}$$

This series reduction is exact only under a specific set of conditions, each of which corresponds to a way the physical hardware could violate it. The bracket and the tendon must carry the same force, which holds for two massless elements coupled in series under quasi-static loading; an additional compliant coupling would require its own term in the series compliance. Their displacements must be kinematically additive along the line of action, with no displacement lost to off-axis geometry. Within a single iteration of the force balance the line of action $\hat{\mathbf{u}}$ is treated as fixed, which is a local approximation: once the balance returns the compliance displacement, the calculation updates the attachment locations with the deflected bracket positions and re-evaluates $\hat{\mathbf{u}}$ and the moment arm at the deflected geometry, as described below. The elements are assumed to carry no moment, so rotational compliance and force–moment coupling at the attachments are neglected. Both elements are linear, with stiffness independent of deflection, and the tendon is modeled without preload, so its tangent stiffness does not stiffen with load state. Finally, compliance deflection shortens the actuator, as Eq. (4.21) assumes, because a BPA produces force only in tension: under its own pulling force the compliant attachments give way in the direction that relieves that tension. A deflection that would lengthen the actuator can be produced by an antagonistic pull, but that loading lies outside this

treatment.

Since δ is the scalar compliance displacement along the BPA line of action, then the compliance-corrected active BPA length used in the force balance is

$$l_m(\delta) = l_m(0) - \delta \tag{4.21}$$

where $l_m(0)$ is the initial muscle length calculated from (4.5).

For a specified set of correction parameters, the following calculation solves the attachment-compliance displacement at each knee angle; it is distinct from the optimization that identifies those parameters from experimental torque data (Section 4.1.4). The corresponding relative contraction is $\varepsilon^*(\delta)$. Since the BPA force depends on contraction, and contraction depends on the compliance displacement, δ is the root of the nonlinear force-balance equation

$$F_{\max} \cdot F_{\mathrm{BPA}}(\varepsilon^*(\delta), P^*) - k_{eq}\,\delta = 0 \tag{4.22}$$

The root is found with a bracketed scalar solver (`fzero` in MATLAB) on the physically admissible interval between zero deflection and the deflection that would bring the actuator to its maximum usable contraction. The evaluator supplies no custom solver options, so termination uses the default displacement tolerance. This numerical tolerance does not establish the accuracy of the physical model. Two safeguards act as secondary stop conditions. If the actuator already sits at or beyond its maximum usable contraction at zero deflection, no root is sought and δ is set to zero; if the residual shows no sign change across the interval, the solver likewise returns $\delta = 0$ rather than a spurious root.

After solving Eq. 4.22, the corresponding force magnitude was used in Eq. 4.12 to compute the vector bracket deflection. This deflection was transformed back into the limb reference frame and used to update the BPA attachment location. The routed BPA length, force direction, moment arm, and torque were then recalculated using the updated attachment location. In the present implementation, the optimized stiffness terms χ_1 and χ_2 were treated as effective compliance parameters for the attachment system rather than direct measurements of an isolated bracket. This allowed unmodeled compliance in the surrounding hardware to be represented by the reduced-order stiffness model.

The actuator path can carry compliant hardware at both ends: the insertion-side bracket of the preceding development together with a second, origin-side bracket. Figure 4.5 distinguishes the three-dimensional displacement of each bracket from the scalar displacement along the BPA force path. For bracket j, the displacement resolves into components $\eta_{1,j}$, $\eta_{2,j}$, and $\eta_{3,j}$ along the local bracket-frame basis, whereas only its projection $\delta_{br,j}$ along the corresponding force direction $\hat{\mathbf{u}}_j$ changes the actuator length. The two bracket projections act in series with the tendon elongation δ_{tendon}:

$$\begin{aligned} \boldsymbol{\eta}_{br,j} &= \sum_{i=1}^{3} \eta_{i,j} \hat{\mathbf{e}}_i^{(j)}, \\ \delta_{br,j} &= \hat{\mathbf{u}}_j^T \boldsymbol{\eta}_{br,j}, \qquad j \in \{1,2\}, \\ \delta_{\text{series}} &= \delta_{br,1} + \delta_{\text{tendon}} + \delta_{br,2}. \end{aligned} \tag{4.23}$$

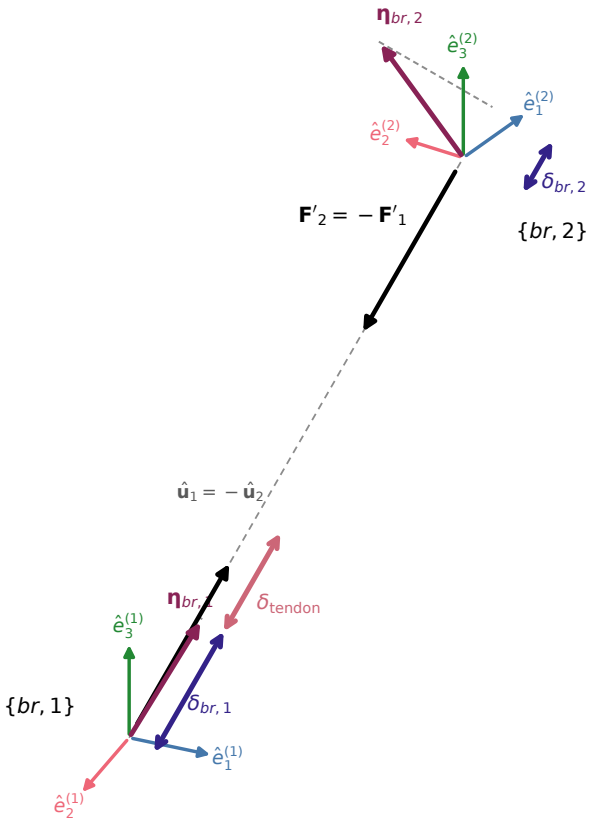


Figure 4.5: Projection of bracket displacement onto the actuator force path. Each bracket has a local basis $\hat{\mathbf{e}}_1^{(j)}$, $\hat{\mathbf{e}}_2^{(j)}$, and $\hat{\mathbf{e}}_3^{(j)}$. The displacement $\boldsymbol{\eta}_{br,j}$ has an axial projection $\delta_{br,j}$ along $\hat{\mathbf{u}}_j$; the two bracket projections and tendon elongation contribute to the change in actuator length. Displacements are exaggerated.

BPA Fictive Contraction

As the knee flexes, the extensor BPA contacts the anterior joint surface and bends around it (Figure 4.6). The rigid routed-length calculation does not account for the resulting change in usable contraction. I represent this effect with fictive strain: an empirical increase in the contraction seen by the force model, beyond the actuator's geometric contraction. The correction is phenomenological; it does not resolve wall stresses or the deformation of the braided actuator. A wrapping-loss term χ_3 scales the associated reduction in usable length:

$$\Delta l = \begin{cases} \chi_3 \; R \; |\theta_{\text{wrap}} - \theta_{\mathbf{k}}| \; \zeta^2 & \theta_{\mathbf{k}} \leqslant \theta_{\text{wrap}} \\ 0 & \theta_{\mathbf{k}} \geqslant \theta_{\text{wrap}} \end{cases} \tag{4.24}$$

where Δl is the change in usable length of the BPA, R is the wrapping radius, $|\theta_{\text{wrap}} - \theta_{\mathbf{k}}|$ is the angle through which the BPA is bent once contact begins, and θ_{wrap} is the knee flexion angle at which the BPA starts to bend over the knee joint, determined from solid modeling. ζ is the additive complement to relative strain, such that

$$\zeta = 1 - \varepsilon^* \tag{4.25}$$

so the fitted wrapping loss grows quadratically toward resting length and vanishes at the normalized free-contraction limit. Combining (4.25) into (4.24), and subtracting Δl from (4.5) yields

$$l_{\text{m}} = l_{LMT} - 2 \, l_{fitting} - \chi_0 - \Delta l \tag{4.26}$$

Figure 4.6 shows a sagittal schematic of an extensor BPA crossing a flexed robot knee, with its braided body following an anterior contact arc between the proximal attachment and distal tendon. Dashed radii bound the contact arc; R is the BPA-path wrapping radius, and $\alpha = |\theta_{\text{wrap}} - \theta_{\mathbf{k}}|$ is the bend angle after contact begins, expressed in radians in Equation (4.24). The drawing illustrates the local circular-wrap approximation, not the full multi-contact route. The fitted usable-length loss Δl is not an arc-minus-chord length.

Identification of the Torque-Model Correction Terms

I identified the correction terms in two stages using `gamultiobj` in MATLAB. The flexor stage fits (χ_0, χ_1, χ_2) with compliance at both attachments. The adopted fold trains on the $l_0 = 48.5$ cm and 41.0 cm tests and validates on the 45.7 cm and 40.6 cm tendon tests. The extensor stage locks (χ_1, χ_2) to that same flexor-front member and fits only (χ_0, χ_3). Its adopted fold trains on the 43.6 cm, 43.2 cm-tendon, and 45.7 cm tests and validates on the 40.0 cm plain, 40.0 cm-tendon, and 46.5 cm tests. The extensor length offset is allowed to be negative. Appendix C records the test constants, bounds, stiffness-axis assignments, and adopted values in Tables C1, C4, C7, C6, and 4.9.

The effective stiffnesses combine deformation in the brackets, fixtures, and loading apparatus, so beam theory alone does not determine them. I therefore identify them from measured torque curves; numerical optimization accommodates the coupled force–length and attachment-displacement calculation.

Each candidate is evaluated using RMSE, FVU, and maximum absolute residual. These metrics jointly measure torque error; none isolates a unique source such as level offset,

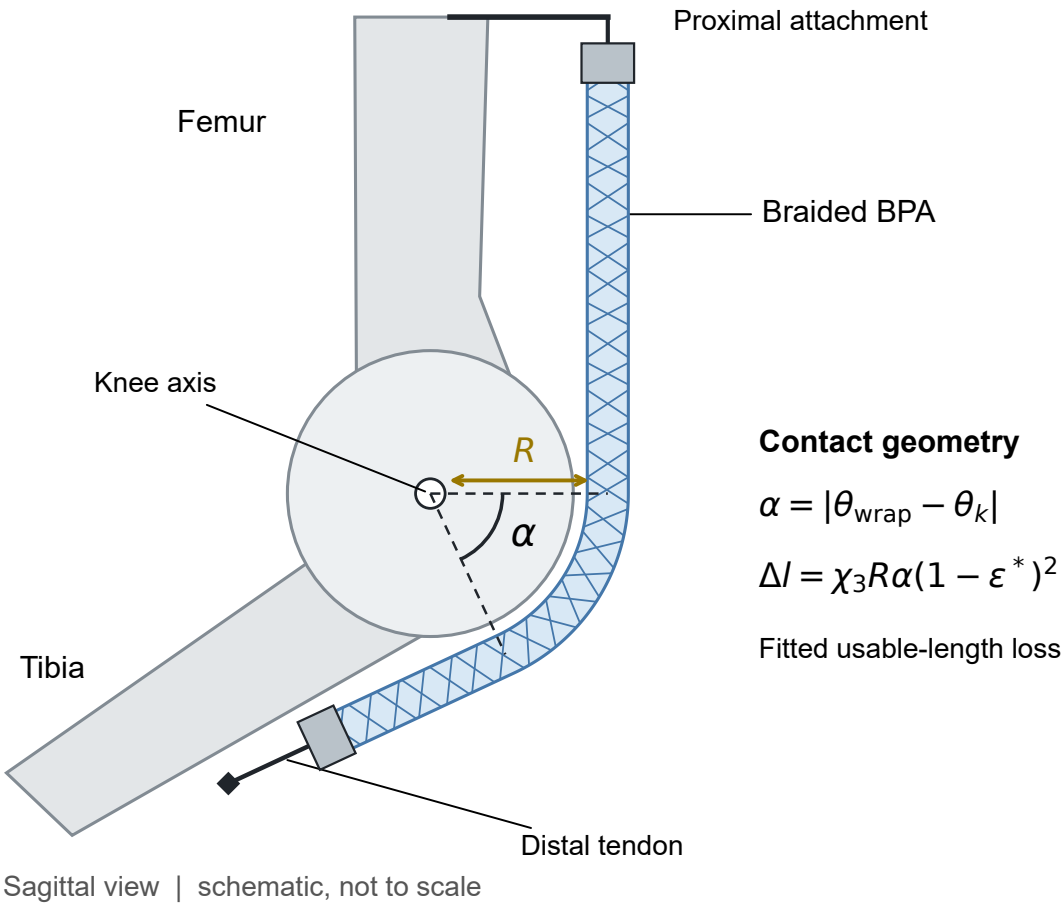


Figure 4.6: Extensor BPA wrapping around a flexed knee. Dashed radii bound the local contact arc, with wrapping radius R and bend angle α.

waveform shape, or angular misalignment.

The objective vector of the identification is defined explicitly. For a candidate correction vector $\boldsymbol{\chi}$, (χ_0, χ_1, χ_2) in the flexor stage and (χ_0, χ_3) on the locked stiffness pair in the extensor refit, the predicted torque curve of test i is interpolated to its n_i measured knee angles θ_{ij}, giving residuals $e_{ij} = M_{ij} - \hat{M}(\theta_{ij}; \boldsymbol{\chi})$ against the measured torque M_{ij}, and the three goodness-of-fit components of that test are

$$\mathrm{RMSE}_i = \sqrt{\frac{1}{n_i}\sum_j e_{ij}^2}, \qquad \mathrm{FVU}_i = \frac{n_i - 1}{n_i} \cdot \frac{\sum_j e_{ij}^2}{\sum_j (M_{ij} - \bar{M}_i)^2}, \qquad e_i^{\max} = \max_j |e_{ij}|. \tag{4.27}$$

RMSE summarizes the overall error magnitude. FVU expresses the residual sum of squares relative to the sum of squared deviations of the measured torque from its mean $\bar{M}_i$, with the factor $(n_i - 1)/n_i$ carrying the small-sample adjustment of the evaluators. The maximum absolute residual records the largest pointwise error. RMSE and the maximum residual have torque units, whereas FVU is dimensionless. For a given test, FVU is proportional to squared RMSE, so the two metrics are not independent measures of shape and level. Across tests, their different scaling changes the relative weighting of the errors. The solver minimizes the objective vector

$$\mathbf{g}(\boldsymbol{\chi}) = \frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} \begin{bmatrix} g_{1,i}(\boldsymbol{\chi}) / g_{1,i}^{(0)} \\ g_{2,i}(\boldsymbol{\chi}) / g_{2,i}^{(0)} \\ g_{3,i}(\boldsymbol{\chi}) / g_{3,i}^{(0)} \end{bmatrix}, \tag{4.28}$$

where $g_{1,i}$, $g_{2,i}$, and $g_{3,i}$ are the RMSE, FVU, and maximum absolute residual of Equation (4.27), the superscript (0) marks the zero-correction baseline value of the same compo-

nent, the same kinematics and force model with every correction term disabled, and $\mathcal{T}$ is the training fold. The fold mean of the baseline-normalized components makes the three objectives dimensionless and the training tests commensurate. The search carried no constraints beyond the parameter bounds of each stage, and the returned Pareto fronts were filtered to members scoring at or below the zero-correction baseline on every component of every test in that stage's identification pool.

The solver is the genetic multiobjective solver `gamultiobj`, an implementation of the elitist nondominated-sorting genetic algorithm NSGA-II [Deb et al., 2002, The MathWorks, Inc., 2025], which returns a Pareto front of candidate correction terms. Predictions are evaluated on the fitting and validation tests of each fold; the adopted rows and complete test accounting are given in Appendix C. The production runs use adaptive-feasible mutation and scattered crossover at a fraction of 0.8, with MATLAB's default Pareto fraction of 0.35; the population settings are stage-specific. The flexor two-bracket fit evolves a population of 150 for up to 600 generations and terminates early once the geometric average relative change in the spread of the Pareto front falls below 4×10^{-3}; the extensor refit uses a population of 50 for up to 150 generations with a spread tolerance of 1×10^{-3}. In both stages the parameters are searched in a normalized space, with length offsets in centimeters and stiffnesses on a base-ten logarithmic scale, so that the genetic search samples the stiffness bounds, which span an order of magnitude, uniformly by decade, and the initial population is seeded around the expected solution region. The search bounds and the adopted values are recorded in Appendix C.

4.1.5 Actuator Placement and Route Redesign

The actuator attachment locations determine both moment arm and the change in muscle length through a joint's range of motion. During this dissertation, Morrow and I developed a placement search that began from the manually selected routes of the earlier robot model [Bolen and Hunt, 2019, Morrow et al., 2020]. The initial study evaluated the erector spinae, internal oblique, and external oblique about the lumbosacral joint. Its human targets were joint-torque surfaces calculated from the Gait2392 muscle paths and maximum isometric forces, sampled at 100 positions along each of the two joint coordinates.

For each pair of attachment points spanning a joint, the search displaced each point by $\pm\epsilon$ along each Cartesian coordinate. This produced eight candidates around each attachment and 64 candidate pairs. The algorithm evaluated the torque surfaces for these pairs, retained the pair with the lowest cost, and repeated the search around the accepted locations. When the cost could no longer be reduced at that scale, it reduced ϵ by a factor of 0.5. The reported search began with $\epsilon = 0.1\,\text{m}$ and ended when the step fell below 1 mm [Morrow et al., 2020].

The original cost combined the summed absolute torque mismatch over the sampled poses, actuator length, displacement of the attachments from their starting locations, and a diameter-dependent penalty. The published weights for the first three terms were 10^{-4}, 10^{-2}, and 1, respectively; the penalties for 10 mm, 20 mm and 40 mm BPAs were 1, 5, and 15. These were numerical weights chosen to balance differently scaled cost terms. The geometric terms discouraged impractical placements but did not impose collision avoidance: attachments could lie away from the modeled bones, and routes could intersect

them. This study established the placement-search approach; it did not establish physical torque accuracy or satisfaction of the human torque requirement at every pose.

A subsequent knee implementation restricted the initial candidates to points on the femur and tibia meshes. For the biceps femoris short-head replacement, every candidate origin–insertion pair was evaluated over 100 knee poses from approximately -120° to 10°, including neutral. The BPA calculation used a fixed 20 mm diameter, determined actuator and tendon lengths from the route geometry, and rejected candidates outside its allowable contraction range. It retained the lowest-cost usable pair. This discrete mesh search was distinct from the earlier cube-perturbation search. Both used rigid attachment geometry without the experimentally identified compliance corrections.

The knee-flexor mesh search also used a different objective from the original torque-mismatch cost. It rewarded torque surplus in the human torque's direction while penalizing disagreement between the human and robot torque-vector directions:

$$C_{\text{mesh}} = -\frac{1}{3N}\sum_{j=1}^{N}\left[w_{\mathbf{k}}\,D_{\mathbf{k},j} + \sum_{a\in\{x,y\}} w_a\,D_{a,j}\right] + \frac{w_u}{N}\sum_{j=1}^{N}\left(1 - \hat{\mathbf{M}}_j^{\mathrm{H}}\cdot\hat{\mathbf{M}}_j^{\mathrm{R}}\right). \tag{4.29}$$

Here N is the number of evaluated poses, and $D_{a,j} = M_{a,j}^{R} - M_{a,j}^{H}$ when $M_{a,j}^{H} \geqslant 0$, with the subtraction reversed otherwise. The implementation assigns weights $w_{\mathbf{k}} = 5$ to the knee-axis component and $w_x = w_y = 0.25$ to the two off-axis components. It averages these weighted component surpluses over $3N$ entries. The direction term averages over N torque vectors, with weight $w_u = 10$; it is zero for aligned vectors and contributes 20 for opposite directions. A zero robot-torque vector is assigned the opposite human direction in this term. The equation assumes a nonzero human torque vector at the evaluated pose.

Torques are supplied in newton meters; this is a numerically weighted score, rather than a sum of normalized physical errors. Minimizing it favors torque capacity in the required direction, but an average surplus does not guarantee that the requirement is met at every pose. Table 4.1 summarizes this knee-specific objective.

Table 4.1: Terms of the knee-flexor mesh-placement cost. The cost is minimized; a surplus reduces it, while a direction mismatch increases it.

Term	**Weight**	**Meaning**
Knee torque surplus	5	Signed surplus $D_{\mathbf{k},j}$, summed with the off-axis terms and divided by $3N$; enters with a negative sign.
Off-axis torque surplus	0.25 each	Signed surplus $D_{x,j}$ and $D_{y,j}$, using the human torque's sign on each axis; enters with a negative sign.
Direction mismatch	10	Mean $1 - \hat{\mathbf{M}}_j^H \cdot \hat{\mathbf{M}}_j^R$; zero for aligned vectors and two for opposing vectors.

Constrained Knee-Route Redesign

The dissertation's route redesign extends the placement search as a constrained continuous optimization over the design vector $\mathbf{x} = [\mathbf{p}_1,\ \mathbf{p}_{\text{end}},\ l_0,\ l_{\text{tendon}}]$, the BPA origin $\mathbf{p}_1$, insertion $\mathbf{p}_{\text{end}}$, resting length, and tendon length (Appendix C), so that the flexor and extensor configurations meet or exceed human knee torque capability over the range of motion. Because the route prediction (contact activation, wrapping, and compliance correction) is expensive to evaluate and only moderately smooth, the redesign minimizes a single scalar cost assembled from weighted penalty terms. The cost has the form

$$J = w_w\, J_{\text{worst}} + w_t\, J_{\text{torque}} + \textstyle\sum_a J_a + J_{\text{geom}} + J_{\text{len}}, \tag{4.30}$$

where J_{worst} is the largest fraction by which robot torque falls short of the human target at any angle across the range of motion, weighted heavily ($w_w = 10^5$); J_{torque} is the mean-squared shortfall fraction ($w_t = 10^3$), weighted two orders of magnitude below the worst-angle term so that no shortfall at a single angle can be traded away for a better average; the J_a are configuration-specific auxiliary penalties; and J_{geom} and J_{len} are regularizers that keep the endpoints and actuator lengths near their seed values. The two leading terms are built from the normalized shortfall

$$s_j = \frac{\max\Big(0,\ (1+m_r)|M_j^{\text{H}}| - |M_j^{\text{R}}|\Big)}{\tau}, \qquad J_{\text{worst}} = \max_j s_j, \qquad J_{\text{torque}} = \frac{1}{N}\sum_j s_j^2, \quad (4.31)$$

where $m_r = 0.05$ is the required torque margin and $\tau = \max(1, \max_j |M_j^{\text{H}}|)$ is the human torque scale. The flexor and extensor configurations carry the remaining terms of Equation (4.30) as

$$\begin{aligned} J_{\text{flx}} = {} & 10^5 J_{\text{worst}} + 10^3 J_{\text{torque}} + 10^{-2} J_{\text{shape}} + J_{\text{off}} + J_{\text{rest}} + J_{\text{strain,hi}} \\ & + J_{\text{strain,lo}} + J_{\text{wrap}} + J_{\text{geom}} + J_{\text{len}} + J_{\text{anchor}}, \end{aligned} \quad (4.32)$$

$$\begin{aligned} J_{\text{ext}} = {} & 10^5 J_{\text{worst}} + 10^3 J_{\text{torque}} + J_{\text{over}} + J_{\text{rest}} + J_{\text{strain,hi}} \\ & + J_{\text{strain,lo}} + J_{\text{contr}} + J_{\text{geom}} + J_{\text{len}}. \end{aligned} \quad (4.33)$$

Tables 4.2–4.4 define the normalized penalties and their weights. A candidate is rejected if its route cannot be constructed or its torque or strain calculation is not finite.

Human torque targets are computed from the Gait2392 lower-extremity model: biceps femoris short head for the flexor and vastus medialis for the extensor [Delp et al., 2007,

Table 4.2: Torque penalties in the constrained knee-route search. FLEX and EXT identify configuration-specific terms.

Term	**Weight**	**Meaning**
J_{worst} worst-angle shortfall	10^5	Largest fraction by which predicted torque magnitude falls short of the required envelope (human curve plus margin m_r) at any angle; dominates so that no single-angle shortfall can be traded for a better average.
J_{torque} mean-square shortfall	10^3	Mean squared shortfall fraction across all angles.
J_{shape} shape agreement (FLEX)	10^{-2}	Small mean-square tracking error of the full torque curve beyond the requirement.
J_{over} overshoot (EXT)	10^{-3}	Mean-square fraction by which torque exceeds the plain human curve, discouraging over-designed moment arms.
J_{off} off-axis excess (FLEX)	10^3	Worst and mean squared excess of the off-axis (x, y) torques beyond the original single-BPA route's off-axis torques, each axis normalized separately.

Table 4.3: Feasibility and geometric penalties in the constrained knee-route search.

Term	Weight	Meaning
J_{rest} rest-length feasibility	10^6	Squared violation of resting length plus tendon plus fittings covering the longest path over the range of motion.
$J_{\text{strain,hi}}$ upper strain limit	10^7 FLEX, 10^4 EXT	Squared violation of the maximum admissible relative strain and contraction.
$J_{\text{strain,lo}}$ stretch limit	10^7 FLEX, 10^4 EXT	Squared violation of the minimum strain, forbidding a stretched (slack-free) actuator.
J_{contr} contraction cap (EXT)	10^6	Squared violation of the measured maximum usable contraction; also enforced as a hard constraint.
J_{wrap} wrap release (FLEX)	10^3	Penalty when the moving wrap contact fails to release during the extension sweep, scaled by the remaining turn of the wrap.
J_{geom} endpoint regularizer	10^{-2}	Keeps origin and insertion near their seed values; the flexor scales each axis separately, allowing large proximal motion of the origin.
J_{len} length regularizer	10^{-3}	Keeps the resting length near its seed value (40 mm norm).
J_{anchor} second-BPA anchor (FLEX)	10	Keeps the second actuator's derived endpoints near the build line of the loaded design (10 mm norm); a score term only, never a constraint.

Table 4.4: Additional cam-profile penalties and rejection values.

Term	Weight	Meaning
J_{arm} moment-arm reward (CAM)	-10^2, capped	Negative reward on the worst-angle margin against the required envelope, capped at $+100\%$ of the torque scale, so the solver keeps buying moment arm instead of stopping at the requirement.
J_{armStep} arm-step penalty (CAM)	10^3	Penalizes adjacent-pose moment-arm steps beyond the accepted release discontinuity of 2.2 mm.
J_{reg} cam regularizer (CAM)	10^{-2}	Tie-break toward the first-cut cam neighborhood (25.4 mm norm).
Infeasible sentinels	10^{11}, 10^{12}	Non-finite predictions, or a route the predictor cannot build, are rejected outright.

Seth et al., 2018]. At each knee angle, the OpenSim moment arm is multiplied by the muscle's maximum isometric force [Delp et al., 1990, Hoy et al., 1990, Sherman et al., 2013]. Appendix A gives the model parameters and exported values. The search also enforces bone clearance, feasible wrapping, actuator contraction limits, and tendon geometry; a candidate that violates these requirements cannot be accepted.

Both configurations use a global search followed by local refinement. MATLAB's `surrogateopt` approximates the expensive route calculation from sampled designs and uses that approximation to choose further candidates [The MathWorks, Inc., 2025]. The evaluation budgets are 3,000 for the direct extensor and 1,000 for the flexor. The best feasible design is refined with `patternsearch`, which compares nearby candidates without requiring derivatives, for up to 15,000 further evaluations [Audet and Dennis, 2006, The MathWorks, Inc., 2025]. The cam variant uses 1,000 surrogate evaluations and 1,500 refinement evaluations. The budgets limit computation; local refinement can also stop when its search-step size falls below the specified tolerance and the constraints are satisfied.

Although both configurations share this pipeline, their cost functions differ in ways that reflect the mechanics. The flexor BPA routes around a single moving wrap point, so clearance from the tibia and femoral condyles is enforced by continuous signed-distance constraints, and its cost adds an off-axis torque penalty (torques about the joint axes other than the knee axis, relative to the original single-BPA route) and a penalty if the wrap fails to release cleanly as the knee extends. The extensor BPA wraps over the front of the joint through as many as seven sequential contacts, so clearance is enforced instead by the contact-elimination rule that governs the route, and its cost penalizes torque overshoot above the human curve and excessive actuator contraction, which risks bottoming out

at high flexion. The flexor and extensor targets include a 5 % torque margin as a design factor for additional effects not represented by the model. Thus, the required prediction is 1.05 times the human target at every design angle. This prescribed margin is a design allowance, rather than an experimentally established error bound. With two BPAs per joint, the flexor route is solved as two separately evaluated paths, while the extensor model scales the force of a single shared route by the actuator count.

The redesign retains the objective of obtaining adequate torque from anatomically constrained routes, but changes both the prediction and the search problem. Actuator and tendon lengths become independent design variables, geometry is constrained continuously, and pointwise torque shortfall is penalized using the corrected force model. Section 4.3.2 discusses the consequences for hardware design.

4.2 Results

The following results compare measured knee torque with the rigid-body prediction, the improved actuator characterization applied to rigid kinematics, and the corrected torque model. The hybrid calculation substitutes measured muscle length and moment arm to help distinguish force-model error from geometric error.

4.2.1 Flexor on Pin Joint

Figure 4.7 shows the measured flexor torque on the pinned knee next to the pre-optimization prediction series for the four cross-validation tests. Both pre-optimization force models overpredicted the magnitude of the measured flexor torque, and on the

angles where the two predictions are both defined the improved BPA characterization fit the measurements better than the original rigid-body prediction on every test, although it degraded near full extension, where the rigid-body prediction is undefined altogether (Table 4.5). The back-calculated (hybrid) torque reproduced the measurements closely, which localizes the misprediction to the modeled kinematics, the muscle length l_{m} and the moment arm r, rather than to the force model.

Figure 4.7 shows pre-optimization pinned-knee torque versus knee angle for the four cross-validation flexor tests. Knee angle is negative in flexion, and flexion torque is negative. Gold circles are the hybrid calculation, which combines the measured muscle length and moment arm with the modeled force; indigo filled circles are experimentally measured torque; the gray dashed line is the rigid-body prediction; and the light orange dash-dot line is the unoptimized prediction, the improved BPA force characterization applied to the same rigid kinematics. Panel titles give the nominal resting length; the actual resting lengths are $l_0 = 48.5\,\mathrm{cm}$ **(A)**, $l_0 = 41.0\,\mathrm{cm}$ **(B)**, $l_0 = 45.7\,\mathrm{cm}$ **(C)**, and $l_0 = 40.6\,\mathrm{cm}$ with tendon **(D)**. Panels **(A)** and **(B)** are the training tests and panels **(C)** and **(D)** the validation tests of the adopted cross-validation fold.

For Table 4.5, the rigid-body and improved-characterization predictions are scored on the same points where the rigid-body strain is defined: 75 of 90, 17 of 20, 14 of 19, and 15 of 22 points in panels A–D, respectively. The hybrid calculation is scored on all measured points. These pre-optimization scores therefore use a different point set from the full-test corrected-model scores in Table 4.6.

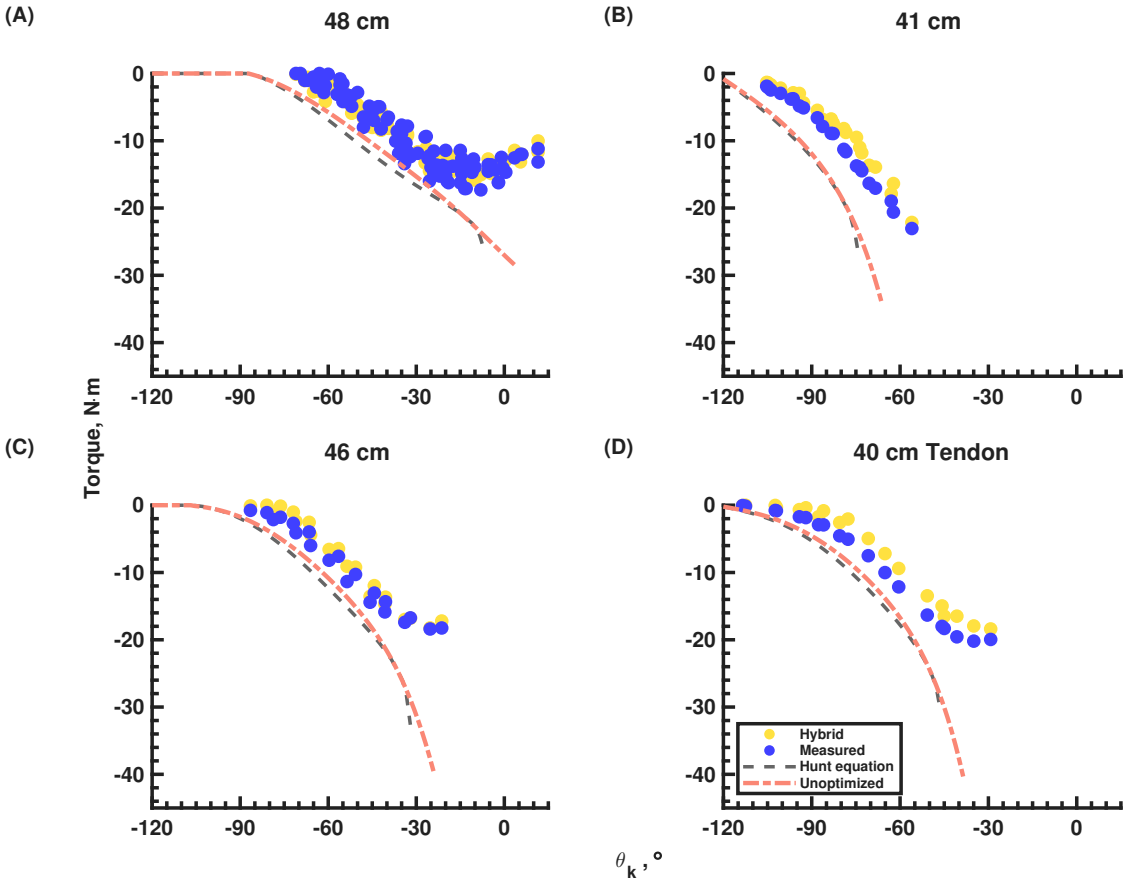


Figure 4.7: Pinned-knee flexor torque before design-parameter optimization: **(A)** $l_0 = 48.5\,\text{cm}$, **(B)** $l_0 = 41.0\,\text{cm}$, **(C)** $l_0 = 45.7\,\text{cm}$, and **(D)** $l_0 = 40.6\,\text{cm}$ with a 35 mm tendon. Panels **(A)** and **(B)** are training tests; **(C)** and **(D)** are validation tests. The legend distinguishes measurements, the Hunt model, the improved BPA model, and the hybrid calculation.

Table 4.5: Pre-optimization pinned-knee flexor torque-prediction goodness of fit per test, comparing the three pre-optimization torque predictions of Figure 4.7: the Hunt-equation rigid-body prediction, the unoptimized prediction of the improved BPA characterization, and the back-calculated (hybrid) torque. RMSE and maximum absolute residual (Max.) are in N m, and FVU denotes the fraction of variance unexplained.

		Hunt equation			**Improved BPA**			**Back-calculated (hybrid)**		
l_0 (cm)	l_{ten} (mm)	**RMSE**	**FVU**	**Max.**	**RMSE**	**FVU**	**Max.**	**RMSE**	**FVU**	**Max.**
48.5	–	5.779	1.1761	9.798	4.871	0.8354	9.865	1.635	0.0991	3.842
41.0	–	6.633	3.2313	11.409	6.074	2.7090	9.262	2.187	0.1184	4.243
45.7	–	5.315	0.9408	9.991	4.465	0.6640	9.536	1.246	0.0409	2.275
40.6	35	3.945	0.6750	7.527	3.226	0.4512	7.376	2.085	0.0755	3.073

The flexor adopted identification is the two-bracket fit of Section 4.1.4 on the four cross-validation tests: the $l_0 = 48.5$ cm and 41.0 cm tests in training and the $l_0 = 45.7$ cm and 40.6 cm-tendon tests held out. Its adopted terms are a length offset $\chi_0 = 3.94$ mm and effective stiffnesses $\chi_1 = 39.98\,\text{kN m}^{-1}$ and $\chi_2 = 14.73\,\text{kN m}^{-1}$ (Table 4.9). At these terms the optimized model follows the measured torque across the range of motion, including the loss of torque near full extension that the rigid-body prediction does not produce (Figure 4.8). The training-set RMSE is 2.165 N m for the $l_0 = 48.5$ cm test and 1.084 N m for the $l_0 = 41.0$ cm test, and the held-out validation tests give 1.368 N m at 45.7 cm and 1.611 N m at 40.6 cm with tendon (Table 4.6). The complete five-test comparison before and after correction is provided in Table C8.

Figure 4.8 shows optimized pinned-knee torque versus knee angle for the four cross-validation flexor tests, at the adopted correction terms (Table 4.9). Knee angle is negative in flexion, and flexion torque is negative. Blue filled circles are experimentally measured

torque; the dashed light orange line is the unoptimized prediction of the improved BPA characterization; the magenta line is the optimized prediction of the optimized model. Panel titles give the nominal resting length; the actual resting lengths are $l_0 = 48.5\,\text{cm}$ **(A)**, $l_0 = 41.0\,\text{cm}$ **(B)**, $l_0 = 45.7\,\text{cm}$ **(C)**, and $l_0 = 40.6\,\text{cm}$ with tendon **(D)**. Panels **(A)** and **(B)** are the training tests and panels **(C)** and **(D)** the validation tests of the adopted cross-validation fold; the goodness-of-fit values are given in Table 4.6.

Table 4.6: Pinned-knee flexor goodness of fit before and after correction, using the adopted fold. RMSE and maximum absolute residual (Max.) are in N m; FVU is dimensionless.

		Improved BPA			**Optimized model**		
l_0 (cm)	l_{ten} (mm)	**RMSE**	**FVU**	**Max.**	**RMSE**	**FVU**	**Max.**
Training							
48.5	–	7.617	2.1278	25.531	2.165	0.1718	5.467
41.0	–	9.694	2.1149	21.076	1.084	0.0277	1.766
Mean		8.656	2.1213	23.303	1.624	0.0998	3.617
Validation							
45.7	–	8.803	1.9380	26.264	1.368	0.0468	3.614
40.6	35	6.373	0.5935	16.859	1.611	0.0427	5.891
Mean		7.588	1.2657	21.562	1.490	0.0447	4.752

Table 4.6 completes this comparison at the adopted correction terms of the full re-identification (Section 4.1.4). Across the four tests of the two-bracket identification, the

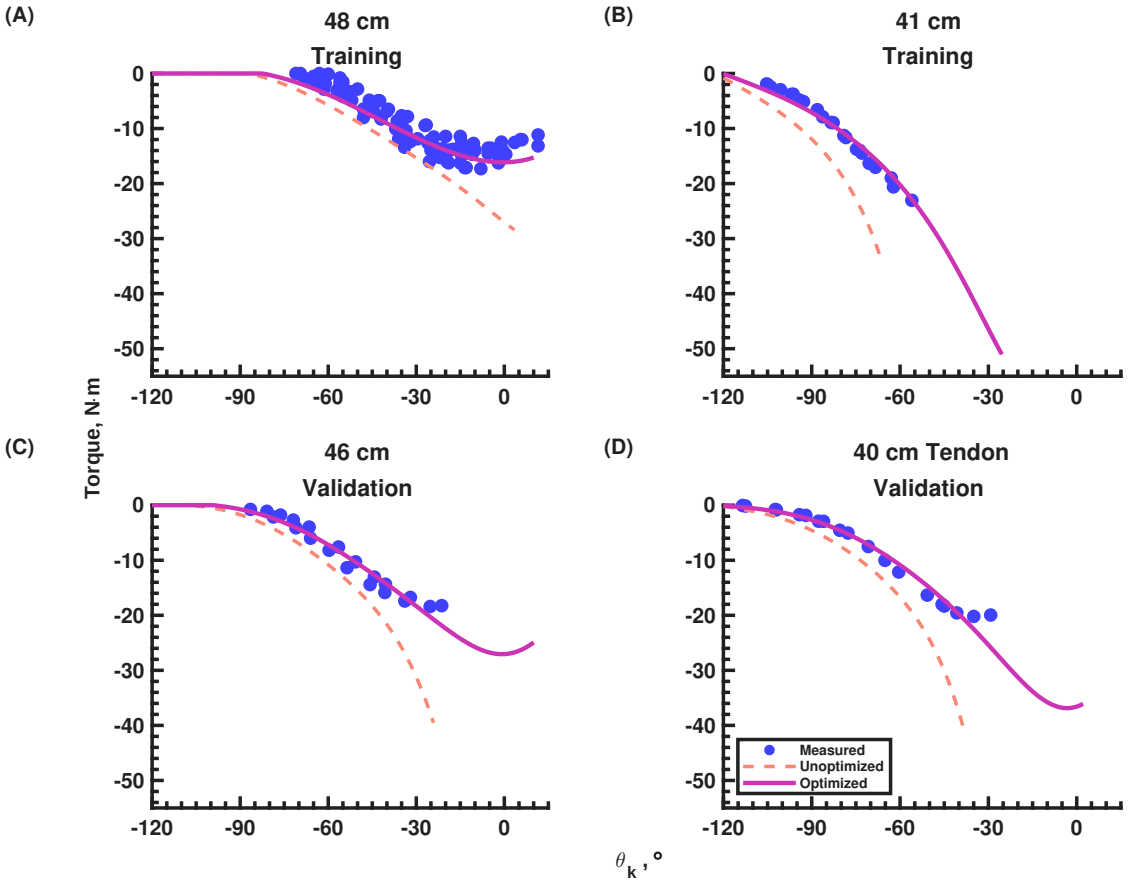


Figure 4.8: Optimized pinned-knee flexor torque: **(A)** $l_0 = 48.5\,\text{cm}$, **(B)** $l_0 = 41.0\,\text{cm}$, **(C)** $l_0 = 45.7\,\text{cm}$, and **(D)** $l_0 = 40.6\,\text{cm}$ with a 35 mm tendon. Panels **(A)** and **(B)** are training tests; **(C)** and **(D)** are validation tests. Circles show measurements, dashed curves the improved BPA model, and solid magenta curves the optimized model.

optimized model reduces the training-mean RMSE, over the $l_0 = 48.5\,\text{cm}$ and $41.0\,\text{cm}$ training tests, from $8.66\,\text{N}\cdot\text{m}$ under the zero-correction prediction, the same kinematics and force model with every correction term disabled, to $1.62\,\text{N}\cdot\text{m}$, and the validation mean over the held-out $l_0 = 45.7\,\text{cm}$ and $40.6\,\text{cm}$ tendon tests improves from $7.59\,\text{N}\cdot\text{m}$ to $1.49\,\text{N}\cdot\text{m}$. The behavior of the original rigid-body torque model on these tests is shown graphically in Fig. 4.7.

4.2.2 Extensor on Pin Joint

When the new stiffness and length offset model was used to predict torque for the knee extensor, measured torque was slightly higher than the original rigid-body model predicts for the 48.0 cm resting length BPA, slightly lower than expected for the 45.7 cm BPA, and much lower for the 41.5 cm BPA. Only the χ_1 and χ_2 terms from the previous optimization results were used with the extensor configuration. By inspection, the zero point of the measured torque was shifted to the right of the original model, indicating an offset opposite in sign to the χ_0 term in the previous section. The BPA with $l_0 = 45.7\,\text{cm}$ illustrates the pattern: the hybrid-method calculated torque correlates well with the originally predicted torque, while the actual measured torque diverges.

Figure 4.9 shows pinned-knee torque versus knee angle for the extensor BPAs: the six identification tests at the adopted correction terms, with the extensor-specific terms (χ_0, χ_3) on the shared stiffness pair (Table 4.9). Knee angle is negative in flexion, and extension torque is positive. Gold circles are the hybrid calculation; indigo filled circles are experimentally measured torque; the dashed light orange line is the unoptimized predic-

tion of the improved BPA characterization; the magenta line is the optimized prediction of the optimized model. Panel titles give the nominal resting length; the actual resting lengths are $l_0 = 43.6\,\text{cm}$ **(A)**, $l_0 = 43.2\,\text{cm}$ with tendon **(B)**, $l_0 = 45.7\,\text{cm}$ **(C)**, $l_0 = 40.0\,\text{cm}$ **(D)**, $l_0 = 40.0\,\text{cm}$ with tendon **(E)**, and $l_0 = 46.5\,\text{cm}$ **(F)**. Panels **(A)–(C)** are the training tests and panels **(D)–(F)** the held-out validation tests of the adopted cross-validation fold; the legend applies to all panels.

The adopted two-stage extensor identification uses the shared flexor stiffness pair and the six-test fold defined in Section 4.1.4. Figure 4.9 and Table 4.7 report that adopted fit.

At the adopted correction terms of the full re-identification (Section 4.1.4), Table 4.7 reports the per-test goodness of fit on the six extensor tests used by the identification, three for training and three for validation: the optimized model improves the RMSE of every used test over the zero-correction prediction and holds FVU below unity on all but the $l_0 = 43.2\,\text{cm}$ tendon test. The original rigid-body model on these tests is shown graphically in Fig. 4.9.

Table 4.7 gives the per-test extensor fit, including training and validation means. The full nine-test before/after comparison is in Table C9; the separate biomimetic-knee fits are in Table 4.8.

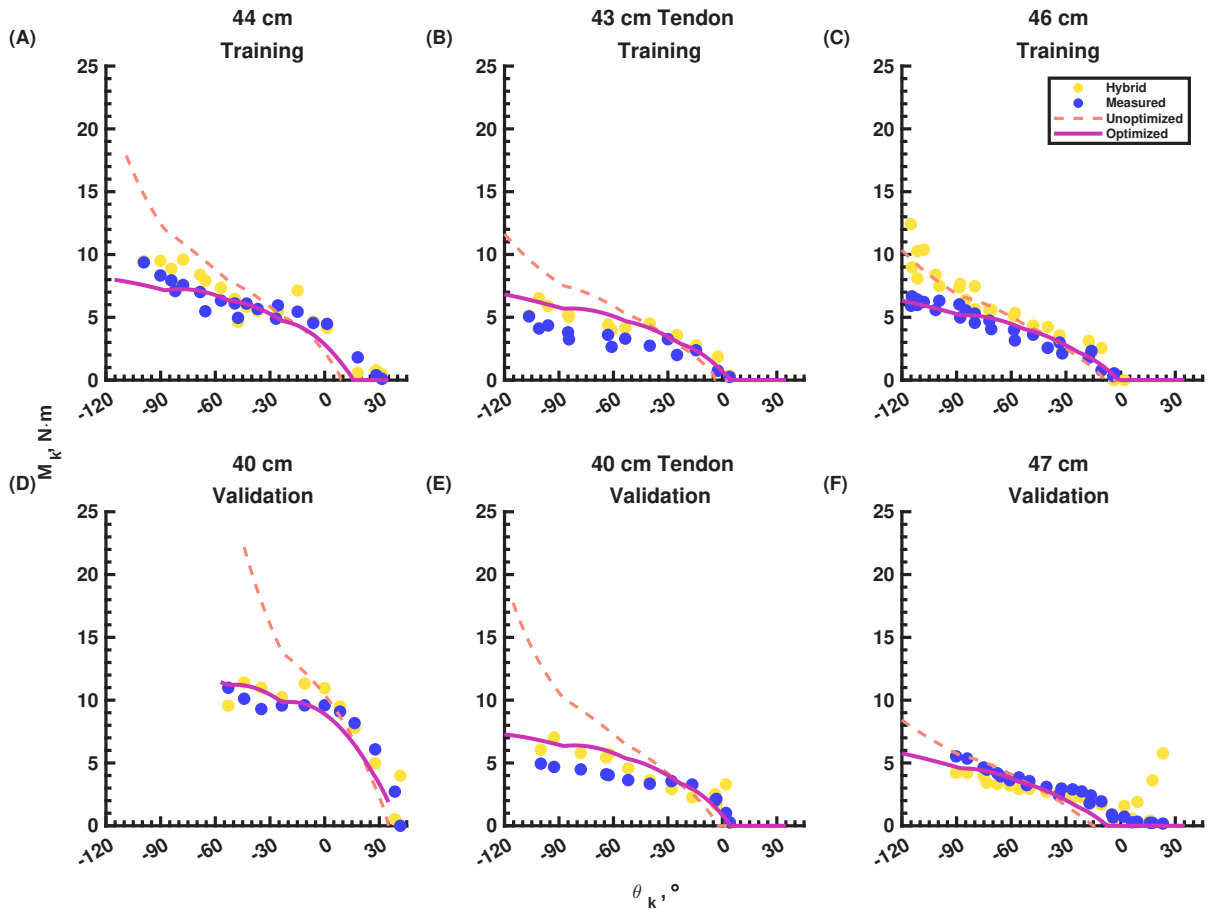


Figure 4.9: Optimized pinned-knee extensor torque: **(A)** $l_0 = 43.6\,\text{cm}$, **(B)** $l_0 = 43.2\,\text{cm}$ with a 22 mm tendon, **(C)** $l_0 = 45.7\,\text{cm}$, **(D)** $l_0 = 40.0\,\text{cm}$, **(E)** $l_0 = 40.0\,\text{cm}$ with a 40 mm tendon, and **(F)** $l_0 = 46.5\,\text{cm}$. Panels **(A)**–**(C)** are training tests; **(D)**–**(F)** are validation tests. The legend applies to all panels.

Table 4.7: Pinned-knee extensor goodness of fit before and after correction, with stiffness shared with the flexor. RMSE and maximum absolute residual (Max.) are in N m; FVU is dimensionless.

		Improved BPA			**Optimized model**		
l_0 (cm)	l_{ten} (mm)	**RMSE**	**FVU**	**Max.**	**RMSE**	**FVU**	**Max.**
Training							
43.6	–	2.437	0.9618	4.284	0.978	0.1634	1.818
43.2	22	2.878	4.7304	4.817	1.525	1.3270	2.481
45.7	–	2.178	1.0586	4.974	0.490	0.0535	1.206
Mean		2.498	2.2503	4.692	0.997	0.5146	1.835
Validation							
40.0	–	3.092	0.5638	4.842	1.244	0.1303	2.196
40.0	40	3.622	6.3919	7.845	1.350	0.8878	1.890
46.5	–	0.949	0.3152	2.272	0.699	0.1714	1.559
Mean		2.554	2.4236	4.986	1.098	0.3965	1.882

4.2.3 Extensor on Biomimetic Knee

The biomimetic knee was tested with a ∅10 mm BPA in the extensor configuration. The bracket frame origin is the centroid of the cantilevered segment that holds the muscle origin bracket to the proximal femur (see Fig. 4.4**C**, right side).

The torque results for a $l_0 = 51.8$ cm BPA are shown in Fig. 4.10. The torque values measured are significantly smaller than both the original prediction and back-calculated torque using the hybrid method. The results from the previous extensor configuration lead to a better torque prediction. GoF values can be found in table C10. Figure 4.10A

shows the BPA path passing between the femoral condyles close to the instantaneous center of rotation. In panel B, indigo points are measurements, gold points are hybrid calculations, the dashed gray line is the rigid-body prediction, and the magenta line is the corrected prediction using the pinned-knee extensor terms. Using the optimization results for extensor BPAs on the pinned knee joint, the $l_0 = 51.8\,\text{cm}$ BPA has an RMSE $= 0.8920$, FVU $= 0.3396$, and a maximum absolute residual of 1.4188.

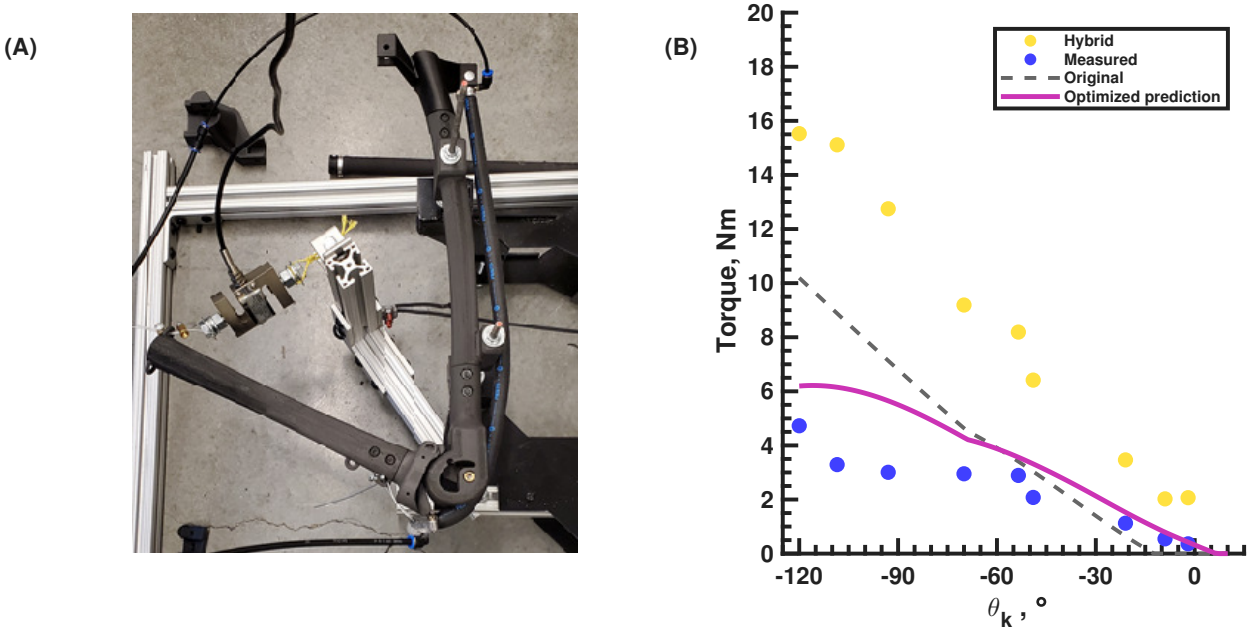


Figure 4.10: Biomimetic-knee extensor test: (A) apparatus with the BPA passing between the femoral condyles; (B) measured torque and model comparisons for the 10 mm-diameter, 51.8 cm BPA. The shared legend identifies the four series.

4.2.4 Flexor on Biomimetic Knee

Fig. 4.11 shows the biomimetic-knee flexor measurements and the corrected-model predictions using the adopted shared terms (Table 4.8); the bracket frame for the ∅10 mm BPA is shown in Fig. 4.4**C**, and for the ∅20 mm BPA in Fig. 4.4**D**. Panel **A** presents the ∅10 mm actuator at $l_0 = 41.5$ cm with a 21 mm artificial tendon (∅1.5 mm Shimano bicycle brake cable wrapped four times), pressurized to 604 kPa; the optimized model reached an RMSE of 2.8002 N m, an FVU of 0.8835, and a maximum absolute residual of 4.1201 N m. Panel **B** presents the ∅20 mm actuator at $l_0 = 41.5$ cm with a 14 mm artificial tendon made out of the same brake cable, wrapped three times so that six cable segments share the load, each wrap contributing two tendon lengths and cross-sectional areas. Panel **B** was tested at 620 kPa, for which the optimized model reached an RMSE of 2.2070 N m, an FVU of 0.0748, and a maximum absolute residual of 3.7320 N m.

The torque of the single ∅20 mm flexor actuator of panel **B** did not exceed the magnitude of the maximum human isometric flexion torque between 60° and 100° of knee flexion, and within a ±20% band about the human mean it still fell short of human-level torque between 78° and 103° of flexion. Because operating at 620 kPa across the full desired range of motion would load the printed femur beyond its ultimate strength, single points were also recorded at the reduced pressures 560 kPa, 421 kPa, 325 kPa and 281 kPa, which show that the desired range of motion is achievable at reduced pressure. This shortfall in flexion torque defines the target that the attachment-route redesign of Section 4.3.2 addresses. Panel **C** shows the outcome for the redesigned two-BPA flexor route of Section 4.3.6: across its fifteen tested measurements from −119° to 5°, the measured torque magnitude met or

exceeded the human biceps-femoris short-head maximum at every point, with margins from 0.42 N m at -105° to 17.0 N m at -14°.

Table 4.8: Biomimetic-knee goodness of fit of the optimized model at the adopted design parameters. RMSE and maximum absolute residual are in N · m.

Configuration	RMSE	FVU	Max. residual
Extensor, biomimetic knee, $l_0 = 51.8$ cm	1.17	0.67	2.21
Flexor, biomimetic knee, ∅10 mm, $l_0 = 41.5$ cm	2.8002	0.8835	4.1201
Flexor, biomimetic knee, ∅20 mm, $l_0 = 41.5$ cm (620 kPa)	2.2070	0.0748	3.7320

4.3 Discussion

4.3.1 Reasons for the Correction Terms

The correction terms of Section 4.1.4 exist because the physical transmission between the actuator and the joint is neither rigid nor lossless, and each term names one family of physical reasons. A constant length offset χ_0 exists because of manufacturing and positioning tolerances: the printed bone segments, the machined knee links, and the end-cap fittings each carry dimensional tolerance, the actuator is pinned at a measured location on that stack of tolerances, and the accumulated discrepancy appears as a length that is consistently shorter than the predicted musculotendon path at every joint angle, including at zero load where no compliance can act. The stiffness terms χ_1 and χ_2 exist because loaded components displace: the isometric system is not rigid, and there is compliance in the winch cable of the test stand, the artificial tendons, and the cantilevered attachment brackets that bend and stretch under actuator load. The wrapping-loss term χ_3 exists because a BPA that bends over the joint loses usable contraction length that no straight-line

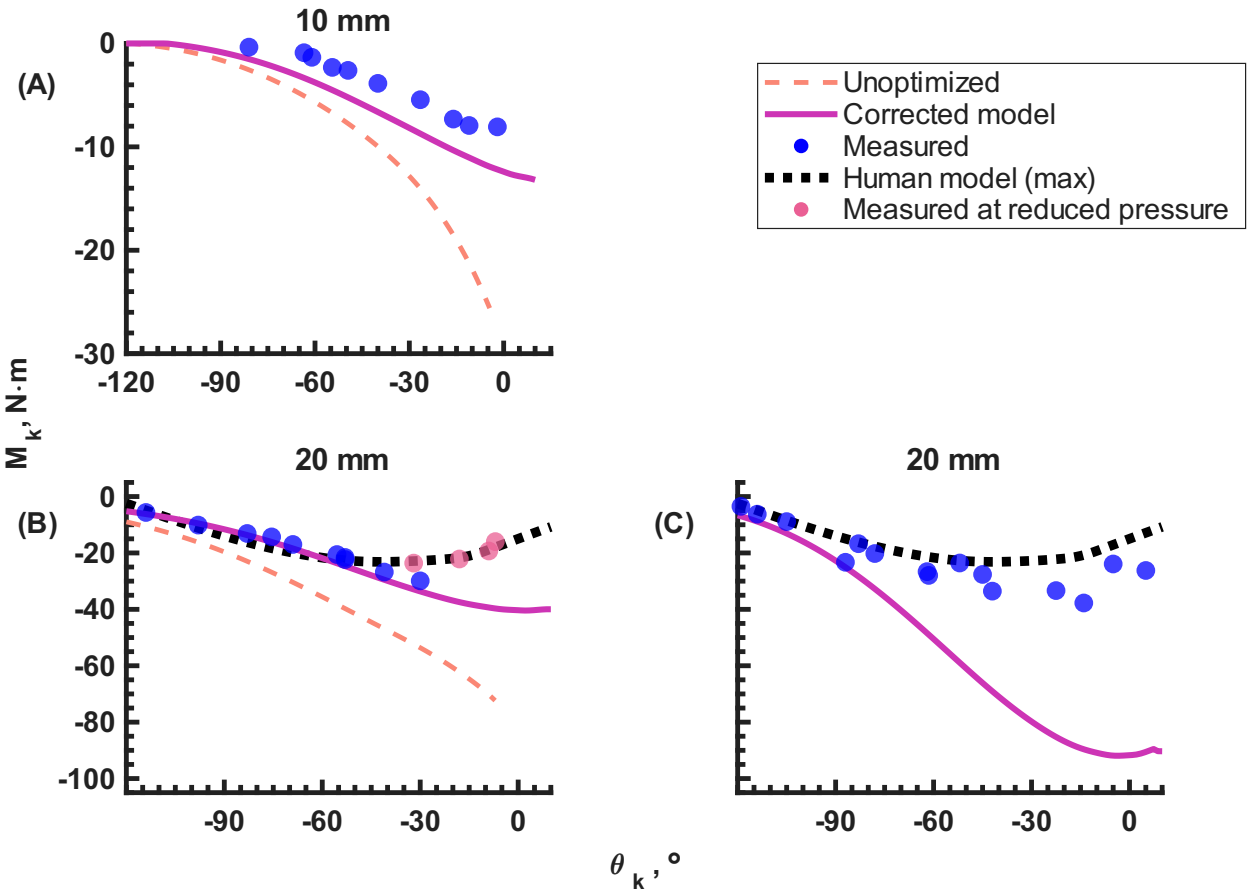


Figure 4.11: Biomimetic-knee flexor torque: single 10 mm BPA (A), single 20 mm BPA (B), and redesigned two-BPA route (C). Measurements, predictions, and human targets follow the shared legend; B and C use the same axis ranges.

model accounts for, and because the same loading could displace the actuator off its intended seat, as the observed slip off the anterior head of the femur bolt shows.

Our improved BPA characterization predicts torque more accurately than the original BPA model (see Fig. 4.7). This only accounts for part of the error. Many factors affect the torque results. Compliance in the testing system, positioning tolerances, flexibility of brackets, imperfect path modeling, and variation in individual BPAs are other factors.

The adopted identification uses four pinned-flexor tests and six pinned-extensor tests, with separate fitting and validation subsets. Sharing the effective stiffness pair between the two configurations tests whether the same representation of attachment compliance can support both torque predictions.

The modeling of extensor BPA paths that wrap around the knee joint is an approximation with inherent error (Fig. 4.10). When the moment arm is short, small changes to it can have a large effect on the resultant torque [Young et al., 2019]. It was assumed that wrapping the BPA around the joint would be easier to account for than adding a patella, deterministically designing its location as a function of knee angle, and accounting for friction. At greater knee flexion angles, shorter l_0 BPAs were observed to stretch and compress (see Fig. 4.1**C**). The same loading sometimes caused the BPA to slip off the anterior head of the bolt that holds the femur to the knee. This produced a $\pm 20\,\text{mm}$ displacement in the Z direction, changing both the force vector and the torque. Fig. 4.9 shows a deviation between measured and originally predicted results that widens as expected torque increases. This indicates system compliance. However, the hybrid-calculated torque matches or exceeds the predicted torque, except at greater flexion angles for the $l_0 = 41.5\,\text{cm}$ BPA, indicating a loss of functional l_m; this is why the χ_3 term was created in (4.24). Fig. 4.9

shows the improved fit from running the optimization that includes this term.

The biomimetic knee joint with an extensor BPA was used to validate the extensor optimization results. The corrected prediction more closely matches the measured torque values, although it remains higher than the measurements.

The nonlinear biomimetic robotic system presented here replaces rigid-body simplifications commonly used in other robotic and biomechanical systems. Instead, progressing from simple to complex systems elucidated compliance mechanisms that can lead to more accurate first-iteration designs. Technologies have been developed to predict system bending and axial compliance, a constant length offset, and a loss of usable length that occurs as artificial muscle bends around a joint.

The isometric force characterization supplies a static force–length relation for later dynamic actuator models, including the pressure–force formulation of Elzein et al. [2025]. It does not identify velocity dependence or damping. Isokinetic tests, with controlled joint speed, would measure how movement changes delivered torque; isotonic tests, under a controlled load, would measure shortening speed and the ability to sustain that load. Loading and unloading sequences would also test hysteresis and pressure-response effects that a static torque curve cannot establish.

4.3.2 Optimizer Updates

The results figures use the adopted two-stage identification. The procedure of Section 4.1.4 yields $\chi_0 = 3.94\,\mathrm{mm}$, $\chi_1 = 3.998 \times 10^4\,\mathrm{N\,m^{-1}}$, and $\chi_2 = 1.473 \times 10^4\,\mathrm{N\,m^{-1}}$ for the flexor, and $\chi_0 = -6.38\,\mathrm{mm}$ with $\chi_3 = 0.158$ for the extensor on that same locked stiffness

pair (Table 4.9); the search bounds and the adopted values are recorded in Appendix C (Tables C7 and 4.9).

Table 4.9: Adopted transmission corrections. The stiffness pair is shared. The wrapping factor was identified on the extensor and is also used in the flexor redesign; it was not fitted to the pinned-flexor tests.

Configuration	χ_0 (m)	χ_1 (N/m)	χ_2 (N/m)	χ_3
Flexor	0.00394	3.998×10^4	1.473×10^4	0.158
Extensor	-0.00638	3.998×10^4	1.473×10^4	0.158

What the correction terms do to the prediction is quantified in Tables 4.6 and 4.7. The corrected flexor model holds FVU at or below 0.18 on the four tests of the identification, the worst case being the 48.5 cm training test at 0.172. On the extensor identification the optimized model improves the RMSE on every used test, cutting the training-triple mean from 2.50 N · m to 1.00 N · m and the validation-triple mean from 2.55 N · m to 1.10 N · m, and holds FVU below unity on every used test except the 43.2 cm tendon test. On the biomimetic extensor test ($l_0 = 51.8$ cm) the corrected prediction improves RMSE from 3.03 N · m to 1.17 N · m and FVU from 4.49 to 0.67, so the shared-pair model explains the bulk of that configuration's measured torque variance as well as its level. The genetic multiobjective solver is retained for the identification, where the Pareto front is required for holdout validation, and the surrogate-plus-pattern-search pipeline serves the route redesign (Section 4.1.5).

The original placement methods selected routes by their predicted torque profiles under rigid attachment assumptions. The isometric tests show why those predictions require compliance corrections before they can guide hardware selection. Measured muscle lengths were shorter than predicted, and torque error depended on load. The redesigned

routes therefore use the corrected force balance when evaluating candidate geometries.

The biomimetic-knee tests assess the corrected torque model on a four-bar joint and on different actuator sizes. On the biomimetic extensor, evaluating the shared adopted terms reduces FVU from 4.49 to 0.67. The biomimetic flexor comparisons use the same adopted stiffness pair and give FVU values of 0.8835 for the 10 mm actuator and 0.0748 for the 20 mm actuator (Table 4.8). The larger error for the smaller actuator limits the accuracy of transfer across configurations. The redesigned two-BPA routes provide a further physical test: their measured torque meets or exceeds the human-reference magnitude at every tested angle (Section 4.3.6).

The final mesh optimizers close the loop between the two families of Section 4.1.5, and their cost functions record the evolution explicitly. The original mesh optimization scored rigid-body torque surplus with the reward of Equation (4.29). The route redesign replaced the reward with the penalized requirement of Equations (4.32) and (4.33), shortfall against a required envelope, evaluated through the stiffness-aware force balance. The extensor's final form, the cam-profile redesign, adds three terms to the family objective,

$$J_{\text{cam}} = J_{\text{ext}}(\mathbf{x}_{\text{full}}) + J_{\text{arm}} + J_{\text{armStep}} + J_{\text{reg}}, \qquad J_{\text{arm}} = -10^2 \, \min\left(1, \, \min_j m_j^{\text{req}}\right), \tag{4.34}$$

where $m_j^{\text{req}} = (|M_j^{\text{R}}| - (1 + m_r)|M_j^{\text{H}}|)/\tau$ is the margin against the required envelope of Equation (4.31), and the moment-arm step and cam regularizer carry the weights of Table 4.2. The reward becomes a capped payment for margin beyond the requirement, so the solver keeps buying moment arm after the envelope is met, and the step term holds the release schedule smooth while it does.

The initial route searches bounded the BPA origins and insertions near the corresponding human muscle attachments to preserve biomimetic routing. Those anatomical bounds had to be relaxed where the available BPA contraction could not cover the knee range of motion. The resulting routes then required a second tradeoff: reducing the actuator forces carried by the attachment hardware after parts failed during testing. The cam redesign changes the moment arm and contact geometry to retain the torque requirement at lower actuator force.

The final flexor optimizer applies the wrapping-loss term χ_3, identified on the extensor, because the flexor route can also bring the BPA into contact with the tibia and the femoral condyles at high flexion, and an actuator that a possible contact could bend must be scored with the length that bending removes. Its search box was deliberately expanded as an acknowledged tradeoff with biomimicry: the muscle origin was freed to move more than 100 mm proximal of the femoral shaft, far from the human biceps femoris origin, purely for reserve contraction, the extra actuator length, and therefore force and torque, available at high flexion where the single-actuator tests fell short. Two 20 mm BPAs are carried, both to reach the required torque and to limit the off-axis torque that the single-mirrored-actuator arrangement was found to load parts past their breaking point. A mirrored two-actuator solution, the two routes exact reflections of one another about the mid-sagittal plane, was found and rejected: it is not bio-realistic if a pelvis is ever introduced, since the medial origin would sit where pelvic structure belongs. The adopted solution is a staggered-offset pair, with a separately solved route for each actuator; the second insertion mirrors the first across the sagittal plane, while its origin preserves the first route's lateral endpoint offset, so both origins sit off the same side of the knee and the two routes do not cross. Each route

is given its own wrapping calculation and its own collision checks.

The final extensor optimizer faces a requirement of a different order. The extensor target is set by vastus medialis, the quadriceps muscle with the smallest maximum isometric force in the reference set, and even that weakest-quad envelope demands roughly twice the peak torque of the flexor requirement, 50.32 N m against 23.16 N m. The two 20 mm extensor BPAs nominally share one path, a single route whose force is scaled by the actuator count, because the extensor route wraps over the front of the joint through as many as seven sequential contacts and duplicating that chain would double the interference problem. That shared-path economy has a programming cost: the large solution space of the wrapped route makes the contact and release angles, the knee angles at which each wrap contact activates and releases, difficult to schedule by inspection, and the calculation that produces them is formalized in Appendix C.7. The routing was additionally modified to reduce the force the actuators exert, because the direct route's higher actuator forces broke the printed and machined parts during testing; reducing moment-arm demand and contact severity at lower force is what led into the cam algorithm of Equation (4.34).

The route redesign built on these values addresses the single-muscle shortfall at high flexion angles documented in Section 4.2: each configuration now carries two 20 mm BPAs per route. The flexor solution meets the flexor torque target with a minimum margin of $+5.01\,\%$ at -107°, with a separately solved route for each actuator. The second insertion mirrors the first across the sagittal plane, while its origin preserves the first route's lateral endpoint offset; its wrapping and clearance are evaluated independently. The direct extensor route meets or exceeds its human-scaled target with the required 5 % overshoot as the thinnest margin at every angle, $+5.02\,\%$ at worst, while respecting the wrap-release

schedule and contraction limits of the route-elimination rule (Appendix C); it was later superseded by the cam redesign of Section 4.3.2. These are mechanical design predictions. Their complete incorporation into the converted neuromechanical models remains future integration work.

Redesigned-Route Results on the Gait2392 Robot-Body Model

This subsection records the quantitative outcomes of the redesigned routes on the Gait2392-derived robot-body model, whose muscle force and torque targets define the redesign objectives. All three result families were produced by the surrogate-plus-pattern-search pipeline of Section 4.1.5 under the adopted correction terms (Appendix C, Table 4.9) with one shared actuator plan: two 20 mm BPAs per route, a 620 kPa pressure ceiling, and a maximum usable contraction of 0.255 in the force model. Each family minimizes its own objective function, a unitless weighted sum of normalized penalty terms in which zero corresponds to no violated penalty, led by the worst-angle and mean-square torque shortfall against the required 5 %-overshoot human envelope, followed by overshoot, rest-length, strain, contraction, and mild design-change penalties. The three functions differ in penalty families and normalizations (the flexor and pulley objectives add off-axis-excess, wrap-release, and second-BPA anchor terms, and the pulley objective evaluates its torque requirement on the insertion-side torque after the tackle's force loss), so their numerical values are not commensurable with one another. The human-scaled requirements are the OpenSim isometric torque profiles of the reference human muscles over the same design-grid angles, the vastus medialis for the extensor and the biceps femoris short head

for the flexor.

The direct extensor route exceeds the human envelope in magnitude at every angle, with zero shortfall points: across the design range the predicted two-BPA extensor torque spans 35.45 N · m to 109.4 N · m against the human-scaled requirement of 20.25 N · m to 50.32 N · m, and the required 5 % overshoot is consumed as the thinnest margin exactly as designed, at a minimum margin of +5.02 % at the worst-case angle of -20.2°.

Allowing the tibial-side route points to trace a cam profile produced the current extensor design (Figure 4.12). The final cam geometry includes the optimized route and the verified adjustment of the distal route point and tendon length. Its minimum torque margin is +30.83 % with respect to the required envelope, the human target plus the 5 % overshoot, and +39.77 % with respect to the plain human target. The pressure required to meet the envelope spans 269 kPa–507 kPa across the range of motion, comfortably above the 150 kPa regulated floor of the VTUG valve manifold that supplies the hardware. The geometry confirms that the headroom is not bought with routing risk: the tibial wrap follows a bend radius of 96.9 mm, the guiding route point clears the femoral-condyle envelope by 43.1 mm at 10° of knee extension, and the moment arm spans 59.4 mm–102.8 mm (mean 72.4 mm) over the range of motion.

The flexor route, built from two separately evaluated BPA paths, meets the flexor torque target with a minimum margin fraction of 0.0501 against the required 0.05. Its predicted knee-axis torque spans -81.23 N · m to -3.864 N · m against the human-scaled requirement of -23.16 N · m to -2.461 N · m (peak magnitude 23.16 N · m), again exceeding the human envelope at every angle. The flexor redesign applies the adopted correction terms of Table 4.9, including the wrapping factor identified on the extensor configuration.

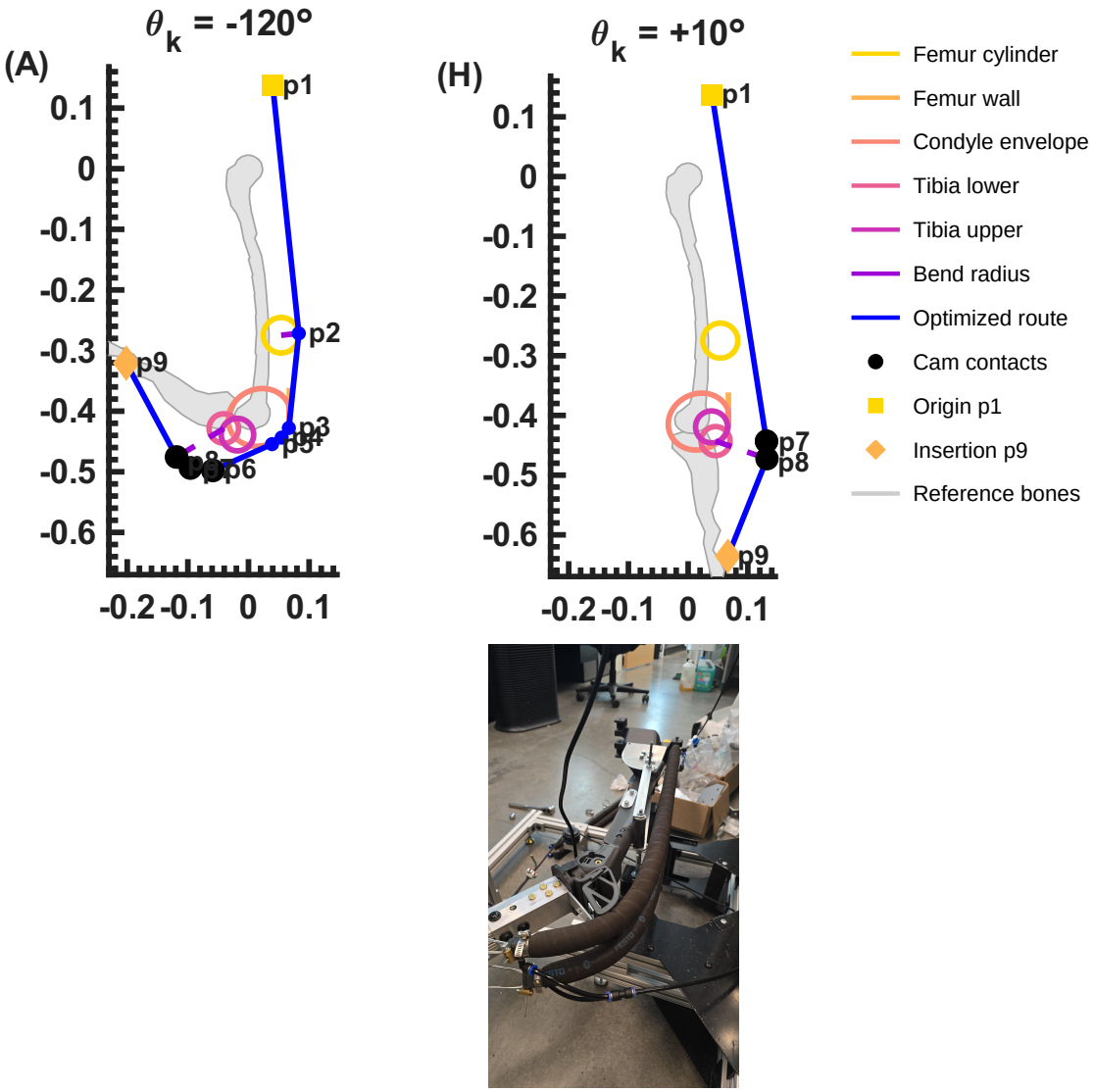


Figure 4.12: Cam-extensor geometry at $-120°$ (A) and $10°$ (H), with the assembled two-BPA hardware below. Panel letters match the complete sequence in Figures C5 and C6.

The Mono/Bi-pulley algorithms extend the route search space by allowing the artificial tendon to be reeved over pulleys before insertion. On the knee-flexor route, the pulley variant converged to a collision-feasible layout with zero bound violations and a peak predicted flexion torque of 85.56 $\mathrm{N} \cdot \mathrm{m}$ on the insertion-side measure, the torque applied at the insertion after the tackle's force loss; Section 4.3.3 compares the optimized configurations using knee-axis and off-axis torque.

4.3.3 Mono- and Bi-Pulley Tendon Routing

Every optimizer of the preceding sections scores its routes with the MonoPamDataExplicit-type calculation: a fixed chain of via points carrying one segment across from the parent body into the child, its contacts wrapped and released by the elimination rule, its length and moment arm read from that chain. The Mono- and Bi-pulley algorithms are the generalization of exactly that calculation. The artificial tendon is reeved over pulleys before insertion, forming a reverse pulley, a block-and-tackle transmission: with G tackle line parts supporting the moving block, the tendon travel at the insertion is multiplied by G while the force delivered there is divided by G, and at $G = 1$ the reeved route reduces to the straight-tendon route of the direct redesign. The tendon exit is either body-fixed on the proximal body so that its direction rotates with the joint, or anchored through a housing to the distal body, and the tackle's increased internal force and the pulley-block bracket deflections enter the same stiffness-aware force balance as the direct route's brackets. Read this way the pulley variants are not a new actuator concept but the general routing case of the mesh-optimization machinery, of which every direct route is the $G = 1$ special case.

At unity transmission, the knee-flexor pulley route increased peak predicted knee-axis torque from 81.23 N m to 85.74 N m, while limiting off-axis torque to 0.803 N m. This comparison measures the effect of the two optimized routing configurations; their different objective functions do not isolate the pulley alone. The pulley serves as a redirect in this configuration rather than a force multiplier. Appendix C.8 records the optimizer diagnostics and the broader route searches.

A subsequent transmission study re-solved actuator sizes for each tackle configuration at five anatomical muscle sites. A two-BPA pair through a two-part tackle preserves net actuator force while doubling insertion travel; increasing travel can extend the angle range over which a route meets its torque requirement. The corresponding torque and margin surfaces are shown in Figures C3 and C4. These calculations use the moving-force-point routing, with the adjacent joint fixed for the biarticular evaluation planes; they do not establish simultaneous performance across both joints. Physical verification of the pulley configurations remains future work.

4.3.4 Actuator-Count and Transmission Study for Vastus Medialis

The vastus medialis study evaluates one to three parallel 20 mm BPAs and one to three tackle line parts at 620 kPa, with actuator and tendon lengths re-solved for each combination. The attachment geometry is fixed by the study's routing family. Both OpenSim and cable-path moment arms are tested, with and without the compliance correction. In the corrected mode the calculation uses the pulley model directly; this model does not include the wrapping-loss term χ_3, so it is not the same prediction as the cam redesign.

No tested configuration meets the target at all five graded angles. The best coverage is four of five, obtained with three BPAs through a two-part tackle using the compliance correction and the cable-path moment arm. Its four passing-angle margins are $+56.3$, $+40.1$, $+35.4$, and $+1.0$ N m. It fails at 10°, where this routing family approaches zero torque. The cable path uses a ring point 0.75 mm off the bone surface. These results identify a limitation of the tested routing family; they do not establish a minimum actuator count capable of meeting the complete envelope. The separate cam redesign meets its own target over its design grid with two BPAs (Section 4.3.2).

4.3.5 Whole-Leg Transmission Capability

The configuration survey grades each actuator against the same-named single muscle of the Gait2392 model [Delp et al., 2007, Seth et al., 2018] on a 5×5 grid of two joint coordinates. The hip adduction domain is clipped by its flexion-dependent envelope. A cell meets the requirement when the BPA is taut and its torque has the human torque's sign and at least its magnitude. The coverage metric is the fraction of included grid cells meeting that condition; the aggregate is the unweighted mean of the per-actuator fractions, not a continuous-area integral or a whole-leg torque sum.

The BPA force model is evaluated at 620 kPa [Bolen et al., 2026]. The comparison includes all 27 actuator sites (Table C11). The baseline uses the master's-era attachment routing at unity transmission; its eight nominal 40 mm sites are evaluated with four parallel 20 mm BPAs, while the other sites use the survey's base configuration. The updated side selects the saved transmission-ladder winner at each site, with actuator length and tendon

length re-solved per configuration. It retains the same underlying attachment routing and the original diameter specification, including 40 mm actuators at those eight sites. The comparison therefore combines transmission, actuator sizing, and the baseline diameter substitution; it does not isolate an effect of attachment placement or compare two all-20 mm assemblies.

Active-force coverage is the fraction of included grid cells with positive insertion tension, excluding slack or infeasible configurations. It does not require the torque to match the human reference. Both coverage measures are calculated for each actuator and then averaged equally across the 27 sites. Table 4.10 reports the resulting means.

The mean fraction meeting the human torque requirement increases from 37.28 % for the baseline to 65.19 % for the saved transmission configurations, improving at 23 sites and tying at four. Mean active-force coverage increases from 92.44 % to 98.81 %. The higher active-force coverage shows that the transmission configurations maintain tension over more of the graded domain, while the lower torque coverage shows that positive tension alone is insufficient to meet the human torque requirement.

Table C12 gives the coverage at each actuator site.

Table 4.10: Mean torque and active-force coverage across 27 actuator sites at 620 kPa.

Configuration	**Human torque met (%)**	**Active force (%)**
Baseline	37.28	92.44
Transmission configurations	65.19	98.81

4.3.6 Physical Tests of the Redesigned Routes

The redesigned cam-extensor route was tested at nine isometric measurements, and the two-BPA flexor route at fifteen. Both test sets span knee angles from -119° to 5°. The analysis transforms the measured load-cell wrench to the moving knee center and evaluates the model at each measured angle and pressure. Figures 4.13 and 4.14 compare these results with the selected human-muscle targets and the optimized 620 kPa predictions.

All nine extensor measurements meet or exceed the vastus medialis reference torque at the measured angles. Measured extension torque ranges from 29.02 N m to 51.17 N m, with margins of 0.24 N m to 3.52 N m, at pressures of 256 kPa to 488 kPa. The measured torque exceeds the pressure-specific prediction at eight angles, by up to 13.75 N m; at -86.5°, it is 2.06 N m below that prediction. This prediction uses the recorded angle and pressure with the modeled route. It is not the hybrid calculation, which additionally requires measured muscle length and moment arm.

All fifteen flexor measurements meet or exceed the biceps femoris short-head reference magnitude, with margins from 0.42 N m at -105° to 17.0 N m at -14°. Agreement with the pressure-specific prediction remains uneven. These measurements support torque capacity at the sampled configurations; variable test pressure and discrete sampling do not establish a measured full-pressure envelope between every pair of angles.

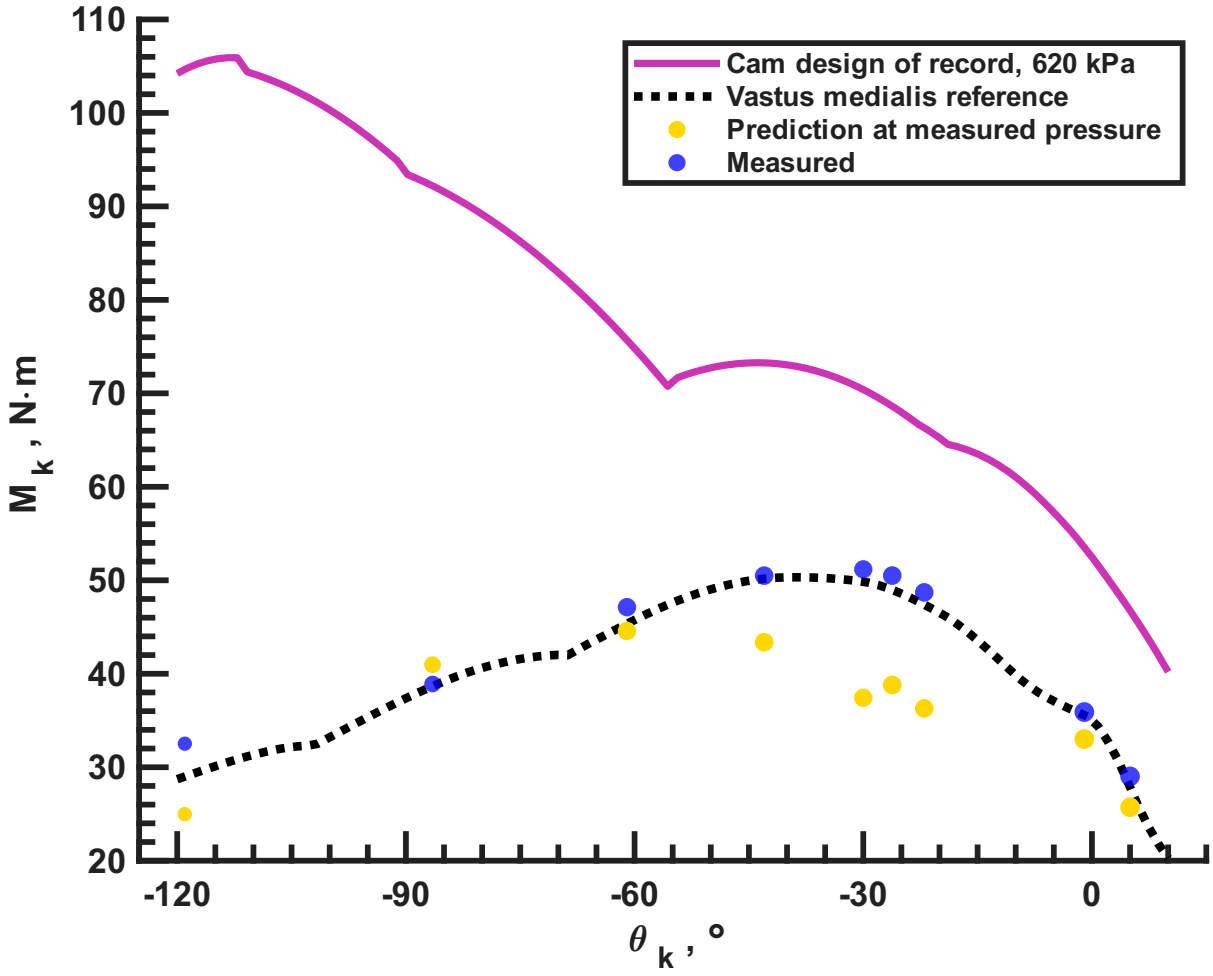


Figure 4.13: Cam-extensor measurements, pressure-specific predictions, the optimized 620 kPa curve, and the vastus medialis reference. Marker area scales with pressure. Series follow the plot legend.

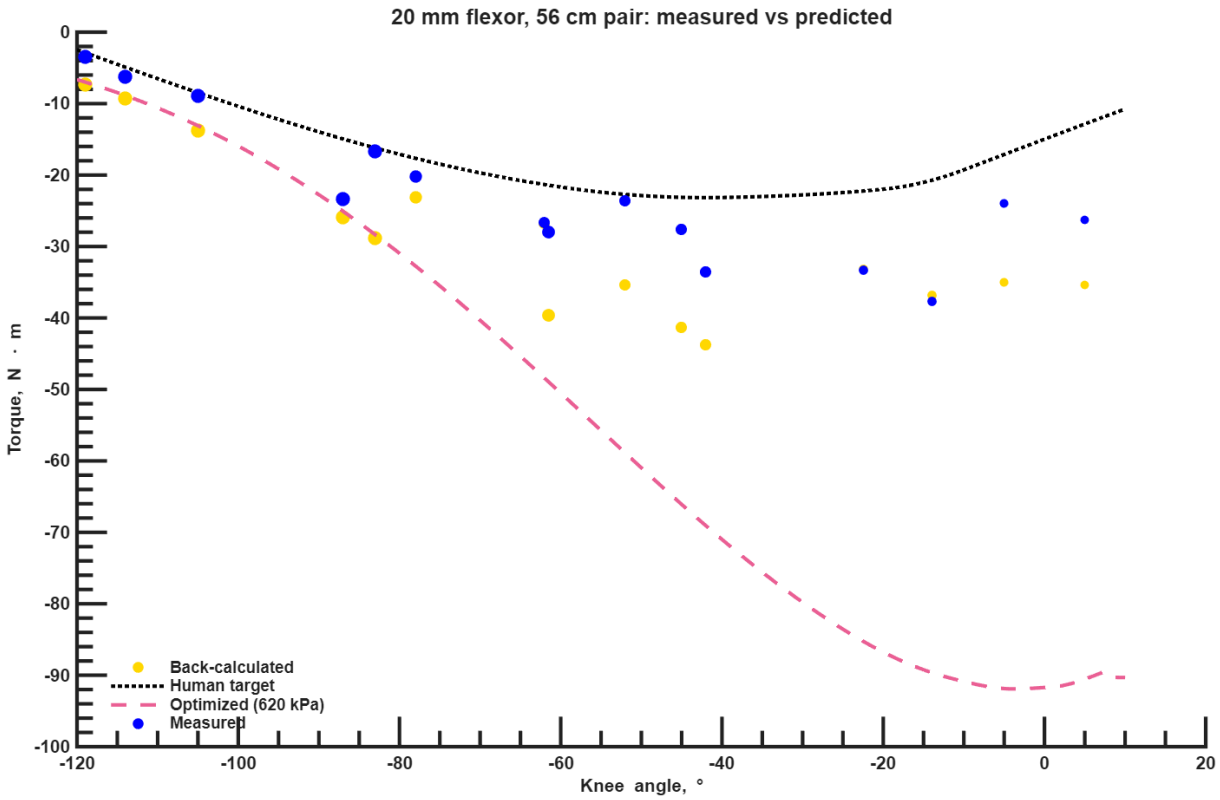


Figure 4.14: Two-BPA flexor tests with 56 cm actuators. Blue points are measured torques; yellow points are predictions at the measured angles and pressures. The dashed pink curve is the optimized 620 kPa prediction, and the black dotted curve is the human target. Point area scales with pressure.

5 The Sensory Afferent Database

5.1 The Resource

The Sensory Afferent Database (SADb) is a curated literature resource for studying sensory feedback in locomotion and posture. It connects papers to afferent classes, experimental preparations, feedback pathways, and computational models. I developed it within the AARL's NeuroNex Communication, Coordination, and Control in Nervous Systems (C3NS) effort and presented the resource at Neuroscience 2023 [Bolen et al., 2023].

When I began this work, selecting sensory feedback for the robot required finding measured afferent properties and feedback pathways across a dispersed literature. SADb addresses the need to organize those findings by their experimental conditions and potential role in a controller. Reviews organized parts of this evidence [Van de Crommert et al., 1998, Windhorst, 2007, 2022], but selecting a feedback rule still required recovering its experimental conditions and separating measurements from modeling assumptions. I built SADb to connect those findings to their sources, afferent classes, and pathway roles so that controller choices could be traced to the evidence and remaining unknowns could be identified.

The primary deliverable is a browser application for searching paper records, counting them by selected classifications, and exploring citation neighborhoods (Figure 5.1). Its embedded data support offline use; its separate online-search tab requires a network connection. The distributed snapshot contains 1,393 records: 943 papers and 450 linked

model, review, feedback-pathway, and recorder records. The application includes 436 records outside the smaller Airtable working base, so the operating cap does not limit access to the collected literature. The pivot and citation views operate on papers; the table view searches across record types.

The working base in Airtable contains linked Papers, Feedback, Models, Review Papers, and Recorders tables. Papers hold bibliographic information and curation notes: short summaries of the reported mechanism, experimental conditions, and relevance to feedback design, grounded in the source text. Feedback records describe a sensory pathway, such as Ib feedback influencing the stance-to-swing transition, and link it to supporting papers; Models link computational implementations to their source literature. The other tables organize review literature and recording methods. Paper matching uses DOI where available and normalized titles otherwise. The paper-table view is shown in Appendix E (Figure E1).

5.2 Curation Methods

The initial collection drew on the literature libraries used for previous AARL papers on locomotion and postural balance, together with records already curated in Airtable. Records were reconciled by DOI and title, then expanded through citation searches. Reviews of spinal and proprioceptive feedback supplied an initial organization of the literature [Van de Crommert et al., 1998, Windhorst, 2007, 2022]. This was a curated research collection, rather than a registered systematic review or an exhaustive search of every relevant publication.

Records were retained when they reported sensory-afferent measurements, investigated feedback in locomotion or posture, or modeled a relevant sensory pathway. Reviews were retained as sources for organizing and locating that evidence. Duplicate records and material outside this feedback scope were removed from the working base. Classification used controlled vocabularies for preparations and afferent classes. Each mechanism summary was grounded in available source text, using the abstract when full text was unavailable; missing text was flagged rather than supplied from recollection.

5.3 Results: Corpus and Curation Coverage

At the close of the curation effort the working base held 957 records: 561 papers, 39 feedback-pathway records, 244 model records, 109 review records, and 4 recorder entries, 43 records below the laboratory's 1,000-record operating cap, with room to grow. Coverage of the retained papers is concentrated where the laboratory's design and experimental work needs it. Of the 561 retained papers, 523 carry curation notes, 426 carry animal-preparation tags, 240 carry notes assessing each paper's animal-study potential, 118 are linked to named feedback pathways, 86 carry an afferent classification, and 258 carry notes translating the paper's findings toward robot or simulation use; 550 carry DOIs, and 38 remain bare pending a findable text.

The citation graph covers all 943 papers in the application, including the 382 papers outside the Airtable working base. Within this corpus, 730 papers are connected by 11,130 directed citation pairs. A Louvain community partition resolves 18 named topic clusters, including afferent locomotor control, spinal interneuron circuit modeling, human balance

and posture, bioinspired flow sensing, and deep-learning methods. Of the 943 papers, 833 carry curation notes and 318 have attached PDFs. The application distributes the corpus, citation graph, cluster assignments, and a flat export together. Figure 5.1 shows the citation map, in which each paper is drawn as a neuron with soma size proportional to citation count. Appendix E shows the table view (Figure E1).

The explorer supports three complementary ways of examining the saved corpus (Figure 5.1A). The table view searches titles, authors, years, DOIs, and curation notes, and combines filters for animal preparation, feedback pathway, afferent type, record type, archive status, year range, and the availability of notes or a PDF. Selecting a record opens its detailed notes, classification tags, DOI link, animal-study assessment, and robot-or-simulation translation. This connects a literature search to the observations and modeling choices recorded during curation. The table includes papers and the linked model, review, feedback-pathway, and recorder records; the pivot and citation-map views operate on papers. These three views use the embedded snapshot offline. The separate online-search tab queries OpenAlex and identifies returned DOIs already represented in the snapshot; suggested additions are downloaded for later curation.

The bubble map supports both publication-year versus citation-count axes and a topic landscape based on the citation-network layout. Its controls select the layout, display style, record grouping, citation-neighborhood depth, and click action (Figure 5.1B). Selecting a paper as the focus identifies it in the left panel (C), which exposes its curation note and proposed robot-or-simulation translation (D). The plotted neighborhood can extend through one, two, or three citation links in either direction (E). Direct connections retain color, while more distant papers are gray. In the illustrated Grillner (1985) neighborhood,

57 in-corpus papers cite the focus, the focus cites 13 in-corpus papers, and 555 papers lie within two links. Green triangles identify incoming citations and red circles identify the focus paper's references. These synapse-like symbols encode bibliographic direction; they do not assert excitatory or inhibitory physiological effects.

The connection list separates papers citing the focus from its references (Figure 5.1F). Selecting a listed paper makes it the new focus, allowing a reader to follow the development of an idea without repeatedly searching for individual titles. The reference-only and cited-by views isolate either direction, and the detail action opens the underlying record. A researcher can therefore start from a known experiment, inspect earlier studies supporting it, follow subsequent models or experiments citing it, and read the associated translation notes before choosing a candidate pathway for simulation.

5.3.1 AI-Assisted Literature Triage

I evaluated AI-assisted literature identification, summarization, and categorization in work presented at Neuroscience 2024 [Bolen and Hunt, 2024]. The comparison included ResearchRabbit, Litmaps, txyz, SciSpace, and MirrorThink. Some tools helped identify related papers or summarize straightforward material, but categorization remained inadequate: none received a good rating, and only txyz was rated adequate. Complex models and nuanced physiological findings required substantial human correction. The tools therefore offered limited help with the central curation task of accurately assigning papers to database categories. I retained responsibility for inclusion, classification, and deletion decisions after checking the source.

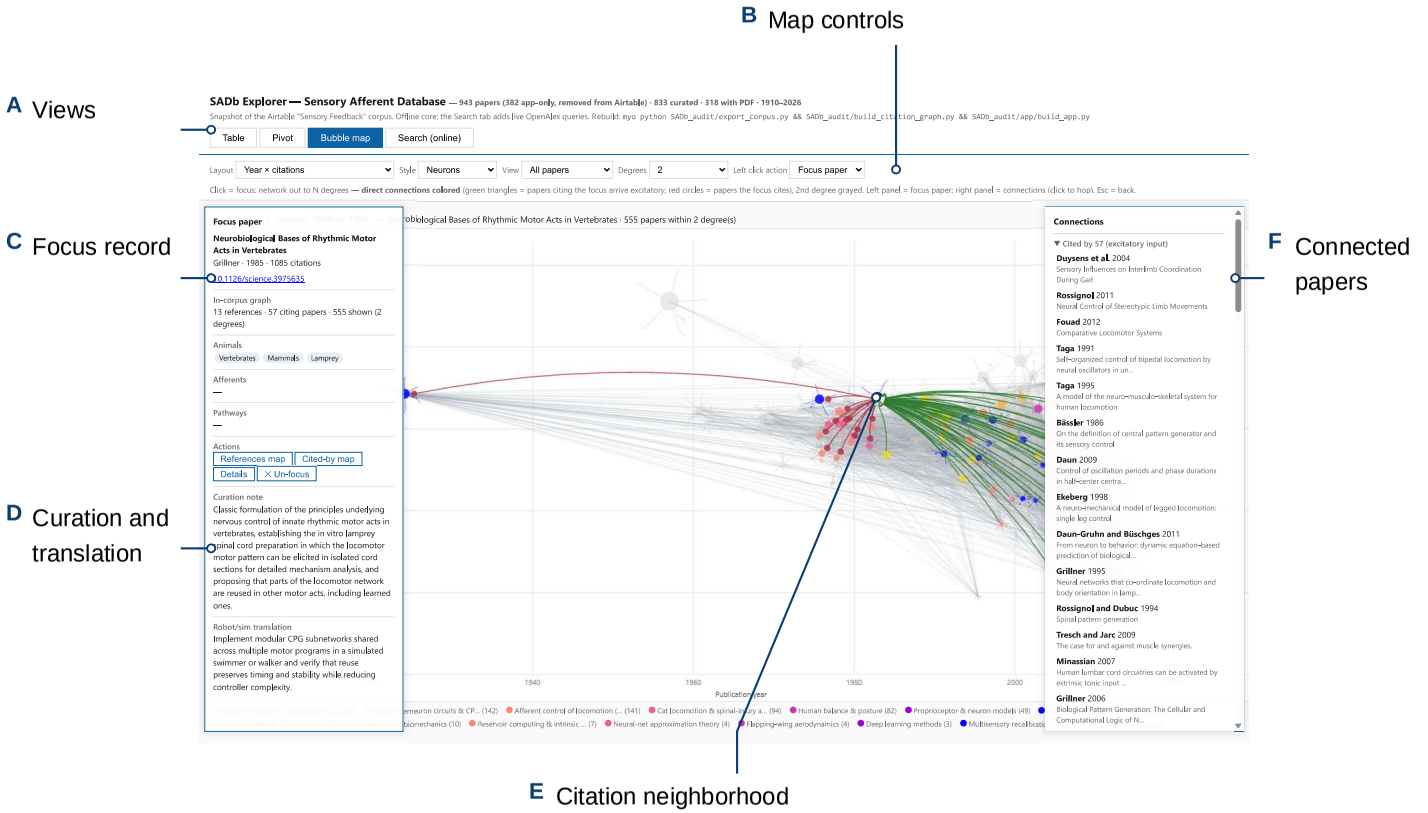


Figure 5.1: SADb Explorer focused on Grillner (1985): views (A), map controls (B), focus record (C), curation and translation notes (D), citation neighborhood (E), and connected papers (F).

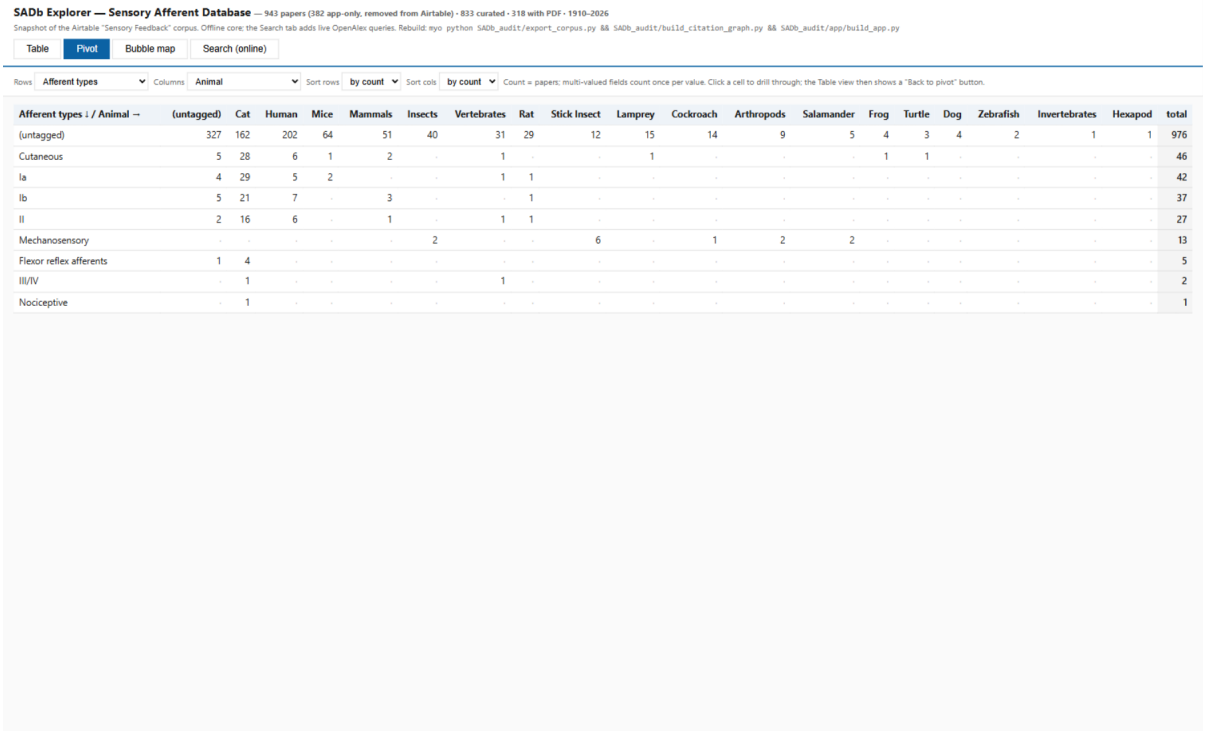
5.4 Discussion

5.4.1 Robot and Simulation Use

The database exists to feed the sensorium of Chapter 6. Its pathway records enumerate the signal taxonomy the controller's sensory channels implement: Ia afferents carrying length and velocity, Ib afferents carrying force, II afferents carrying length, and contact receptors marking the stance–swing boundary (Section 6.3); Table E1 in Appendix E summarizes what the corpus establishes for each class. The pathway naming grammar maps directly onto circuit edges, because a record identifies a candidate sensory signal, gating phase, and connection sign. Receptor dynamics, gains, delays, and species-specific assumptions still require explicit selection and testing; Table E2 lists the principal pathway records with the papers that demonstrate them. The Models table carries the lineage the controller instantiates, from the two-level rhythm-generator and pattern-formation architecture [Rybak et al., 2006] through muscle-reflex walking [Geyer and Herr, 2010] to the two-layer neuromechanical walker the laboratory's controller descends from [Deng et al., 2019], and the 258 robot-and-simulation translation notes exist precisely to shorten the step from paper to wiring decision. Figure 5.2 shows the two-dimension count the explorer's pivotable tables produce over the full paper snapshot, the afferent classes against the animal preparations: the classical cat preparation carries the largest share of the Ia, Ib, II, and cutaneous records, and the arthropod records enter through the mechanosensory class.

The pivot view counts papers by two selected dimensions, including topic cluster, decade, primary author, animal preparation, feedback pathway, and afferent classification.

Figure 5.2 crosses afferent classes with animal preparations: each cell counts papers carrying both tags. Rows and columns can be ordered alphabetically or by count. Selecting a cell opens the corresponding records in the table view, and a return control restores the pivot. The pivot is computed over all 943 papers in the embedded snapshot, including archived papers; it is not restricted to the 561 retained Airtable papers. Because a paper can carry several animal or afferent tags, it can contribute to several cells. The counts describe the distribution of curated records, not independent experimental observations or the strength of evidence for a pathway.



Afferent types ↓ / Animal →	(untagged)	Cat	Human	Mice	Mammals	Insects	Vertebrates	Rat	Stick Insect	Lamprey	Cockroach	Arthropods	Salamander	Frog	Turtle	Dog	Zebrafish	Invertebrates	Hexapod	total
(untagged)	327	162	202	64	51	40	31	29	12	15	14	9	5	4	3	4	2	1	1	976
Cutaneous	5	28	6	1	2	·	1	·	·	1	·	·	·	1	1	·	·	·	·	46
Ia	4	29	5	2	·	·	1	1	·	·	·	·	·	·	·	·	·	·	·	42
Ib	5	21	7	·	3	·	·	1	·	·	·	·	·	·	·	·	·	·	·	37
II	2	16	6	·	1	·	1	1	·	·	·	·	·	·	·	·	·	·	·	27
Mechanosensory	·	·	·	·	·	2	·	·	6	·	1	2	2	·	·	·	·	·	·	13
Flexor reflex afferents	1	4	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	5
III/IV	·	1	·	·	·	·	1	·	·	·	·	·	·	·	·	·	·	·	·	2
Nociceptive	·	1	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	1

Figure 5.2: SADb Explorer pivot table by afferent classification and animal preparation. Each cell counts papers carrying both tags and links to those records.

The 39 pathway records extend beyond the mammalian core of Table E2: paired

excitation and inhibition records cover the chordotonal organs and trochanteral hair plates of insect legs, a load-feedback record covers the campaniform sensilla, and further records hold fictive locomotion without sensory feedback (12 linked papers), group III/IV fatigue, and one paper each for somatosensory, vestibular, and visual contributions to postural control. The controller of Chapter 6 implements the mammalian subset.

The corpus also reaches ahead to the sensorium extensions planned in Section 7.2.5. The human balance and posture cluster and the multisensory recalibration cluster are the entry points for the vestibular and ocular feedback work, already curated and already connected to the locomotor core through the citation graph. The knowledge base makes this reach practical for new students as well: the cluster, pathway, and animal pages compress a literature that took months to curate into an onboarding document measured in pages, and the animal pages let an experimentalist start from a preparation rather than a topic.

5.4.2 Further Improvements

Classification remains incomplete. Only 86 of the 561 papers in the Airtable working base carry an afferent classification, and 118 link to named pathways. Completing these fields would improve comparisons across preparations and expose where a proposed pathway depends on a small or indirect body of evidence. Missing source text should remain explicit until the paper can be checked; a model or review should link to the primary experiments supporting its assumptions.

Further curation should connect pathway records to the papers that establish their signals, connection signs, and task dependence, and connect model records to the pathways

they implement. Vestibular and visual contributions to posture require expanded classifications and source coverage before the database can support the additional sensory channels proposed in Section 7.2.5. New records should include a source identifier and a substantive curation note, with incomplete classifications flagged for review. These improvements would make the resource more useful for selecting experiments and constructing testable neural models.

6 Synthetic Nervous System Simulation

The actuator and torque models of Chapters 3 and 4 describe the robot's mechanics. Studying their interaction with neural control also requires a simulation that connects the mechanical model to a synthetic nervous system (SNS). I first attempted this integration in AnimatLab by combining a published walker with a spinal controller. The model produced stepping-like motion in air, but hand tuning did not produce walking. This chapter explains how that attempt motivated the development of software tools for the assembly, testing, and comparison of neuromechanical models.

6.1 The AnimatLab Bipedal Walker

For a course project in ME 610: Neuromechanical Modeling, Morrow and I combined the bipedal mechanics of Li et al. [2017] with the two-level central pattern generator (CPG) of Deng et al. [2019] in AnimatLab [Cofer et al., 2010]. The mechanical model had a pelvis and bilateral thigh, shank, foot, and toe segments, with sagittal-plane joints and antagonist Hill-type muscles at the hip, knee, and ankle. I added a passive toe joint. The initial controller used one rhythm generator (RG) to control both sides, with hip and knee/ankle pattern-formation (PF) groups for each leg. We added afferent feedback at the motoneuron (MN) level. Figure 6.1 shows the model and controller organization.

We adjusted applied currents, synaptic weights, and muscle parameters by hand. The suspended model developed alternating muscle activity and stepping-like motion, presented at the 2021 NIH BRAIN Initiative Investigators Meeting [Bolen and Hunt, 2021].

It did not achieve weight-bearing walking.

I subsequently replaced the shared rhythm generator with one rhythm generator on each side and linked the two through commissural pathways. The bilateral pattern-formation and motor organization was already present in the original model. I added afferent feedback to the RG and PF layers, and heel- and toe-contact signals during the subsequent development. This later model is referred to as the Walker Two-Layer (W2L) model. Further circuit schematics and implementation records are provided in Appendix D.

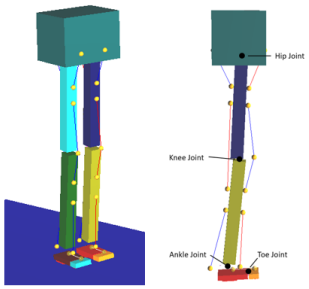
The AnimatLab model established that the combined circuit could generate alternating activity and move the suspended limbs. To investigate the contribution of sensory feedback and actuator mechanics, I needed to vary those components while retaining a specified neural organization and mechanical support condition. Extending that approach to the human-scale Gait2392 model and its BPA-routed robot counterpart also required checking joint directions, muscle paths, actuator forces, and the signals exchanged with the controller.

Figure 6.1 precedes the later split into one rhythm generator per side and the addition of afferent and foot-contact feedback to the RG and PF layers.

6.2 Knowledge Gaps and the New Tools

Researchers studying neuromechanical control desire to investigate movement generated by a neural circuit and how the control system is shaped by sensory feedback, actuator mechanics, and external support. For the humanoid models studied here, some unresolved

(A)



(B)

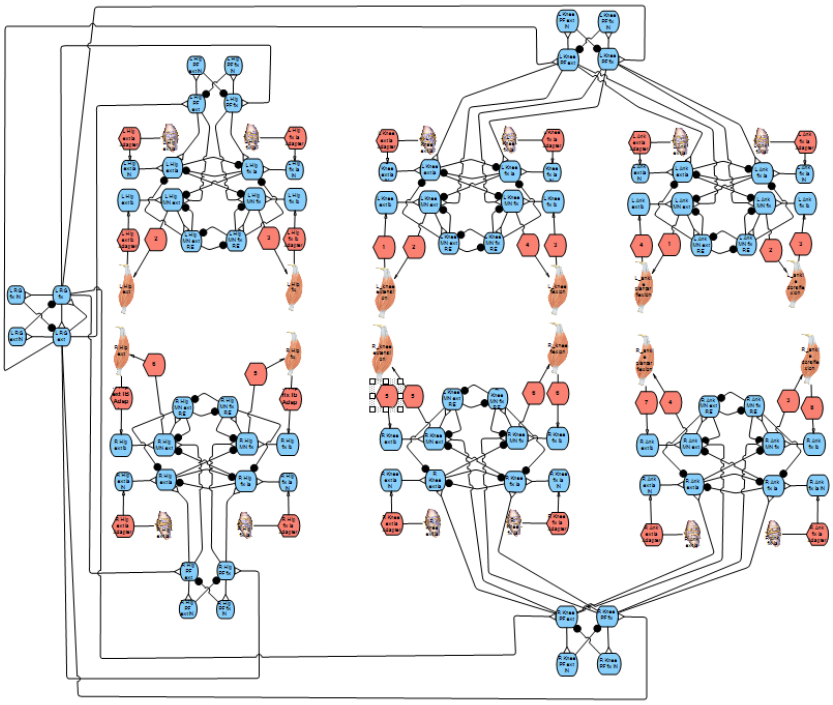


Figure 6.1: Original AnimatLab walker (A) and shared-rhythm-generator controller (B), developed by Morrow and Bolen. The four-neuron RG drives hip and knee/ankle pattern-formation groups on both sides.

questions were whether a specified circuit retained its behavior when implemented in a different simulator, and how its output changed when coupled to human musculotendon mechanics or artificial muscles. A successful neural trace alone could not answer the mechanical question, and a moving model alone could not establish that the intended neural pathways caused the movement.

I addressed this gap by constructing models in which the neural pathways, actuator characteristics, sensory inputs, and support conditions can be specified and varied, together with tests that distinguish circuit behavior from implementation and coupling errors. The tools include the Li–Deng/W2L and Gait2392 model implementations, the SNS-Simscape library and GUI, neural and BPA-actuator coupling to a SolidWorks-derived Simscape assembly, the MuJoCo–Simulink interface, and experiment drivers for staged tuning and controlled interventions. Literature-derived and spiking-neuron implementations support comparisons of circuit mechanisms and coding assumptions. Table 6.1 gives selected functional-verification tests and proof-of-concept demonstrations. These establish that the tools can construct, exercise, and compare the specified models. The feedback experiments also provide a limited mechanistic result: removing particular sensory pathways can eliminate stepping in a fixed supported configuration, identifying a dependence that would not be apparent from its intact motion alone.

Existing software supplied the underlying simulation capabilities. OpenSim provided musculoskeletal geometry, MyoConverter translated models to MuJoCo, and SNS-Toolbox provided neuron and synapse formulations [Delp et al., 2007, Caggiano et al., 2022, Todorov et al., 2012, Nourse et al., 2023]. Simulink provided graphical block programming, and Simscape Multibody provided mechanical modeling and import of SolidWorks-derived

mechanisms. My contributions are the particular model implementations, custom blocks, actuator integrations, and evaluation procedures built with those packages. The SNS-Simscape library makes the neural pathways explicit in a block diagram, while the MuJoCo implementations support repeated scripted evaluations and parameter searches. Matched-input tests check selected circuits before their outputs are interpreted as neural control of a mechanical model.

6.3 Methods: Software Tools

The tools provide three complementary capabilities: constructing a neuromechanical model, testing its component behavior and signal exchange, and evaluating controlled changes in neural or mechanical conditions. Table 6.1 summarizes representative demonstrations. Figure 6.2 places the simulation tools within the longer-term robot-control workflow.

6.3.1 Model Conversion and Actuator Coupling

My first conversion attempt used the robot model, which combined BPA routing on the Gait2392 skeleton with unchanged human muscle parameters. MyoConverter's muscle-equilibrium checks rejected infeasible combinations of routed length and muscle parameters. I therefore separated conversion of the human reference from reconstruction of the robot routing. This distinction matters because an error caused by an artificial-actuator route should not be interpreted as a property of the original human model.

The human-model conversion also required specific repairs. Three small hip rotators

Table 6.1: Tools developed and representative verification tests. Each demonstration applies to the named circuit or mechanical model.

Tool developed	**Function**	**Representative demonstration**
Li–Deng/W2L model implementations	Combine the walker mechanics and neural architecture; transfer selected controller components between environments	Matched contact-driven rhythm cores reproduce 1.000 s periods with trace correlation at least 0.99996
Gait2392/SNS model implementations	Evaluate alternative controllers on the human reference and custom actuator/routing models	Individual-actuator tests check torque directions; selected human-model controllers produce bilateral stepping-like activity under support
SNS-Simscape library and GUI	Construct neural pathways graphically and connect sensory, neural, and actuator blocks	The loaded-arm reflex demo gives 2.00° sag with feedback and 7.85° without it; actuator blocks agree with reference force calculations
W2L BPA-force integration	Apply the Festo force model within the coupled simulation	The 12-actuator BPA model demonstrates air stepping and retained-support standing, with peak pressure 372 kPa
SNS-Simscape/MuJoCo integration	Exchange activation commands and mechanical measurements through the MathWorks bridge	Single-muscle and control-rate tests reproduce the Python-reference sensor readback over the scored windows
Curriculum and experiment drivers	Automate staged tuning, support changes, neural stimuli, and mechanical perturbations	Saved stage-specific trials, candidate replays, and fixed-controller pathway and body-change comparisons demonstrate the workflow
Literature-circuit and spiking implementations	Test selected published neural behaviors and compare graded and spiking formulations	The reduced Rybak circuit passes 13 drive/deletion checks; selected spiking counterparts have paired implementation tests

per side, the quadratus femoris, gemelli, and piriformis, were disabled at model load. The converted model retains 92 actuator entries, but these six contribute no force. The converted knee retains the OpenSim flexion-negative sign convention. Spatial frame transformations and individual-actuator tests check joint motion and torque directions; the CAD-derived model uses the anatomical frame specified in Appendix D.5.2.

The source represents patella motion through muscle insertion locations that vary with knee angle. In the converted model, limits on the joints representing these moving path points conflicted with their coupling constraints. Removing those conflicting limits allowed the coupled path points to follow their prescribed trajectories. The rectus femoris path also required a patella-tracking via point to restore its knee-extension action. Explicit collision pairs introduced unwanted contacts, so contact was restricted to the foot–ground pairs. The anatomical joint limits also had to be re-enabled; these driver joints were treated separately from the coupled path-point joints. Passive joint torques supplemented the joint behavior. These repairs explain why conversion needed checks beyond successful loading: a model could load while producing an incorrect torque direction or unintended contact force.

Comparisons with the OpenSim attachment paths and single-actuator tests checked the transferred geometry and force directions. They do not establish that every muscle reproduces the source force throughout its range or that the controller produces walking. The neural and mechanical models exchange signals at a 2 ms solver time step. Appendix D.6 records the individual repairs and their scope.

For artificial-muscle simulations, I implemented an actuator-force calculation that evaluates the quasi-static BPA model from Chapter 3 at the current simulated actuator length

and commanded pressure. The calculated tensile force acts along the model's specified tendon route. Length, velocity, force, and foot-contact measurements are then returned to the neural controller through sensory channels. This closes the neural-command–actuator-force–sensory-feedback loop. The calculation represents the measured static force dependence; it does not add valve-flow or pressure-transient dynamics. Routing geometry and dynamic force transmission therefore require their own checks beyond the isometric force fit.

The SADb of Chapter 5 supplies the physiological context for selecting Ia, II, Ib, and contact pathways. A database link identifies the supporting observation; it does not by itself determine a numerical gain or guarantee the same function in another model. The implemented encoding, connection sign, phase gating, and tested response therefore remain explicit parts of the controller specification.

6.3.2 Block Programming Neural Pathways

The SNS-Simscape library implements neuron, synapse, afferent-encoder, activation, and muscle-force elements as editable Simulink blocks. The library retains the conductance-based neuron and synapse formulation supplied by SNS-Toolbox; it does not introduce a new neuron model. Verification uses selected circuits under matched inputs. Figure 6.3 shows the core blocks and graded-neuron signal flow.

I imported OpenSim-derived and SolidWorks-derived mechanisms into Simscape Multibody and coupled the SolidWorks robot assembly to a neural controller and BPA actuators using my force model. The block library also supports reduced-order demonstra-

tions in which the sensory pathway and mechanical response can be inspected directly. These smaller models provide interpretable checks of the components used in larger assemblies.

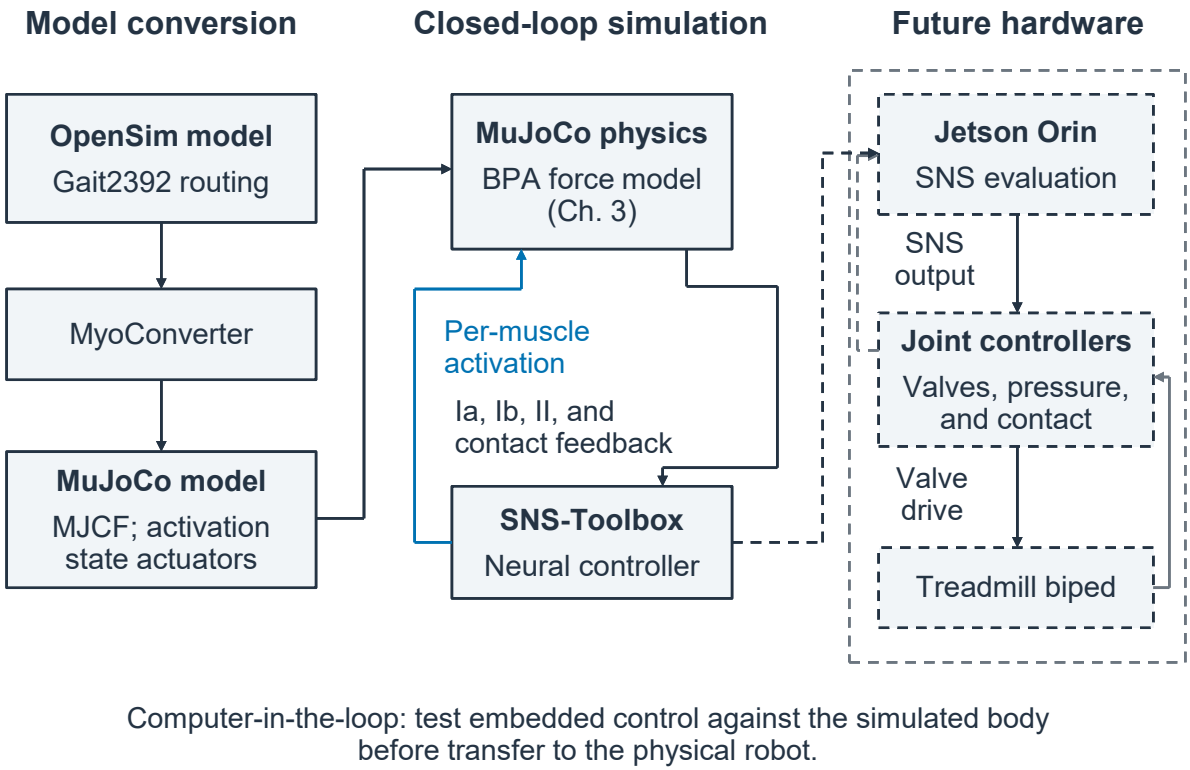
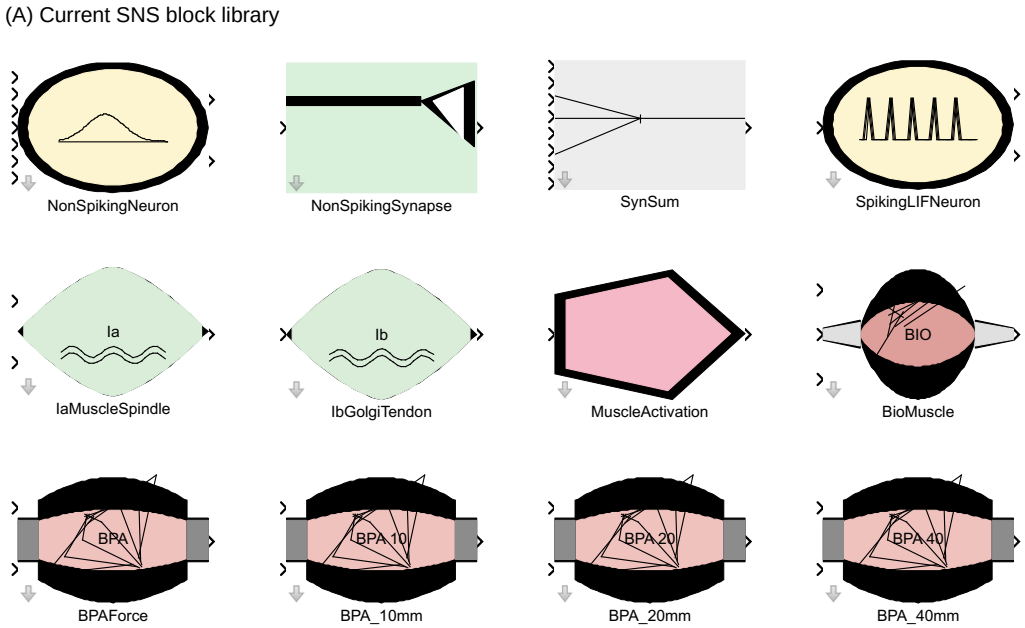


Figure 6.2: Model conversion, closed-loop simulation, and proposed hardware deployment. Solid simulation interfaces connect neural activation to BPA force and return afferent signals. The embedded valve-control and network-evaluation system is future work (Chapter 7).



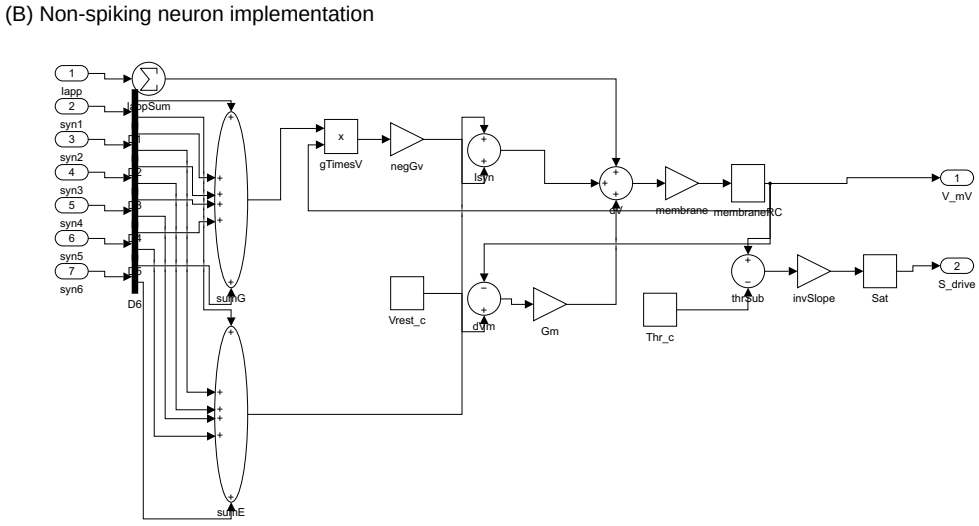


Figure 6.3: Current SNS-Simscape library (A) and native non-spiking neuron implementation (B). The twelve library blocks include biological-muscle and diameter-specific BPA elements. Summed currents drive membrane integration; the threshold and saturation bound neural output.

6.4 Results: Verification of the Implemented Interfaces

The tests address three distinct questions: whether a circuit produces its expected response, whether implementations agree under matched inputs, and whether the neural and mechanical models exchange signals in a closed loop. The larger evaluation sets and their limitations are collected in Appendix D; neuron equations and calibration details remain in Appendix B.

6.4.1 Reflex Response

The library assembles Ia stretch excitation, reciprocal inhibition, and Ib autogenic inhibition in a reduced-order antagonist knee model (Figure 6.4C). A second demonstration tests feedback in an arm holding a beverage container as it is loaded. With the reflex enabled, arm sag was 2.00°, compared with 7.85° when it was disabled. The reduced sag demonstrates that the implemented sensory pathway changes motor activation in a direction that opposes the applied load. It is a functional test of the coupled neural and actuator blocks, rather than an experimental measurement of human-arm behavior.

The current demonstration figures show the mechanical arrangement together with the measured simulation outputs (Figures 6.4 and 6.5). In the knee example, the Simulink and NumPy implementations settle near 44.4° and 43.0°, respectively; the maximum angular difference is 3.6°. Thus, the coupled example reproduces the qualitative response with a measurable numerical discrepancy.

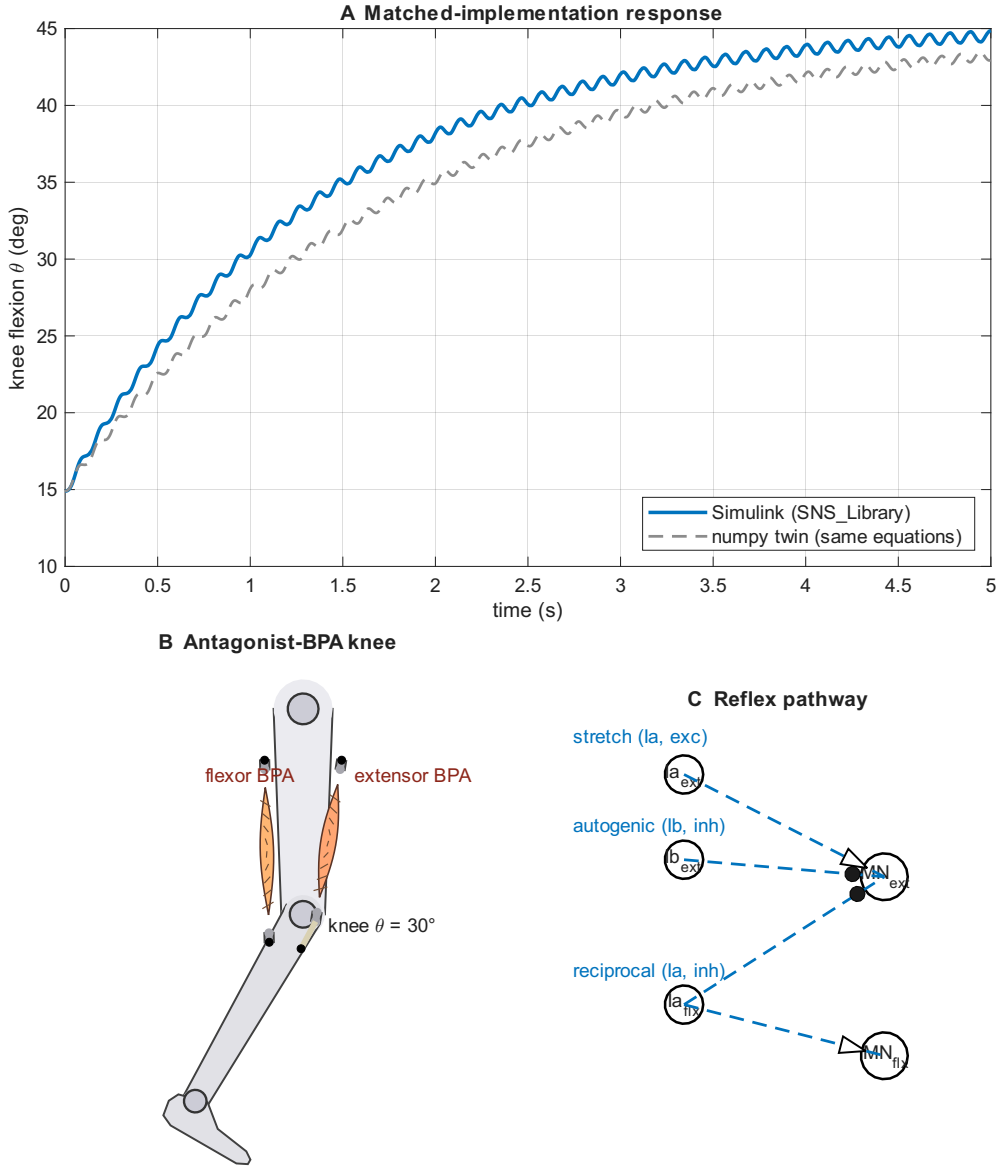


Figure 6.4: Knee-reflex demonstration: matched-implementation kinematics (A), antagonist-BPA leg model (B), and sensory pathways (C).

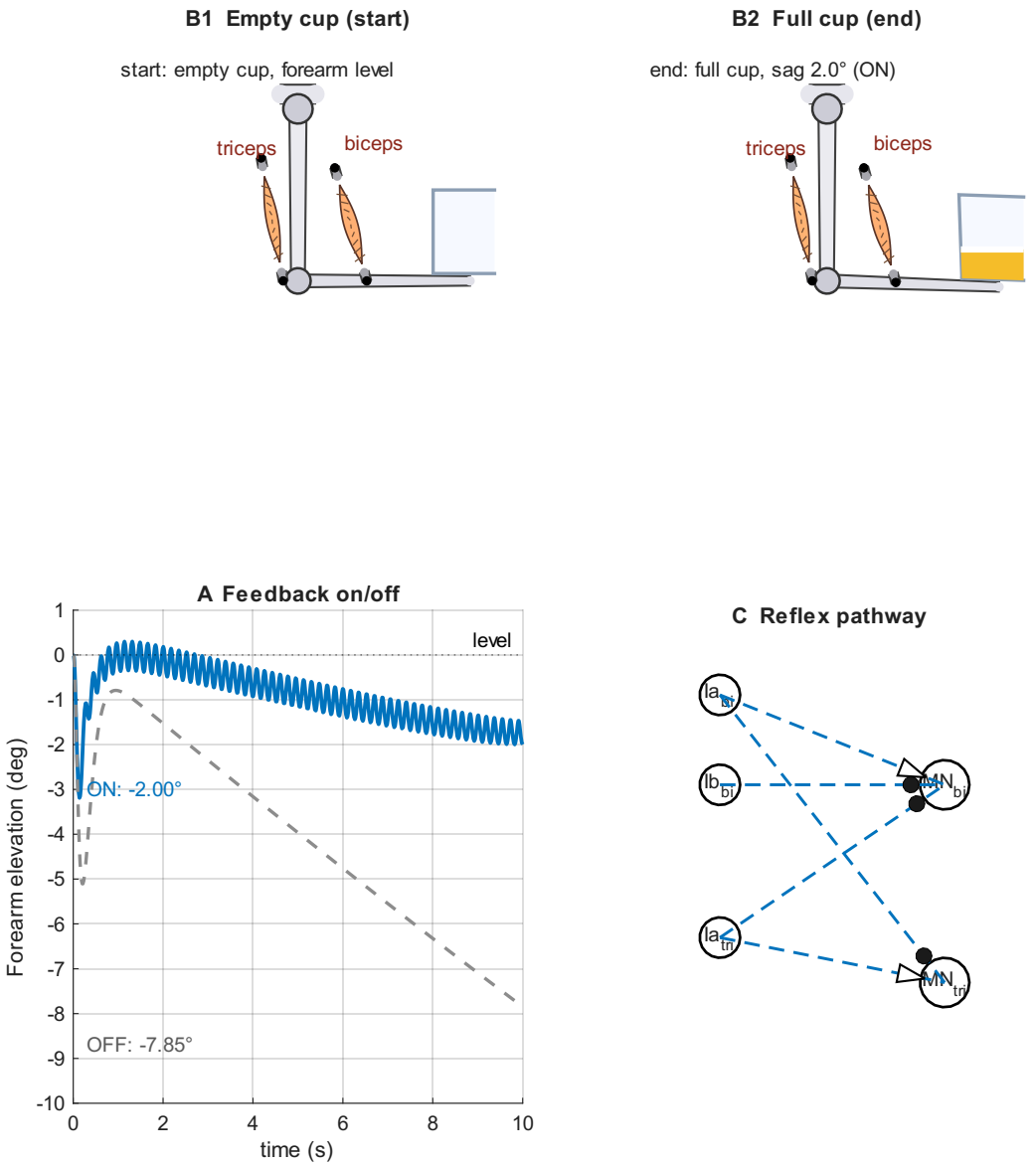


Figure 6.5: Loaded-arm demonstration: feedback-on/off kinematics (A), empty and loaded cup positions (B1, B2), and reflex pathways (C).

6.4.2 Matched Neural Implementations

The contact-driven W2L rhythm core was implemented in NumPy and native Simulink blocks from a common connectivity specification. Both implementations received the same input protocol. Burst periods matched at 1.000 s, all ten expected bursts were reproduced, and voltage-trace correlation was at least 0.99996 (Figure 6.6). The agreement is specific to this circuit and input protocol.

Removing the phasic heel-contact inputs caused the circuit to latch in both implementations. The alternating activity therefore depends on the specified sensory drive.

6.4.3 Conversion Checks and Muscle-Force Agreement

The conversion checks test the quantities needed to interpret a neural-control experiment: the direction of each muscle's joint action, the coupled motion of path points, the force scale, and the intended ground contacts. They exposed reversed hip actions, inconsistent path-point limits, unintended contact pairs, and incorrect force scales in individual actuator entries. Correcting those faults made the converted human model usable for the subsequent controller tests. Appendix D.6 gives the individual repairs and the full force comparison.

At matched poses and OpenSim static-optimization activations, eight of twelve sampled muscles had force-prediction R^2 between 0.87 and 0.98 and mean forces within 10 % of the reference. Rectus femoris and biceps femoris short head had negative R^2, indicating errors greater than a constant mean-force prediction. These results support selected force predictions while identifying remaining muscle-specific differences. Because the

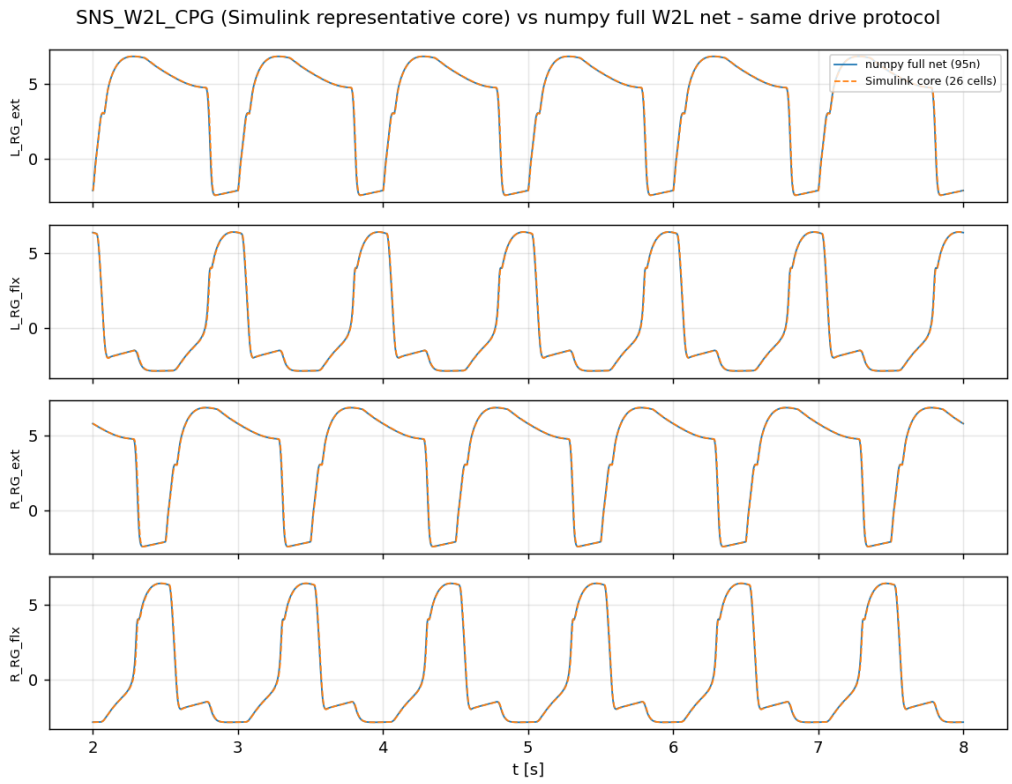


Figure 6.6: W2L rhythm-generator voltages under matched inputs in NumPy and Simulink. Burst periods and counts agree; full-trace correlation is at least 0.99996. The residual 2 ms lag is one discrete solver step. The test concerns the driven rhythm core, not walking mechanics.

comparison sets pose and activation frame by frame without reproducing trajectory velocities or tendon dynamics, it does not establish dynamic equivalence between the muscle formulations. That distinction matters when interpreting feedback based on muscle force.

6.4.4 Closed-Loop Model Coupling

The MathWorks MuJoCo–Simulink bridge exchanges activation commands and mechanical measurements between a Simulink controller and an existing MuJoCo mechanical model. I paired this bridge with SNS-Simscape and tested single-muscle activation, different control rates, and sensor readback against a Python integration. Readback was bit-exact over the scored window. An exported neural network also drove all 92 actuator channels during a 5 s run without afferent return; a separate closed-loop reflex experiment exercised both commands and sensory measurements. These tests verify signal exchange through the interface. Under tonic drive, one Simulink circuit realization reached a fixed point while its NumPy reference oscillated, showing why successful signal exchange must be followed by circuit-specific dynamical checks before transferring tuned parameters.

6.4.5 Feedback Contribution at Fixed Controller Settings

At fixed controller and drive settings, removal of contact and Ib force-feedback pathways reduced stepping in the supported W2L model from 33 left and 28 right cycles to one left and no right cycles. Forward displacement decreased from 1.16 m to 0.09 m. These measures cover the 25.499 s scored window after settling within each 30 s simulation. The harness configuration was unchanged, although its measured support contribution differed

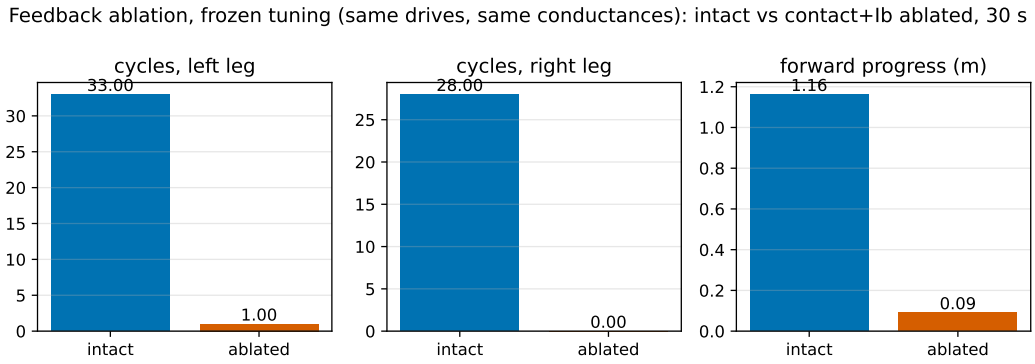


Figure 6.7: Contact and Ib force-feedback ablation at fixed controller and drive settings in the supported W2L model. Counts and displacement refer to the post-settling scored window. The retained Ia and II pathways distinguish this test from complete deafferentation.

between conditions. Ia and II pathways remained active. This comparison demonstrates the contribution of the removed pathways at this operating point; it uses the standard muscle-actuated W2L model.

6.4.6 Repeatability and Mechanical Support

A separate W2L configuration was independently replayed four times after verifying that the saved synaptic gains reached their intended connections. All four 30 s runs produced identical trajectories. Over the 4.5 s to 30 s scoring window, the model produced 28 left and 21 right contact events, a median bilateral phase of 0.496 cycles, and 2.246 m forward displacement. The fitted forward speed was $0.0918\,\mathrm{m\,s^{-1}}$. All twelve reruns with 1 % parameter perturbations retained supported stepping. This configuration is distinct from the fixed-controller ablation in Figure 6.7; the two experiments test different questions.

These results establish repeatability under the specified assistance. Setting the constant

weight-support term to zero still left a vertical position-control harness supplying approximately 16.76 % of body weight. Contact duty also remained below the strict walking criterion. Reduced nominal support therefore cannot be reported as harness-free walking. The detailed support accounting is in Appendix D.5.1, and the compiled network's connectivity and conductance values are in Appendix B.5.

A separate CAD-derived robot with four-bar knees and spherical hips completed a 20 s supported stance test. Its controller contributed to maintaining the configuration: disabling muscle forces caused lumbar failure after 0.202 s. However, the rig carried 94.68 % of body weight, and the actuators supplied reference tensions rather than calibrated BPA forces. This establishes a coupled stance demonstration, with independent balance and BPA-controlled walking still unverified (Appendix D.5.2).

6.5 Discussion: What the Tools Establish

The principal contribution is a set of executable neuromechanical models and experimental tools that make the roles of neural pathways and actuator mechanics accessible to controlled tests. The initial AnimatLab model established the combined walker and CPG organization. The Gait2392 implementations extend controller development to a human-scale model with many joints and actuator channels, while the BPA implementations connect the measured actuator-force model to neural control. SNS-Simscape provides a graphical representation of the pathways, and the curriculum and experiment drivers provide repeatable changes in drive, support, feedback, and perturbation conditions.

The representative results answer different parts of the knowledge gap. Agreement be-

tween the NumPy and Simulink rhythm cores establishes that the specified circuit response can be preserved across those implementations under matched inputs. The loaded-arm comparison establishes a corrective effect of feedback in the coupled block model. The fixed-controller W2L ablation shows that its tested stepping behavior depends on contact and Ib-force feedback. The bridge tests establish the command and measurement exchange needed to carry out such experiments across software environments. Together, these results provide an operational method for separating implementation agreement, sensory contribution, and mechanical response.

The supported robot evaluations produced stepping-like motion and, in some configurations, forward progression, but no harness-free walking. Reducing tonic neural drive did not remove mechanical support. Spiking-neuron comparisons preserved a continuous motor-activation interface while changing rhythm timing and, in some runs, limb coordination (Appendix D).

Researchers can use these tools to compare candidate neural pathways, actuator descriptions, and support conditions while recording their effects on the same defined outputs. The supported demonstrations establish this experimental capability; they do not establish balance without the support rig or transfer of every controller between models. Chapter 7 discusses extensions to weight-bearing control and hardware. Those extensions build on the models and tests already demonstrated here.

7 Discussion

This chapter considers how the actuator models, torque corrections, sensory-afferent resource, and simulation tools support biomimetic robot design, and identifies the experiments and integration work still required.

7.1 The Work as a Whole

The outcome that organizes this discussion is that improved first-pass design of a BPA-actuated biomimetic leg is now achievable: a designer can predict actuator force, joint torque, and range of motion accurately enough to commit to an actuator set, a resting length, and a route before the hardware exists (Figure 7.1). At the center of that outcome are the identified design factors that remove the rigid-body assumptions from the torque model (Chapter 4): a constant length offset χ_0, effective stiffnesses χ_1 and χ_2 that lump the series compliance of the attachment brackets, fixtures, and artificial tendons into the bracket frame's principal directions, and a wrapping-loss factor χ_3 that removes usable actuator length as an extensor bends over the joint. The terms were identified on the pinned knee. Applied to the biomimetic knee, they reduce prediction error, although the remaining error depends on the configuration. The biomimetic extensor's fraction of variance unexplained falls from 4.49 to 0.67 with the shared terms (Section 4.3.2). Whether the set is complete is itself now an experimental question: a fourth term χ_4, a stiffness acting along the force path independently of force orientation and presently absorbed by the stiffness pair (χ_1 and χ_2), is arguably the next candidate, and the identification

machinery delivered here is the instrument for settling it.

Improved BPA characterization is the first enabler (Chapter 3). The maximum isometric force of Festo BPAs was shown to depend strongly on resting length, a previously unreported characteristic that corrects the manufacturer's tool at exactly the short lengths a leg design requires, and a nondimensional force surface predicts isometric force from relative strain and relative pressure with only three diameter-dependent coefficients. Because resting length is a primary degree of freedom in actuator selection, this model is what makes that selection rational. The quasi-static surface is complemented at the dynamic end by the state-space, optimization-based model of pressure and force dynamics in pulse-modulated BPAs [Elzein et al., 2025], which supplies the actuator behavior that the isometric maps cannot.

Attachment and routing optimization is the second enabler. Morrow, Bolen, and Hunt developed a mesh search for three lumbosacral muscle replacements, balancing torque matching against actuator length, attachment displacement, and diameter [Morrow et al., 2020]. This provided a systematic response to the finding that human attachment locations do not transfer directly to a BPA-actuated robot [Bolen, 2019, Bolen and Hunt, 2019]. The later knee-route optimization uses the corrected torque model and a continuous search over attachment, routing, and actuator variables, with explicit mechanical constraints (Section 4.3.2). In the route-design predictions, with two 20 mm BPAs per route and routes chosen with the actuators' bending in mind, the redesigned flexor and extensor each meet or exceed the magnitude of their human-scaled torque targets across the full knee range of motion, and the cam-profile extensor converts the direct route's thin margin into genuine design headroom (Section 4.3.2). Stated as a design program, the generalization is one or

more BPAs replacing one or more human muscles across one or more joints, serving the two degrees of freedom that matter most at each joint, with routes grown outward from positions close to the bone and held to the spatial constraints of the candidate actuators and of the BPAs already mounted; the broader route-search diagnostics are collected in Appendix C.8. Where the attachments are fixed and only the route and the tendon length can change, the cam-shape solver applies: the tibial-side route points trace a designed cam profile, trading biomimicry for torque-profile optimization and reduced component forces.

The Sensory Afferent Database (SADb) is the third enabler (Chapter 5). It organizes the sensory-afferent literature into a structured afferent resource [Bolen et al., 2023]: of the 561 papers retained after the audit and the prune, 523 carry structured curation notes, 86 carry an afferent classification, and 258 carry notes translating each paper toward robot or simulation use, and the full corpus of 943 papers, including the 382 papers outside the Airtable working base, is available in the application. The toolset matters as much as the corpus. Its detailed table drills from any filter down to a paper's full notes; its pivot view counts the corpus along any two dimensions at once; its bubble map draws every paper as a neuron, soma sized by citations, with citation links rendered as excitatory and inhibitory synapses in the laboratory's convention and a focus mode that walks the citation neighborhood. The monthly literature-scraping extension is under development; it is separate from the completed curation and application described here. The evaluation of AI-assisted literature triage found limited help with paper discovery and summarization, but inadequate categorization; inclusion and classification therefore remain dependent on human review [Bolen and Hunt, 2024].

The fourth contribution is the set of neuromechanical modeling tools in Chapter 6. The

Li–Deng walker, Gait2392-based controller implementations, and BPA-force integrations provide defined combinations of neural and mechanical models. SNS-Simscape makes the neural pathways editable as block diagrams, and the experiment drivers vary support, neural drive, sensory pathways, and external perturbations. These tools address the difficulty of determining whether a change in movement arises from neural organization, actuator mechanics, or implementation differences. Matched rhythm-core tests establish agreement between selected NumPy and Simulink implementations; the loaded-arm demonstration shows corrective feedback; and the fixed-controller W2L ablation shows that the tested contact and force-feedback pathways sustain its supported stepping. The contribution is both the reusable experimental capability and these specific demonstrations of what it can distinguish.

The coupled-model results must be interpreted under their stated support conditions. Supported stepping and forward progression establish functioning neural–mechanical loops, while removal of tonic drive tests a different dependence from removal of mechanical support. The balance-platform studies provide a separate hardware demonstration of a BPA-actuated limb under pseudo-vestibular inertial feedback [Bolen et al., 2024b,a,c]. Together, the simulation tools and hardware precedents support further study of balance and weight bearing without treating those capabilities as already demonstrated by the stepping tests.

Figure 7.1 shows the first-pass design workflow the delivered enablers make possible. Human torque and range-of-motion targets from the OpenSim benchmark and the validated BPA force model of Chapter 3 drive actuator selection; route design runs under the corrected torque model of Chapter 4; the predicted torque and range of motion are

checked against the targets, with route redesign or actuator reselection as the loop when they fall short; and the candidate configuration is checked in closed-loop simulation with the tools of Chapter 6 before hardware is cut.

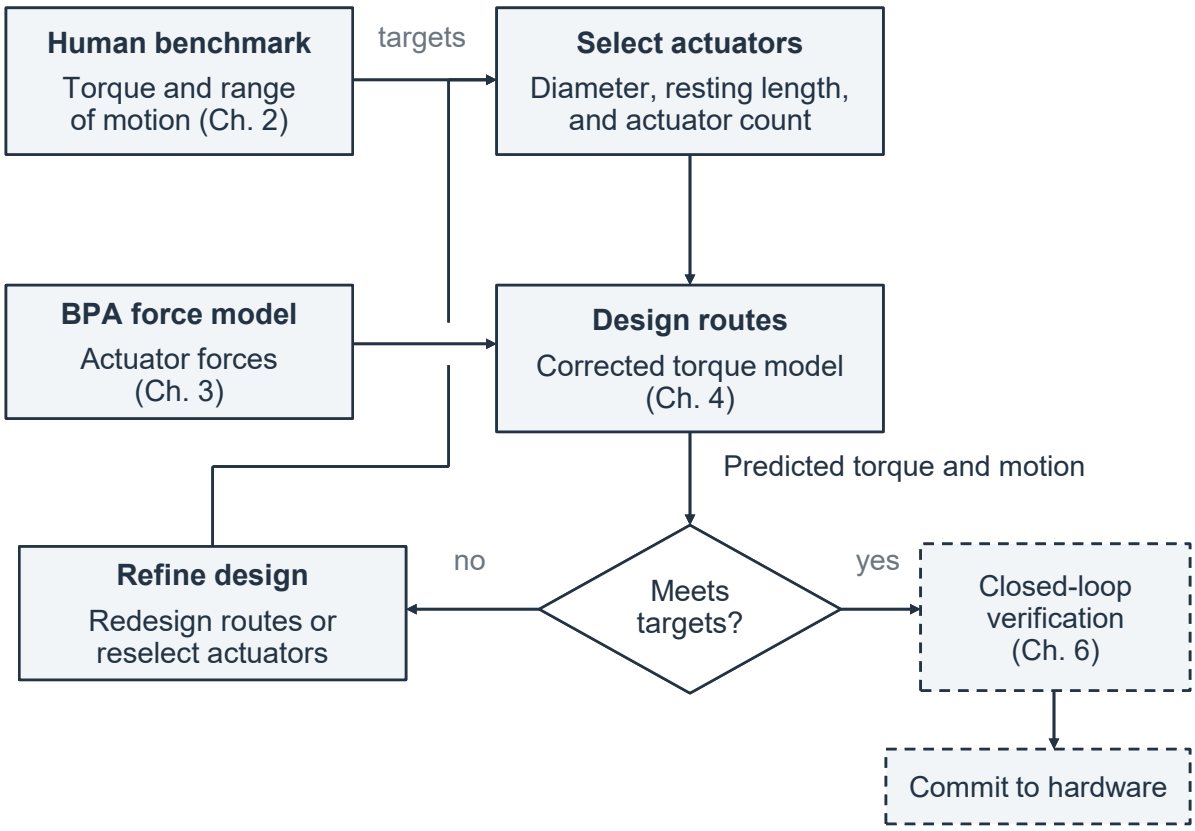


Figure 7.1: Proposed workflow connecting human torque targets, BPA characterization, route design, neuromechanical simulation, and hardware development.

7.2 Future Work

The subsections that follow are options for the researchers who take this program up, ordered from the actuator outward to the whole body. Each draws directly on the delivered models, the database, and the simulation tools.

7.2.1 Expanding the BPA Force Characterization

The isometric characterization of Chapter 3 leaves one actuator curve unidentified and one regime unmeasured, and both are tractable next experiments. The unidentified curve is the maximum-contraction function: ε_{620} could not be pinned to a predictive relationship with resting length, so each actuator's free contraction is recorded individually before use. Controlled tests designed for that relationship, across multiple product batches of the 10 mm, 20 mm and 40 mm diameters, would either produce general fit surfaces or establish batch variation as the binding uncertainty. The 20 mm actuators characterized here came from a single product batch, so batch-to-batch variation is unquantified and those fits in particular carry that limitation.

The unmeasured regime is everything off the isometric plateau. Biological muscle models separate active and passive force and include activation, contraction, and tendon dynamics [Thelen, 2003, Millard et al., 2013]. Appendix A presents these relations and explains their parameters. A corresponding BPA model would require separate characterization of active pressurized-braid force, passive resistance of the unpressurized actuator and sleeve, and contraction dynamics. Isokinetic and isotonic or isobaric characterizations could then be assimilated into the same framework rather than fitted separately, with

the pressure–force dynamics model of Elzein et al. [2025] supplying the actuator-side dynamics.

Two experimental programs would complete the mechanical picture. Instron tensile testing of actuators with their Onyx and aluminum end caps, unpressurized and pressurized, would measure the end-effect region that the characterization identifies as the mechanism behind the resting-length dependence of maximum force. A finite-element program sketched from the construction disclosed in Festo's patent literature would give the actuator a first-principles model to check the empirical surfaces against [Festo AG & Co. KG, 2026]: a symmetry half-model of the actuator, a half circle fixed at the outer and inner diameter at one end with the diameter free to expand at the other and an optional lateral force or fixity, assembled from thin-shell or membrane layers in inter-layer contact, with the two handed aramid weaves of the braided sleeve represented as beam or truss elements. Figure 7.2 shows the construction such a model would represent.

In Figure 7.2, longitudinal symmetry permits in-plane expansion. At the half-length boundary, axial displacement can be fixed or axial motion allowed under an optional load. Internal pressure loads the contact interfaces; cap details are schematic.

7.2.2 Expanding Attachment and Routing Design

Each routing tier of Section 7.1 leaves a named next step. The knee's adopted flexor redesign evaluates two staggered paths separately. Physical verification of the two paths' clearances, load sharing, and off-axis loading remains necessary before extending that configuration to a complete leg. The bi-pulley sweep showed that reeved routes do not yet

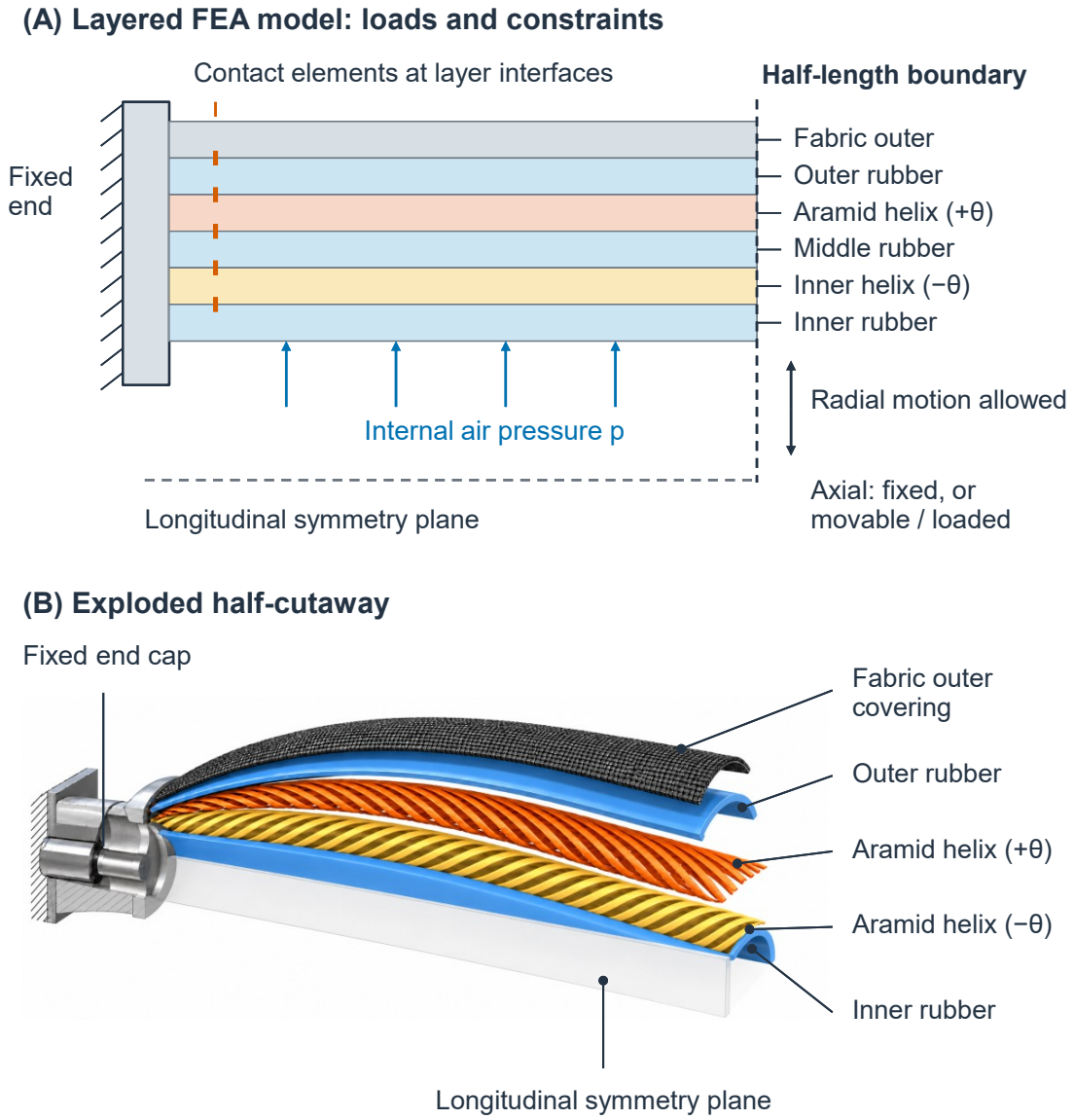


Figure 7.2: Proposed layered BPA finite-element model: contact and boundary conditions (A), and illustrative material cutaway (B). Geometry and layer spacing are schematic; the middle rubber layer is separated only in A.

fit the anatomical corridors, where geometry rather than torque was the binding constraint, so clearance-relaxed attachment sites are where that freedom should pay. The whole-body pass needs its route-builder failure fixed and either larger actuators or relaxed corridors on the two routes that satisfy geometry but fall short of torque.

The design element this program has so far declined is a patella. The torque model wraps the extensor BPA over the joint surface and accounts for the loss with χ_3, which was easier than adding a patella, deterministically designing its location as a function of knee angle, and accounting for friction (Chapter 4). A designed sesamoid is the biomimetic answer: an artificial patella, its location optimized over the range of motion the way route points now are, would buy back moment arm at high flexion and shield the actuator from the joint, at the cost of a friction model and a contact schedule the optimizer must respect. The cam-profile solver is the proof that a moving routing surface can be a design variable; a patella is its biomimetic limit case.

7.2.3 Expanding the Sensory Afferent Database

A monthly literature-scraping extension is being implemented to identify candidate additions to the database. Its retrieval and classification must be evaluated against human decisions before it can support routine updates. Further curation can also target missing afferent classifications, pathway links, and preparation metadata in the existing records. The AI-assisted triage study establishes the relevant division of responsibility: automated tools draft and retrieve, while I retain the inclusion and interpretation decisions [Bolen and Hunt, 2024].

7.2.4 Closing the Loop in Simulation

The simulation laboratory of Chapter 6 is assembled, audited, and in active use. The converted models support individual checks of actuator geometry, force direction, and signal exchange, and the BPA force interface evaluates the isometric force model inside the loop; the implemented interfaces and their limits are described in Section 6.3. The 410-neuron SNS-Toolbox instantiation of the spinal network, with per-leg rhythm-generator half-centers, phase-shifted pattern-formation groups, motoneuron pools for every actuator, and Ia, Ib, and II afferent populations, couples to the converted model at a 2 ms neural step, and supported stepping configurations have been tuned against a walking reference and evaluated on the ground (Section D.3.1). A drive-weaning study has since produced a feedback-sustained bilateral configuration whose stepping persists with the hand-drive withdrawn entirely, still in place under the support rig, and the robot-side model progresses under its harness at weaned drive (Section D.5). What those same evaluations localize as missing defines the work remaining here: the balance and weight-bearing layers that unsupported, bilateral walking requires; transfer from a single tuning reference to the diversity of the locomotion reference library; and the systematic verification series that follows.

The verification series comes first because it is cheap and decisive. The circuits' signs, gains, and phase structure are checked against the experimental literature with the stimulus protocols the physiology itself uses: a tonic stimulus applied at the motoneuron, pattern-formation, and rhythm-generator layers, with the resulting outputs compared against the tonic central and sensory stimulation that facilitates involuntary air-stepping in humans

[Selionov et al., 2009] and the deletion and stimulation phenomena reproduced by the two-level CPG models and their neuromechanical implementations [Rybak et al., 2006, Markin et al., 2010, Shevtsova et al., 2016]. Isolated rhythm-generator tests checked antiphase responses under tonic drive; the motoneuron-level and pattern-formation-level series remain to be run, together with the motoneuron skipping-deletion phenomenon of the reference models.

The laboratory's hypothesis-testing program is stimulus, deletion, and addition. The stimulus and deletion families have their first sweeps (Section D.3.2). Pathway addition is their complement and the one the tools have not yet exercised: adding a candidate circuit to a tuned configuration, a commissural class, a sensory gain, or a modulatory layer, and re-tuning, which turns connectome hypotheses into falsifiable experiments. The closed-loop comparison of spiking and non-spiking controllers of the same topology belongs to the same program, with the spiking mirror's untuned bilateral ground gait as the standing lead (Section D.4).

The existing integrations provide a starting point for extending the model set. A full-body OpenSim model with arms and trunk would require its own geometry and actuator checks after conversion. Its controller could then be assembled or inspected with the neural block tools and evaluated with the relevant simulation interface. The matched-circuit tests and pathway-intervention procedures provide a reusable verification method, but each new model and controller combination must establish its own behavior.

7.2.5 Extending the Sensorium

Once stepping is stable, three supraspinal contributions will be added, and a fourth is in line behind them (Figure 7.3). First, vestibular feedback will be simulated with IMU-like sensing of body orientation and angular velocity, extending the balance-platform work, in which an inverted pendulum with pseudo-vestibular IMU feedback maintained balance on an oscillating platform [Bolen et al., 2024b,a,c] and analytic neural networks reproduced human balance dynamics [Hilts et al., 2019]. Second, ocular feedback will be modeled to investigate how visual stabilization cues can help reset IMU drift, a well-known practical problem for inertial sensing on robots, by treating visual measurements of the horizon and surroundings as a low-frequency correction to the vestibular estimate. Third, a cerebellar layer will be explored as an adaptive gateway between higher-level locomotion commands and the CPG: a structure that can gate between gaits or movement patterns (e.g., walking, running, and turning) and tune the phase relationships among them, in line with the cerebellum's established role in adaptation and coordination. Multilevel spinal-plus-cerebellar synthetic nervous systems have already been demonstrated for insect-like legs [Grummer et al., 2024], and hierarchical cortico-cerebellar-spinal models have been proposed for adaptive step-length control in bipedal locomotion [Chen et al., 2025]. The fourth addition, basal-ganglia circuits realizing the inhibitory and excitatory gating that selects among actions and locomotor modes, would give the controller a substrate for choosing between behaviors rather than only executing one. The database already holds the entry points for this program: the human balance and posture cluster and the multisensory recalibration cluster are curated and connected to the locomotor

core through the citation graph (Chapter 5), and the proprioceptive–vestibular interaction literature will guide how the signals are weighted [Akay and Murray, 2021].

Figure 7.3 shows the control stack with the planned supraspinal additions. The spinal network tier, the two-level central pattern generator with motor-neuron pools and Ia, Ib, and II afferent plus heel and toe contact feedback, and the BPA-actuated body tier are implemented and demonstrated (Chapter 6), with the balance-platform studies as the demonstrated precursor of the vestibular channel. The dashed supraspinal tier is the extension program of this section: basal-ganglia circuits for action and mode selection, a cerebellar adaptive gateway between high-level commands and the pattern generator, and vestibular and ocular sensing closing body-orientation and visual feedback onto the same loop.

7.2.6 Real-Time Hardware Realization

The final stage returns to hardware, and its near version is a step-by-step integration problem for which the laboratory's existing equipment already stocks the parts. The stack is fixed: Teensy-class microcontrollers handle the low-latency per-joint loops (valve control, pressure sensing, and contact sensing), and an NVIDIA Jetson Orin serves as the companion computer for SNS network evaluation and activation computation. The integration steps then run in order. Execute the SNS-Toolbox network in real time on the treadmill biped. Close feedback from the sensors the lab already uses: the joint encoders, the actuator pressure sensors, the liquid-wire strain gauges, the pressure manifolds, and the IMUs. Compute the afferent signals from those streams exactly as the simulation

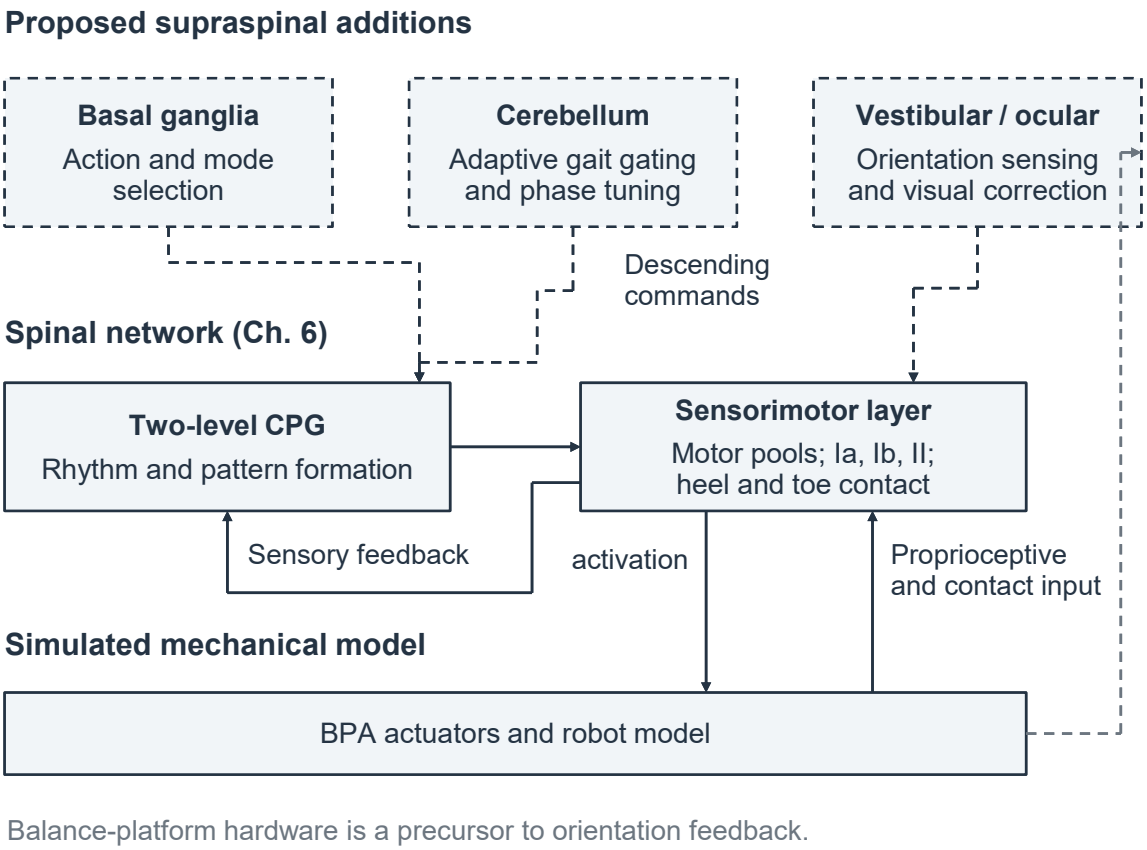


Figure 7.3: Control hierarchy connecting spinal pathways and the BPA-actuated model to proposed supraspinal and sensory extensions. Dashed elements indicate future work.

tools compute them, Ia length and velocity and II length from the encoders, and Ib force from pressure-based force estimates checked against the strain gauges. Tune computer-in-the-loop, with the embedded stack executing the network against the simulated body before it executes against the physical one, so that gain changes are tested at simulation speed and validated on hardware only when they survive. Keep the serial connection to the host computer for data logging and high-level commands only (e.g., run, walk, or stimulate a named neural pathway), so the control loop stays fully embedded. The wiring schematics, the neural-pathway schematics, and the step-by-step commissioning procedure that this integration produces belong in a hardware appendix rather than in this chapter; the connectome figures of Appendix B already serve as the pathway schematics for the spinal network.

The precedent and the margins both support feasibility. Walking control of a physical musculoskeletal biped with McKibben muscles has been demonstrated with spring-loaded inverted-pendulum controllers [Li et al., 2024], and the AARL's quadruped has already traveled part of this road: a MuJoCo model of the BPA quadruped was coupled to SNS-Toolbox, produced closed-loop coordinated stepping, and was transferred to hardware with potentiometer and pressure feedback on a treadmill [Lutz, 2025]; the present program extends that approach from the quadruped to a leg built on human anatomy. An actuation-level design lever also remains open: the BPA's limited contraction can force long musculotendon routings, a burden visible in the extensor redesign bounds of Appendix C and encountered in my master's routing work [Bolen and Hunt, 2019]. A block-and-tackle "reverse pulley" arrangement, demonstrated on the AARL's self-balancing inverted-pendulum platform, addresses this by trading transmitted force for travel: paired BPAs pulled through com-

pound pulleys on a tendon coupled to the limb, increasing range of motion at the cost of reduced transmitted force and increased reaction force on the structure [Bolen et al., 2024b,a,c]. Because joint torque is computed in the body frame, configurations that turn that reaction force to advantage may exist, making the arrangement a candidate for joints whose optimized routes demand impractically long actuators.

Because the static force and torque models of Chapters 3 and 4 are computationally inexpensive lookup-table candidates, they will serve as the feedforward model inside the real-time controller, while the dynamic pressure–force behavior under pulse-modulated control will draw on the state-space model of Elzein et al. [2025]. Embedded execution of the complete CPG-plus-reflex network on this stack is the intended demonstration of the dissertation project: a robot that walks with human-like biomechanics under a biologically grounded neural controller, and whose nervous system, unlike a living animal's, can be instrumented, stimulated, and lesioned at will.

7.2.7 Toward Dynamic Whole-Body Maneuvers

Beyond steady walking, the long-term goal of this line of research is dynamic, whole-body maneuvering: transitions between gaits, reactive balancing, and ultimately ballistic maneuvers that require coordinating the full actuator set through brief, precisely timed activation bursts. My aspiration for this program is a robot that can perform a kickflip, a maneuver that, like its skateboard namesake, demands explosive coordinated extension, precise timing, aerial body orientation, and controlled landing. It is exactly the kind of behavior that exercises every layer of the architecture described above, from contact

receptors and spinal pattern generation to vestibular orientation sensing and cerebellar coordination, and it would demonstrate that biomimetic actuation and neuromechanical control scale from steady locomotion to dynamic whole-body movement.

These subsections are options, not prerequisites; each can be pursued on the delivered framework independently of the others. What they all draw on, the actuator and torque models, the routing machinery, the afferent database, and the simulation laboratory, is consolidated in Chapter 8.

8 Conclusion

This dissertation moved the biomimetic humanoid robot project from an unvalidated design concept to a quantitatively characterized actuator set and experimentally tested knee-torque models, and it defined the path from those experimental foundations to neuromechanical control.

At the actuator level, the analysis in Chapter 3 produced novel equations for calculating isometric force in Festo 10 mm and 20 mm-diameter BPAs. The work elucidated the relationship between maximum BPA force at 620 kPa and resting length, $F_{620}(l_0)$, a previously unreported characteristic of these actuators that the manufacturer's tool predicts with the opposite trend, and it produced a nondimensional force surface, $F^*(\epsilon^*, P^*)$, with only three diameter-dependent coefficients. Taken together, these fits predict isometric BPA force across resting lengths from a small number of measurements, correct the large overpredictions of the manufacturer's tool at short resting lengths, and outperform published single-length models. In a design workflow, these equations make it possible to select actuator diameter and resting length and to evaluate whether a proposed muscle routing can meet force and range-of-motion requirements, before committing to a prototype.

At the joint level (Chapter 4), the BPA model was combined with transmission corrections for effective length, series compliance, and wrapping. Identification on the pinned knee improved torque prediction, and the shared correction terms also improved predictions on the biomimetic four-bar knee. The corrected model supports route redesign: the

optimized two-BPA flexor and cam-extensor configurations are predicted to meet their human-scaled torque targets across the design range. The isometric tests demonstrate that the redesigned configurations meet or exceed their selected human-muscle reference torques at all fifteen flexor and nine extensor measurement points. This establishes capacity at the sampled angles and pressures, while agreement with pressure-specific predictions remains uneven.

The Sensory Afferent Database in Chapter 5 organizes physiological evidence by properties and experimental context, making it easier to select and assess candidate feedback pathways. Chapter 6 addresses the complementary problem of testing those pathways in coupled neural and mechanical models. I developed the Li–Deng walker implementations, Gait2392-based neuromechanical models, SNS-Simscape neural blocks and actuator integrations, and staged tuning and experiment procedures. Selected demonstrations establish implementation agreement for a contact-driven rhythm core, corrective feedback in a loaded-arm model, and feedback-dependent stepping in a fixed supported W2L configuration. These results show that the tools can distinguish circuit behavior, sensory contribution, and mechanical response under specified conditions.

The demonstrated contributions support controlled neuromechanical experiments with explicit neural pathways and actuator descriptions. Extending these results to unsupported whole-body behavior and hardware requires additional balance, dynamic-actuation, and transfer studies (Chapter 7). The present findings concern the measured actuator and joint behavior and the specified simulation tests.

The dissertation provides experimentally grounded actuator and joint models, a curated afferent resource, and functioning neuromechanical modeling and evaluation tools.

Together, they make actuator selection, torque prediction, sensory-pathway selection, and controlled simulation tests more explicit and reproducible. These are usable contributions to biomimetic robot design and to studying how neural control interacts with mechanical structure.

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APPENDICES

Appendix A: Muscle Force and Joint Torque Calculations

This appendix documents the reference implementation of the force, torque, and compliance equations used in Chapters 3 and 4. When present in MATLAB code, file names are given so that any number in this dissertation can be traced to the code that produced it.

A.1 Human muscle force calculations

A.1.1 OpenSim Thelen2003 and Millard2012 Formulations

The Thelen2003 and Millard2012 equilibrium muscle models describe active and passive fiber force, pennation, and a series tendon [Thelen, 2003, Millard et al., 2013]. Their common force balance is

$$F_T(l_T) = F_M(l_M, \dot{l}_M, a)\cos\alpha, \qquad l_{MT} = l_T + l_M\cos\alpha, \tag{A1}$$

where a is activation, α is pennation, and l_M, l_T, and l_{MT} are fiber, tendon, and musculotendon lengths. The cosine projects fiber force onto the tendon direction. When tendon compliance is enabled, fiber length is a state determined by the equilibrium calculation rather than the full musculotendon path length.

In the damped Millard formulation, fiber force has the form

$$F_M = F_0\left[a f_L(\tilde{l}_M) f_V(\tilde{v}_M) + f_{PE}(\tilde{l}_M) + \beta\tilde{v}_M\right], \tag{A2}$$

with normalized fiber length $\tilde{l}_M$, normalized velocity $\tilde{v}_M$, maximum isometric force F_0, and damping coefficient β. The active, passive, and tendon curves and the treatment of singular configurations differ between the Thelen and Millard models. The OpenSim implementation can also disable tendon compliance or activation dynamics, so the model class alone does not establish which dynamics were exercised in an analysis [Millard et al., 2013].

Thelen activation and contraction dynamics

The following seven relations correspond, in order, to Equations (1)–(7) of Thelen [2003]. They specify the activation state, passive and active force–length curves, tendon force, and fiber velocity. Bars denote normalization by maximum isometric fiber force F_0^M or optimal fiber length L_0^M, as appropriate.

Activation follows

$$\frac{da}{dt} = \frac{u - a}{\tau_a(a, u)}, \tag{A3}$$

with an activation-dependent time constant

$$\tau_a(a, u) = \begin{cases} \tau_{\text{act}}(0.5 + 1.5a), & u > a, \\ \tau_{\text{deact}}/(0.5 + 1.5a), & u \leqslant a. \end{cases} \tag{A4}$$

Here $u \in [0, 1]$ is neural excitation, a is muscle activation, and τ_{act} and τ_{deact} set the activation and deactivation time scales.

The normalized passive fiber force and active force–length multiplier are

$$\bar{F}^{PE} = \frac{\exp\left[k^{PE}(\bar{L}^M - 1)/\varepsilon_0^M\right] - 1}{\exp(k^{PE}) - 1}, \tag{A5}$$

$$f_l = \exp\left[-\frac{(\bar{L}^M - 1)^2}{\gamma}\right]. \tag{A6}$$

The tendon relation is

$$\bar{F}^T = \begin{cases} \dfrac{\bar{F}^T_{\text{toe}}}{\exp(k_{\text{toe}}) - 1}\left[\exp\left(\dfrac{k_{\text{toe}}\varepsilon^T}{\varepsilon^T_{\text{toe}}}\right) - 1\right], & \varepsilon^T \leqslant \varepsilon^T_{\text{toe}}, \\ k_{\text{lin}}(\varepsilon^T - \varepsilon^T_{\text{toe}}) + \bar{F}^T_{\text{toe}}, & \varepsilon^T > \varepsilon^T_{\text{toe}}. \end{cases} \tag{A7}$$

The fiber-velocity relation and its shortening/lengthening denominator are

$$V^M = (0.25 + 0.75a)V^M_{\max}\frac{\bar{F}^M - af_l}{b}, \tag{A8}$$

$$b = \begin{cases} af_l + \bar{F}^M/A_f, & \bar{F}^M \leqslant af_l, \\ \dfrac{(2 + 2/A_f)(af_l\bar{F}^M_{\text{len}} - \bar{F}^M)}{\bar{F}^M_{\text{len}} - 1}, & \bar{F}^M > af_l. \end{cases} \tag{A9}$$

Here $\bar{F}^M$ is the normalized active fiber force entering the force–velocity relation, $V^M = \dot{L}^M$ is negative during shortening, and $V^M_{\max}$ is the maximum shortening-speed scale. Table A1 defines the curve parameters. A shape factor is a dimensionless parameter controlling a curve's curvature or width; it is not a stiffness in N/m. For example, increasing γ broadens the active force–length curve while preserving its peak at $\bar{L}^M = 1$. Changing k^{PE} changes the passive curve's shape while retaining unit force at strain ε_0^M.

Table A1: Length, strain, and shape parameters in the Thelen formulation.

Symbol	Definition
$\bar{L}^M = L^M / L_0^M$	Fiber length relative to optimal fiber length.
ε_0^M	Passive fiber strain at force F_0^M.
k^{PE}	Passive force–length exponential shape factor (5 in the paper).
γ	Active force–length width parameter (0.45 in the paper).
ε^T	Tendon strain relative to tendon slack length.
ε_0^T	Tendon strain at force F_0^M.
$\varepsilon_{\text{toe}}^T$	Strain at the end of the nonlinear tendon region; $0.609\varepsilon_0^T$.
$\bar{F}_{\text{toe}}^T$	Normalized force at that transition; 0.33.
k_{toe}, k_{lin}	Tendon exponential shape factor (3) and linear slope ($1.712/\varepsilon_0^T$).
A_f	Force–velocity shape factor (0.25 in the paper).
$\bar{F}_{\text{len}}^M$	Maximum normalized lengthening-force factor.

The passive expressions describe tensile loading; they do not imply that a slack muscle or tendon transmits compression. These are the original paper's constitutive relations. Numerical implementations must also handle slack states, minimum activation, pennation limits, and the inversion of the force–velocity relation. Consequently, the original equations alone do not specify every numerical safeguard in an OpenSim implementation.

Millard equilibrium and damping

The equilibrium relation of Millard et al. [2013] can be written in normalized form as

$$\left[a f_L(\tilde{l}_M) f_V(\tilde{v}_M) + f_{PE}(\tilde{l}_M)\right] \cos\alpha - f_T(\tilde{l}_T) = 0. \tag{A10}$$

The damped formulation adds $\beta\tilde{v}_M$ inside the brackets, as in Equation A2. The functions f_L, f_V, f_{PE}, and f_T describe active force–length, force–velocity, passive fiber force, and

tendon force. Millard's smooth characteristic curves and damping address numerical behavior while retaining the series force balance. Figure 3 of that paper illustrates the equilibrium, damped-equilibrium, and rigid-tendon formulations. A rigid-tendon model removes tendon stretch as a state; it does not simply set the passive fiber force to zero.

A.1.2 OpenSim Moment Arms and Human Torque Targets

The human torque targets of Section 4.1.5 are OpenSim tendon-excursion moment arms: for each muscle, the moment arm about the knee is the partial derivative of muscle-tendon length with respect to the knee coordinate, with positive tensile force and torque positive in the coordinate direction, $r(\theta_k) = -\partial l_{\mathrm{MT}} / \partial \theta_k$, evaluated across the range of motion. The exported analysis tables archived with the test data (`OpenSim_Vasti_MomentArm.txt`, `OpenSim_Vasti_Results.txt`, `OpenSim_Bifem_MomentArm.txt`, `OpenSim_Bifem_Results.txt`) carry the per-muscle moment arms and the corresponding maximum-isometric torque curves (moment arm multiplied by each muscle's maximum isometric force) for vastus intermedius, lateralis, and medialis, and for the biceps femoris short head; the optimization loads the vastus medialis and biceps femoris short-head columns as its targets (`buildKneeExtContext20mm.m` lines 119–140; `Knee_Flexor_Data_20mm.m` line 546). For an unconstrained planar rotation, this derivative definition and the geometric perpendicular-foot construction of Equation (A14) below are equivalent for a straight muscle path crossing a planar revolute joint; the constrained and moving-path-point geometry is evaluated by OpenSim rather than by substituting the robot-side straight-path construction.

A.2 Converted MuJoCo Muscle Formulation

A.2.1 MuJoCo Muscle Actuator

The converted model uses MuJoCo muscle actuators, which combine activation-dependent active force with passive tensile force. Let L denote the routed musculotendon length, L_T the constant biological-tendon length, and L_0 the optimal fiber length inferred from the actual and normalized operating ranges. The normalized muscle length is $\bar{L} = (L - L_T)/L_0$. The signed actuator force is

$$F_{\text{act}} = -F_0 \left[a f_L(\bar{L}) f_V(\dot{L}/(L_0 v_{\max})) + f_P(\bar{L}) \right]. \tag{A11}$$

Here $v_{\max}$ is the maximum shortening speed in optimal fiber lengths per second. The negative sign denotes a pulling actuator force; both active and passive terms contribute to tension. The rigid tendon has a constant length, but no elastic state. MuJoCo also omits pennation, so its differences from the OpenSim equilibrium formulations are not limited to tendon compliance [MuJoCo Developers, 2023].

A.3 BPA Force Model Implementation

A.3.1 Strain Measures

The musculotendon path length l_{LMT} is the sum of straight segments between route via points, transformed into a common frame at each frame crossing. The contractile strain used by the force model subtracts the artificial tendon and both end-cap fittings from the

path length before normalization by the resting length,

$$\varepsilon \;=\; \frac{l_0 - \left(l_{LMT} - l_{ten} - 2\,l_{fitting} - \chi_0\right)}{l_0}, \tag{A12}$$

where χ_0 is the constant length offset of Section 4.1.4 (zero in the uncorrected model). [Provenance: `MonoPamDataExplicit.m` get.Contraction, lines 210–217; `minimizeFlxPin.m` lines 415–426. End-cap fitting length $l_{fitting} = 25.4\,\text{mm}$ is a constructor default, `MonoPamDataExplicit.m` line 69.]

The maximum free-load contraction at 620 kPa is $\varepsilon_{620} = (l_0 - l_{620})/l_0$, where l_{620} is the BPA length at 620 kPa with no external load (code variable `KMAX`; `MonoPamDataExplicit.m` line 278). The relative strain is $\varepsilon^* = \varepsilon/\varepsilon_{620}$ (line 281).

A.3.2 Normalized Force Surface

The normalized isometric force surface of Equation (3.4) is implemented in closed form in `Functions/f_festo.m` (lines 1–26),

$$F^* = c_0\left(e^{-c_1\varepsilon^*} - 1\right) + P^*\,e^{-c_2(\varepsilon^*)^2}, \qquad F^* \leqslant 0 \;\Rightarrow\; 0, \tag{A13}$$

with $P^* = P/620\,\text{kPa}$. The zero-clamp $F^* \leqslant 0 \Rightarrow 0$ is applied by the callers (e.g. `Functions/festo4.m` line 35), not inside `f_festo.m` itself. The coefficients were fitted with `fittype` nonlinear least squares and least-absolute-residual robustness (`Testing_Data/bpaFits.m`, lines 45–60 and 115–140, lower bounds $[0, 0, 0]$) and are reported in Table 3.2. The code additionally holds a 40 mm fit not used in this dissertation ($c_0 = 0.1224$, $c_1 = 10.47$,

$c_2 = 2.023$; `f_festo.m` lines 17–20).

At runtime the torque model may alternatively evaluate a spline interpolation of the manufacturer's datasheet curves (`Functions/festo4.m`, backed by `FestoLookup.mat`), with force clamped to zero for $\varepsilon^* > 1$ or $F^* < 0$. Equation (A13) replaced this lookup in the corrected model.

A.3.3 Maximum Force

The resting-length-dependent maximum force of Equation (3.5) is implemented in `Functions/maxBPAforce.m` (10 mm: line 46; 20 mm: line 57; duplicate for 20 mm in `Functions/Fmax20.m`) with the fixed 7.5 mm end-contact offset. The fitted 20 mm coefficients carry confidence intervals $a_1 \in [1.4745, 1.5011]\,\mathrm{N\,kPa^{-1}}$ and $a_2 \in [0.0238, 0.0258]\,\mathrm{kPa^{-1}\,m^{-1}}$ (`Fmax20.m` lines 26–27). Total force is the product $F = F^*(\varepsilon^*, P^*)\, F_{620}(l_0)$, resolved into a vector along the force direction (`MonoPamDataExplicit.m` lines 283–287).

A.4 Muscle Path Length and Moment Arm

Path length accumulates the Euclidean norm of each segment between via points, applying the body-frame transformation whenever the path crosses between the femur and tibia frames (`MonoPamDataExplicit.m` lines 78–95, total at lines 122–133).

The vector moment arm is implemented as the perpendicular foot of the joint origin onto the line of action,

$$\mathbf{r} \;=\; \mathbf{p} \;-\; \hat{\mathbf{u}}\,(\hat{\mathbf{u}} \cdot \mathbf{p})\,, \tag{A14}$$

where $\mathbf{p}$ is the vector from the joint origin to the muscle-path crossing point and $\hat{\mathbf{u}}$ is the unit force direction (`MonoPamDataExplicit.m` lines 156–168; identical implementations in `MonoMuscleData.m` lines 111–123, the `Mom` function of `minimizeFlxPin.m` at line 539, and `MonoPam_mult.m` lines 186–201). This is the same quantity as the cross-product projection of Equation (4.2); the planar scalar moment arm about the joint axis is the magnitude of the in-plane components, computed as `hypot(mA(:,1), mA(:,2))` (`predictKneeFlexor20mm.m` line 217). Torque about the joint z-axis is $\mathbf{m} = \mathbf{r} \times \mathbf{f}$, evaluated only where the BPA is in tension (values are discarded for $\varepsilon < -0.02$, the excessive stretching region; `MonoPamDataExplicit.m` line 308).

A.5 Compliance Model Implementation

The compliance-corrected muscle length entering Equation (4.5) is computed as $l_m = l_{LMT} - \chi_0 - l_{ten} - 2\,l_{fitting}$ (`minimizeFlxPin.m` lines 415–426; identically `minimizeExtX3.m` line 345). In the two-bracket fit of record, the insertion-bracket stiffness matrix and the projected scalar stiffness along the force direction are

$$\mathbf{K}_{br} = \mathrm{diag}(\chi_1,\, \chi_2,\, \chi_1), \qquad k_{\hat{u},br} = \left(\hat{u}^{\,T}\mathbf{K}_{br}^{-1}\hat{u}\right)^{-1}, \tag{A15}$$

matching Equation (4.8), and the origin-side second bracket contributes an additional scalar compliance c_{b2} in series (Appendix C, Table C6). As executed by the evaluator of record, both bracket frames are assembled pitch-only, at `minimizeFlxPin.m` line 289 (insertion) and line 306 (origin), with the two-rotation assemblies carried as commented alternatives; the insertion array is $[X_1, X_2, X_1]$ (line 387) and the origin array $[X_1, X_1, X_2]$

(line 400). The two-rotation-capable twin of the same block is `minimizeFlxPin2brk.m` lines 387–418. An earlier arm-1 array $[X_1, X_2, X_2]$ is superseded and survives only in the provenance comment at line 387, and the legacy single-bracket evaluator uses $[X_1, X_2, X_1]$ (`minimizeFlxPinX3.m` line 267).

The artificial tendon is modeled as an axial spring with stiffness

$$k_{ten} \;=\; \frac{n\,A_c\,E}{L_{ten}}, \tag{A16}$$

with $n = 2$ parallel cable strands ($\times 6$ for the 20 mm assembly), $A_c = 1.51 \times 10^{-6}\,\mathrm{m}^2$ (19-strand cable), and $E = 193 \times 10^9$ Pa (`MonoPam_mult.m`, function `Spr()`, lines 627–637; the evaluators obtain the spring rate through `Spr(bpa_i)`, `minimizeFlxPin.m` line 156). Series combination with the projected bracket compliance gives $k_{eq} \;=\; \left(k_{\hat{u},br}^{-1} + k_{ten}^{-1} + c_{b2}\right)^{-1}$ (`minimizeFlxPin.m` lines 394–409).

The scalar compliance displacement δ of Equation (4.22) is found numerically with `fzero` on the bracket $[0,\, l_m - l_{620}]$,

$$F_{620}(l_0)\,F^*(\varepsilon^*(\delta), P^*) - k_{eq}\,\delta = 0, \tag{A17}$$

with $\delta \equiv 0$ when the uncorrected relative strain already exceeds unity (`minimizeFlxPin.m` lines 434–441). The vector bracket deflection is then $\boldsymbol{\epsilon}_{br} = \mathbf{K}_{br}^{-1}\,(\|\mathbf{f}\|\,\hat{\mathbf{u}})$ (line 461), the cable stretch is $\|\mathbf{f}\|/k_{ten}$ (line 478), and the attachment point, path, moment arm, and torque are recomputed. For the two-BPA flexor grouping, the coupled balance shares one common force between both muscles and is solved analogously on $[0, F_{620}]$ (`MonoPam_mult.m` lines 556–612).

A.6 Wrapping-Loss Implementation

For the pinned-knee extensor, Equation (4.24) is implemented segment-wise over the wrap arc with radii $R_1 = 12\,\text{mm}$ and $R_2 = 40\,\text{mm}$ and segment boundaries $\theta_1 = 27^\circ$ and $\theta_2 = -23^\circ$ (`minimizeExtX3.m`, nested `Contraction`, lines 327–370). For the biomimetic-knee 20 mm cell, the arc-length factor is the routed bend measure $b(\theta_k)$ (arc length swept by the wrap, built with the `wRap` bend radius in `buildKneeFlexorRoute20mm.m` lines 265–268), giving $\Delta l = \chi_3\, b(\theta_k)\,(1-\varepsilon^*)^2$ (`MonoPam_mult.m` lines 398–441), with a hardcoded-geometry fallback for degenerate routes (lines 428–434). The loss enters only the force-producing strain; the kinematic (prediction) strain excludes it (`minimizeExtX3.m` lines 236–240).

Appendix B: Neuron Model Equations

This appendix defines the neuron and synapse equations and records the parameters used in the paired graded/spiking comparisons. The archived AnimatLab and 410-neuron Gait2392 comparisons are distinct from the independently replayed W2L configuration in Chapter 6. The final section documents that W2L configuration directly from its compiled network.

B.1 Spiking Neuron Model

Each spiking neuron is a conductance-based leaky integrator with threshold and adaptation,

$$C_m \dot{V} \;=\; -\,g_{\text{leak}}\,(V - E_{\text{rest}}) \;+\; I_{\text{syn}} \;+\; I_{\text{app}}, \tag{B1}$$

where V is the membrane potential, C_m the membrane capacitance ($g_{\text{leak}} = C_m/\tau_m$ with membrane time constant τ_m), E_{rest} the resting potential, I_{syn} the summed synaptic current, and I_{app} any applied (tonic stimulation) current. A spike is emitted when V exceeds the firing threshold, after which V resets and an after-hyperpolarization (AHP) of amplitude A_{AHP} with time constant τ_{AHP} is applied; the threshold itself accommodates with the firing history (relative accommodation r_{acc}, time constant τ_{acc}). Optional calcium channels are defined but disabled in the archived AnimatLab model ($G_{\text{max,Ca}} = 0$).

The parameter values shared by the neurons in the walker model are given in Table B1.

Table B1: Spiking neuron parameters in the AnimatLab bipedal walker model (`Biped_2xCPG_wSubs`). All neurons share these values except the motor neuron threshold noted below.

Parameter	Symbol	Value
Resting potential	E_{rest}	$-60\,\text{mV}$
Membrane time constant	τ_m	$5\,\text{ms}$
Initial firing threshold	V_{th}	$-55\,\text{mV}$ (motor neurons: $-59\,\text{mV}$)
Relative accommodation	r_{acc}	0.3
Accommodation time const	τ_{acc}	$10\,\text{ms}$
AHP amplitude	A_{AHP}	$1\,\text{mV}$
AHP time constant	τ_{AHP}	$3\,\text{ms}$

B.2 Non-Spiking (Graded) Chemical Synapses

Pattern-formation and motoneuron-drive connections in the walker use graded chemical synapses, whose conductance is a clipped linear ramp of the presynaptic potential,

$$g_{\text{syn}}(V_{\text{pre}}) \;=\; g_{\max}\;\text{clamp}\!\left(\frac{V_{\text{pre}}-V_{\text{th}}}{V_{\text{sat}}-V_{\text{th}}},\,0,\,1\right), \qquad I_{\text{syn}} = g_{\text{syn}}\left(E_{\text{syn}}-V_{\text{post}}\right), \tag{B2}$$

Here inward current is positive, consistently with Equation (B1); an excitatory reversal potential above the postsynaptic voltage therefore produces depolarization. The quantities V_{th} and V_{sat} are the synaptic threshold and saturation potentials and E_{syn} the reversal potential. The walker model instantiates three non-spiking synapse types:

- **Nicotinic ACh (excitatory drive):** $E_{\text{syn}} = -10\,\text{mV}$, $g_{\max} = 0.5\,\mu\text{S}$, $V_{\text{th}} \to V_{\text{sat}} = -65\,\text{mV} \to -20\,\text{mV}$.
- **Hyperpolarizing IPSP (inhibition):** $E_{\text{syn}} = -70\,\text{mV}$, $V_{\text{th}} \to V_{\text{sat}} = -55\,\text{mV} \to -40\,\text{mV}$.

- **Depolarizing IPSP:** $E_{\text{syn}} = -55\,\text{mV}$, $V_{\text{th}} \rightarrow V_{\text{sat}} = -60\,\text{mV} \rightarrow -40\,\text{mV}$.

Spiking chemical synapses and electrical synapses (rectifying coupling weight 0.1, non-rectifying 0.2) are also defined in the model and are available for the sensory afferent pathways of the MuJoCo/SNS implementation.

B.3 Relation to the SNS-Toolbox Implementation

These are the standard AnimatLab firing-rate/conductance models. SNS-Toolbox [Nourse et al., 2023] implements corresponding classes of non-spiking and spiking neurons with conductance-based synapses, and the AnimatLab parameters above served as the physically motivated initialization for the first SNS-Toolbox instantiation of the spinal network: a 410-neuron network with per-leg rhythm-generator half-centers, phase-shifted pattern-formation groups, motoneuron pools for all 92 muscle-tendon actuators of the converted model, and Ia, Ib, and II afferent populations. In that implementation the non-spiking membrane and graded-synapse equations take the same forms as Equations (B1) and (B2) with the spiking threshold and accommodation terms removed, and the afferent resting-tone offsets are zeroed so that Ia and Ib signals are pure activity measures; the rhythm-generator layer has been verified to oscillate at a human-like period with the two legs in antiphase under tonic drive (Section 7.2.4). The matched circuit checks in Chapter 6 verify selected ports, but do not establish full equivalence of these larger networks; the tonic stimulation tests of Section 7.2.4 correspond to constant I_{app} injections into Equation (B1) at the motoneuron, pattern-formation, and rhythm-generator layers respectively.

B.4 Spiking-Mirror Implementation, Calibration, and Verification

This appendix documents the numeric details behind the spiking-neuron mirrors of Section D.4, in the same file-traceable form as Appendix A: the cell and synapse equations as implemented, the calibration factors, the full verification-gate table of the MuJoCo mirror, and the variant-by-variant AnimatLab conversion table. The artifacts live under `Code/MuJoCo_SNS/spinal/reports_spiking_20261002/` (reports, gate scripts, logs, and metric files) and `Code/MuJoCo_SNS/spinal/` (the mirror builder `build_network_spiking.py` and its calibration script `calibrate_spiking.py`); the Simulink blocks are built by `Code/Matlab/SNS_Simscape/sns_build_spiking.m` and tested by `sns_units_test_spiking.m` in the same folder.

B.4.1 Adapting Integrate-and-Fire Cells and Hybrid Synapses

Between spikes, every spiking cell of the MuJoCo mirror obeys the membrane equation

$$C_m \dot{V} \;=\; -\,g_m\,(V - V_{\text{rest}}) \;+\; I_{\text{syn}} \;+\; I_{\text{bias}}, \tag{B3}$$

with a spike emitted when V reaches the firing threshold θ, after which V resets to V_{reset} and θ increases by the increment $\Delta\theta$; between spikes θ decays back toward its initial value with time constant τ_θ, bounded by the adaptation floor. The tonic bias I_{bias} parks the equilibrium a few millivolts below threshold, reproducing the analog cells' near-threshold operating point. Table B2 lists the per-class values as coded (SNS-Toolbox's `SpikingNeuron` class; the numeric membrane update used by this archived implementation is consistent

with its specified current units at $g_m = 1\,\mu S$, which the mirror therefore uses throughout). The rhythm-generator adaptation time constant reuses the analog network's persistent-sodium inactivation value, so that tuned adaptation values transfer between builds; at the rhythm gate's operating point it is 0.35 s. The mechanism behind that value, recorded here rather than in the chapter body, is as follows. In the analog rhythm cells of the contact-driven cores, burst termination is carried by a persistent sodium current, a slowly inactivating inward current; in the contact-driven implementations described here its inactivation time constant is held fixed rather than voltage-dependent, because the tested voltage-dependent form quenched burst termination and leaves the network latched in a static state, the failure documented with the archived-model results of Chapter 6.

Table B2: Spiking cell parameters of the MuJoCo mirror (real millivolts; `build_network_spiking.py`, `SPIKE` table, with the rhythm-generator increment superseded by the calibration of record). All cells rest at -70 mV with membrane conductance 1 µS and the tonic bias of Table B3.

Cell class	θ (mV)	V_{reset} (mV)	$\Delta\theta$ (mV/spike)	τ_θ (s)
Rhythm generator E/F	-50	-60	4.0 (calibrated)	0.35 (analog value)
Pattern-formation	-50	-60	0.3	0.15
Interneurons, commissurals	-50	-60	—	—
Afferent encoders	-52	-58	—	—
Renshaw	-52.5	-58	—	—

Spiking chemical synapses, in both the toolbox class used in the MuJoCo mirror and the Simulink `SpikingSynapse` block, carry a conductance that decays between events and

jumps by a fixed increment per presynaptic spike, saturating at its maximum,

$$\dot{g} \;=\; -\,g/\tau_{\text{syn}} \;\text{ between spikes,} \qquad g \leftarrow \min(g_{\max},\, g + g_{\text{inc}}) \;\text{ per spike,} \tag{B4}$$

with $I_{\text{syn}} = g\,(E_{\text{syn}} - V_{\text{post}})$ as for the graded synapse of Equation (B2). The Simulink `HybridSpikingSynapse` implements the same dynamics with the spike detected from the presynaptic voltage crossing a detection threshold that must sit strictly between the presynaptic resting and firing potentials (a detection threshold equal to the firing potential fires nothing, because the presynaptic voltage resets at threshold). Reversal potentials are 0 mV (excitation, above rest) and -70 mV (inhibition, shunting at rest) in the spiking frame, and $+8/-5$ mV in the motor neurons' 0–5 mV frame. The graded-to-spiking command pathways (descending drive, posture, and bias surrogates) keep the graded synapse of Equation (B2).

B.4.2 Calibration of Record

Table B3 lists every calibration factor produced by `calibrate_spiking.py` (`spiking_calibration.json`), with the measured residual of each. The spike-to-motor-neuron factors are bisected so that a steady 40 Hz presynaptic train reproduces the non-spiking steady-state motor-neuron voltage; the spike-to-spiking factors are rate-calibrated (a 40 Hz source drives a relay cell at 40 Hz, and a calibrated inhibitory factor silences a 40 Hz-driven cell); the readout-tap factor maps a 50 Hz train to the full-scale analog level. The rhythm-generator threshold increment was set by an isolated-pair sweep followed by a full-network sweep over alternation quality and period target.

Table B3: Calibration factors for the spiking mirror (`spiking_calibration.json`). Residuals are the measured deviation at each factor's calibration target.

Factor	Value	Calibrated against	Residual
Graded saturation reference	0.9567	analog network measurement	—
Spike-to-MN, excitatory ($k_{sn,exc}$)	5.4087	MN steady voltage	1.0×10^{-4} mV
Spike-to-MN, inhibitory ($k_{sn,inh}$)	1.2284	MN steady voltage	2.6×10^{-4} mV
Graded-to-spiking drive (k_{ns})	0.35	relay firing rate	-0.67 Hz
Spike-to-spiking, excitation	1.8754	40 Hz rate transfer	0.0 Hz
Spike-to-spiking, inhibition	0.9996	silences 40 Hz cell	0.0 Hz
Readout tap (g_{inc})	1.7093	50 Hz → full scale	5.00 mV check
RG threshold increment	4.0 mV/spike	pair + full-net sweeps	period target
Tonic bias current	16 nA	equilibrium -54 mV	—
Synaptic time constants	5 ms (exc), 20 ms (inh)	analog values	—

B.4.3 Verification Gates of the MuJoCo Mirror

Table B4 records the complete gate set of Section D.4 with verdicts and headline numbers. The two failures are reported as failures: the constant-drive period misses its tolerance with the tuning knobs identified, and the air-regime cadence clause misses while every structural clause passes. Gate scripts, logs, and machine-readable outputs are archived in the reports folder (`tools/`, `logs/`, and the `json` metric files respectively).

B.4.4 AnimatLab Conversion Variants

Table B5 catalogues every spiking copy built from the AnimatLab standalone exports. The conversion rules, applied by `tools/make_spiking.py` to copies (no original file touched), are: thresholds of neurons with unreachable (positive) thresholds move into the spiking regime, either $-55\,\mathrm{mV}$ with $-59\,\mathrm{mV}$ for motor-neuron-named cells in the v1 copies, or per-class calibrated values from the measured graded peaks in the v2 copies (rhythm-generator/pattern-formation/interneuron $-53.5\,\mathrm{mV}$, motor neurons $-44\,\mathrm{mV}$, Renshaw $-44.5\,\mathrm{mV}$); relative accommodation is zeroed; every graded chemical synapse type is re-emitted as a per-spike type with the same identifier and amplitude, $10\,\mathrm{ms}$ decay, and maximum conductance $\max(5,\ 2\times\text{amplitude})$; and each connection's own class tag is flipped to the spiking enum. That last rule is a documented format trap: the per-connection `<Type>` tag is stored independently of the synapse type, and converting only the type kills the run at load with no diagnostic detail. Structural validation confirms for every copy that neuron, connection, synapse-type, chart, and stimulus sets are identical to the original and that every class tag matches its synapse type.

Table B4: Complete verification-gate table of the spiking mirror. Baseline scores reproduce digit for digit; "split" denotes a gate whose clauses pass and fail in part.

Gate	Verdict	Headline numbers
Pipeline compile	PASS	runner, curriculum, parameters, and network builder all compile.
Defaults gate	PASS	(410, 376, 1186) populations, inputs, and connections after the selector edit; variant build (888, 382, 7430).
Non-spiking bit-identity	PASS	default baseline score -161.56754173676563 identical before and after the edit (all 17 digits); other baselines -144.46440311078322 and -200.37955834081458.
Topology mirror	PASS	directed edge multisets identical in three gain configurations (1,186 / 7,908 / 7,762); inputs and motor-neuron maps identical; readout taps 12 / 12 / 16.
Constant-drive rhythm	SPLIT	sustained bilateral antiphase (E 4.9 Hz, F 3.2 Hz per leg; swing 5.13 mV in every window) = PASS; period 0.621 s versus analog 0.399 s ($+55.6\,\% > 20\,\%$) = FAIL.
Perturbation basin	PASS	12/12 trials under 1 % jitter on every scalar; final swing minimum 4.58 mV (margin $0.89\times$); zero explosions.
Air-regime smoke	FAIL (cadence)	finite, 53 envelope rises, knee range 106.0° of twin's 117.9°, hip 50.4° of 71.1°; period 0.215 s versus 2.108 s.
Untuned ground evaluation	ran	score -236.35; bilateral (8 right / 9 left cycles); cycle knee minimum -79.8°; no non-finite states; stayed up. Single run, no statistics.
Simulink regression	PASS	all committed demonstrations and units tests reproduce exactly with the spiking blocks in the library (two-neuron test 6.07×10^{-4} mV).
AnimatLab baselines	PASS	five headless baseline runs, exit 0, periods and antiphase reproducing the records (0.443 s, 0.462 s, 1.305 s).

Table B5: AnimatLab spiking-conversion variants of the spiking mirror. "Connections" counts flipped per-connection class tags; the ablation converts thresholds only. All copies validate structurally and run headless to their configured end; rhythm outcomes as in Table D5.

Variant	**Thresholds**	**Neurons**	**Synapse types**	**Connections**	**Rhythm outcome**
Two-layer v1	−55/−59 mV	80	17	150	dead (2 spikes / 5 s)
Two-layer v2	calibrated	80	17	150	dead
Two-layer v3	v1, 30 ms decay	80	17	150	dead
Bilateral v1	−55/−59 mV	96	20	208	latch, one side 442 Hz–450 Hz
Bilateral v2	calibrated	96	20	208	latch
Project export v1	−55/−59 mV	122	50	243	synchronous 11.8 Hz co-bursting
Project export v2	calibrated	122	50	243	dead (passive settle)
Ablation	spiking, graded synapses	80	0	0	oscillates, near-baseline ranges
2023 reference (baseline)	spiking from birth	56	3 (spiking)	84	1.305 s bursts, antiphase +0.494 cycle

Figure B1 shows the ablation result at the joint level for the two-layer family: with graded synapses kept, spiking-regime thresholds leave the joint oscillation essentially intact, while the full conversion pins the joints. This is the visual form of Table D5's conclusion that the synapse conversion is the breaking change.

The spiking copies are headless-run artifacts only: none has been opened in the AnimatLab graphical environment, and graphical verification (or an equivalent acceptance ruling) remains outstanding for every `.asim` listed in Table B5. The mirror work also scoped, but did not attempt, two further conversions: the full 410-population Simulink spinal network (whose every one of 1,392 gains is tuned for graded voltages, making a spiking conversion a full re-tune rather than a parameter port) and the spiking mirrors of the alternative MuJoCo network variants (separate architectures whose fresh tuning is future work).

B.5 Connectome of the Verified W2L Model

Figures B2 and B3 document the W2L configuration used for the independent replays in Section D.5.1. The corrected bilateral template compiles to 96 neurons, 204 chemical synapses, and 22 input channels. The node and connection lists were read from the compiled SNS-Toolbox network using the saved configuration `w2l_verified_config.json`. This is a different network from the 410-neuron Gait2392 model used in the earlier spiking comparisons; their neuron counts and results are not interchangeable.

The two sides each contain 48 neurons. RG denotes rhythm-generator cells; RG-IN, their local interneurons; PF, pattern-formation cells; PF-IN, their local interneurons; MN, motor

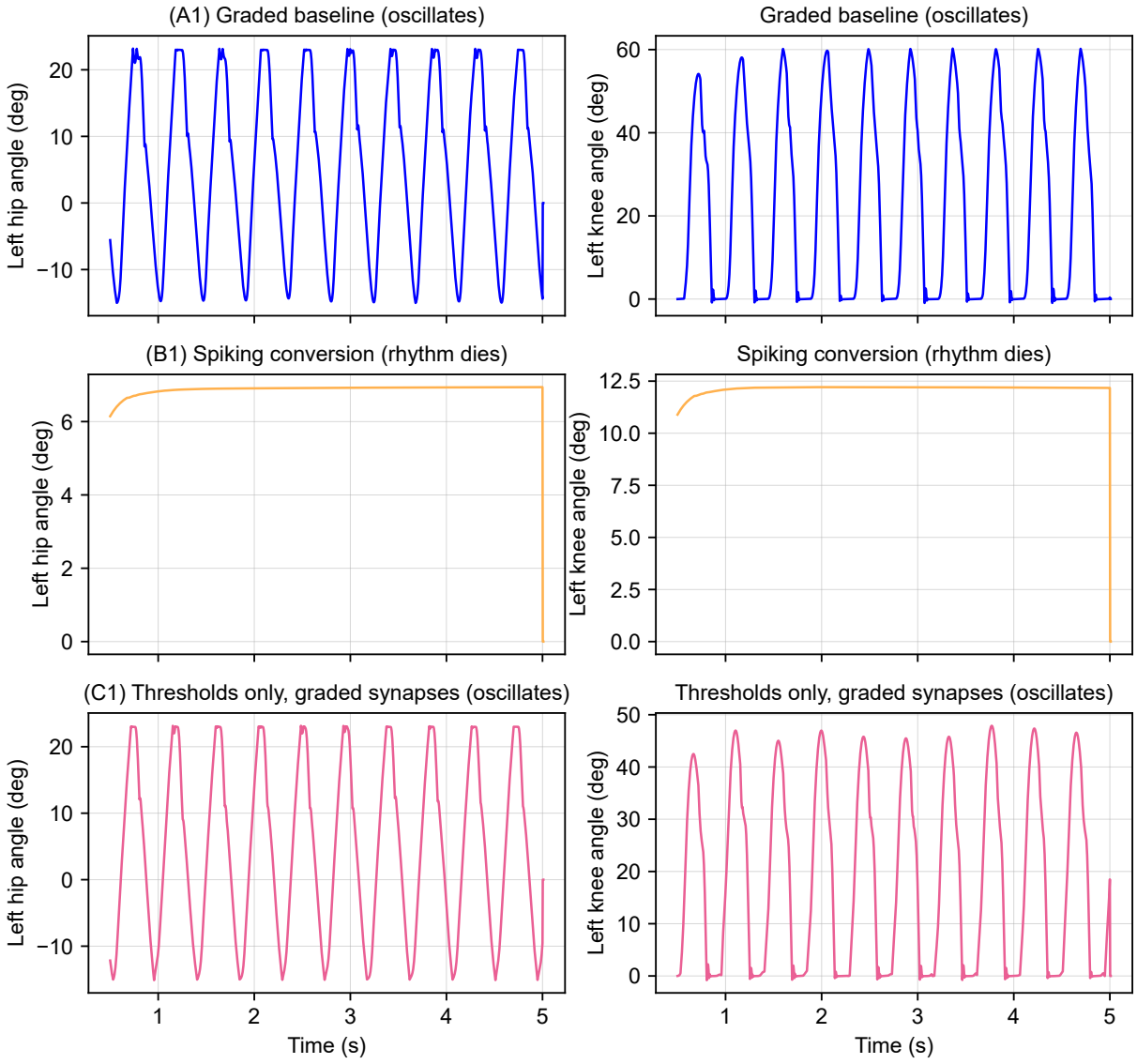


Figure B1: AnimatLab left hip (left column) and knee (right) angles over 0.5 s to 5 s: graded baseline (A, B), full spiking conversion (C, D), and spiking thresholds with graded synapses retained (E, F).

neurons; RC, Renshaw cells; Ia, Ib, and II, muscle-afferent encoders; and Ia-IN, reciprocal-inhibition interneurons. The c1 and V3 populations provide commissural pathways, and Contact comprises heel and toe encoders. The motor outputs drive six actuator channels per side. External inputs and actuator outputs are interfaces, not additional neurons.

Figure B2 groups individual connections by their presynaptic and postsynaptic populations. A symbol indicates at least one connection of that sign; a population pair can contain both excitatory and inhibitory connections. The diagonal quadrants contain within-side pathways, and the other quadrants show interleg pathways. The matrix exposes the sensory projections to both RG and PF populations as well as to motor circuitry.

Figure B3 shows the actual individual maximum conductance values, grouped by postsynaptic population. Equal values overlap. This representation retains the spread of synaptic strengths without implying that every connection in a pathway family shares a single gain. Conductance values alone do not predict the closed-loop contribution of a pathway; that requires the controlled interventions described in Chapter 6.

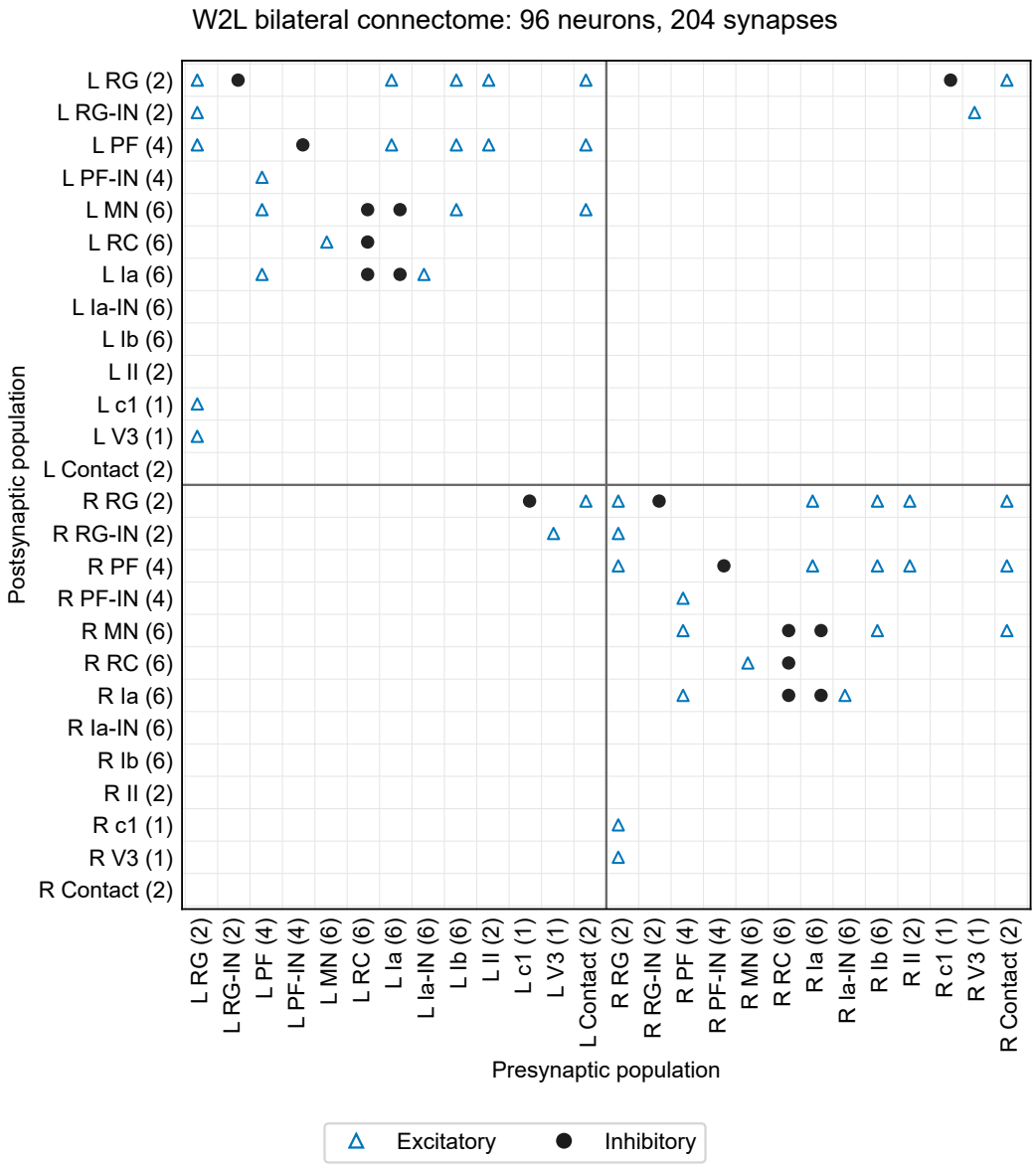


Figure B2: Population connectivity of the verified W2L network. Parentheses give neuron counts. Open triangles denote excitation, and filled circles inhibition.

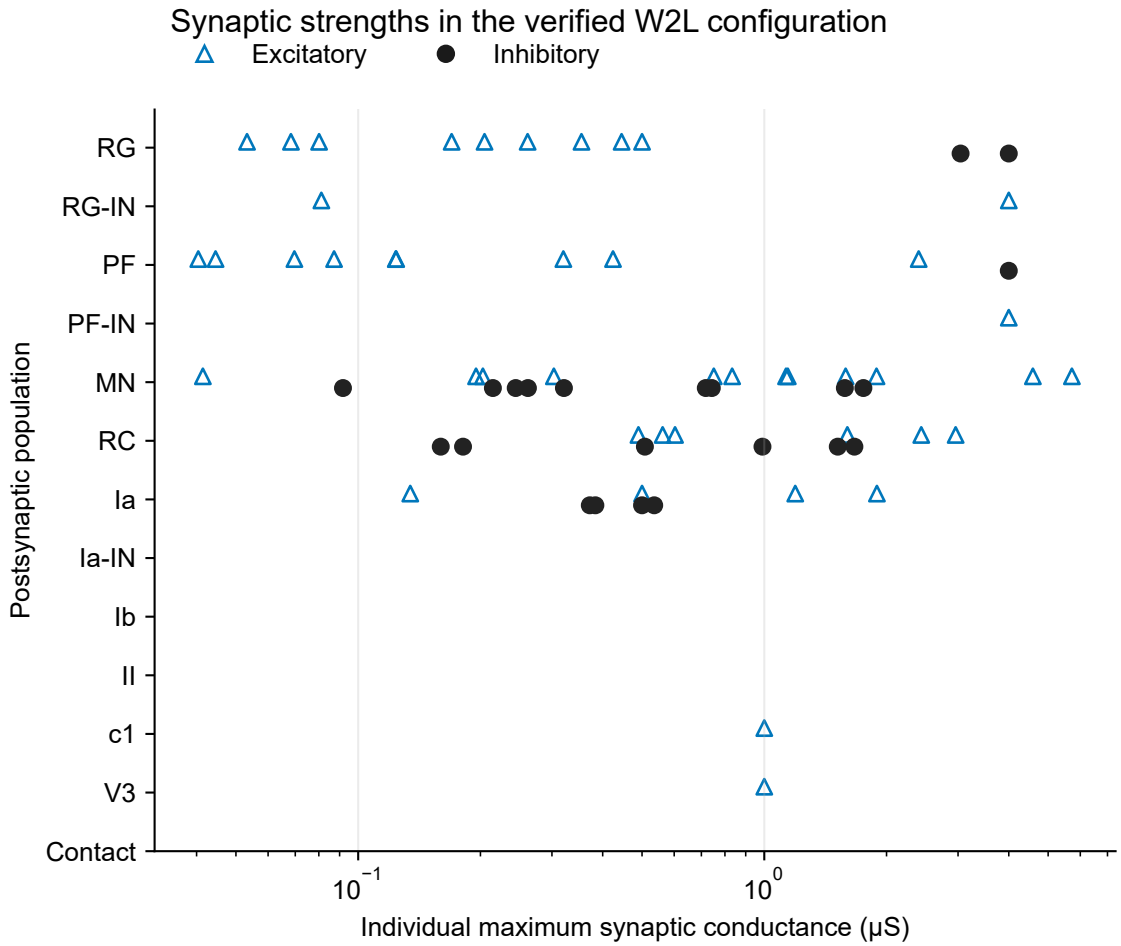


Figure B3: Synaptic conductances in the verified W2L configuration, grouped by target population. The horizontal axis is logarithmic; coincident values overlap.

Appendix C: Model Geometry, Bracket Frames, and Compliance Parameters

This appendix documents the numeric geometry and compliance parameters behind the force and torque models of Chapters 3 and 4 as they exist in the project's MATLAB code, with file names given so that any value can be traced to the code that produced it. Route and bracket coordinates are coded in meters and reported here in millimeters; stiffnesses are coded and reported in newtons per meter. The bracket reference points are interpretation choices for the 3D-printed (Onyx FDM) attachment hardware, set from CAD by the author; moving a reference point changes the identified stiffness values, so the points are recorded here alongside the values they produced.

C.1 Static BPA Pathway Data

The static BPA model routes each actuator as a straight-segment polyline through via points stored in a per-test `Location` array of dimensions $M \times 3 \times N$, where N is the number of joint frames in the motion sweep and M the number of via points; the first row in the distal link frame is marked by a cross-point index. There are no binormal or surface-frame quantities: at each frame the force direction is the normalized vector from the cross-point row to the previous row transformed into the crossing frame,

$$\mathbf{d} = \mathbf{R}_{ft}^{\mathsf{T}}(\mathbf{p}_{C-1} - \mathbf{t}_{ft}) - \mathbf{p}_C, \qquad \hat{\mathbf{u}} = \frac{\mathbf{d}}{\|\mathbf{d}\|}, \tag{C1}$$

where $\mathbf{p}_C$ is expressed in the tibia frame, $\mathbf{p}_{C-1}$ in the femur frame, and $\mathbf{T}_{ft} = [\mathbf{R}_{ft}, \mathbf{t}_{ft}; \mathbf{0}^\mathsf{T}, 1]$ maps tibia coordinates to femur coordinates. The inverse transform includes both rotation and translation, as in the implemented point transformation [Provenance: `minimizeFlxPin.m` lines 160 and 524–537 (`UD`) and 201–235 (`computeForceVector`); `minimizeExtX3.m` lines 687–700; BPA struct fields, `minimizeFlxPin.m` lines 28–35]. The pinned-knee test routes are archived as `FlxPinBPASet.mat` (5 flexor tests, $M = 2$, knee angle -120° to $+10^\circ$) and `ExtPinBPASet.mat` (9 extensor tests, $M = 7$, knee angle -125° to $+35^\circ$).

Table C1 lists the five pinned-knee flexor tests; their routes are two points, femur-frame origin $\mathbf{p}_1 = [-75.0,\ 100.0,\ 32.8]$ mm and tibia-frame insertion $\mathbf{p}_2 = [-50.1,\ -45.0,\ 32.3]$ mm (test 1; `FlxPinBPASet.mat`, kf(1), frame 1).

Table C3 lists the extensor pinned-knee route of test 1 at $\theta_\mathbf{k} = -125^\circ$; rows 1–6 are femur-frame and row 7 is tibia-frame, and as the knee extends, rows 2–6 release onto $\mathbf{p}_1$ (wrap released at $+35^\circ$).

Table C4 lists the per-test constants of the nine extensor tests; the correction terms of Section 4.1.4 used tests {1, 2, 5, 6, 7, 8}, of which the adopted fold trained on {5, 6, 7} and held out tests 1, 2, and 8 for validation; tests 3, 4, and 9 were not used in identification; their retrospective predictions are reported below [Provenance: fold columns of `minimizeExt10mmX3_results.mat`; `Dig_crossPredict.m` line 46].

Table C1 lists pinned-knee flexor test BPA constants, read directly from `FlxPinBPASet.mat` (kf(1)–kf(5)). Labels are the laboratory shorthand used in the fitting logs (`minimizeFlxPin10mm_2brk.m` line 51); l_0, tendon, and pressure are the values used by the model. All tests use ∅10 mm BPAs; the 47 cm test's measured angles carry the $+5.3^\circ$ encoder correction at load time (`minimizeFlxPin.m` lines 77–80). The test was not used in the

two-bracket identification, and the corrected-model fit reported for it in Table C8 is a post hoc evaluation; of the remaining four, the adopted cross-validation fold trained on kf(1) and kf(5) and held out kf(2) and kf(4) for validation.

Table C1: Pinned-knee flexor test constants from `FlxPinBPASet.mat`. All actuators have 10 mm diameter; test names use laboratory shorthand.

Test	**Label**	l_0 (m)	Tendon (m)	P (kPa)	F_m (N)	**Use**
kf(1)	48 cm	0.485	0	605.7	443.4	training
kf(2)	46 cm	0.457	0	614.2	441.4	validation
kf(3)	47 cm	0.479	0	623.2	443.0	not used
kf(4)	40 cm-tendon	0.406	0.035	623.4	436.9	validation
kf(5)	41 cm	0.410	0	615.4	437.3	training

The seed in Table C2 follows my femoral-condyle ray construction. The saved direct-route elimination schedule releases p5 at -89.8°, p4 at -55.7°, p6 at -47.8°, p7 at -20.2°, p3 at $+6.1^\circ$, and p8 at $+7.4^\circ$. These values come from `routeInfo.eliminatedAngleD` in `Vas_Pam_20mm_Result_20260925.mat`; the seed construction is in `buildKneeExtContext20mm.m`, lines 431–440.

Table C4 lists pinned-knee extensor test BPA constants, read directly from `ExtPinBPASet.mat` (ke(1)–ke(9)). Labels are the laboratory shorthand used in the fitting logs (`Dig_resultsTable.m` line 32); l_0, tendon, and pressure are the values used by the model; the shorthand rounds several lengths (e.g. "42 cm" is $l_0 = 0.415$ m and "43 cm-tendon" is $l_0 = 0.432$ m). All tests use ∅10 mm BPAs. The correction terms of Section 4.1.4 used tests $\{1, 2, 5, 6, 7, 8\}$: the adopted fold trained on $\{5, 6, 7\}$ and held out tests 1, 2, and 8 for validation; tests 3, 4, and 9 were not used in identification; their retrospective predictions are included in Table C9.

Table C2: Direct-route seed for the 20 mm biomimetic-knee extensor. Points p1–p5 use the femur frame; p6–p9 use the tibia frame.

Point	x (mm)	y (mm)	z (mm)	**Frame**	**Role**
p1	40.00	35.00	0	femur	BPA origin (attachment)
p2	83.90	−274.76	0	femur	BPA contacts mounting base
p3	64.06	−426.80	0	femur	femoral contact
p4	52.83	−441.88	0	femur	femoral condyle contact
p5	37.77	−452.14	0	femur	femoral condyle contact
p6	69.85	34.33	0	tibia	tibial contact (upper)
p7	73.50	19.13	0	tibia	tibial contact (mid)
p8	71.98	−12.23	0	tibia	tibial contact (lower)
p9	34.06	−66.87	0	tibia	patellar ligament ring (distal, pEnd)

Table C3: Pinned-knee extensor BPA route of test 1 at $\theta_{\mathbf{k}} = -125^{\circ}$ (`ExtPinBPASet.mat`, ke(1), frame 1; mm).

Point	x (mm)	y (mm)	z (mm)
p1	30.00	−50.00	0
p2	65.00	−350.00	0
p3	42.40	−408.50	0
p4	28.20	−421.60	0
p5	10.50	−425.20	0
p6	−8.60	−422.80	0
p7	42.50	−75.90	0

Table C4: Pinned-knee extensor test constants from `ExtPinBPASet.mat`. All actuators have 10 mm diameter; test names use laboratory shorthand.

Test	**Label**	$\boldsymbol{l_0}$ (m)	Tendon (m)	P (kPa)	F_m (N)	**Use**
ke(1)	40 cm	0.400	0	624.9	436.3	validation
ke(2)	40 cm-tendon	0.400	0.040	618.4	436.3	validation
ke(3)	42 cm	0.415	0	605.2	437.8	not used
ke(4)	42 cm-tendon	0.415	0.040	605.2	437.8	not used
ke(5)	43 cm	0.436	0	621.8	439.7	training
ke(6)	43 cm-tendon	0.432	0.022	622.9	439.3	training
ke(7)	46 cm	0.457	0	605.9	441.4	training
ke(8)	47 cm	0.465	0	605.9	442.0	validation
ke(9)	48 cm	0.480	0	605.9	443.1	not used

The biomimetic flexor route (`buildKneeFlexorContext20mm.m`) is a three-point path (femur-frame origin $\mathbf{p}_1$, a moving wrap point $\mathbf{p}_{\text{wrap}}$ on the tibia, and tibia-frame insertion $\mathbf{p}_{\text{end}}$), with the wrap point held on a standoff surface of radius `bpaRb` about the joint and its z placed on the projected $\mathbf{p}_1$–$\mathbf{p}_{\text{end}}$ line; the optimizer seed is $\mathbf{p}_{1,0} = [-50.0,\ 35.0,\ 50.0]$ mm and $\mathbf{p}_{2,0} = [-12.24,\ -8.87,\ 27.87]$ mm with $l_0 = 415$ mm and 25 mm tendon [Provenance: `buildKneeFlexorContext20mm.m` lines 209–218; `buildKneeFlexorRoute20mm.m` lines 98–186]. The biomimetic extensor seed route is given in Table C2, including the release schedule of the route-elimination rule.

C.2 Joint Transformation Matrices

The knee transform is planar: every joint frame is a rotation about z by the knee angle $\theta_{\mathbf{k}}$ composed with a translation fitted to robot CAD tables of the migrating instantaneous center of rotation. With $\mathbf{R} = \mathbf{R}_z(\theta_{\mathbf{k}})$, the robot (femur→ICR) transform is $\mathbf{T}_{\text{Pam}} = [\mathbf{R},\ [f_3(\theta_{\mathbf{k}}),\ f_4(\theta_{\mathbf{k}}),\ 0]^\top;\ \mathbf{0}^\top\ 1]$ with f_3, f_4 cubic-spline fits (ICR x: 23.30 to 10.47 mm; y: -416.65 to -395.66 mm over $\theta_{\mathbf{k}} = 0.17$ to -2.62 rad), and the ICR→tibia transform is the pure translation $[f_{13}(\theta_{\mathbf{k}}),\ f_{14}(\theta_{\mathbf{k}}),\ 0]^\top$ (x: 29.66 to -7.58 mm; y: 25.97 to 22.33 mm), composed as $\mathbf{T}_{t1\to f} = \mathbf{T}_{t1\to\text{ICR}}\ \mathbf{T}_{\text{Pam}}^{-1}$ over 100 frames spanning $\theta_{\mathbf{k}} \in [-120^\circ, +10^\circ]$ [Provenance: `buildKneeFlexorContext20mm.m` lines 34–80; `buildKneeExtContext20mm.m` lines 30–78; `Functions/ModernRobotics/RpToTrans.m`]. In the archived test sets the per-frame

transform maps tibia to femur/ICR and takes the form

$$\mathbf{T}_{\mathbf{k}}(\theta_{\mathbf{k}}) \;=\; \begin{bmatrix} \cos\theta_{\mathbf{k}} & -\sin\theta_{\mathbf{k}} & 0 & d_x(\theta_{\mathbf{k}}) \\ \sin\theta_{\mathbf{k}} & \cos\theta_{\mathbf{k}} & 0 & d_y(\theta_{\mathbf{k}}) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \tag{C2}$$

with rotation block $\mathbf{R}_z(\theta_{\mathbf{k}})$ and translation components d_x, d_y in meters. In the archived flexor test set the translation is constant across the sweep, $[d_x,\, d_y,\, 0] = [4.5,\, -395.8,\, 0]$ mm, and the rotation sweeps from $\mathbf{R}_z(-120^\circ)$ at full flexion to $\mathbf{R}_z(+10^\circ)$ at full extension (`FlxPinBPASet.mat`, kf(1)).

C.3 Bracket Reference Points and Frames

Table C5 lists the bracket reference points. Each compliant bracket carries an instantaneous body-fixed frame anchored at its reference point, constructed by two rotations (yaw about z, then pitch about the yawed y). With $\mathbf{p}_b$ the vector from the reference point to the bracket-to-BPA attachment target, the yaw and pitch angles are

$$\theta_{br} = \operatorname{atan2}(p_{by},\, p_{bx}), \qquad \phi_{br} = \operatorname{atan2}(p'_{bz},\, p'_{bx}) \quad \text{with} \quad \mathbf{p}'_b = \mathbf{R}_z(\theta_{br})^\top \mathbf{p}_b, \tag{C3}$$

and the bracket-frame pose is

$$\mathbf{R}_{kbr} = \mathbf{R}_z(\theta_{br})\, \mathbf{R}_y(\phi_{br})^\top, \qquad \mathbf{T}_{kbr} = [\mathbf{R}_{kbr},\; \mathbf{p}_{br};\; \mathbf{0}\;\; 1]\,, \tag{C4}$$

so the frame's x axis aims along the nominal load path. The angles are computed per configuration from the bracket geometry, not fixed [Provenance: `minimizeFlxPin2brk.m` lines 252–298; `minimizeExtX3.m` lines 461–483]. As executed, however, the flexor evaluator of record assembles *both* of its bracket frames from the yaw rotation alone (the pitch-only, one-transform pose $\mathbf{T}_{kbr} = [\mathbf{R}_z(\theta_{br}), \mathbf{p}_{br}; \mathbf{0}\ 1]$): insertion bracket at line 289 and origin bracket at line 305 of `minimizeFlxPin.m`, with the matching pitch-only point reconstructions at lines 318 and 331; the two-rotation assembly of Eq. (C4) is computed but left commented at lines 288, 304, and 330. For the pinned extensor the two-rotation assembly is live (transMode default `2trans`; `minimizeExtX3.m` lines 3–6 and 479–482). Force is transformed into the bracket frame by $\mathbf{RowVecTrans}(\mathbf{T}_{kbr}^{-1}, \mathbf{F_k})$ and deformed points are mapped back by $\mathbf{RowVecTrans}(\mathbf{T}_{kbr}, \cdot)$. Because the insertion array $[X_1, X_2, X_1]$ is y-symmetric, the assembly choice is numerically immaterial there; for the origin bracket it is not (see Table C6). The pitch-only (yaw-only) variant that the earlier one-transform fit used survives in `minimizeFlxPinX3.m` lines 200–209.

Table C5: Bracket reference points (mm, as coded; divided by 1000 in code). All are author-set CAD interpretation points; frame denotes the frame the vector is expressed in.

Point	x	y	z	**Configuration / meaning**
Pbri	−48.11	−107.81	13.80	pinned-knee flexor, insertion bracket (knee-ICR frame)
Pbr2	−52.61	0	75.06	pinned-knee flexor, origin-side 2nd bracket (hip frame)
Pbr	−3.84	−46.44	62.50	pinned-knee extensor, rib midpoint, medial side (hip frame)
Pbr	8.38	20.75	25.10	biomimetic extensor 10 mm, cantilever centroid
Pbr	9.48	−33.38	33.90	biomimetic flexor 20 mm, bolt-pattern centroid
Pbr	−19.00	22.00	27.60	biomimetic flexor 10 mm, cantilever centroid

C.4 Stiffness Arrays and Compliance

Every configuration models bracket compliance as a diagonal Hooke-law spring in the bracket frame, $\mathbf{K}_{br} = \mathrm{diag}(k_1, k_2, k_3)$ newtons per meter. The scalar compliance along the force path and its series combinations are

$$c_b = \hat{\mathbf{u}}^\top \mathbf{K}_{br}^{-1} \hat{\mathbf{u}}, \qquad c_{\text{eff}} = c_b + c_{\text{ten}} \ (+\, c_{b2} \text{ for the second bracket}), \tag{C5}$$

exactly as in Equations (4.8)–(4.18); when an artificial tendon is present its axial spring ($\times 2$ cables, effective steel area 1.51×10^{-6} m^2, $E = 193$ GPa) adds in series, and the force balance of Equation (4.22) is solved per frame by `fzero` on the contraction interval [Provenance: stiffness arrays: `minimizeFlxPin.m` lines 387 and 400; `minimizeExtX3.m` lines 575–584; `minimizeExt.m` line 352; `minimizeFlx.m` line 314. `fzero` force balance: `minimizeFlxPin.m` line 438; `minimizeExtX3.m` line 611; `minimizeExt.m` line 389; `minimizeFlx.m` line 352]. The per-configuration stiffness arrays and their series combinations are given in Table C6.

Table C6 lists bracket stiffness arrays by configuration ($K = [k_1, k_2, k_3]$ in bracket-frame axes, N/m; optimizer solves X_1, X_2 assigned as shown), as executed by the evaluators of record. The flexor evaluator has no frame switch: both brackets use the pitch-only assembly (see the Bracket Reference Points and Frames section above). The pinned-extensor evaluator carries a `transMode` argument (default `2trans`, env `EXTX3_TRANS`); its axis assignment was revised by the author, with the previous assignment kept commented in the file.

The wrapping-loss term X_3 (Equations 4.24–4.26) is evaluated as $X_3\, R\, |\Delta\theta|\, \zeta^2$ per contact arc. On the pinned extensor the two physical contact arcs use $R_1 = 12$ mm (27° to -23°)

Table C6: Bracket-frame stiffness arrays by configuration, in N/m. The fitted variables X_1 and X_2 are assigned to the axes shown.

Configuration	$\boldsymbol{K}$ array	**Source**
Pinned flexor, single bracket (legacy)	$[X_1,\ X_2,\ X_1]$	`minimizeFlxPinX3.m` line 267
Pinned flexor, insertion bracket (pitch-only frame)	$[X_1,\ X_2,\ X_1]$	`minimizeFlxPin.m` line 387 (frame line 289)
Pinned flexor, origin bracket (pitch-only frame)	$[X_1,\ X_1,\ X_2]$	`minimizeFlxPin.m` line 400 (frame line 305)
Pinned extensor, 2trans (with X_3)	$[X_1,\ X_2,\ X_2]$	`minimizeExtX3.m` line 575
Biomimetic extensor	$[X_1,\ X_2,\ X_1]$	`minimizeExt.m` line 352
Biomimetic flexor	$[X_1,\ X_2,\ X_2]$	`minimizeFlx.m` line 314

and $R_2 = 40$ mm, with arc-length normalizers $27.75/(ang_1 - ang_2)$ and $92.46/(ang_2 + 120^\circ)$; on the biomimetic configurations the bend measure is $w_{\text{rap}}|\theta_{\text{turn}}|$ with $w_{\text{rap}} = 25$ mm [Provenance: `minimizeExtX3.m` lines 333–359; `buildKneeFlexorRoute20mm.m` line 268; `MonoPamDataExplicit_balanceX3.m` lines 489–494]. Table C7 records the optimizer parameterization and bounds of the adopted identification: the two-bracket flexor fit solves all three terms, and the extensor refit locks (X_1, X_2) to the adopted flexor-front values (the extensor front was generated with the pair fixed to the adopted values of Table 4.9 and only (X_0, X_3) searched), enforcing a single stiffness pair across configurations.

The initial flexor population in Table C7 used $X_0 \in [5, 15]$ mm, $X_1 \in [5 \times 10^4, 5 \times 10^5]$ N/m, and $X_2 \in [7 \times 10^3, 1.5 \times 10^4]$ N/m.

[Provenance: `minimizeFlxPin10mm_2brk.m` lines 82–83 (bounds) and 424–432 (parameterization); `minimizeExt10mmX3.m` lines 23–26 (lock load) and 67–68 (bounds); `Collect_ExtPinX3_sweep.m` lines 59–60.]

Table C7: Identification parameters and bounds. Length offsets use centimeters; effective stiffnesses use base-10 logarithms.

Identification	Solved terms	Bounds	Lock
Two-bracket pinned flexor	X_0, X_1, X_2	$X_0 \in [0, 20]$ mm; $X_1 \in [3 \times 10^4, 10^6]$, $X_2 \in [5 \times 10^3, 2 \times 10^4]$ N/m	—
Pinned extensor refit	X_0, X_3	$X_0 \in [-20, +5]$ mm; $X_3 \in [0, 1]$	X_1, X_2 = flexor front

C.5 Adopted Correction Terms and Complete Pinned-Knee Results

The adopted correction terms are listed in Table 4.9. The flexor uses row 77 of `filtered_results` in `minimizeFlxPin10_results.mat`; the extensor uses row 32 in `minimizeExt10mmX3_results.mat`. The extensor identification holds (χ_1, χ_2) at the flexor values and solves (χ_0, χ_3), preserving one effective stiffness pair across both configurations.

Tables C8 and C9 report every recorded pinned-knee test before and after applying the adopted corrections. "Before" uses the characterized BPA force model with zero length and wrapping corrections and rigid attachments; "after" uses the adopted correction terms. Both predictions use the same measured angles and test inputs. RMSE and maximum absolute residual have units of N m, and FVU is dimensionless. The sample-size convention for FVU is given in Section C.9.

The adopted flexor fold fitted tests 1 and 5 and held out tests 2 and 4. Test 3 (47 cm) was not used in identification. The extensor fold fitted tests 5, 6, and 7 and held out tests 1, 2, and 8. Tests 3, 4, and 9 (42 cm, 42 cm with tendon, and 48 cm) were not used

because the attachment bracket shifted during those early tests. Results for unused tests are retrospective predictions, not held-out validation results.

Table C8: Complete pinned-knee flexor test results before and after correction. Test names follow the laboratory shorthand.

		Before			After		
Test	Use	RMSE	FVU	Max.	RMSE	FVU	Max.
1: 48 cm	Fit	7.617	2.1278	25.531	2.165	0.1718	5.467
2: 46 cm	Held out	8.803	1.9380	26.264	1.368	0.0468	3.614
3: 47 cm	Unused	7.670	1.8463	21.650	2.287	0.1641	7.818
4: 40 cm+tendon	Held out	6.373	0.5935	16.859	1.611	0.0427	5.891
5: 41 cm	Fit	9.694	2.1149	21.076	1.084	0.0277	1.766

Table C9: Complete pinned-knee extensor test results before and after correction. Test names follow the laboratory shorthand.

		Before			After		
Test	Use	RMSE	FVU	Max.	RMSE	FVU	Max.
1: 40 cm	Held out	3.092	0.5638	4.842	1.244	0.1303	2.196
2: 40 cm+tendon	Held out	3.622	6.3919	7.845	1.350	0.8878	1.890
3: 42 cm	Unused	3.528	2.2190	7.665	1.299	0.3418	2.700
4: 42 cm+tendon	Unused	2.910	6.2250	6.367	1.289	1.2209	2.003
5: 43 cm	Fit	2.437	0.9618	4.284	0.978	0.1634	1.818
6: 43 cm+tendon	Fit	2.878	4.7304	4.817	1.525	1.3270	2.481
7: 46 cm	Fit	2.178	1.0586	4.974	0.490	0.0535	1.206
8: 47 cm	Held out	0.949	0.3152	2.272	0.699	0.1714	1.559
9: 48 cm	Unused	0.578	0.1085	1.569	0.784	0.1996	2.282

The adopted corrections reduce all three errors on every test used in identification. The unused 48 cm extensor test has larger errors after correction, as shown in Table C9. The 43 cm-tendon extensor test retains an FVU above unity despite improvement over its baseline. These outcomes bound the accuracy of the shared correction model.

The values were evaluated with `minimizeFlxPin.m` and `minimizeExtX3.m`. On the biomimetic extensor test ($l_0 = 51.8\,\text{cm}$), the shared adopted terms give RMSE 1.17 N m, FVU 0.67, and maximum residual 2.21 N m, compared with the rigid-model values 3.03 N m, 4.49, and 5.59 N m. Table C10 collects the biomimetic comparisons at the adopted terms.

C.6 Redesigned Extensor Route Geometry

This section documents the direct redesigned extensor route, the first closed redesign of the extensor configuration. It remains the reference geometry for the route-elimination machinery and the seed from which the later work proceeded, but it is no longer the extensor design of record: when hardware testing of the direct route began, the printed and machined parts started breaking under the route's higher actuator forces, so the operating force had to be lowered, and the cam-profile redesign of Section 4.3.2, which converts the direct route's thin torque margin into headroom at reduced force, was developed in response and supersedes it as the current design. The geometry below is recorded as saved by the optimization results file (`Vas_Pam_20mm_Result_20260925.mat`, the run cited in Table C2). Figure C1 reproduces the optimizer's tiled route-geometry view from that results file: eight pose panels sweep the 100-frame solution, drawing full flexion ($\theta_{\mathbf{k}} = -120.0^\circ$), the solver frame immediately before each of the six contact-elimination events, and full extension ($+10.0^\circ$). The panel angles are the last poses at which the released row is still active; the release angles of Table C2 are the first poses at which each row is off, one pose spacing of 1.31° later. Beyond the release schedule, the results file also

carries a top-level `removedText` field reading '`-p8`'. This field is a plotting leftover rather than a route statement: the plotting block of `Opt_run_Ext.m` (lines 730–736) computes `removedText` only as part of each interior pose-panel title, naming the point released in the next frame, and the value left in the results file is the one from the last interior panel, the p8 release-event panel at $+6.06^\circ$ in Figure C1. The route of record does contain p8: like p3, released at $+6.1^\circ$, p8 stays active from full flexion through 97 of the 100 design-grid frames until its release at $+7.4^\circ$ near full extension (Table C2), so the seed table and release schedule document the route as run. Each panel draws the active p1:p9 route in the femur frame (tibia-side rows transformed through the ICR) over the femur-cylinder, femur-wall, corrected-condyle, and tibia clearance envelopes of the route-elimination rule, with the local bend-radius checks at p2 and p8 drawn dashed. The optimized endpoint values of this design are $\mathbf{p}_1 = [40.0,\ 137.0,\ 0]$ mm (femur frame; the square marker of Figure C1) and $\mathbf{p}_{\mathrm{End}} = [34.4,\ -197.1,\ 0]$ mm (tibia frame; the diamond marker), with resting length 827.7 mm and 36.8 mm of tendon.

Figure C2 places the same optimized route on the OpenSim bone-surface point clouds (`Open_Sim_Bone_Geometry`) at exactly zero knee angle: the full skeleton (femur, tibia, pelvis and hip, sacrum, lumbar spine, and foot) is drawn behind the three human vastus reference paths in light orange, the optimized p1-to-pEnd route is drawn in magenta with its endpoints circled, and the optimized route carries a total BPA-plus-tendon path length of 0.784 m at this pose. Animated sweeps of both redesigned routes over the full flexion range are archived with the optimization results (`Knee_Extensor_20mm.gif`, 2.3 MB, and `Knee_Flexor_20mm.gif`, 4.7 MB, in `Code/Matlab/Mesh_Optimization/Results`).

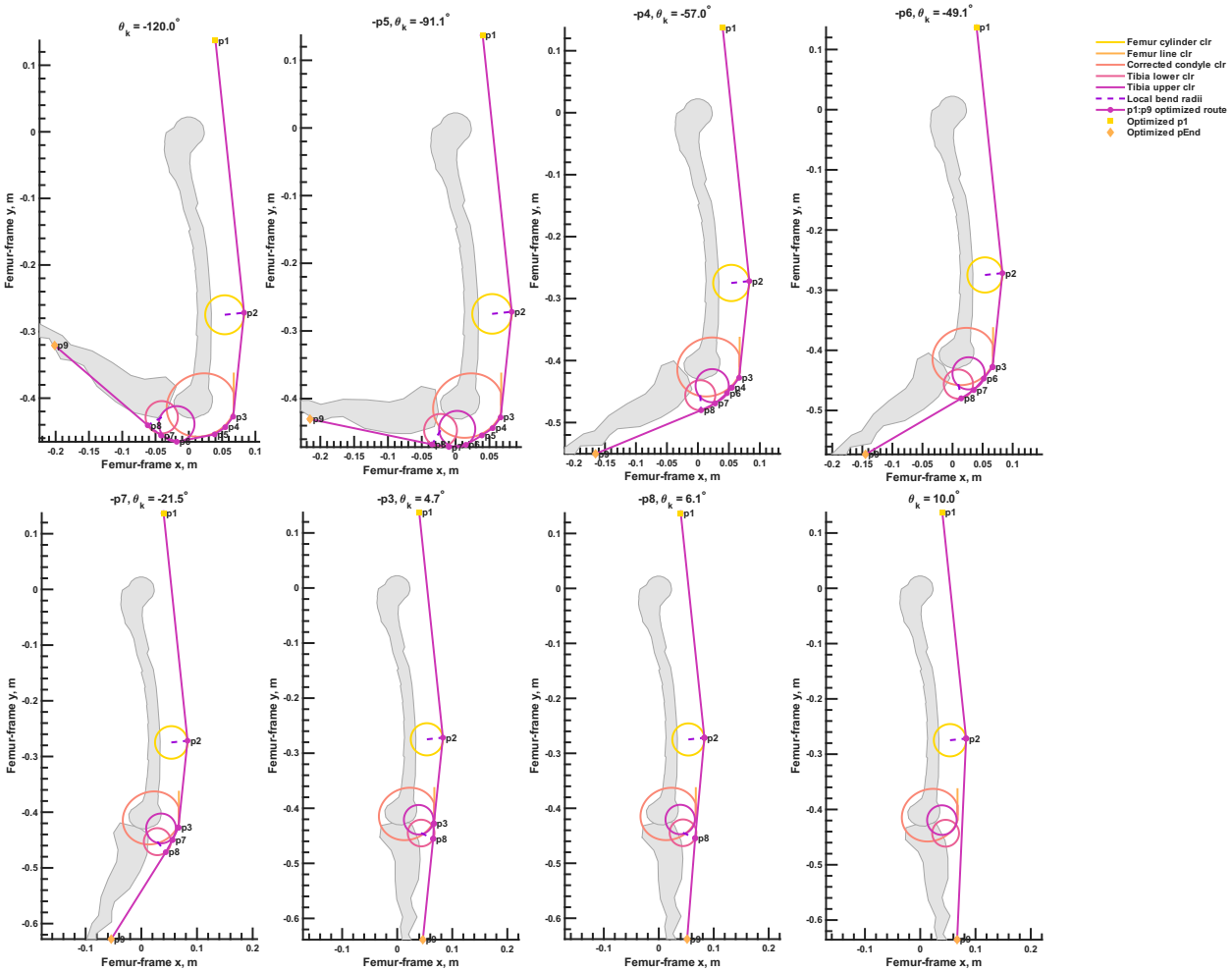


Figure C1: Eight-panel route-geometry view of the optimized 9-point extensor route of record (`Vas_Pam_20mm_Result_20260925.mat`), drawn in the femur frame. Gray silhouettes show the OpenSim femur and tibia as anatomical context; the colored curves are the robot route and clearance envelopes.

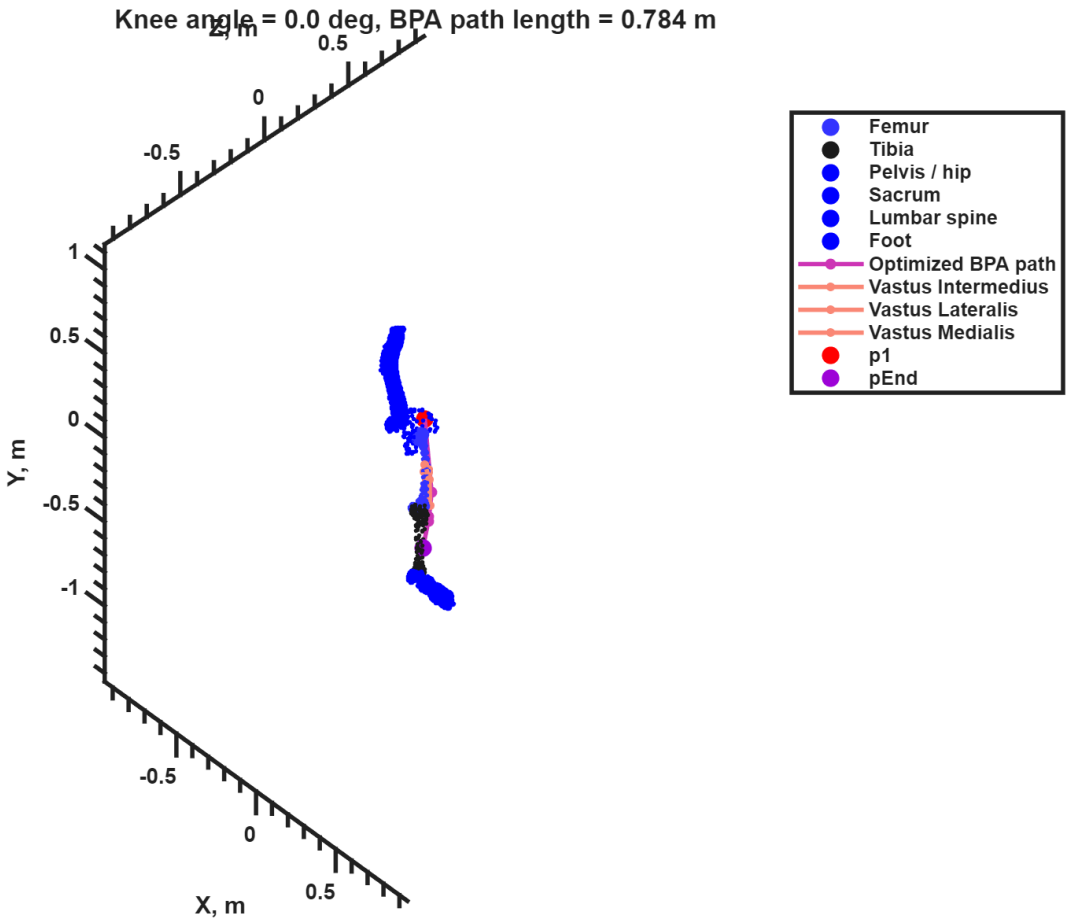


Figure C2: Static bone-mesh view of the optimized 20 mm extensor route at $\theta_k = 0^\circ$, rendered by `Export_AppendixBoneMesh.m`.

C.7 Contact and Release Angles of the Wrapped Route

The route-elimination rule that schedules the wrapped extensor route is a route-topology state machine rather than a shortest-path solver: the fully flexed route carries every candidate contact, and each contact is released at the first more-extended knee angle at which the straight chord between its surviving neighbors can replace it without entering the exclusion envelopes. This section states the calculation in generic terms, so that it applies to any wrapped actuator route on a rotating joint pair; the extensor route of Table C2 is the worked example, and the release schedule the calculation produces is the one quoted there.

The route is stored as a fixed number of rows $p_1, \ldots, p_9$: rows p_1 through p_5 are femur-frame points, rows p_6 through p_9 are tibia-frame points, and the segment from p_5 to p_6 crosses from one body frame into the other. The endpoints p_1 and p_9 are the actuator origin and insertion, and they are always active. Each interior row is either active, a genuine wrap contact, or eliminated; an eliminated femur-side row repeats the preceding active femur row, and an eliminated tibia-side row repeats the neighboring active tibia row, so the route keeps a fixed size and its segment chain crosses the joint exactly once. Let $\mathcal{A}(\theta_{\mathbf{k}}) \subseteq \{2, \ldots, 8\}$ denote the active interior rows at knee angle $\theta_{\mathbf{k}}$. The sweep over the design grid, here 100 poses from -120° to 10° at a spacing of 1.31°, proceeds from full flexion toward extension, each pose inherits the active set of the previous pose, and rows are only ever removed, so $\mathcal{A}$ shrinks monotonically as the knee extends.

At every pose, every active interior row is tested between its nearest active anchors,

$$i_j^- = \max\{i < j : i \in \mathcal{A} \cup \{1\}\}, \qquad i_j^+ = \min\{i > j : i \in \mathcal{A} \cup \{9\}\}, \tag{C6}$$

with the anchors expressed in the candidate's native frame through the per-angle joint transforms. Write A, B, and C for the planar, sagittal positions of $p_{i_j^-}$, $p_{i_j^+}$, and p_j, and let $s(x, y)$ denote the signed angle from vector x to vector y, positive counterclockwise, evaluated as a single atan2 of the planar cross product and the dot product so that no principal-value seam can corrupt the comparison. The elimination margin of row j is

$$\mu_j(\theta_{\mathbf{k}}) = \begin{cases} s(C - A,\ B - A)\,, & j \leqslant 5, \\ \\ s(A - B,\ C - B)\,, & j \geqslant 6, \end{cases} \tag{C7}$$

a signed angle between the through-contact leg and the bypass chord, measured in the native wrap sense of the candidate's side of the crossing segment; the two cases differ only in that sense of rotation, because the femur-side contacts turn the route one way around the distal femur and the tibia-side contacts turn it the opposite way around the proximal tibia. The rotation rule releases the contact once the margin turns positive, the orientation at which the chord, rather than the contact, provides the outer boundary of the route on that side. The gate is opened slightly early,

$$\mu_j(\theta_{\mathbf{k}}) > -\Delta, \qquad \Delta = 1.5^\circ, \tag{C8}$$

as a release hysteresis: in the last poses before the contact becomes collinear with its chord,

the computed moment arm of the wrapping segment collapses, and releasing the contact slightly before that orientation, with the collision gate below still controlling validity, removes the artifact without changing the route family.

Opening the rotation gate is necessary but not sufficient. A removal additionally requires the straight bypass chord $\overline{AB}$ to clear the exclusion envelopes of the candidate's side,

$$\operatorname{dist}\big(\overline{AB}_{\text{trim}},\, \mathcal{E}_j\big) \;\geqslant\; g, \qquad g = 0.5\,\text{mm}, \tag{C9}$$

where dist is the planar clearance between the trimmed chord and the envelope and tangency is allowed. The chord is trimmed by 2 mm at each end, capped at 45 % of its length, because a legitimate contact departs from the clearance boundary itself. On the femur side the envelopes $\mathcal{E}_j$ are the femoral-cylinder clearance circle, the condyle clearance polygon, and the femoral-wall span; on the tibia side they are the two tibial-cylinder clearance circles and the wall band between them. Every envelope is the normal offset of a bone profile by the fully inflated BPA radius R_{BPA},

$$\mathbf{q}(\vartheta) = \mathbf{e}(\vartheta) + R_{\text{BPA}}\,\hat{\mathbf{n}}(\vartheta), \tag{C10}$$

where $\mathbf{e}(\vartheta)$ traces the physical profile and $\hat{\mathbf{n}}(\vartheta)$ is its outward normal; the offset curve shares the tangent of the physical profile, its radius of curvature is $\rho(\vartheta) = \rho_{\text{phys}}(\vartheta) + R_{\text{BPA}}$, and the candidate contact positions themselves are seeded at tangent points of this curve. Because the envelopes are drawn at the fully inflated radius while the routed BPA is contracted at the release angles, each envelope radius may be relaxed by $\rho = 3\,\text{mm}$ inside the gate, floored at the bone profile plus 0.5 mm; a rotation-gate opening that still fails the

relaxed collision gate indicates that the seed placement, not the gate, must move.

At the first pose of the sweep, full flexion, the seed route is settled to a fixpoint: every contact whose two gates already open there is removed, and removals repeat until no further gate opens, so that the sweep starts from a valid topology; a colinearity guard may additionally remove a seed contact that starts the sweep already aligned with its neighbors, and its tolerance is zero in the design of record, so every seed row of Table C2 starts active. At every later pose at most one contact is released: if several rows open both gates in the same pose, the row with the largest elimination margin, the one whose rotation rule is most decisively open, is released first, and the remaining rows are retested at the next pose. The release angle of row j is recorded as the first grid pose at which the row is inactive. The activation angle is read from the same record with the sweep reversed, as the last pose at which the row is active, which is the pose at which the contact re-engages when the route is driven from extension toward flexion.

The same active-set record drives the geometric bend measure that scales the fictive-strain wrapping loss of Equations (4.24) and (4.26) inside the stiffness-aware force balance. Each wrap family, the femoral-cylinder contact, the femoral-condyle curve, the tibial contact curve, and the crossing-point curve at the ICR, contributes an arc length over its effective radius, and the total

$$b(\theta_{\mathbf{k}}) = \sum_{f} R_f(\theta_{\mathbf{k}})\, \alpha_f(\theta_{\mathbf{k}}) \tag{C11}$$

is the bend signal handed to the force model. Each family's turn angle α_f ramps with knee angle between the two route-state events that bound it, holding zero before the first

event and its plateau after the second, so the topology state and the bending signal are two readings of one record. The formulation is joint-agnostic: it requires only a fixed route representation with one crossing segment, per-angle transforms between the two body frames, and clearance envelopes on each side, so it carries unchanged to the other wrapped routes of this program.

C.8 Supporting Route Searches and Transmission Sweeps

The transmission studies extend the route search by allowing an artificial tendon to pass through a tackle before insertion. A redirect changes the force direction; a multi-part tackle also changes the relation between actuator travel and insertion force. The comparisons below re-solve actuator sizes for each transmission configuration and evaluate insertion-side torque against the same human reference. They remain model predictions; physical tests of the redesigned flexor and cam extensor are reported in Section 4.3.6.

The robot-body model itself carries one update from this work. The adductor magnus attachment point P2 was restored to its full-precision location $(-0.1357, -0.0929, 0.0591)$ m, displacing a rounded copy that was disqualified for producing a zero-length via point that nullified the flexion torque; the regenerated mirrored model passed its validation with zero non-mirrored pairs. At the restored location, the muscle delivers 34.28 N m of flexion torque, $1.12\times$ the human target of 30.61 N m, and 27.62 N m of adduction torque, $0.82\times$ the human target of 33.60 N m. These redesigned routes remain optimizer-stage results: the robot-body OpenSim model contains no pulley entities, and no via points or markers were added to it for these routes (its eight pre-existing moving-path-point vias,

the rectus-femoris and vastii patella-tracking points, one per muscle per leg, belong to the original muscle paths). The MuJoCo conversion used for the simulation work predates the pulley batch and the adductor-magnus restoration, so it carries one stale path point, the pre-restoration rounded adductor-magnus P2. Physical torque tests of the redesigned routes are reported in Section 4.3.6.

The same redesign pipeline was then extended from the knee to all 27 muscle-actuators of the robot-body model, and this whole-body extension is reported as work in progress rather than a completed result. The first pass reached 19 of the 27 actuators before the run stopped, and its chosen solutions showed that the penalty-based formulation itself was the binding limitation: all but one of the chosen routes violated the feasibility constraints, the sole exception being the biceps femoris long-head pair, whose worst-angle margin of $+0.03\,\%$ is indistinguishable from zero. The formulation was then restructured so that the task envelope, the joint-angle window each route must serve, acts as a hard feasible set rather than a penalty. In the restructured pass, twelve of the 27 actuators terminated on a concatenation error in the route builder (the three lumbar-wall actuators, tibialis posterior, the seven distal-foot actuators, and adductor magnus), leaving fifteen finalized routes. Eight of the fifteen were chosen from constraint-feasible configurations: semimembranosus, biceps femoris long head, tensor fasciae latae, soleus, lateral gastrocnemius, tibialis anterior, iliacus, and psoas. Of those eight, six also met the torque requirement at every angle, with worst-angle margins spanning $+0.25\,\%$ (biceps femoris long head) to $+250.7\,\%$ (tensor fasciae latae); the semimembranosus and soleus routes satisfy the geometry but remain short of their torque targets. The crash set and the two torque-short feasible routes define the remaining engineering: the route builder needs a fix for the concatenation failure, and

the short routes need either larger actuators or relaxed corridors.

The unadjusted cam optimizer solution had minimum margins of $+31.30\,\%$ against the required envelope and $+42.34\,\%$ against the plain human target; the adjusted design used in Chapter 4 has margins of $+30.83\,\%$ and $+39.77\,\%$, respectively.

Figure C3 shows predicted insertion-torque magnitude versus joint angle and supply pressure for two BPAs through a two-part tackle (force gain 1, travel gain 2), with actuator sizes re-solved. Panels show soleus, tibialis anterior, biceps femoris short head (BFSH), medial gastrocnemius with ankle fixed at zero, and rectus femoris evaluated at the hip. Dashed curves mark equality with the maximum-isometric OpenSim requirement; no equality curve lies within the soleus panel.

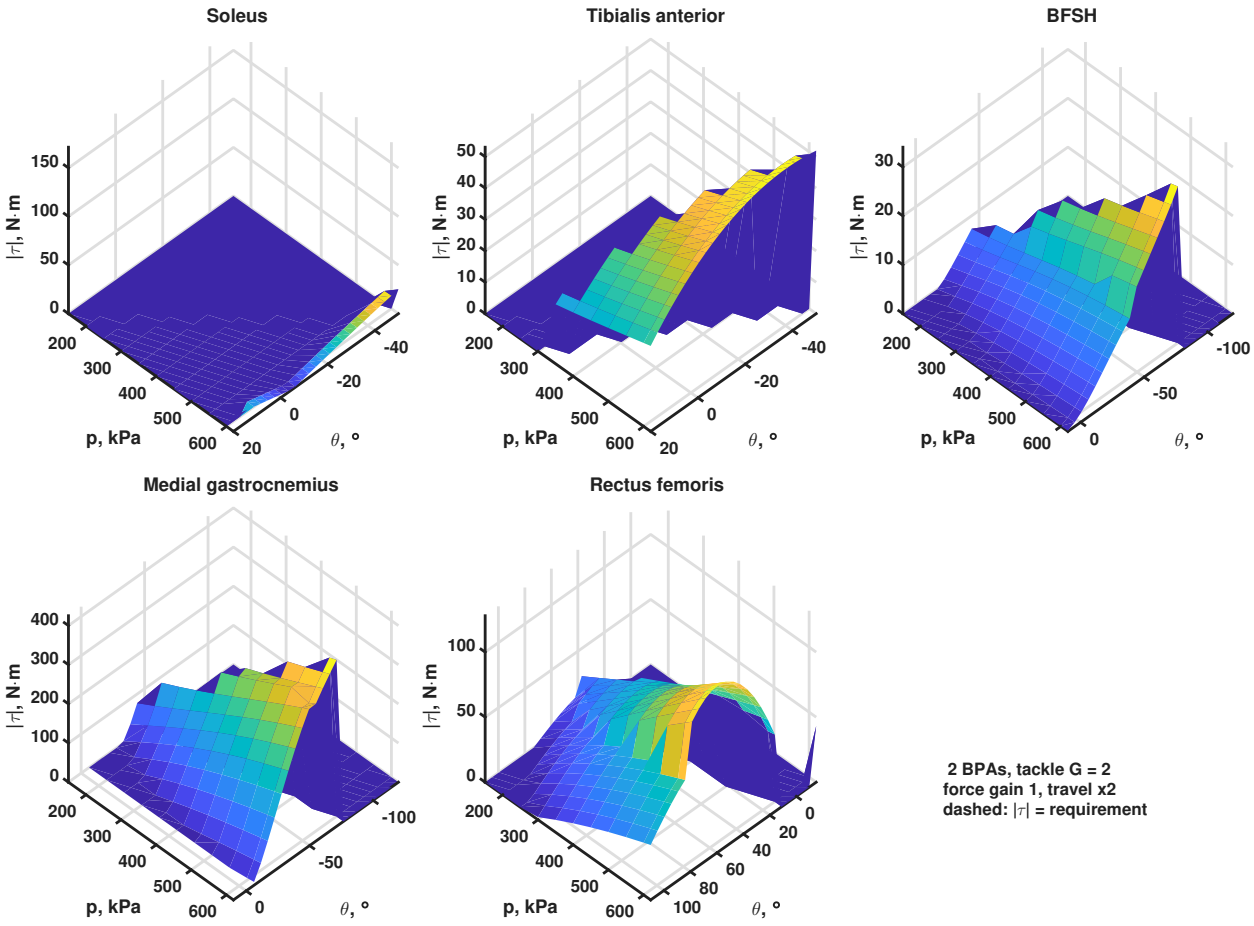


Figure C3: Predicted insertion-torque magnitude for two BPAs through a two-part tackle, with sizes re-solved. Dashed curves mark equality with the maximum-isometric human target.

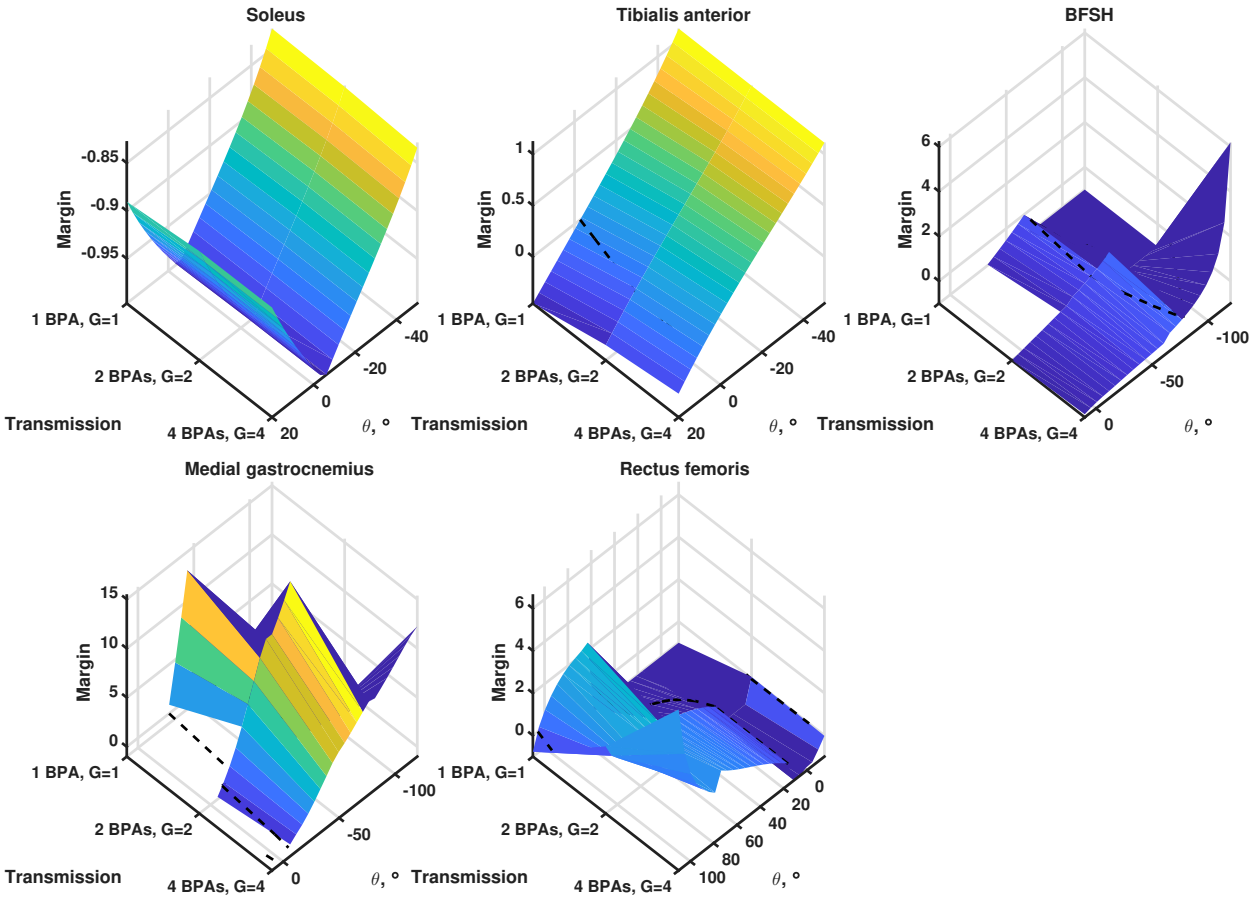


Figure C4: Torque margins at 620 kPa for the single-BPA baseline, two BPAs with a two-part tackle, and four BPAs with a four-part tackle. Sizes are re-solved. Positive values exceed the reference; panels follow Figure C3.

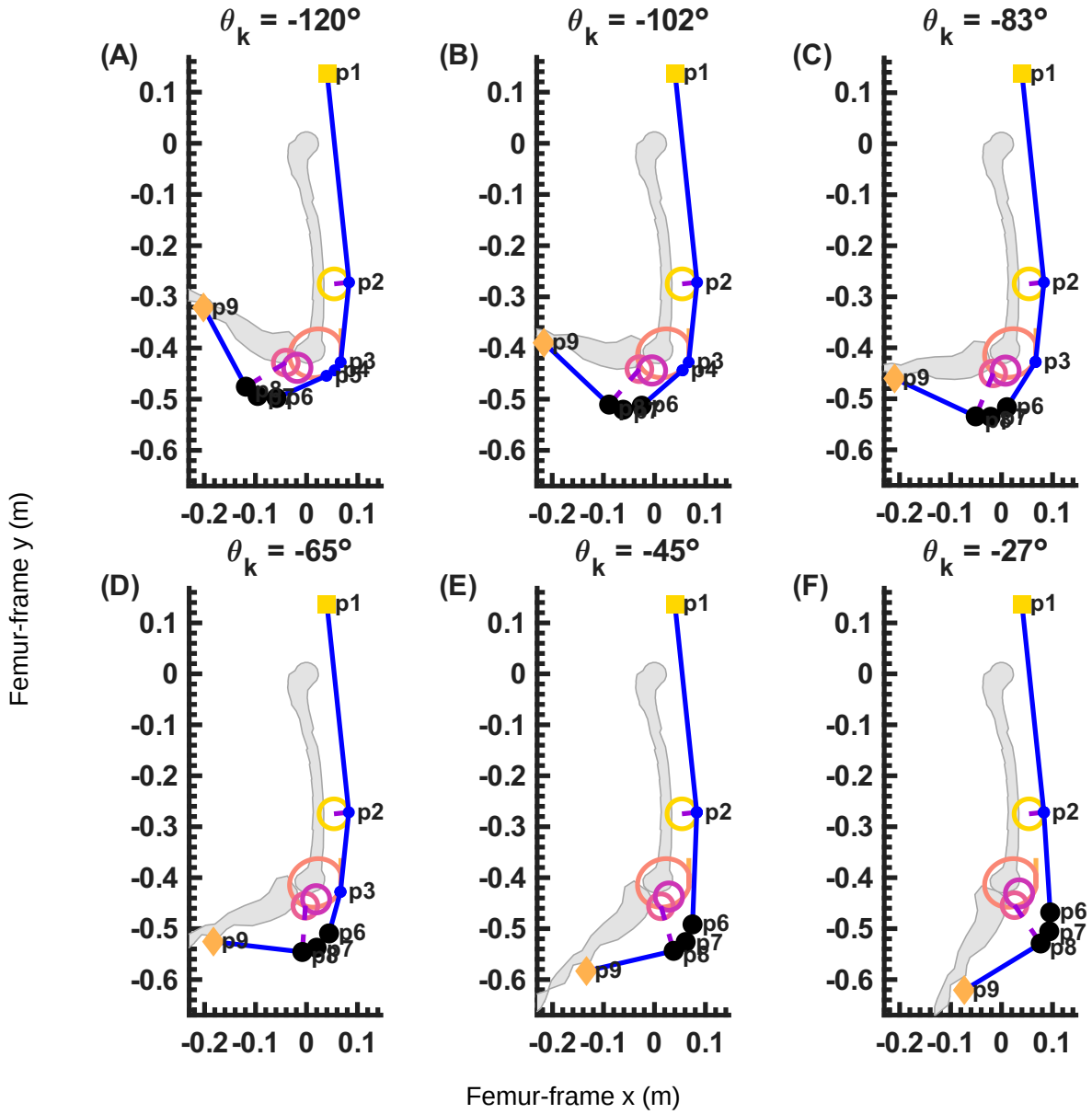
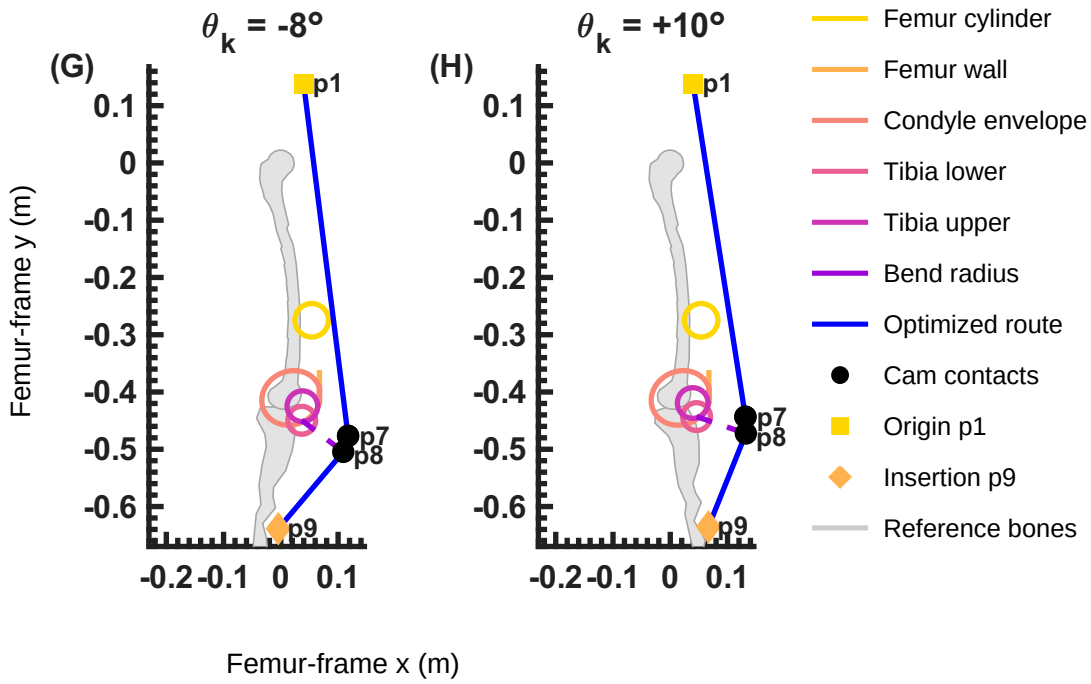


Figure C5: Cam-extensor poses A–F, spanning -120° to -27°. Blue lines show the route; colored outlines show the clearance envelopes. Gray silhouettes show the OpenSim anatomical reference. The remaining poses and legend are in Figure C6.



Cam-drum radius: 96.9 mm

Release angles: p5 -110.8 deg; p4 -89.8 deg; p3 -54.3 deg;
p2 -21.5 deg; p6 -18.9 deg. Gray bones are anatomical references.

Figure C6: Cam-extensor poses G and H at -8° and 10°, with the legend, 96.9 mm cam-drum radius, and contact-release schedule. Panel letters continue Figure C5.

C.9 Supporting Torque-Prediction Fit Statistics

C.9.1 Goodness-of-Fit Metrics

The MATLAB torque-fit comparisons use the shared routine `Functions/Go_OfF.m`: the root-mean-square error RMSE $= \sqrt{\frac{1}{n}\sum_i (y_i - \hat{y}_i)^2}$, the adjusted fraction of variance unexplained FVU $= \frac{n-1}{n}\ \sum_i (y_i - \hat{y}_i)^2\ /\ \sum_i (y_i - \bar{y})^2$ (a sample-size-scaled residual ratio, distinct from the conventional $1 - R^2 = \mathrm{SSE}/\mathrm{SST}$), and the maximum absolute residual $\max_i |y_i - \hat{y}_i|$ (`Go_OfF.m` lines 45–52). Lower is better for all three, which makes the triplet directly usable inside optimization objectives.

Table C10 reports the biomimetic-knee comparisons at the adopted shared stiffness pair, with the configuration-specific length and wrapping terms defined in Chapter 4. The complete pinned-knee results are given in Tables C8 and C9.

Table C10: Biomimetic-knee torque predictions at the adopted correction terms. RMSE and maximum absolute residual are in $\mathrm{N} \cdot \mathrm{m}$.

Configuration	**RMSE**	**FVU**	**Max. residual**
Extensor, biomimetic knee, $l_0 = 51.8\,\mathrm{cm}$	1.17	0.67	2.21
Flexor, biomimetic knee, $\varnothing 10\,\mathrm{mm}$, $l_0 = 41.5\,\mathrm{cm}$	2.8002	0.8835	4.1201
Flexor, biomimetic knee, $\varnothing 20\,\mathrm{mm}$, $l_0 = 41.5\,\mathrm{cm}$ (620 kPa)	2.2070	0.0748	3.7320

C.10 Configuration-Survey Coordinates and Actuator Sizes

The two coordinate columns of Table C11 define the graded surface, rather than every joint crossed by a multiarticular route. Other baseline sites use one actuator of the listed diameter.

Table C11: Graded coordinates and nominal BPA sizes for the 27-site survey. At nominal 40 mm sites, the baseline uses four parallel 20 mm BPAs; the updated configurations retain the inherited diameter.

Site	Primary DoF	Secondary DoF	Dia. (mm)	Baseline (mm)
ercspn_r	lumbar ext.	lumbar bend.	40	4×20
intobl_r	lumbar ext.	lumbar bend.	20	1×20
extobl_r	lumbar ext.	lumbar bend.	20	1×20
bifemlh_r	knee	hip flex.	20	1×20
semimem_r	knee	hip flex.	40	4×20
sar_r	knee	hip flex.	10	1×10
tfl_r	knee	hip flex.	10	1×10
rect_fem_r	knee	hip flex.	20	1×20
grac_r	knee	hip flex.	10	1×10
bifemsh_r	knee	hip flex.	20	1×20
vas_int_r	knee	hip flex.	40	4×20
soleus_r	ankle	knee	40	4×20
med_gas_r	ankle	knee	40	4×20
lat_gas_r	ankle	knee	20	1×20
tib_post_r	subtalar	ankle	40	4×20
tib_ant_r	ankle	subtalar	20	1×20
per_brev_r	subtalar	ankle	10	1×10
per_long_r	subtalar	ankle	20	1×20
per_tert_r	subtalar	ankle	10	1×10
flex_dig_r	MTP	ankle	10	1×10
flex_hal_r	MTP	ankle	10	1×10
ext_dig_r	MTP	ankle	10	1×10
ext_hal_r	MTP	ankle	10	1×10
glut_max_r	hip flex.	hip add.	40	4×20
add_mag_r	hip add.	hip flex.	40	4×20
iliacus_r	hip flex.	hip add.	20	1×20
psoas_r	hip flex.	hip add.	20	1×20

C.10.1 Torque and Active-Force Coverage

Table C12 expands the means reported in Table 4.10. Torque coverage requires the reference torque direction and magnitude; active-force coverage requires positive insertion tension. Both exclude cells outside the graded domain, and slack or infeasible configurations do not count as active.

Table C12: Torque and active-force coverage at each actuator site. Percentages use the included grid cells for that site.

	Baseline (%)		Transmission (%)	
Site	**Torque met**	**Active force**	**Torque met**	**Active force**
ercspn_r	100.00	100.00	100.00	100.00
intobl_r	8.00	96.00	72.00	96.00
extobl_r	0.00	100.00	76.00	92.00
bifemlh_r	64.00	96.00	64.00	100.00
semimem_r	24.00	100.00	32.00	80.00
sar_r	40.00	100.00	60.00	100.00
tfl_r	16.00	64.00	40.00	100.00
rect_fem_r	12.00	68.00	24.00	100.00
grac_r	20.00	100.00	36.00	100.00
bifemsh_r	80.00	80.00	100.00	100.00
vas_int_r	20.00	100.00	20.00	100.00
soleus_r	40.00	100.00	100.00	100.00
med_gas_r	0.00	100.00	20.00	100.00
lat_gas_r	0.00	100.00	20.00	100.00
tib_post_r	96.00	100.00	96.00	100.00
tib_ant_r	56.00	100.00	100.00	100.00
per_brev_r	8.00	24.00	88.00	100.00
per_long_r	4.00	100.00	80.00	100.00
per_tert_r	80.00	96.00	96.00	100.00
flex_dig_r	56.00	80.00	96.00	100.00
flex_hal_r	48.00	100.00	60.00	100.00
ext_dig_r	8.00	92.00	12.00	100.00
ext_hal_r	80.00	100.00	88.00	100.00
glut_max_r	46.67	100.00	53.33	100.00
add_mag_r	26.67	100.00	60.00	100.00
iliacus_r	33.33	100.00	80.00	100.00
psoas_r	40.00	100.00	86.67	100.00
Mean	37.28	92.44	65.19	98.81

Appendix D: Supplementary Neuromechanical Verification

This appendix supplies supporting model records and evaluation sets for Chapter 6. It separates implementation checks, gait comparisons, pathway-removal tests, and spiking conversions because they test different claims. Neuron equations and calibration tables are provided in Appendix B.

D.1 Model Construction and Recording

The archived AnimatLab projects provide reference circuits for implementation comparisons. Figure D1 shows their neural organization. A later reorganized bilateral project added commissural and biarticular pathways; it ran headless but latched under the tested drives. That assembly result is not evidence of stepping. The phase-1 recordings below concern an earlier instrumented configuration, not the reorganized network.

A 10 s instrumented run recorded joint, neural, sensory, and body-motion channels at a 0.2 ms interval. Contact-sensory recordings are membrane voltages, not contact forces. The initial external horizontal force and the limited window prevent the joint traces from establishing stable walking.

D.2 Controller Ports and Reference Checks

The matched W2L test compares implementations from a common circuit specification under identical drive and contact inputs, using burst period, bilateral phase, burst count,

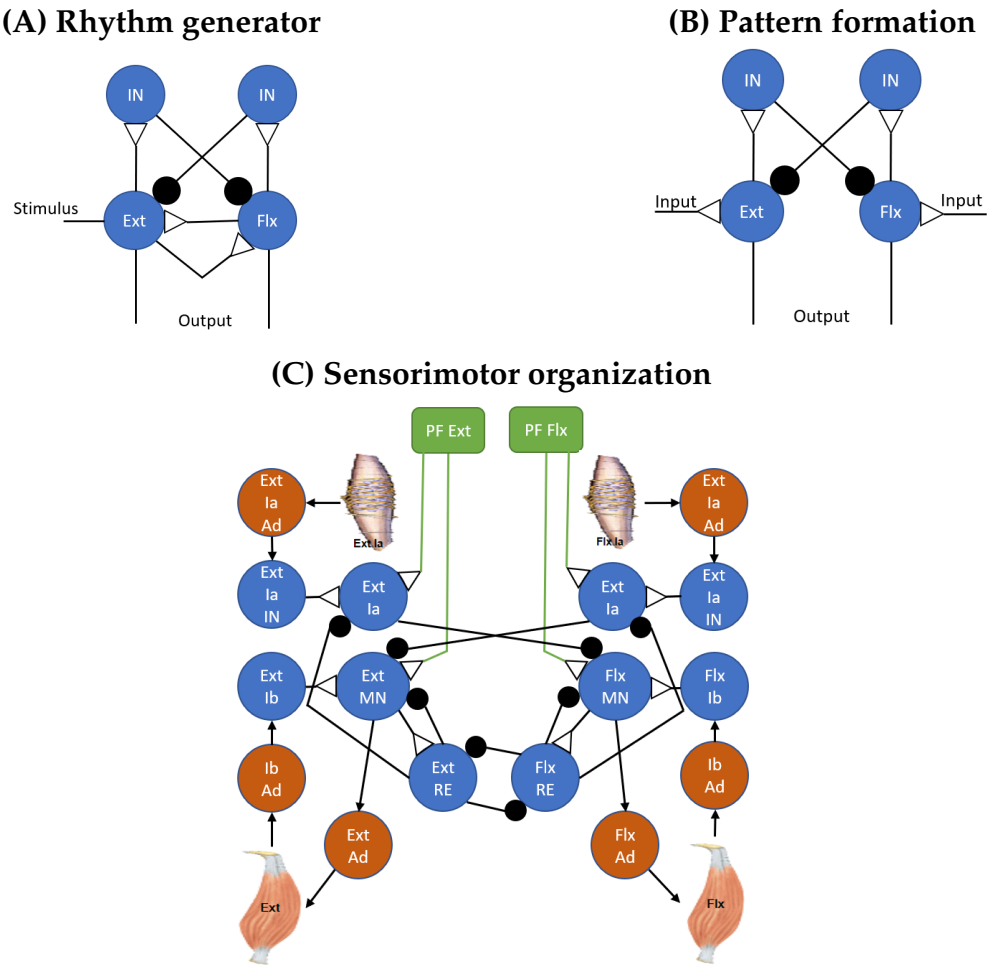


Figure D1: Archived AnimatLab motifs: rhythm generation (A), pattern formation (B), and sensorimotor organization (C). Ext and Flx denote antagonist pathways; IN denotes interneurons, MN motor neurons, and Ia and Ib muscle afferents. Triangles and circles distinguish excitatory and inhibitory connections.

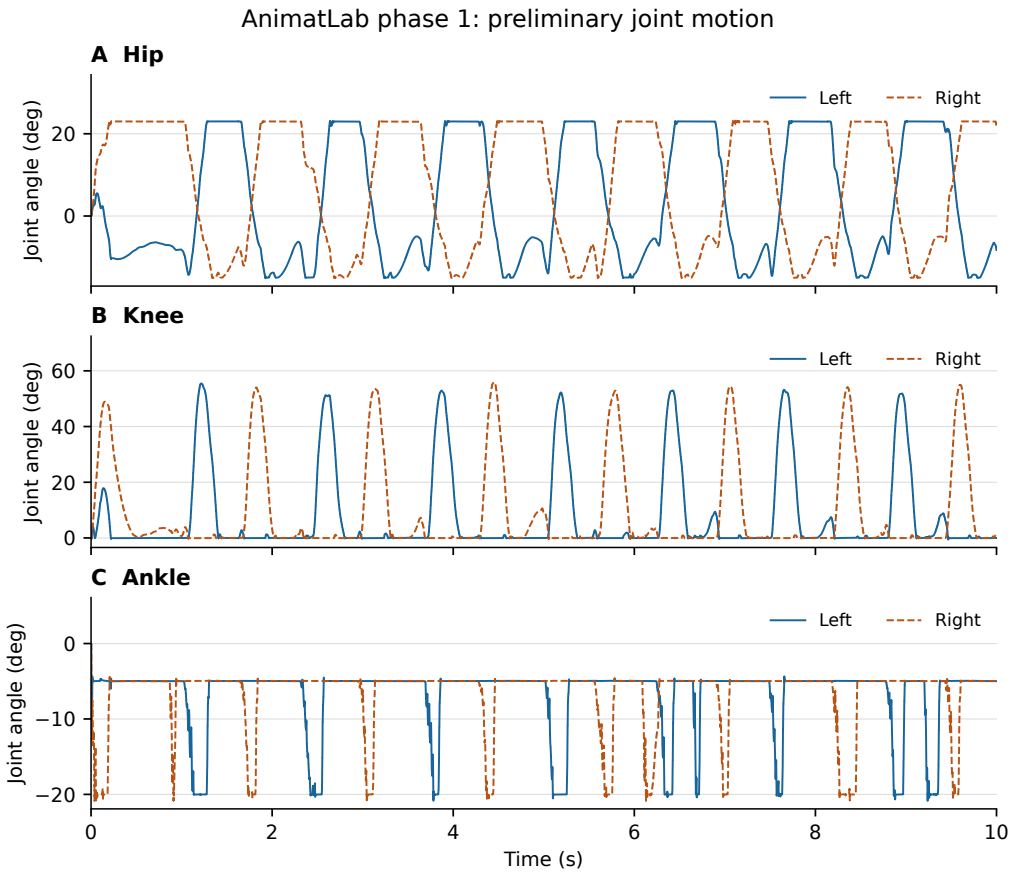
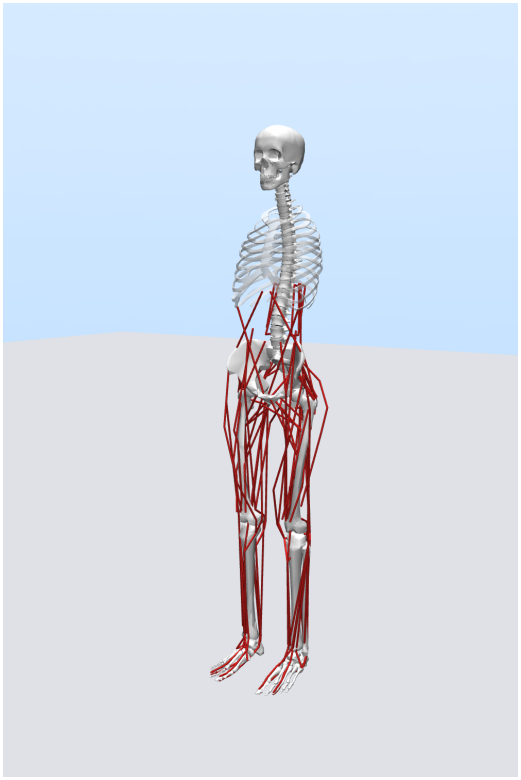


Figure D2: Bilateral hip (A), knee (B), and ankle (C) angles in the recorded joint conventions. All 50,001 samples in the 10 s window are retained. An external horizontal force acts during the first second. The recordings show motion, not validated walking or ground-contact gating.

(A) Converted robot



(B) SNS-Simscape knee model

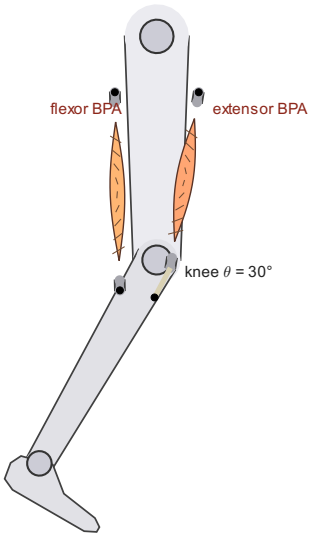


Figure D3: Converted Gait2392 robot (A) and the antagonist-BPA knee illustration used in the current SNS-Simscape demonstration (B). The two panels represent different models; scales differ.

and voltage correlation. The Deng controller port was checked separately: a current-pulse test produced a 1.938 s period against a 1.94 s NumPy reference, with antagonist correlation $r = -0.86$. Agreement on these tests does not establish long-term phase agreement in every network.

The Deng port comparison above establishes agreement between the translated implementations, not quantitative reproduction of the original AnimatLab model. A separate rerun of the archived original produced a period of approximately 0.221 s, compared with 1.938 s in the translated test. Antagonist alternation was present, but the bilateral rhythm was in phase. The original and translated models use different parameterizations, so these results do not establish quantitative reproduction of the original dynamics. Their rhythm-generator perturbation responses do agree qualitatively: activity is perturbed and then recovers without sustained suppression. An earlier suppression interpretation arose from counting no peaks inside a window that also contained no baseline peaks; comparing the perturbed and unperturbed sequences resolves that analysis error.

The full exported network agreed with its NumPy reference to 4.2×10^{-6} mV over the early transient across 410 neurons. Its subsequent divergence and the tonic-drive fixed-point discrepancy reported in Chapter 6 limit parameter transfer. The OpenSim-derived Simscape mechanism contains 1,446 blocks and 149 joints; mechanism compilation is distinct from completing its neural and muscle coupling.

D.2.1 A Published Neural Mechanism as a Verification Target

The reduced Python implementation of the two-level circuit of Rybak et al. [2006] passed 13 checks covering drive-dependent activity, repeatability, and the distinction between resetting and non-resetting deletions. Stimulating the pattern-formation extensor population suppressed flexor motor output while the rhythm generator continued, so flexor activity resumed at its expected phase. Stimulating the rhythm-generator extensor population also suppressed flexor output but shifted the subsequent rhythm phase. This comparison tests the functional separation between rhythm generation and pattern formation: similar missing motor bursts can arise through different neural mechanisms. It is a neural-only verification of the translated circuit, independent of a mechanical model and separate from the Simulink demonstrations.

D.2.2 Archived AnimatLab Models

Headless reruns tested whether archived models reproduced their reference behavior in the original simulator. Eleven targetable rhythm and gait metrics passed their acceptance bands. The bilateral contact-driven model's rhythm generators had periods of 0.450 and 0.467 s; the modern W2L model showed joint oscillations at 2.25 Hz. The Li-model rerun reached $0.637\,\mathrm{m\,s^{-1}}$ with step periods near 1.3 s [Li et al., 2017]. These are reference-model checks, not results for the combined Deng–Li model or evidence of unsupported walking.

D.3 Tuning and Supported Evaluations

Free network gains were searched with a tree-structured Parzen estimator, with each candidate evaluated through simulation. The staged procedure separates suspended rhythm generation, sensory feedback, standing, and ground evaluations (Figure D4). Finalists were checked under 1 % multiplicative parameter perturbations and changes in drive. Such checks probe the neighborhood of a selected configuration; they do not establish robustness to untested loads or removal of the harness.

The cycle-shape metric assigns zero to perfect agreement. A run without detectable cycles returns -25 as a failure sentinel. Because this value is numerically greater than the genuine-gait scores below, failed runs must be identified before ranking scores. The sentinel itself is not a valid measure of gait quality.

D.3.1 Reference-Library Comparison

The 43-cycle library contains 32 running trials from eight subjects at four speeds, ten walking references, and a predicted-walking cycle. The walking references span prescribed speeds of $0.5\,\mathrm{m\,s^{-1}}$ to $2.0\,\mathrm{m\,s^{-1}}$ and self-selected trials. The four configurations were scored without treating a good tuning-reference score as evidence of generalization. Library scores were substantially more negative (Table D1), and the evaluations showed unilateral or asymmetric gait. This comparison bounds transfer to the available references; it does not validate normal human walking.

Table D1 uses the production pipeline's cycle-shape kinematic score. The tuning-reference column reports each configuration on its tuning reference; the library mean

No.	Preparation	Capability	Score
1	**Deafferented air stepping**	Intrinsic rhythm; feedback off	Rhythm and swing (rhythm-gated)
2	**Afferented air stepping**	Heel, toe, and load feedback active	Rhythm and swing (rhythm-gated)
3	**Standing balance**	Posture under push perturbations	Posture maintenance under push
4	**Ground walking**	Ground contact and forward gait	Kinematics vs. human reference cycle
5	**Contact-driven ground walking**	Contact-driven pathway variant	Kinematics vs. human reference cycle

Each winner seeds the next stage. Finalists undergo perturbation and basin checks.
Air stages are suspended; ground evaluations retain the support harness.

Figure D4: Staged tuning procedure. Air-stepping evaluations are suspended, and the reported ground evaluations retain mechanical support. Each stage introduces additional pathways or tasks. Scores require a separate rhythm/failure check because the no-cycle sentinel cannot be ranked as a gait score.

averages that score over all 43 reference cycles.

Table D1: Four controller configurations evaluated on their tuning reference and the 43-cycle reference library. For genuine gait cycles, larger kinematic scores indicate closer agreement; zero is perfect agreement.

	Tuning-ref. score	**Library mean score (43 refs)**	**Most negative reference (score)**
`s3k`	−186.4	−8776	`Case_40_motion` (−8841)
`s3kpruned`	−176.1	−6924	`Case_40_motion` (−6986)
`syn6`	−200.4	−9961	`Case_40_motion` (−10040)
`w2lvar`	−144.5	−5808	`subject08_Run_20002` (−5896)

D.3.2 Pathway-Removal Tests

Individual pathway classes were removed from each selected configuration and the model re-evaluated on its tuning reference (Table D2). Removing Ia feedback from the production configuration eliminated detected cycles. Other removals changed the score without necessarily eliminating movement. The different response of the six-synergy variant shows that these dependencies belong to the tested network configurations, rather than constituting a general claim about biological necessity. The systematic pattern-formation and motoneuron stimulation series was not completed.

D.4 Spiking-Neuron Comparisons

The spiking variants replace afferent, rhythm-generator, and interneuron populations while retaining graded motor neurons. Synapses onto motor neurons are calibrated by mean conductance, preserving the continuous activation interface. This construction

Table D2: Pathway-removal tests on the tuning reference. Full-model rows give unchanged baselines. A score of −25 is the fixed no-cycle fallback and denotes failure, despite exceeding the genuine-gait scores.

Configuration	**Removed pathway**	**Score**
`s3k` (full model)	none	−186.4
	Ia afferents	−25 (no cycles detected)
	all afferent classes	−267.8
	Renshaw cells	−267.3
	group-II afferents	−255.2
	Ib afferents	−180.3
	contact channels	−183.9
	interleg commissural pathways	−174.9
`syn6` (full model)	none	−200.4
	all afferent classes	−190.5
`w2lvar` (full model)	none	−144.5
	Ia afferents	−273.5
	weak rhythm-generator mutual excitation	−279.9
	Renshaw cells	−142.4

does not require the network dynamics or resulting movement to remain unchanged. Calibration and full conversion tables are in Appendix B.

D.4.1 MuJoCo Network and Ground Evaluation

Under constant drive, the non-spiking and spiking rhythm periods were 0.399 and 0.621 s, respectively (Figure D5). The spiking period is 55.6 % longer. Timing agreement therefore requires more than copying the selected parameters. The untuned spiking variant produced bilateral cycling in a ground evaluation where the non-spiking winner cycled predominantly on one side (Table D3; Figure D6). This individual comparison motivates further controlled testing; it does not establish superiority of spiking control.

Table D3: Supported ground evaluation at the same selected parameters (16 s), with no spiking re-tuning. Knee minima and contact duty are computed over detected cycles. The kinematic score is a cycle-shape error metric: zero denotes perfect agreement, and less negative scores indicate closer agreement.

Metric	**Non-spiking winner**	**Spiking mirror (untuned)**
Kinematic score	-161.57	-236.35
Gait cycles, right / left	9 / 0 (left frozen)	8 / 9 (bilateral)
Cycle knee minimum	-64.3°	-79.8°
Contact duty, right / left	0.637 / 0.985	0.799 / 0.856
Maximum pelvis tilt	22.1°	19.2°
Non-finite states / fell	no / up	no / up

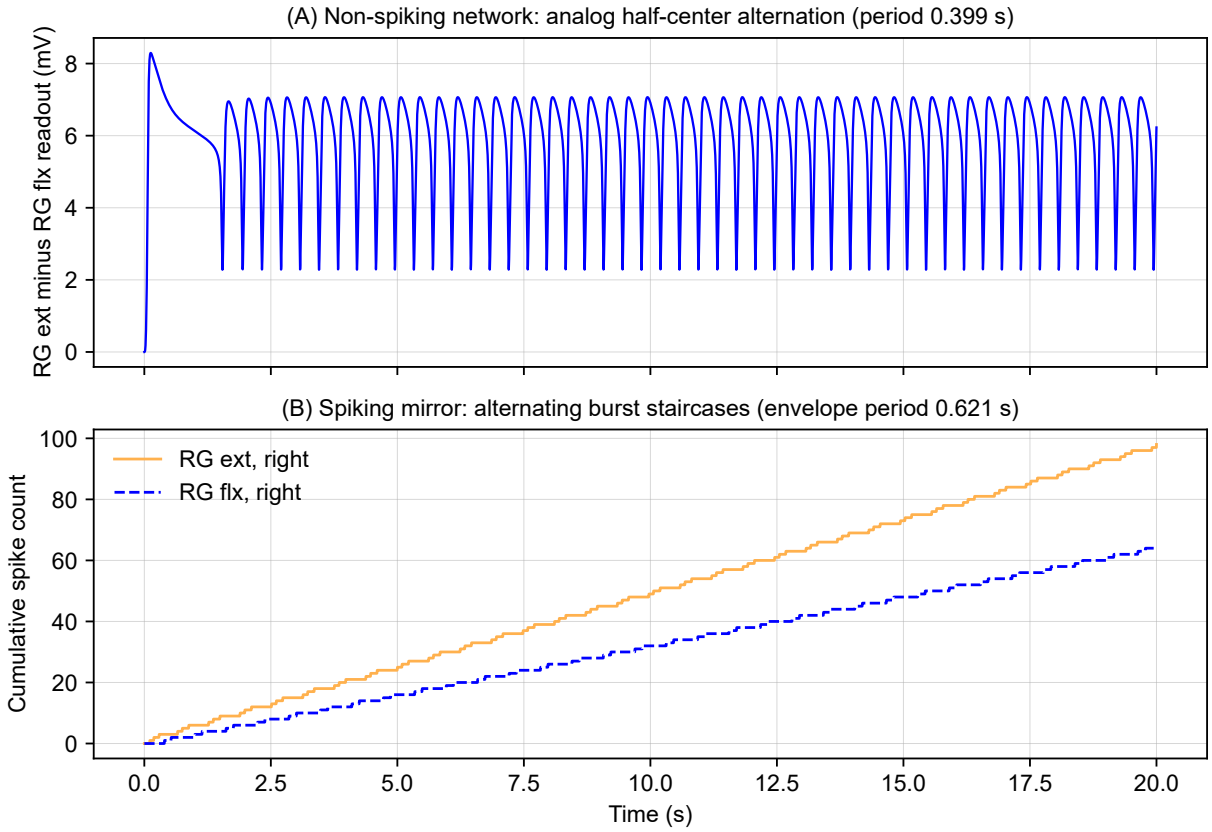


Figure D5: Constant-drive rhythm in the non-spiking network (A) and spiking variant (B). The observed periods are 0.399 and 0.621 s; the latter is 55.6 % longer. The panels use different neural readouts: graded population difference versus individual spiking cells.

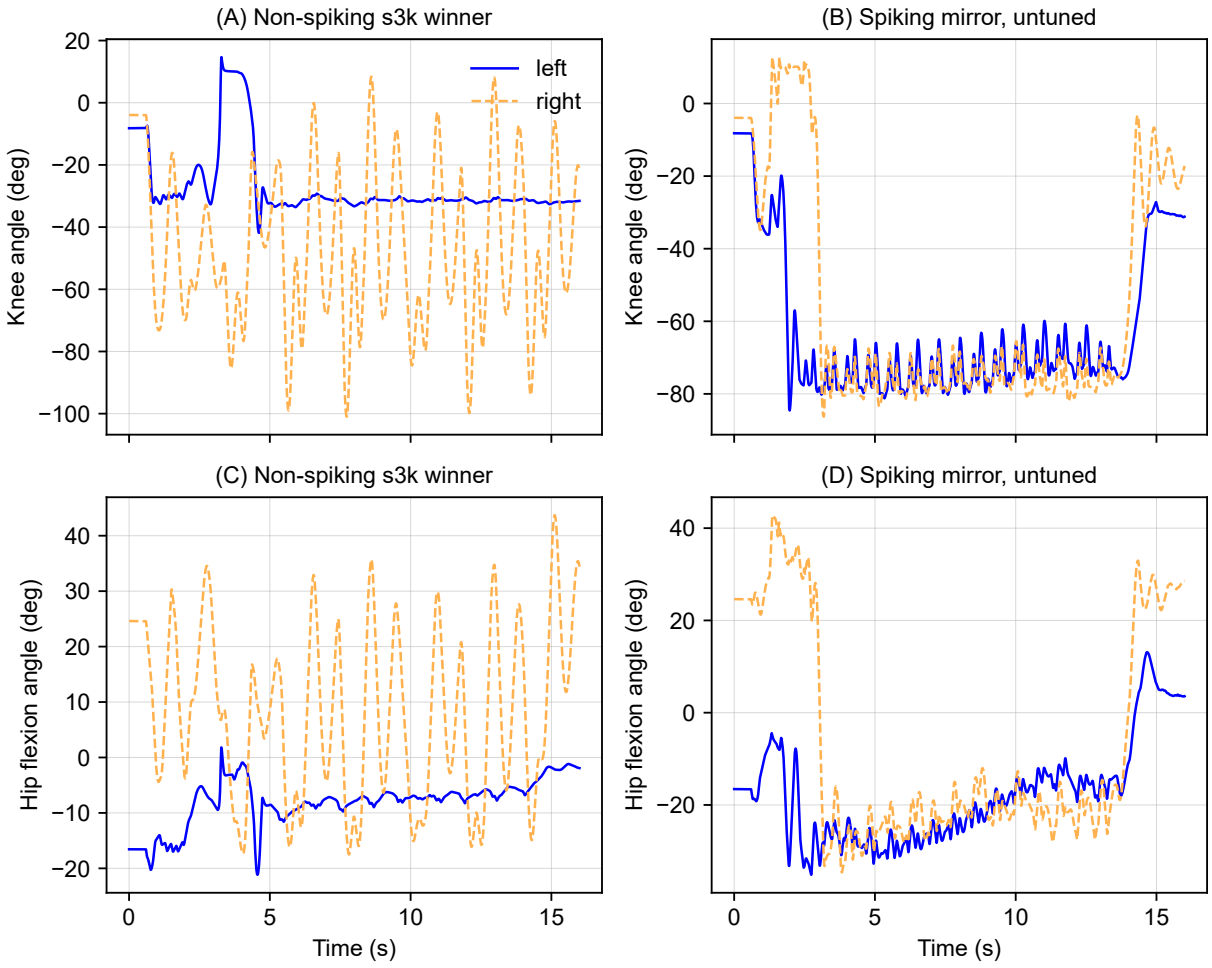


Figure D6: Ground-evaluation joint angles at the non-spiking winner parameters. The non-spiking model cycles predominantly on the right (A, C), while the untuned spiking variant cycles both legs (B, D). Solid indigo denotes left, and dashed orange right. Both use the supported evaluation protocol.

D.4.2 Simulink Demonstrations

The graded and spiking knee-reflex demonstrations were compared over matched 5 s runs (Table D4; Figure D7). The beverage example retained approximately the same sag, 2.02° versus 2.00°, compared with 7.85° without the reflex. These examples demonstrate transfer of particular reflex responses after calibration, not invariance of all controlled behavior to neural coding.

Table D4: Knee-reflex demonstration, graded baseline versus spiking twin at the matched operating point (5 s runs; settle statistics over the last 2 s).

Metric	**Baseline**	**Spiking twin**
Knee angle at 0.05 s	15.79°	15.79°
Settle mean (last 2 s)	43.5°	44.4°
Settle range	41.6°–44.9°	41.9°–46.7°
Extensor / flexor activation (window means)	0.53 / 0.53	0.58 / 0.56
Antagonist alternation	0.5 Hz–2 Hz	7.4 Hz (dominant peak)
Ia interneuron firing at settle	graded, subthreshold	95 Hz; Ib 0 Hz

D.4.3 AnimatLab Conversions

The tested spiking conversions did not preserve the graded models' rhythm (Table D5). Converting thresholds while retaining graded synapses preserved near-baseline oscillations, whereas converting synaptic amplitudes and decay changed or eliminated alternation. These results identify the synapse conversion as a limitation of this mapping. They do not demonstrate that a suitably calibrated spiking implementation cannot reproduce the reference behavior.

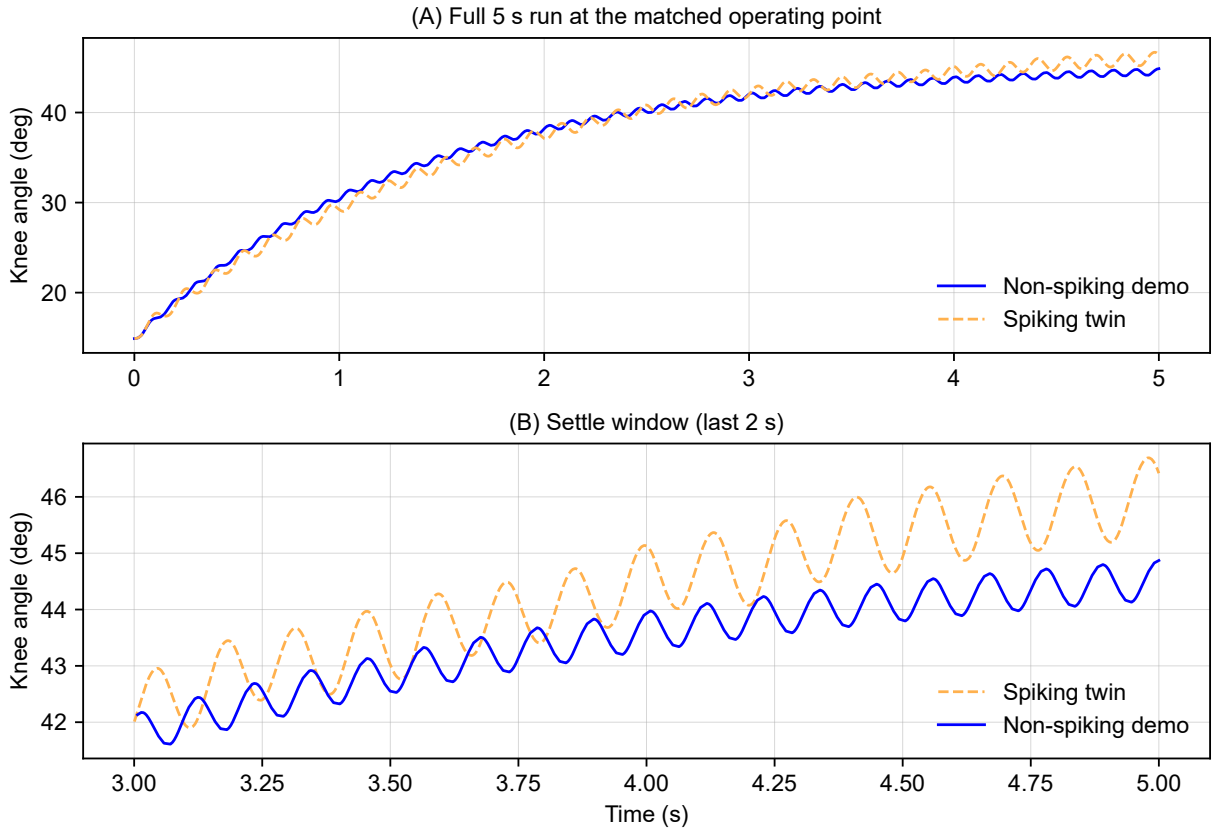


Figure D7: Knee-reflex demonstration at the matched operating point: graded baseline (solid indigo) and spiking variant (dashed orange). Full 5 s run (A) and final 2 s window (B). The spiking response has a slightly wider oscillation band.

Table D5: AnimatLab spiking conversions versus their graded baselines (5.1 s headless runs; joint ranges in degrees where the rhythm survives). The thresholds-only ablation converts the neurons but keeps the graded synapses.

Variant	Baseline (graded)	Spiking copy
Two-layer walker	0.443 s rhythm; hip -15 to $+23$, knee -1 to $+60$	dead (2 spikes; hip pinned near $+6$)
Bilateral variant	0.462 s; antiphase -0.42 cycle	latched; one side at 442 Hz–450 Hz
Reorganized project (standalone export)	latched (tonic drives)	synchronous 11.8 Hz co-bursting; same-leg envelopes $r = 1.000$ at lag 0
Thresholds only (graded synapses)	hip -15 to $+23$, knee -1 to $+60$	oscillates: hip -15.1 to $+23.2$, knee -0.8 to $+47.9$

D.5 Supported BPA-Robot Evaluations

The BPA-actuated walker exercised the chain from neural activation through the actuator force model to multi-joint motion. In the supported air-stepping run, the virtual harness carried 70 % of the 411 N body weight. The period was 1.542 s, knee RoM was 63.5°, and peak forces were 880 and 1372 N, at a peak pressure of 372 kPa.

With split left/right drives and retuned feedback, the W2L robot progressed 1.72 m over 30 s under the support protocol. It retained progression at 60 % drive and passed 12 of 12 perturbed reruns, but did not satisfy the strict contact-duty walking criterion; below that drive the gait ceased. The mechanical support was not weaned.

In the separate human-model drive-weaning study, the selected score improved from -161.57 to -145.07, with ten contact cycles per leg and contact duties of 0.71 and 0.65. It passed eight of eight perturbed reruns and retained bilateral stepping down to zero hand drive. The motion remained in place under support. These two studies distinguish reduced neural assistance from reduced mechanical assistance.

D.5.1 Independent Replays and Remaining Mechanical Assistance

A separate W2L configuration was evaluated after correcting the delivery of saved per-target gains to the evaluator. Four independent 30 s replays produced identical trajectories. Over the 4.5 s to 30 s analysis window, the run had 28 left and 21 right contact events, a median bilateral phase of 0.496 cycles, forward displacement of 2.246 m, and fitted speed of $0.0918\,\mathrm{m\,s^{-1}}$. All twelve independent reruns with 1 % parameter perturbations passed the supported-stepping criterion. These are results for this configuration, distinct from the drive-weaning study above.

Removing nominal constant weight support did not remove the position-control harness: its mean vertical force still supplied approximately 16.76 % of body weight. Contact duty remained below the strict walking criterion. Thus, repeatable supported stepping was demonstrated, while harness-free walking was not. An accelerated neural backend preserved the complete reference trajectories and reduced the isolated neural-step cost by a factor of 1.425; this does not establish the same speed improvement for the complete neuromechanical simulation.

D.5.2 CAD-Derived Robot Integration

A separate CAD-derived robot model retained the four-bar knees and spherical hips, with 27 selected actuator routes per side. Its anatomical frame is x anterior, y superior, and z right; explicit frame transformations distinguish it from older converted models. The controller drives simulated reference tensions rather than calibrated BPA forces, so this model does not yet validate the complete actuator-to-controller integration proposed in

Chapter 7.

The model completed a 20 s supported crouched-stance test. During the final five seconds, knee excursions were 1.35° right and 0.95° left, while the vertical rig carried 94.68 % of body weight and angular stabilization remained active. Disabling muscle forces under the same support settings caused lumbar failure after 0.202 s. This comparison establishes a contribution of the muscle controller under strong assistance, not independent balance. A separate run with rhythmic drive completed its horizon, but alternating gait and progression after settling have not been verified.

The W2L replay metrics and support accounting are archived in `Code/MuJoCo_SNS/spinal/reports_20261027_codex/verification.json` and the accompanying support-ladder records. CAD-model stance metrics are in `Code/MuJoCo_SNS/robot_torso/controller_assessment.json`. These records retain the configurations and evaluation conditions needed to distinguish the two models.

D.6 OpenSim-to-MuJoCo Conversion Repairs

The conversion work in Section 6.3.1 involved two distinct problems: importing the robot's artificial-muscle routing with human muscle parameters, and repairing the converted human reference used for controller development. The first combination failed muscle-equilibrium checks because the routed lengths were incompatible with the retained human parameters. The subsequent human-model repairs described below belong to that reference-model loading path; they are not evidence that the BPA-routed model reproduces every human muscle force.

Joint directions and coordinate conventions. Single-actuator tests exposed reversed hip-flexion and hip-adduction actions. The corresponding hinge-axis directions were negated so that the muscle actions agreed with the adopted OpenSim conventions. The knee required a different treatment: its converted flexion-negative convention was retained. Rotating the knee without updating its coupled translational coordinates gives a misleading geometry comparison, so those coordinates must follow the prescribed coupling during a pose or actuator test. These are specific joint and path checks, rather than a change from a left-handed to a right-handed model.

Small muscles and force scales. The quadratus femoris, gemelli, and piriformis actuators were disabled on both sides. Their six entries remain in the 92-actuator model, with no force contribution. A separate check found eight muscle-actuator entries with an active maximum-force scale of only 1 N. Their active-force scales were restored from the source-model values. Table D6 lists the corrected values; each applies separately to the left and right actuator.

Table D6: Active maximum-force scales restored in the converted human model.

Muscle	Converted scale (N)	Restored scale (N)
Erector spinae	1	2500
Internal oblique	1	900
External oblique	1	900
Extensor hallucis longus	1	162

These are source-model maximum-force parameters, not measured BPA capacities or the realized force at every pose. The passive-force scales already contained the source values. The converted length ranges for these entries remain a limitation, so restoring the

active scale alone does not establish full force–length agreement.

Patella-related paths and coupled joints. The converted moving muscle-path points use joints coupled to knee angle. Their sampled limits conflicted with the coupling constraints away from the sampled pose, producing large constraint forces and restricted motion. The coupled path-point joints were therefore left without those conflicting limits while retaining their prescribed trajectories. The anatomical driver joints were treated separately: limits were re-enabled on 14 hinges using the converted ranges. The rectus femoris route also lacked the patella tracking needed for knee-extension action; its intermediate path point was replaced by the vastus medialis patella-tracking point. The force comparison below quantifies the remaining differences at matched poses and activations.

Contact and virtual geometry. The converted contact section contained 19 explicit ground–body pairs. These predefined pairs are independent of the individual geometry contact flags, so changing those flags alone did not remove ground contact in a suspended test. Suspended tests remove the explicit contact pairs; ground-contact tests retain only the six foot–ground pairs. Virtual muscle-path marker geometry was also prevented from colliding. This separates intended foot loading from unintended contact involving path markers or other body segments.

The repairs establish specific geometry, sign, constraint, and contact corrections. They do not remove the rigid-tendon approximation or establish complete equivalence with the OpenSim model. The appropriate result remains the corresponding test: correct muscle action, a valid coupled trajectory, intended ground contact, or agreement in the measured force comparison. None alone establishes walking performance.

D.6.1 Cross-Platform Force Comparison

The comparison script reads OpenSim static-optimization forces and activations and assigns the corresponding poses and activations to the converted MuJoCo model. It calls the forward calculation at each pose; it does not assign the trajectory velocities or integrate muscle activation and fiber dynamics through time. OpenSim static optimization likewise solves frame-by-frame force allocation rather than a compliant-tendon forward simulation [OpenSim Project, n.d.]. The comparison therefore tests force predictions along sampled poses at matched activation, not dynamic equivalence or tendon-energy storage.

Table D7 retains the saved per-muscle statistics: $R^2 = 1 - \mathrm{SSE}/\mathrm{SST}$, squared Pearson correlation, and the best-aligning lag in analysis frames. The calculation and saved values are in `Code/MuJoCo_SNS/spinal/_muscle_force_compare.py` and `muscle_force_compare.csv`.

Table D7: MuJoCo vs. OpenSim muscle force along the subject01 walk at matched activations. Lag in analysis frames (≈ 17 ms each).

Muscle	R^2	shape corr^2	lag [frames]
soleus	0.742	0.800	1
med. gastrocnemius	0.888	0.894	2
lat. gastrocnemius	0.884	0.887	2
tibialis anterior	0.945	0.945	0
vastus lateralis	0.941	0.945	1
rectus femoris	−1.15	0.384	0
semimembranosus	0.565	0.621	0
biceps femoris short	−0.26	0.868	0
psoas	0.927	0.934	1
gluteus maximus (2)	0.926	0.943	1
tensor fasciae latae	0.933	0.935	0
peroneus brevis	0.977	0.989	0

Eight of the twelve representative muscles have R^2 between 0.87 and 0.98 and mean-force ratios within 10 %. Soleus and semimembranosus have lower agreement, and the negative R^2 values for rectus femoris and biceps femoris short head indicate errors larger than predicting the corresponding mean reference force. Shape correlation alone does not remove those errors. The rectus femoris path was altered during conversion, while the biceps femoris short-head result has a substantial force-scale difference. These observations identify muscles requiring closer inspection; they do not isolate routing, force-law, and velocity effects from one another.

The frame-lag calculation describes alignment of these saved sequences. It cannot establish a physical delay caused by tendon compliance because the comparison does not simulate that mechanism. The results support selected pose-wise force checks, with the reported exceptions, rather than a general claim of equivalence for neural-control studies. A dynamic comparison would require matched velocities, activation histories, and explicitly specified tendon dynamics.

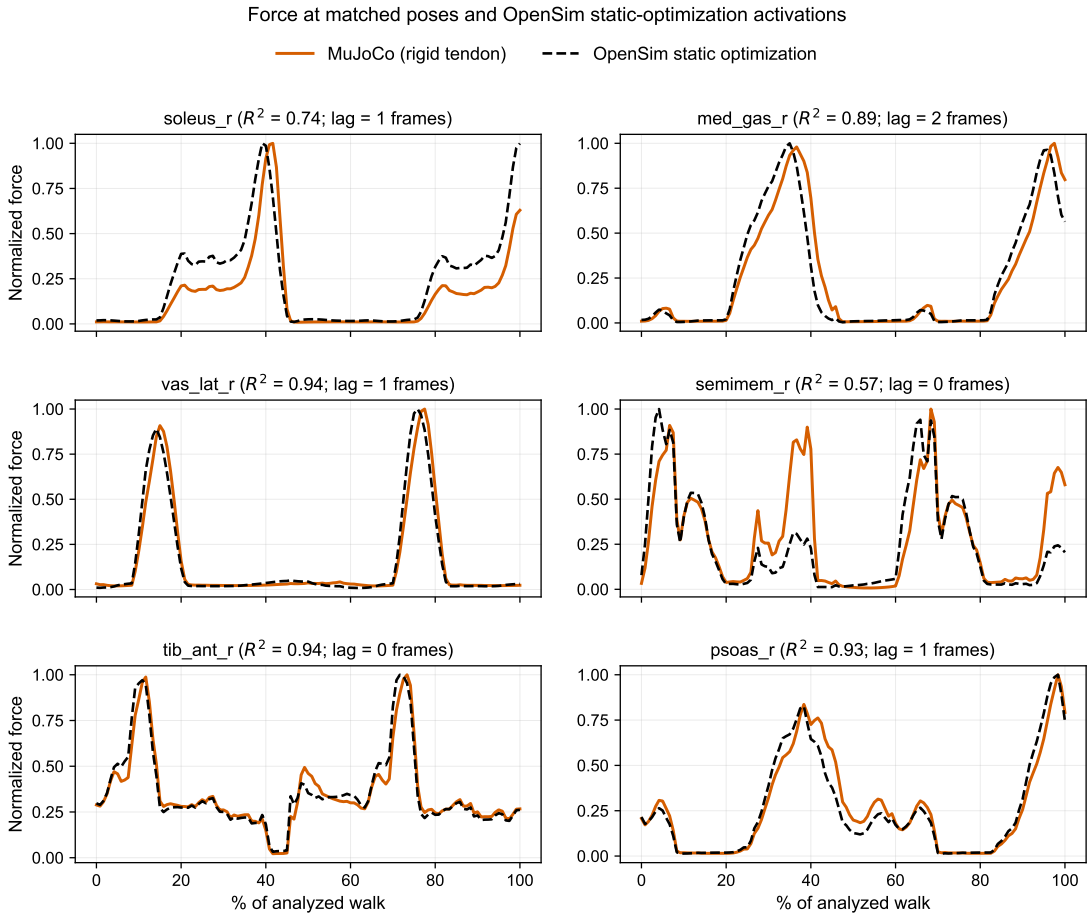


Figure D8: Force sequences at matched poses and activations, normalized to each curve's peak. Absolute-force agreement is reported in Table D7.

Appendix E: Supporting SADb Records

This appendix records the curation snapshot and pathway examples supporting Chapter 5. Counts refer to the stated audited or retained set; they are not estimates of the size of the neuroscience literature.

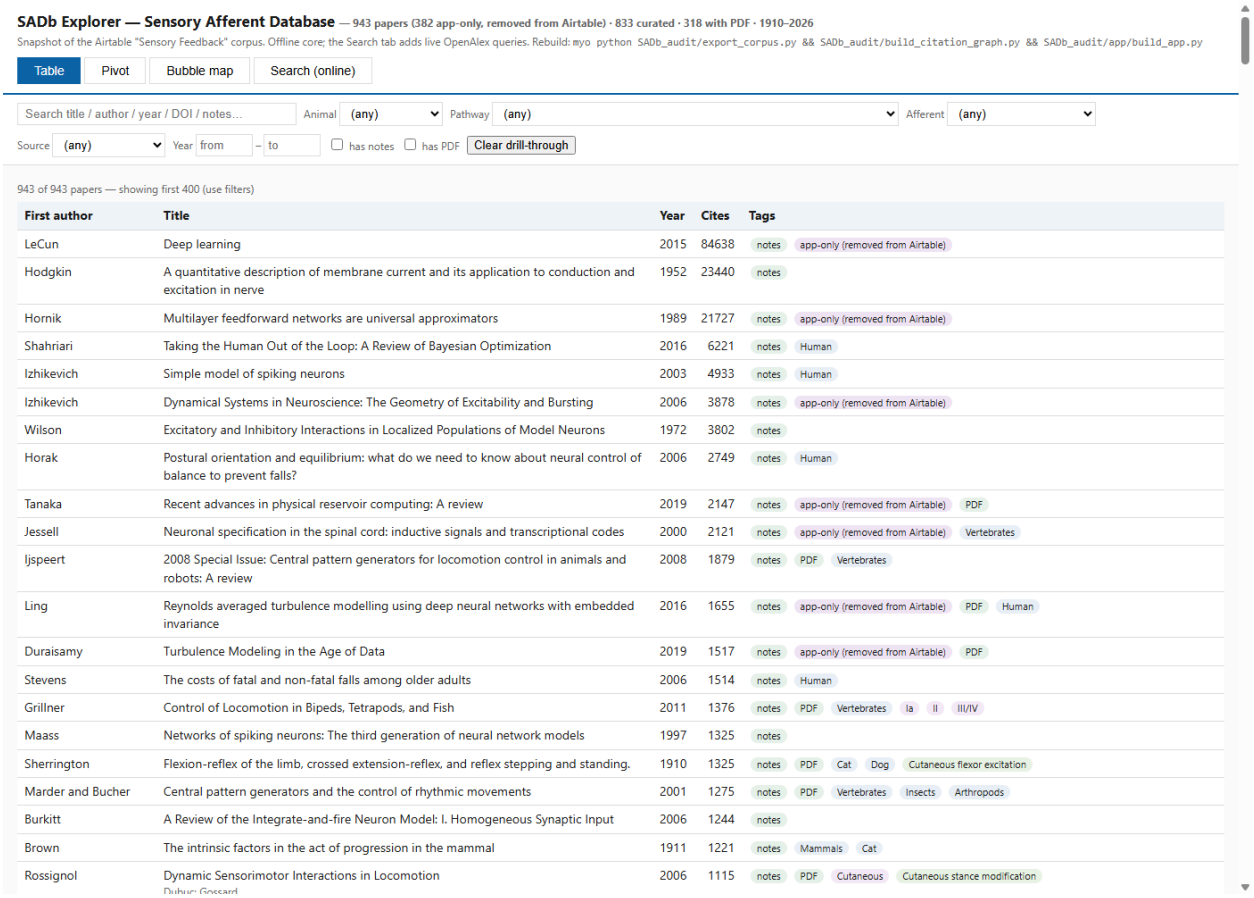
E.1 Record Sets and Coverage

The application contains 1,393 records from the five-table base: 943 Papers, 247 Models, 157 Review Papers, 42 Feedback pathways, and 4 Recorders. Of these, 436 are in the app but not in Airtable due to their 1000 entry limitation. The retained working base in Airtable contains 957 records: 561 Papers, 244 Models, 109 Review Papers, 39 Feedback pathways, and 4 Recorders. The application table includes all record types, while its pivot and citation views use the paper set.

Among the 561 retained papers, 523 carry curation notes, 426 preparation tags, 240 notes assessing animal-study potential, 118 links to named feedback pathways, 86 an afferent classification, and 258 robot/simulation translation notes. There are 550 DOIs; 38 records remain without grounding text. These fields overlap and should not be summed as disjoint categories.

The citation graph was constructed over the 943-paper audit: 730 papers were connected by 11,130 directed in-corpus citation pairs, and the Louvain partition produced 18 named groups. Citation proximity supplies a navigation aid, not evidence that two papers establish the same mechanism. The citation-neighborhood view is shown in Chapter 5

(Figure 5.1); Figure E1 provides the complementary paper-table view.



SADb Explorer — Sensory Afferent Database — 943 papers (382 app-only, removed from Airtable) · 833 curated · 318 with PDF · 1910–2026

Snapshot of the Airtable "Sensory Feedback" corpus. Offline core; the Search tab adds live OpenAlex queries. Rebuild: `myo python SADb_audit/export_corpus.py && SADb_audit/build_citation_graph.py && SADb_audit/app/build_app.py`

Table | Pivot | Bubble map | Search (online)

Search title / author / year / DOI / notes… | Animal (any) | Pathway (any) | Afferent (any)

Source (any) | Year from – to | has notes | has PDF | Clear drill-through

943 of 943 papers — showing first 400 (use filters)

First author	Title	Year	Cites	Tags
LeCun	Deep learning	2015	84638	notes, app-only (removed from Airtable)
Hodgkin	A quantitative description of membrane current and its application to conduction and excitation in nerve	1952	23440	notes
Hornik	Multilayer feedforward networks are universal approximators	1989	21727	notes, app-only (removed from Airtable)
Shahriari	Taking the Human Out of the Loop: A Review of Bayesian Optimization	2016	6221	notes, Human
Izhikevich	Simple model of spiking neurons	2003	4933	notes, Human
Izhikevich	Dynamical Systems in Neuroscience: The Geometry of Excitability and Bursting	2006	3878	notes, app-only (removed from Airtable)
Wilson	Excitatory and Inhibitory Interactions in Localized Populations of Model Neurons	1972	3802	notes
Horak	Postural orientation and equilibrium: what do we need to know about neural control of balance to prevent falls?	2006	2749	notes, Human
Tanaka	Recent advances in physical reservoir computing: A review	2019	2147	notes, app-only (removed from Airtable), PDF
Jessell	Neuronal specification in the spinal cord: inductive signals and transcriptional codes	2000	2121	notes, app-only (removed from Airtable), Vertebrates
Ijspeert	2008 Special Issue: Central pattern generators for locomotion control in animals and robots: A review	2008	1879	notes, PDF, Vertebrates
Ling	Reynolds averaged turbulence modelling using deep neural networks with embedded invariance	2016	1655	notes, app-only (removed from Airtable), PDF, Human
Duraisamy	Turbulence Modeling in the Age of Data	2019	1517	notes, app-only (removed from Airtable), PDF
Stevens	The costs of fatal and non-fatal falls among older adults	2006	1514	notes, Human
Grillner	Control of Locomotion in Bipeds, Tetrapods, and Fish	2011	1376	notes, PDF, Vertebrates, Ia, II, III/IV
Maass	Networks of spiking neurons: The third generation of neural network models	1997	1325	notes
Sherrington	Flexion-reflex of the limb, crossed extension-reflex, and reflex stepping and standing.	1910	1325	notes, PDF, Cat, Dog, Cutaneous flexor excitation
Marder and Bucher	Central pattern generators and the control of rhythmic movements	2001	1275	notes, PDF, Vertebrates, Insects, Arthropods
Burkitt	A Review of the Integrate-and-fire Neuron Model: I. Homogeneous Synaptic Input	2006	1244	notes
Brown	The intrinsic factors in the act of progression in the mammal	1911	1221	notes, Mammals, Cat
Rossignol	Dynamic Sensorimotor Interactions in Locomotion	2006	1115	notes, PDF, Cutaneous, Cutaneous stance modification

Figure E1: SADb Explorer paper-table view. Searches and filters expose bibliographic information, curation notes, preparation tags, and afferent or pathway links. The embedded corpus is available offline.

E.2 Afferent Classes and Linked Pathways

The tables reproduce the curated signal summaries and representative linked pathways.

Their record names are database labels; pathway action depends on the experimental

preparation and locomotor state described in the cited sources. Translating these records into a controller requires an explicit encoding and verification step.

Table E1: Afferent classes, encoded signals, and pathway roles represented in the Sensory Afferent Database. References link to corpus records; Ia, II, and Ib signal summaries follow Prochazka and Gorassini [1998].

Class	**Receptor**	**Signal carried**	**Recorded pathway roles (anchors)**
Ia	muscle spindle primary	muscle length and velocity	homonymous stretch excitation [Eccles and Lundberg, 1958]; reciprocal inhibition of the antagonist pool [Jankowska et al., 1967]
II	muscle spindle secondary	static muscle length	midlumbar disynaptic relay onto motor nuclei [Edgley and Jankowska, 1987]; stance-phase support of extensor activity [Mazzaro et al., 2006]
Ib	Golgi tendon organ	whole-muscle force	autogenic inhibition at rest, stance-phase reversal to excitation, and the load gate on swing onset [Dietz and Duysens, 2000, Domínguez-Rodríguez et al., 2020]
III/IV	free nerve endings	metabolic and nociceptive signals, converging with cutaneous and group II–III afferents as flexor-reflex afferents	phase-dependent resetting of the locomotor rhythm [Schomburg et al., 1998]
Cutaneous	skin and foot-pad receptors	contact and perturbation events	flexor excitation and phase-modulated stance correction [Sherrington, 1910, Forssberg, 1979]

Table E2: Principal feedback pathways, named by afferent, action, and phase, with linked demonstrating papers. Counts are identical in the audited 943-paper corpus and retained 561-paper corpus.

Pathway record	Class	Recorded action in the step cycle	Linked papers	Demonstrating references
Ia monosynaptic excitation	Ia	stretch excitation of the homonymous motor pool, with the pathway gain modulated over the step cycle	7	[Eccles and Lundberg, 1958, Capaday and Stein, 1986]
Ia reciprocal inhibition	Ia	inhibition of the antagonist motor pool through the Ia inhibitory interneurons, rhythmically active in fictive locomotion	6	[Jankowska et al., 1967, Pratt and Jordan, 1987]
Ib disynaptic inhibition	Ib	autogenic non-reciprocal group I inhibition of the loaded muscle's own pool, reduced during stance	5	[Jankowska et al., 1981, Stephens and Yang, 1996]
Ib disynaptic excitation	Ib	the locomotor reversal: group I input excites synergist extensor pools during the extensor phase, replacing the resting inhibition	6	[McCrea et al., 1995, Pearson et al., 1998]
Ib stance to swing	Ib	the load gate: Ib input from ankle extensors prolongs stance, so swing is not initiated while the leg is loaded	17	[Dietz and Duysens, 2000, Domínguez-Rodríguez et al., 2020]
Ia or II stance to swing	Ia, II	flexor-length gate on stance termination: lengthening of flexor muscles in late stance ends extension and resets the cycle	10	[Hiebert et al., 1996, Akay et al., 2014]
type II excitatory	II	stance-phase support of ankle-extensor activity through the midlumbar group II relay	5	[Edgley and Jankowska, 1987, Mazzaro et al., 2006]
Cutaneous flexor excitation	Cutaneous	flexion withdrawal and the stumbling corrective reaction, phase-modulated over the cycle	16	[Sherrington, 1910, Forssberg, 1979]